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|---|---|
| Company / customer | Spacemaker Inc Limited |
| Drawing number | C40572 |
| Drawing Revision | D |
| Project description | SpaceX |

# ELECTRICAL DRAWINGS FOR VTD INBOUND PANELS 1-5

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: TITLE PAGE / COVER SHEET

| REV | DESCRIPTION | DATE |
|---|---|---|
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JOB # C40572

TOTAL PAGES: 175

SHEET A OF 56

| Drawing Sheet | Page Description |
|---|---|
| A | TITLE PAGE / COVER SHEET |
| B | TABLE OF CONTENTS |
| C | TABLE OF CONTENTS |
| 1.0 | INCOMING 460V SUPPLY |
| 2.0 | MAIN EARTH BARS |
| 3.0 | 460V POWER DISTRIBUTION |
| 3.1 | 460V POWER DISTRIBUTION |
| 4.0 | 460V POWER INDICATOR |
| 5.0 | 24V DC POWER INDICATORS |
| 6.0 | SPARE |
| 7.0 | PANEL COOLING |
| 8.0 | CONTROL POWER SUPPLIES MAIN PANEL |
| 8.1 | CONTROL POWER SUPPLIES MAIN PANEL FOR REMOTE COMPONENTS |
| 8.2 | CONTROL POWER SUPPLIES MAIN PANEL UPS I/O |
| 8.3 | CONTROL POWER SUPPLIES MAIN PANEL REDUNDANT 24 VDC |
| 9.0 | 24V ELECTRONIC LOAD MONITORING MAIN PANEL |
| 9.1 | 24V DISTRIBUTION MAIN PANEL |
| 9.2 | 24V DISTRIBUTION MAIN PANEL |
| 9.3 | 24V DISTRIBUTION MAIN PANEL |
| 9.4 | 24V DISTRIBUTION MAIN PANEL |
| 10.0 | 24V ELECTRONIC LOAD MONITORING MAIN PANEL FOR REMOTE COMPONENTS |
| 10.1 | 24V DISTRIBUTION MAIN PANEL FOR REMOTE COMPONENTS |
| 10.2 | 24V DISTRIBUTION MAIN PANEL FOR REMOTE COMPONENTS |
| 11.0 | 0V DISTRIBUTION MAIN PANEL |
| 11.1 | 24V DISTRIBUTION MAIN PANEL |
| 12.0 | 0V DISTRIBUTION MAIN PANEL FOR REMOTE COMPONENTS |
| 12.1 | 24V DISTRIBUTION MAIN PANEL FOR REMOTE COMPONENTS |
| 12.2 | 24V DISTRIBUTION MAIN PANEL FOR REMOTE COMPONENTS |
| 13.0 | SPARE |
| 14.0 | SPARE |
| 15.0 | VTD STATIC AREA DRIVE 15VFD1 |
| 15.1 | VTD STATIC AREA DRIVE POWER FEED |
| 15.2 | VTD STATIC AREA DRIVE |
| 15.3 | VTD STATIC AREA DRIVE SSI CONNECTION |
| 15.4 | VTD STATIC AREA DRIVE SSI CONNECTION |
| 16.0 | VTD MOBILE EXTRACTOR CONVEYOR AREA DRIVE 16VFD1 |
| 16.1 | VTD MOBILE EXTRACTOR CONVEYOR AREA DRIVE |
| 16.2 | VTD MOBILE EXTRACTOR CONVEYOR AREA DRIVE POWER |
| 16.3 | VTD MOBILE EXTRACTOR CONVEYOR AREA MOTOR POWER |
| 16.4 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD 24V DC POWER |
| 16.5 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD STO |
| 16.6 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD EtherCAT INTERFACE |
| 16.7 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE |
| 16.8 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE |
| 17.0 | BUFFER CONVEYORS AREA VFD's SECTION 1 |

| Drawing Sheet | Page Description |
|---|---|
| 17.1 | BUFFER CONVEYORS AREA VFD's SECTION 1 POWER |
| 17.2 | BUFFER CONVEYORS AREA VFD's SECTION 1 POWER |
| 17.3 | BUFFER CONVEYORS AREA VFD's SECTION 1 MOTOR POWER |
| 17.4 | BUFFER CONVEYORS AREA VFD's SECTION 1 MOTOR POWER |
| 17.5 | BUFFER CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER |
| 17.6 | BUFFER CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER |
| 17.7 | BUFFER CONVEYORS AREA VFD's SECTION 1 VFD STO |
| 17.8 | BUFFER CONVEYORS AREA VFD's SECTION 1 VFD STO |
| 17.9 | BUFFER CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE |
| 17.10 | BUFFER CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE |
| 17.11 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE |
| 17.12 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE |
| 18.0 | BUFFER CONVEYORS AREA VFD's SECTION 2 |
| 18.1 | BUFFER CONVEYORS AREA VFD's SECTION 2 POWER |
| 18.2 | BUFFER CONVEYORS AREA VFD's SECTION 2 MOTOR POWER |
| 18.3 | BUFFER CONVEYORS AREA VFD's SECTION 2 VFD 24VDC POWER |
| 18.4 | BUFFER CONVEYORS AREA VFD's SECTION 2 VFD STO |
| 18.5 | BUFFER CONVEYORS AREA VFD's SECTION 2 VFD EtherCAT INTERFACE |
| 18.6 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE |
| 18.7 | VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE |
| 19.0 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 |
| 19.1 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 POWER |
| 19.2 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 POWER |
| 19.3 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 MOTOR POWER |
| 19.4 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 MOTOR POWER |
| 19.5 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD 24VDC POWER |
| 19.6 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD 24VDC POWER |
| 19.7 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD STO |
| 19.8 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD STO |
| 19.9 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD EtherCAT INTERFACE |
| 19.10 | CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD EtherCAT INTERFACE |
| 20.0 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 |
| 20.1 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER |
| 20.2 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER |
| 20.3 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 MOTOR POWER |
| 20.4 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 MOTOR POWER |
| 20.5 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER |
| 20.6 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER |
| 20.7 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD STO |
| 20.8 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD STO |
| 20.9 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE |
| 20.10 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE |
| 21.0 | CAROUSEL CONVEYORS AREA VFD's SECTION 2 |
| 21.1 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER |
| 21.2 | CAROUSEL CONVEYORS AREA VFD's SECTION 2 MOTOR POWER |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: TABLE OF CONTENTS

| REV | DESCRIPTION | DATE |
|---|---|---|
| | | |
| | | |

| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
|---|---|---|
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| DATE | DATE | CURRENT REVISION: |
| 9/17/26 | 9/8/26 | D |

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Saddle Brook, NJ
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JOB # C40572

TOTAL PAGES: 175

SHEET B OF 56

| Drawing Sheet | Page Description |
|---|---|
| 21.3 | CAROUSEL CONVEYORS AREA VFD's SECTION 2 VFD 24VDC POWER |
| 21.4 | CAROUSEL CONVEYORS AREA VFD's SECTION 2 VFD STO |
| 21.5 | CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE |
| 22.0 | SPARE |
| 23.0 | PLC OVERVIEW RACK |
| 23.1 | PLC 24VDC |
| 23.2 | PLC 24VDC |
| 24.0 | DC INPUTS CARD 23.0-1 |
| 24.0 | DC INPUTS CARD 23.0-1 |
| 25.0 | DC INPUTS CARD 23.0-2 |
| 26.0 | DC OUTPUTS CARD 23.0-3 |
| 26.1 | DC OUTPUTS CARD 23.0-3 |
| 27.0 | DRIVE LIFTING TRANSFER ROLLER CONVEYOR#3 |
| 28.0 | SAFETY CPU CONTROLLER 23.0-5 |
| 29.0 | SAFETY INPUTS 23.0-6 |
| 29.1 | SAFETY INPUTS 23.0-6 |
| 30.0 | SAFETY OUTPUTS 23.0-7 |
| 30.0 | SAFETY OUTPUTS 23.0-7 |
| 31.0 | DC SUPPLY FOR NETWORK INTERFACE CARDS 23.0-8 |
| 32.0 | EtherCAT P NETWORK INTERFACE CARD 23.0-9 |
| 32.1 | EtherCAT P NETWORK INTERFACE CARD 23.0-9 |
| 33.0 | EtherCAT P NETWORK INTERFACE CARD 23.0-10 |
| 34.0 | EtherCAT P NETWORK INTERFACE CARD 23.0-11 |
| 35.0 | PROFINET NETWORK INTERFACE CARD 23.0-12 |
| 35.1 | PROFINET NETWORK INTERFACE CARD 23.0-12 |
| 36.0 | EtherCAT NETWORK INTERFACE CARD 23.0-13 |
| 36.1 | EtherCAT NETWORK INTERFACE CARD 23.0-13 |
| 37.0 | SPARE |
| 38.0 | SPARE |
| 39.0 | NETWORK SWITCH 39NS1 POWER 24VDC |
| 39.1 | NETWORK SWITCH 39NS1 PROFINET NETWORK INTERFACE |
| 39.2 | NETWORK SWITCH 39NS2 POWER 24VDC |
| 39.3 | NETWORK SWITCH 39NS2 PROFINET NETWORK INTERFACE |
| 39.4 | NETWORK SWITCH 39NS2 PROFINET NETWORK INTERFACE |
| 40.0 | 40RIO1 NETWORK + I/O + POWER |
| 40.1 | 40BCS1 - 40BCC1 NETWORK + POWER |
| 41.0 | NETWORK SWITCH 41NS1 POWER 24VDC |
| 41.1 | NETWORK SWITCH 41NS1 POWER EtherCAT NETWORK INTERFACE |
| 42.0 | NETWORK SWITCH 42NS1 POWER 24VDC |
| 42.1 | RACK 01 PLC SAFETY DIGITAL IN MODULE 3 |
| 43.0 | 43SRIO1 SAFETY DIGITAL IN MODULE |
| 43.1 | 43SRIO1 SAFETY DIGITAL IN MODULE |
| 43.2 | 43SRIO2 SAFETY DIGITAL IN/OUT MODULE |
| 43.3 | 43SRIO2 SAFETY DIGITAL IN/OUT MODULE |
| 43.4 | LIGHT CURTAIN - CAROUSEL CONVEYORS |

| Drawing Sheet | Page Description |
|---|---|
| 44.0 | Guard lock with guard lock monitoring |
| 44.1 | Guard lock with guard lock monitoring |
| 44.2 | Guard lock with guard lock monitoring |
| 45.0 | EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 45.1 | EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 45.2 | 45RIO1 REMOTE I/O INPUT LAYOUT |
| 45.3 | 45RIO2 REMOTE I/O INPUT LAYOUT |
| 45.4 | 45RIO3 REMOTE I/O INPUT LAYOUT |
| 45.5 | 45RIO4 REMOTE I/O INPUT LAYOUT |
| 46.0 | EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 46.1 | 46RIO1 REMOTE I/O INPUT LAYOUT |
| 46.2 | 46RIO2 REMOTE I/O INPUT LAYOUT |
| 46.3 | 46RIO3 REMOTE I/O INPUT LAYOUT |
| 46.4 | 46RIO4 REMOTE I/O INPUT LAYOUT |
| 46.5 | 46RIO5 REMOTE I/O INPUT LAYOUT |
| 46.6 | 46RIO6 REMOTE I/O INPUT LAYOUT |
| 46.7 | 46RIO7 REMOTE I/O INPUT LAYOUT |
| 47.0 | EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 47.1 | 47RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 47.2 | 47RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 47.3 | 47RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 47.4 | 47RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 48.0 | EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 48.1 | 48RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE |
| 48.2 | 48RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 48.3 | 48RIO3 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 48.4 | 48RIO4 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 48.5 | 48RIO5 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 48.6 | 48RIO6 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT |
| 49.0 | SPARE |
| 50.0 | RELAYS |
| 50.1 | RELAYS |
| 51.0 | RELAYS |
| 51.1 | RELAYS |
| 52.0 | SAFETY RELAYS |
| 53.0 | LEVEL 1 STOPPER MOTORS |
| 53.1 | LEVEL 2 STOPPER MOTORS |
| 53.2 | LEVEL 3 STOPPER MOTORS |
| 53.3 | LEVEL 4 STOPPER MOTORS |
| 53.4 | LEVEL 5 STOPPER MOTORS |
| 53.5 | LEVEL 6 STOPPER MOTORS |
| 54.0 | SPARE |
| 55.0 | SPARE |
| 56.0 | BOM |
| 56.1 | BOM |
| 56.2 | BOM |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: TABLE OF CONTENTS

| REV | DESCRIPTION | DATE |
|---|---|---|
| | | |
| | | |

| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
|---|---|---|
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| 9/17/26 | 9/8/26 | D |

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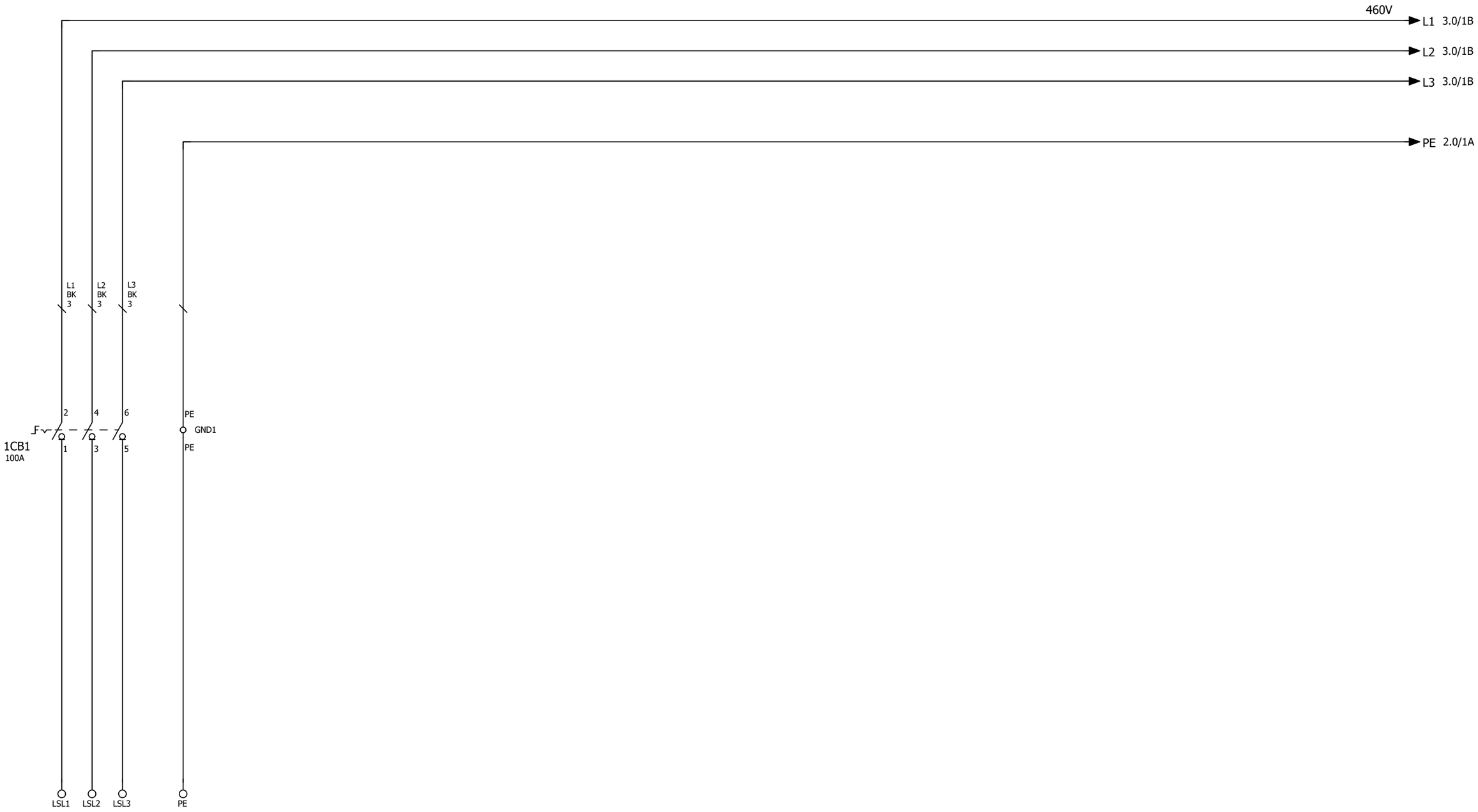
JOB # C40572

TOTAL PAGES: 175

SHEET C OF 56




LINE ENTRY
460V/3PH/60HZ
AVAILABLE FAULT LEVEL
MUST NOT EXCEED 10kA r.m.s.
SCCR 10kA r.m.s.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: INCOMING 460V SUPPLY

| REV | DESCRIPTION | DATE |
|---|---|---|
| | | |
| | | |

| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
|---|---|---|
| PAUL | Carlos A | C40572 |
| DATE | DATE | CURRENT REVISION: |
| 9/17/26 | 9/8/26 | D |

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JOB # C40572

TOTAL PAGES: 175

SHEET 1.0 OF 56

1.0/9B PE

2.1/1B PE

Mains Busbar
(MAIN Panel)

PE
GNYE
14

GND3

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 16 | PE GNYE 16 | PE GNYE 16 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 |
| | PE | PE | PE | PE | PE | PE | PE | PE | PE | PE | PE | PE |
| | 8.0/3E | 8.1/2B | 11.0/ | 12.0/ | 38.0/3E | 38.0/3E | 37/5F | ??.?/?? | ??.?/?? | ??.?/?? | ??.?/?? | ??.?/?? |
| | 8PS1 | 8PS2 | 0V | 0VDC | 38PH1 | 38PH1 | 37KF1 | - | - | - | - | - |

Mains Busbar
(MAIN Panel)

PE
GNYE
14

GND2

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PE GNYE 14 | PE GNYE 18 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 | PE GNYE 14 |
| | PE | PE | PE | PE | PE | PE | PE | PE | PE | PE | PE | PE |
| | 4.0/1F | 7.0/2D | 8.0/3E | 8.1/3E | 8.3/3E | LEFT SIDE ELECTRICAL CABINET DOOR | RIGHT SIDE ELECTRICAL CABINET DOOR | CENTER STRUT ELECTRICAL CABINET | | | | |
| | 4PI1 | 7FAN1 | 8PS1 | 8PS2 | 8PS3 | | | | | | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: MAIN EARTH BARS

| REV | DESCRIPTION | DATE |
|---|---|---|
| | | |
| | | |

| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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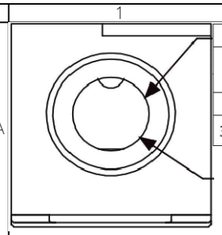
JOB # C40572

TOTAL PAGES: 175

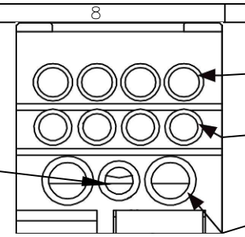
SHEET 2.0 OF 56



| DB250 | | |
|---|---|---|
| Size | Torque | Strip lenght |
| 35...120 mm² | 19.0 Nm | 27 mm |

250A
DISTRIBUTION
BLOCK

| Size | Torque | Strip lenght |
|---|---|---|
| 2,5...16 mm² | 2.7 Nm | 16 mm |



| Size (4x) | Torque | Strip lenght |
|---|---|---|
| 2,5...16 mm² | 2.7 Nm | 11 mm |
| Size (4x) | Torque | Strip lenght |
| 2,5...10 mm² | 2.7 Nm | 10 mm |
| Size (2x) | Torque | Strip lenght |
| 6...35 mm² | 4.4 Nm | 16 mm |

460V

1.0/9A L1 → L1 3.1/1A

1.0/9A L2 → L2 3.1/1A

1.0/9A L3 → L3 3.1/1A

460V

PI1
PRE MADE CABLE

L1 BK 14 | L2 BK 14 | L3 BK 14

L1 BK 10 | L2 BK 10 | L3 BK 10

4.0/1A L1 | 4.0/1A L2 | 4.0/1A L3 → 4PI1

7/1A L1 | 7/1A L2 | 7/1A L3 → 7CB1

8/1A L1 | 8/1A L2 | 8/1A L3 → 8CB1

8.1/1A L1 | 8.1/1A L2 | 8.1/1A L3 → 8CB3

8.3/1A L1 | 8.3/1A L2 | 8.3/1A L3 → 8CB5

15.1/4E L1 | 15.1/4E L2 | 15.1/5E L3 → 15CB1

16.2/2E L1 | 16.2/2E L2 | 16.2/2E L3 → 16CB1

17.1/1E L1 | 17.1/1E L2 | 17.1/1E L3 → 17CB1

18.1/1E L1 | 18.1/1E L2 | 18.1/1E L3 → 18CB1

19.1/1E L1 | 19.1/1E L2 | 19.1/1E L3 → 19CB1

20.1/1E L1 | 20.1/1E L2 | 20.1/1E L3 → 20CB1

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TOTAL PAGES: 175

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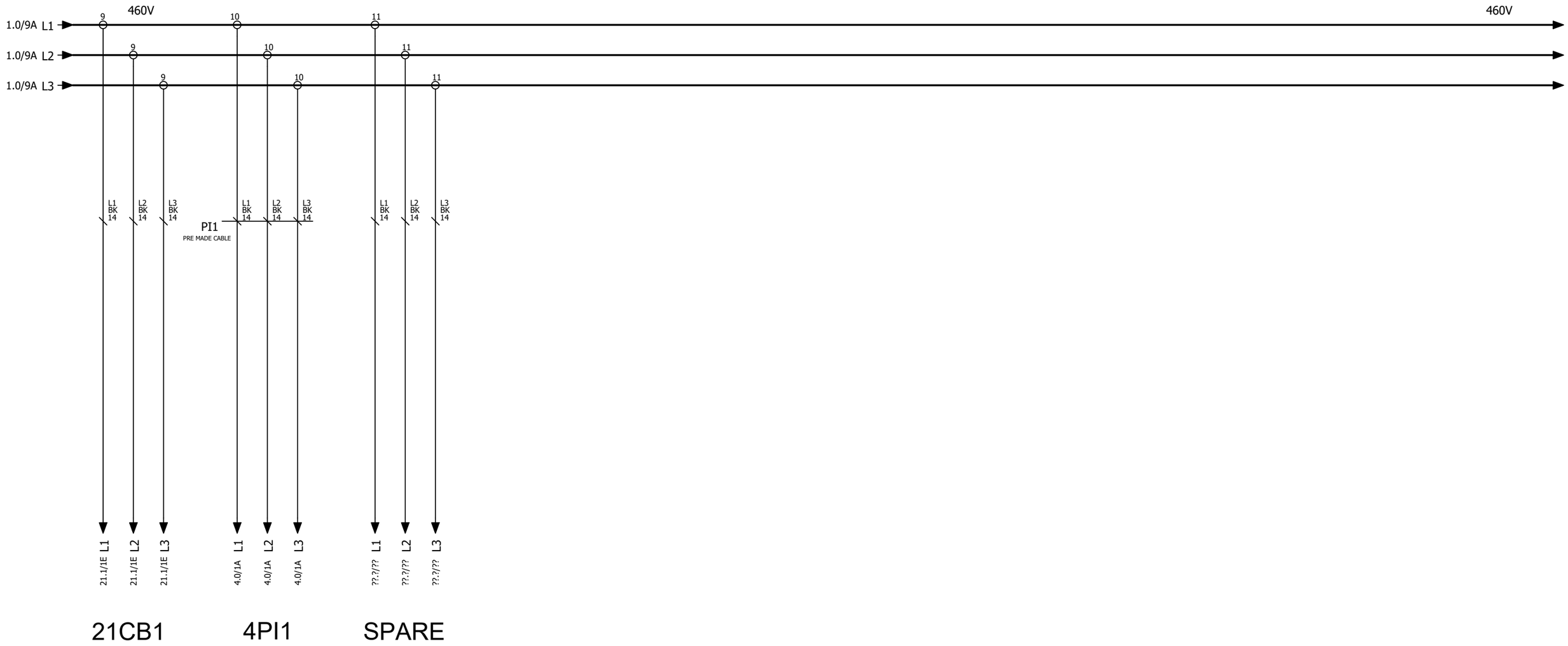


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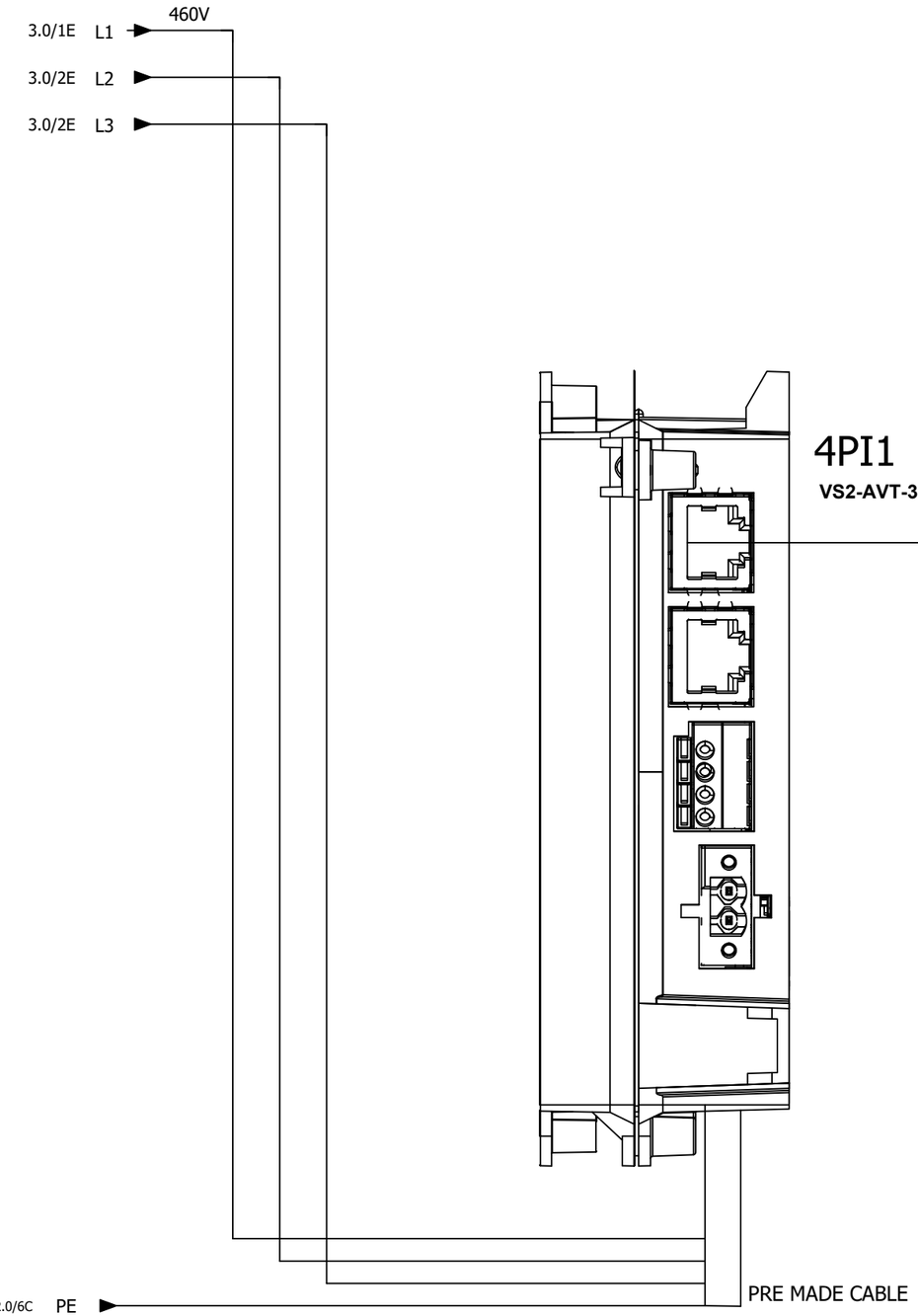





4PI1
**VS2-AVT-3P-ISO**

**PANDUIT**

VOLTAGE
INDICATOR AC

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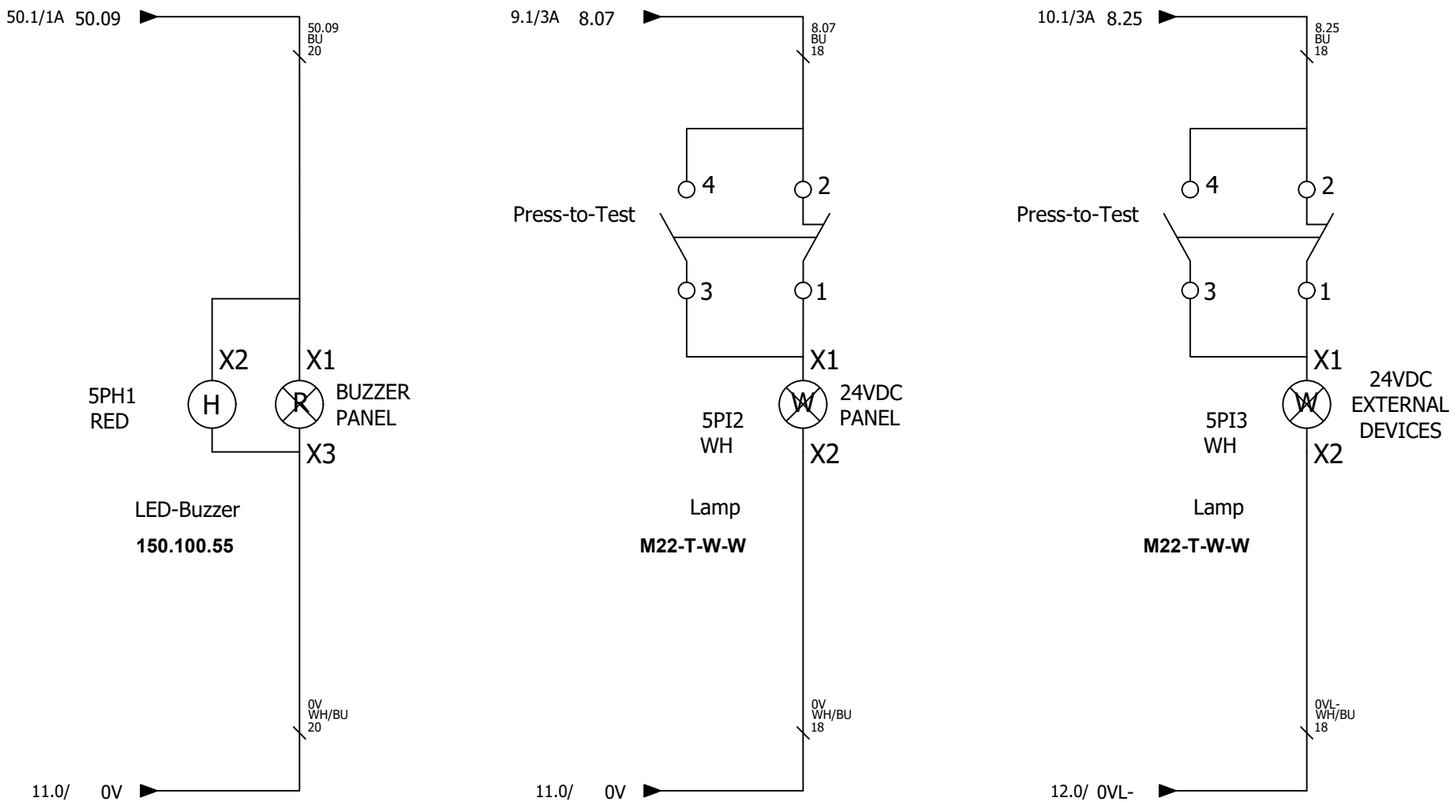


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PAGE DESCRIPTION: 24V DC POWER INDICATORS

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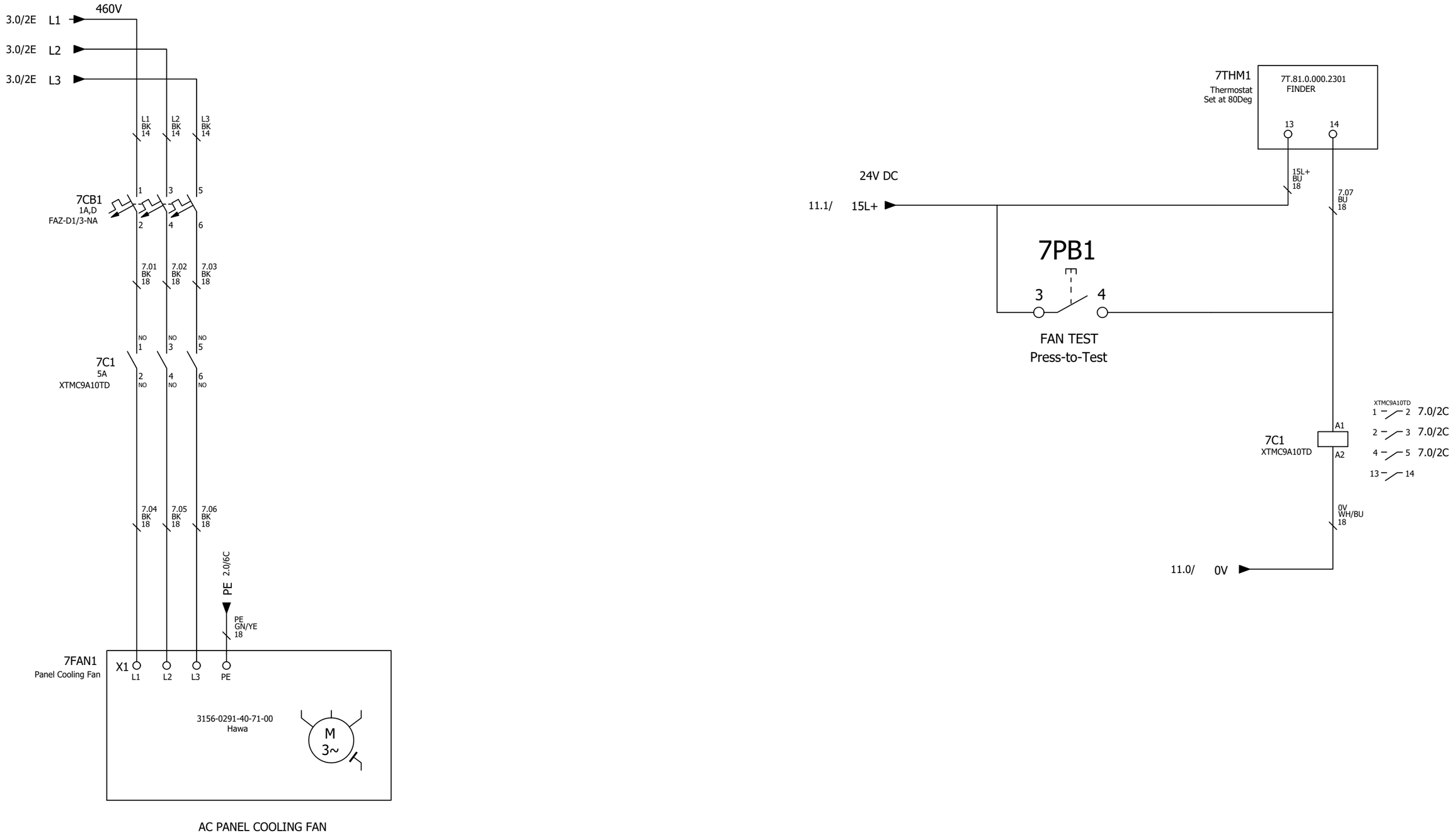


AC PANEL COOLING FAN

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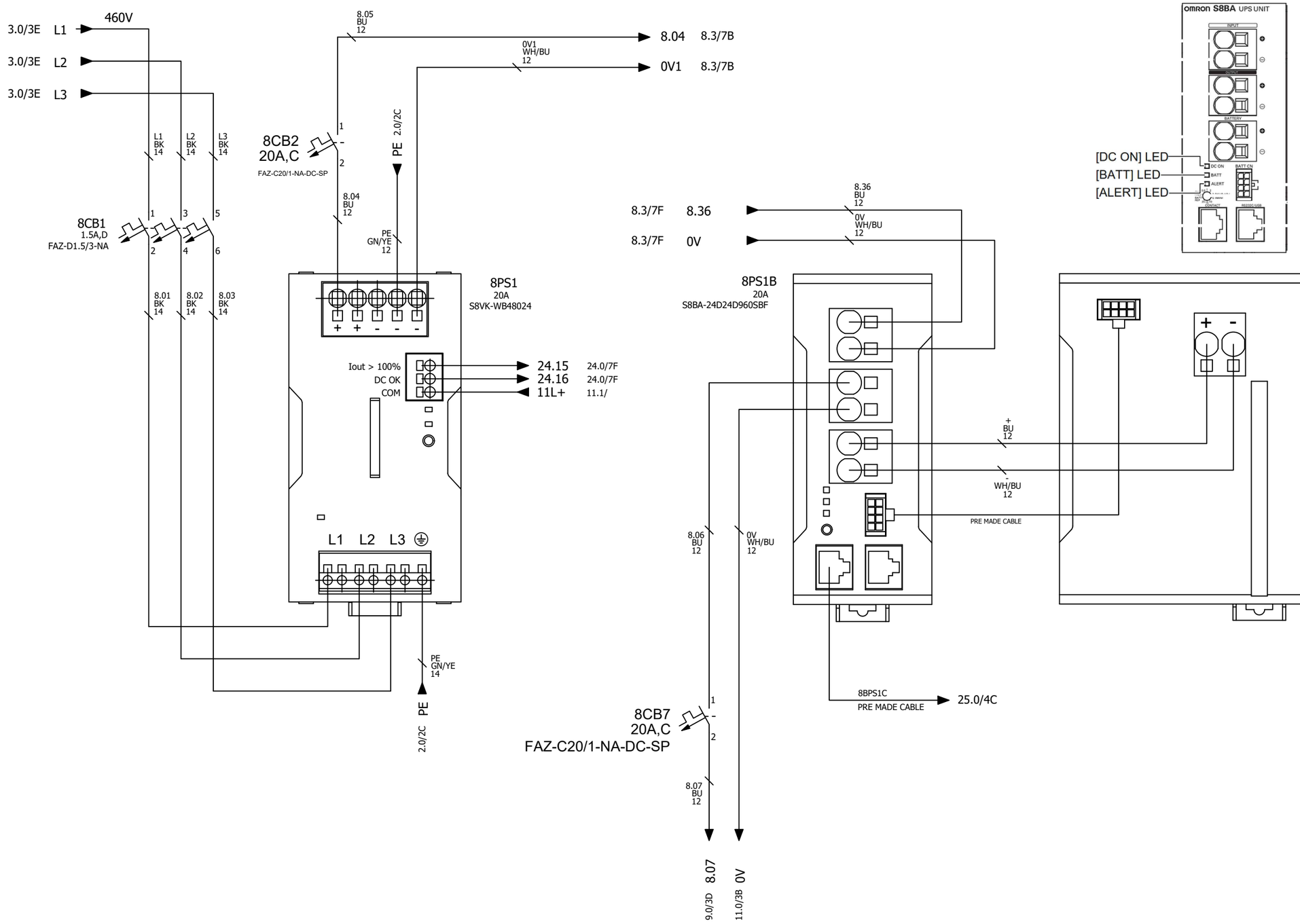


| Pin number | Cable color | Item |
|---|---|---|
| 1 | White/orange | Backup signal output (BU) |
| 2 | Orange | Remote ON/OFF input (-) |
| 3 | White/green | Trouble signal output (TR) |
| 4 | Blue | COMMON (COM) |
| 5 | White/blue | Battery LOW signal output (BL) |
| 6 | Green | Backup stop signal input (BS) |
| 7 | White/brown | Battery Replacement Signal output (WB) |
| 8 | Brown | Remote ON/OFF input (+) |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL

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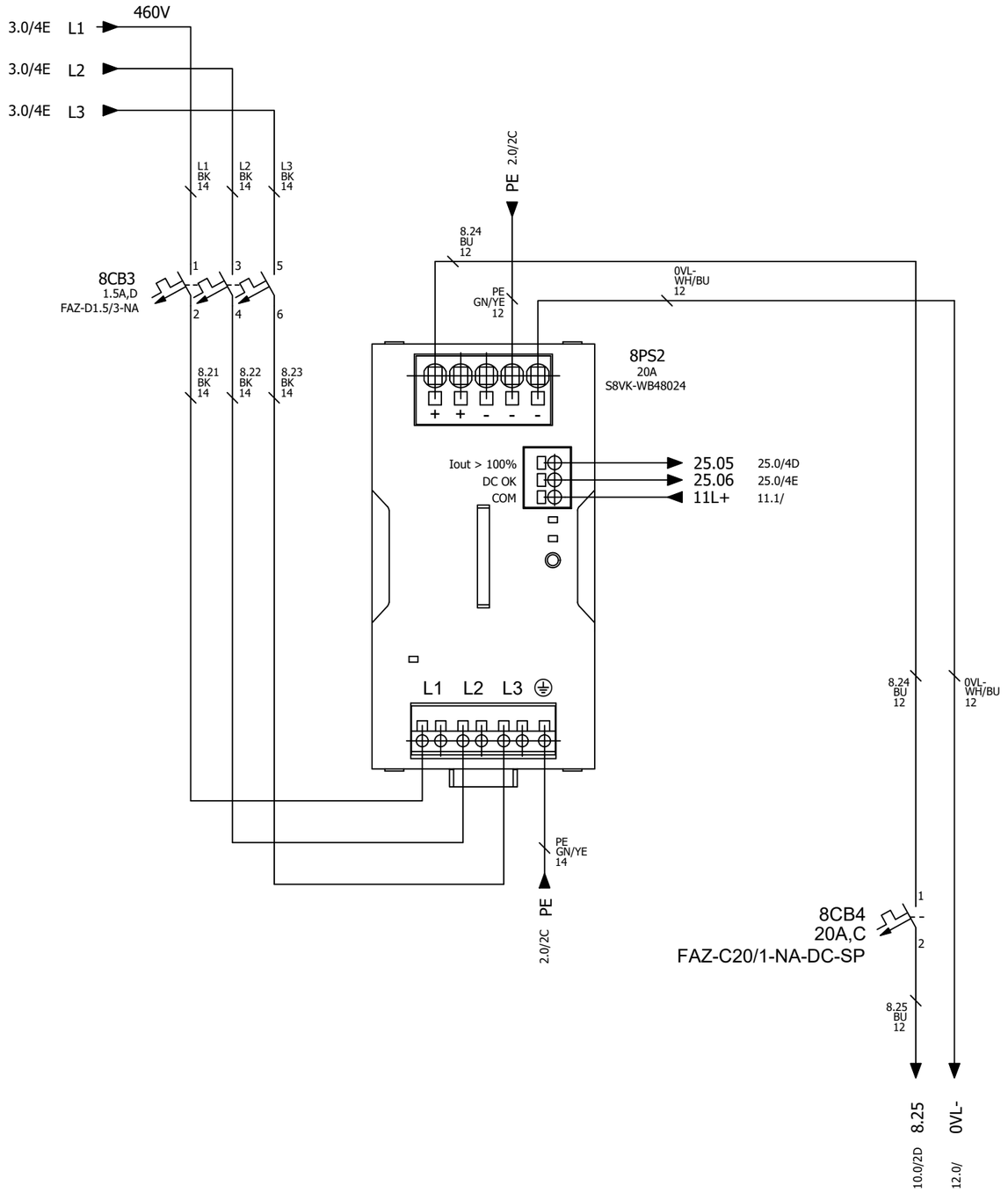


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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL FOR REMOTE COMPONENTS

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**I/O signal port (RJ45 connector)**

| Outlook of the connector | Pin number | Cable color | Item |
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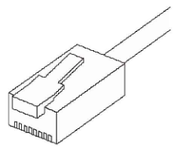
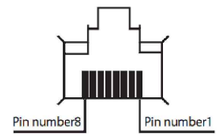
|  | 1 | White/orange | Backup signal output (BU) |
| | 2 | Orange | Remote ON/OFF input (-) |
| | 3 | White/green | Trouble signal output (TR) |
| | 4 | Blue | COMMON (COM) |
| | 5 | White/blue | Battery LOW signal output (BL) |
| | 6 | Green | Backup stop signal input (BS) |
| | 7 | White/brown | Battery Replacement Signal output (WB) |
| | 8 | Brown | Remote ON/OFF input (+) |

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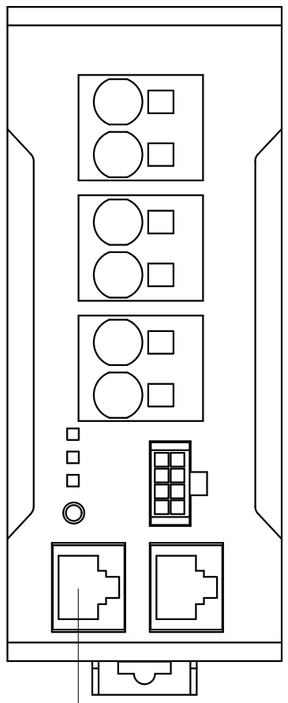

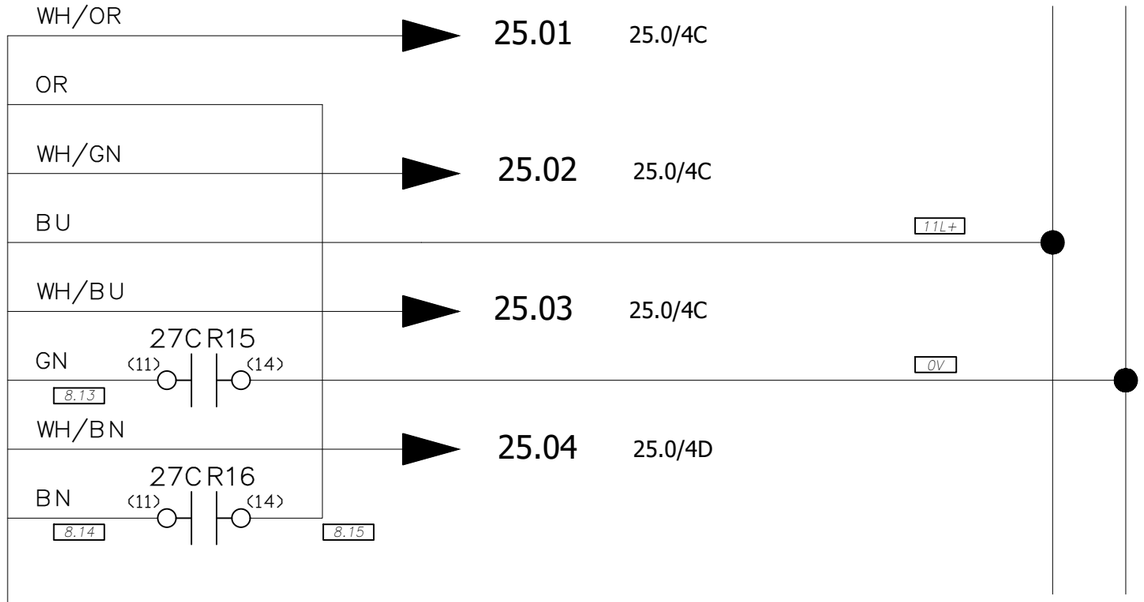


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PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL UPS I/O

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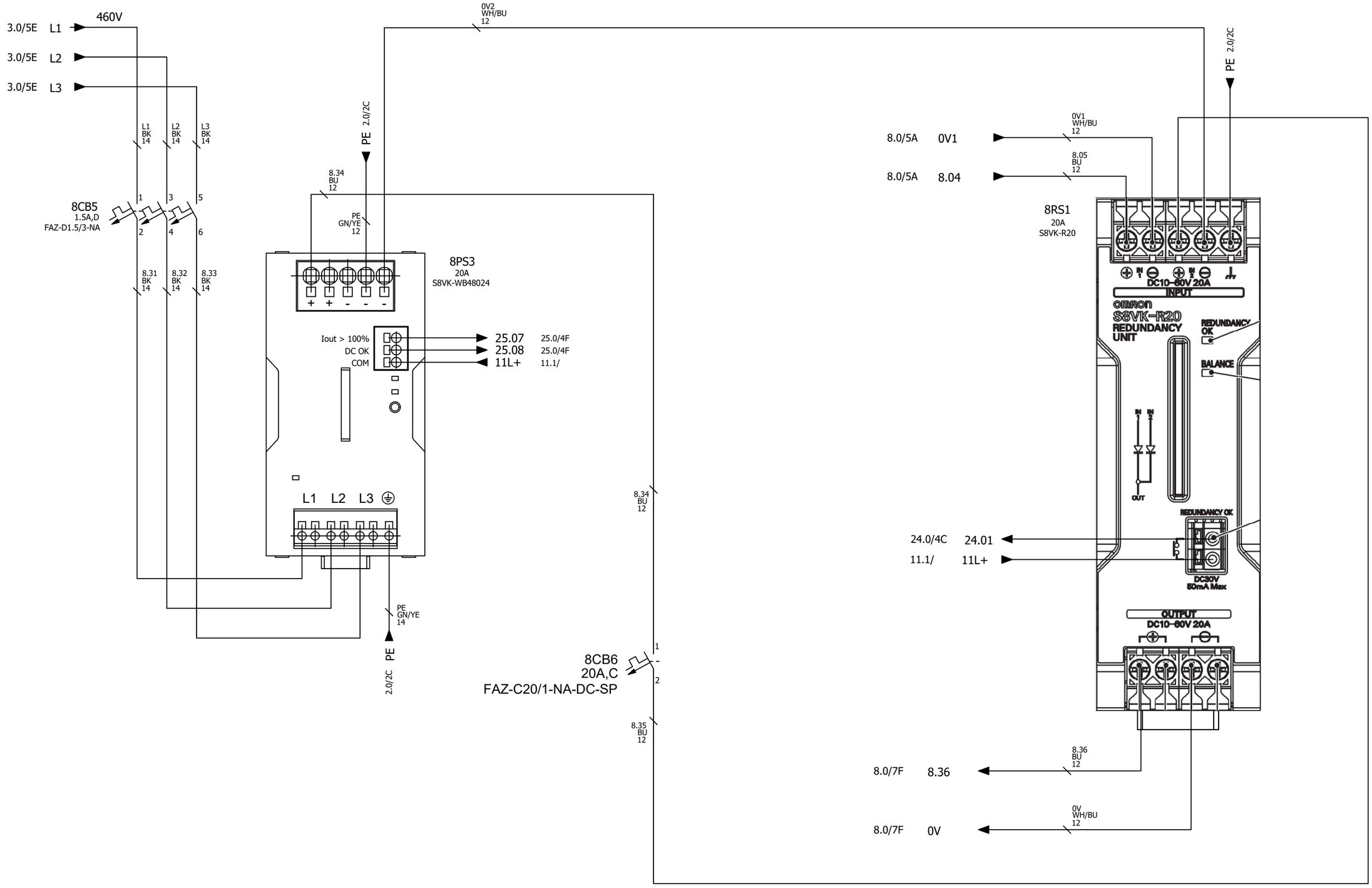


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PAGE DESCRIPTION: CONTROL POWER SUPPLIES MAIN PANEL REDUNDANT 24 VDC

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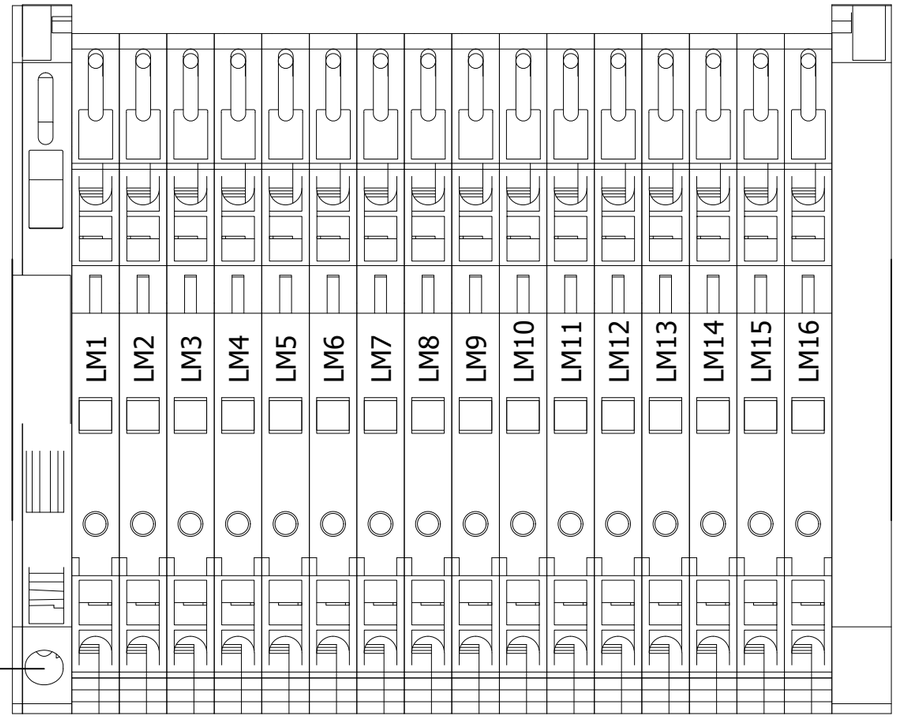


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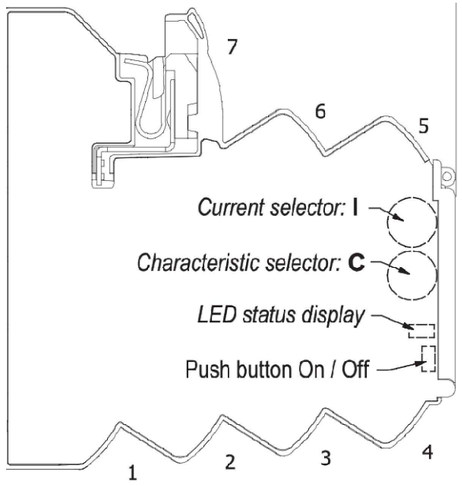
## LOAD MONITORING(LM) PIN ASSIGNMENT

| |
|---|
| PIN 1–OUT+(LOAD) |
| PIN 2–CTRL INPUT (SET/RESET) |
| PIN 3–STATUS OUTPUT |
| PIN 4–NC |
| PIN 5–0V |
| PIN 6–SUPPLY (ALTERNATIVE) |
| PIN 7–SUPPLY |

## TRIP CURVES:

1. CHARACTERISTIC FAST
2. CHARACTERISTIC MEDIUM
3. CHARACTERISTIC SLOW–1
4. CHARACTERISTIC SLOW–2
5. CHARACTERISTIC SLOW–3

## ELECTRONIC CIRCUIT BREAKER (LM)

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PAGE DESCRIPTION: 24V ELECTRONIC LOAD MONITORING MAIN PANEL

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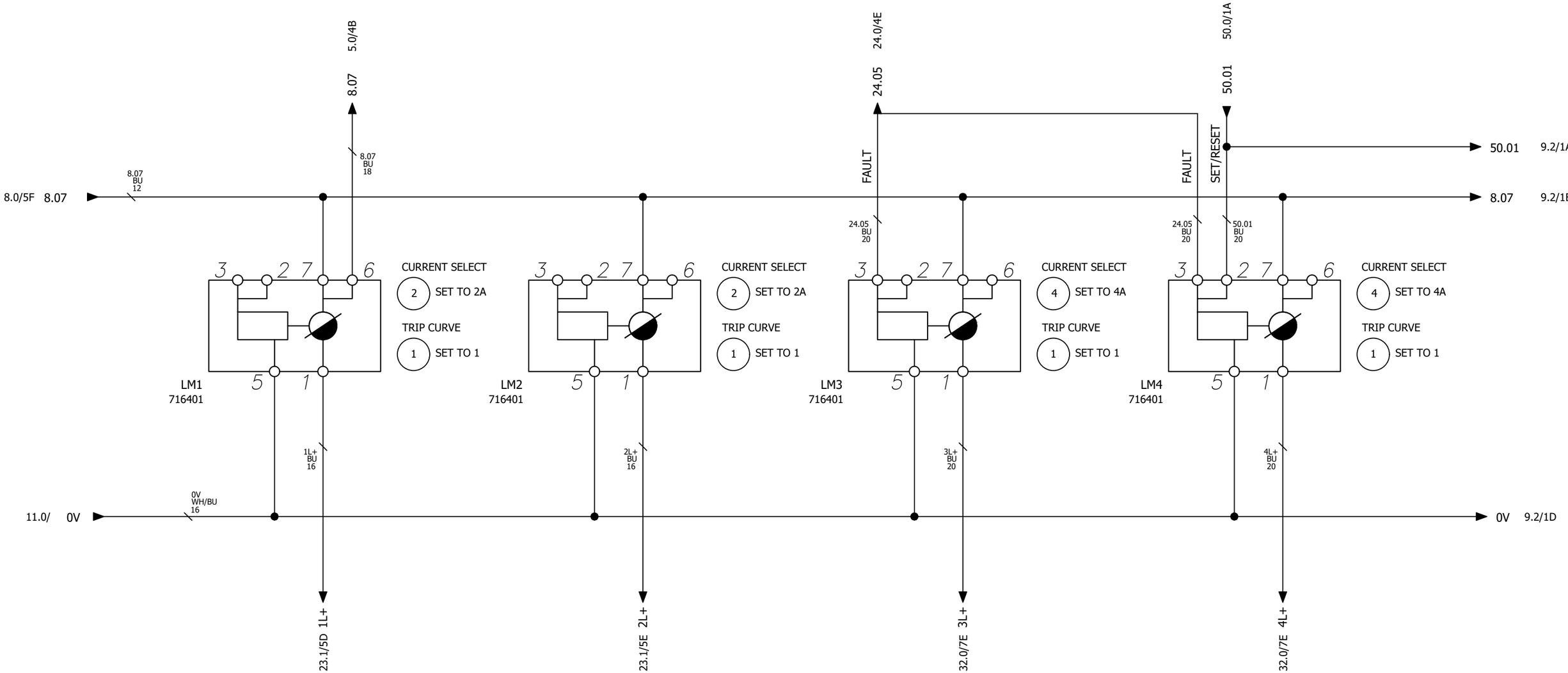


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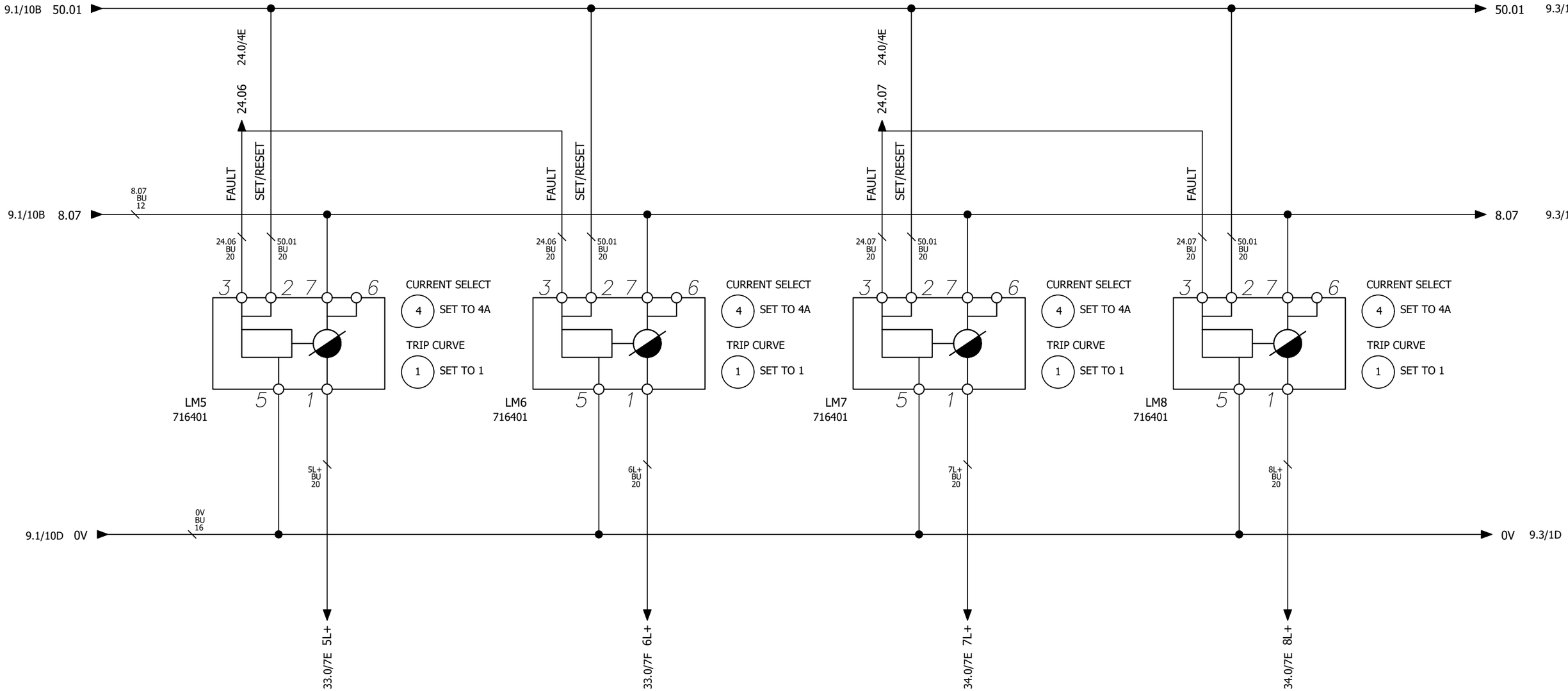


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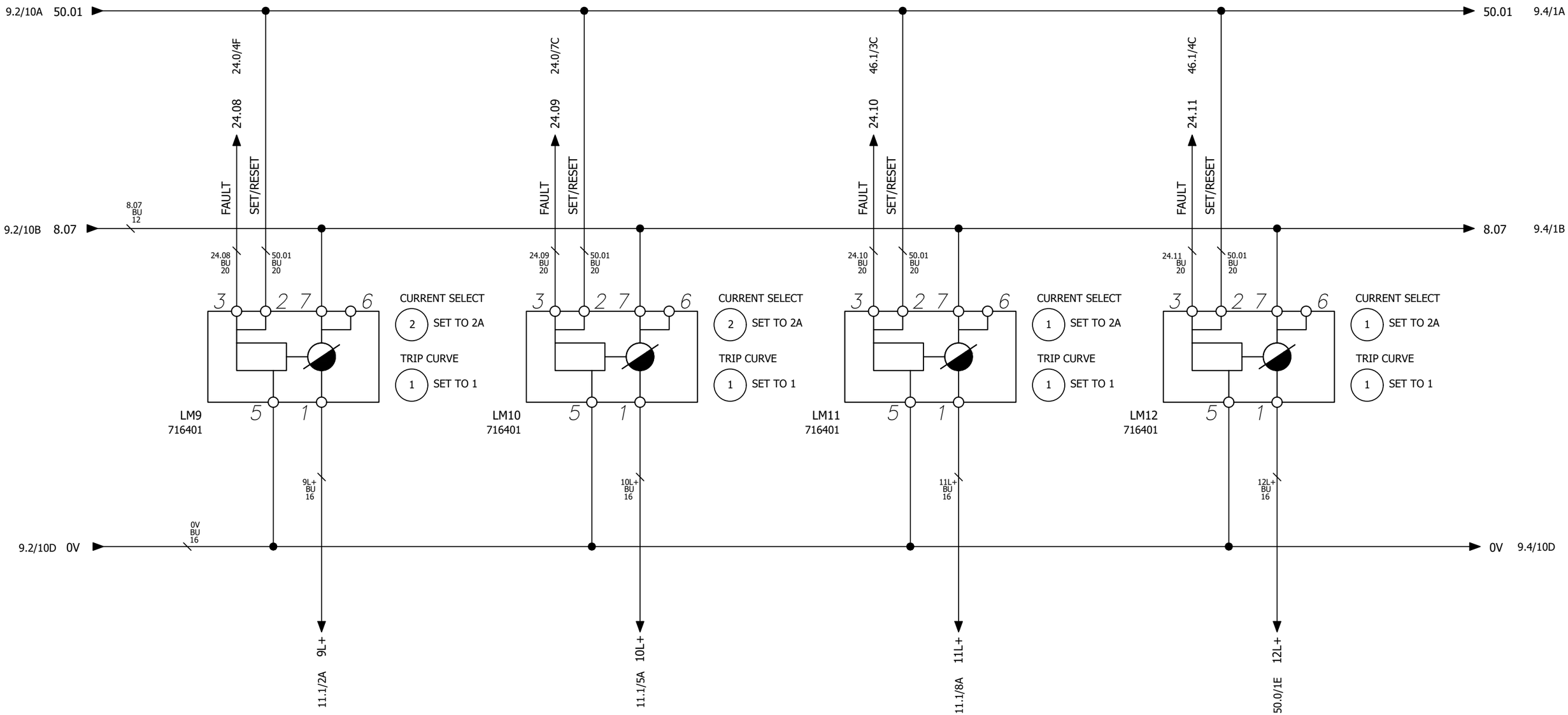


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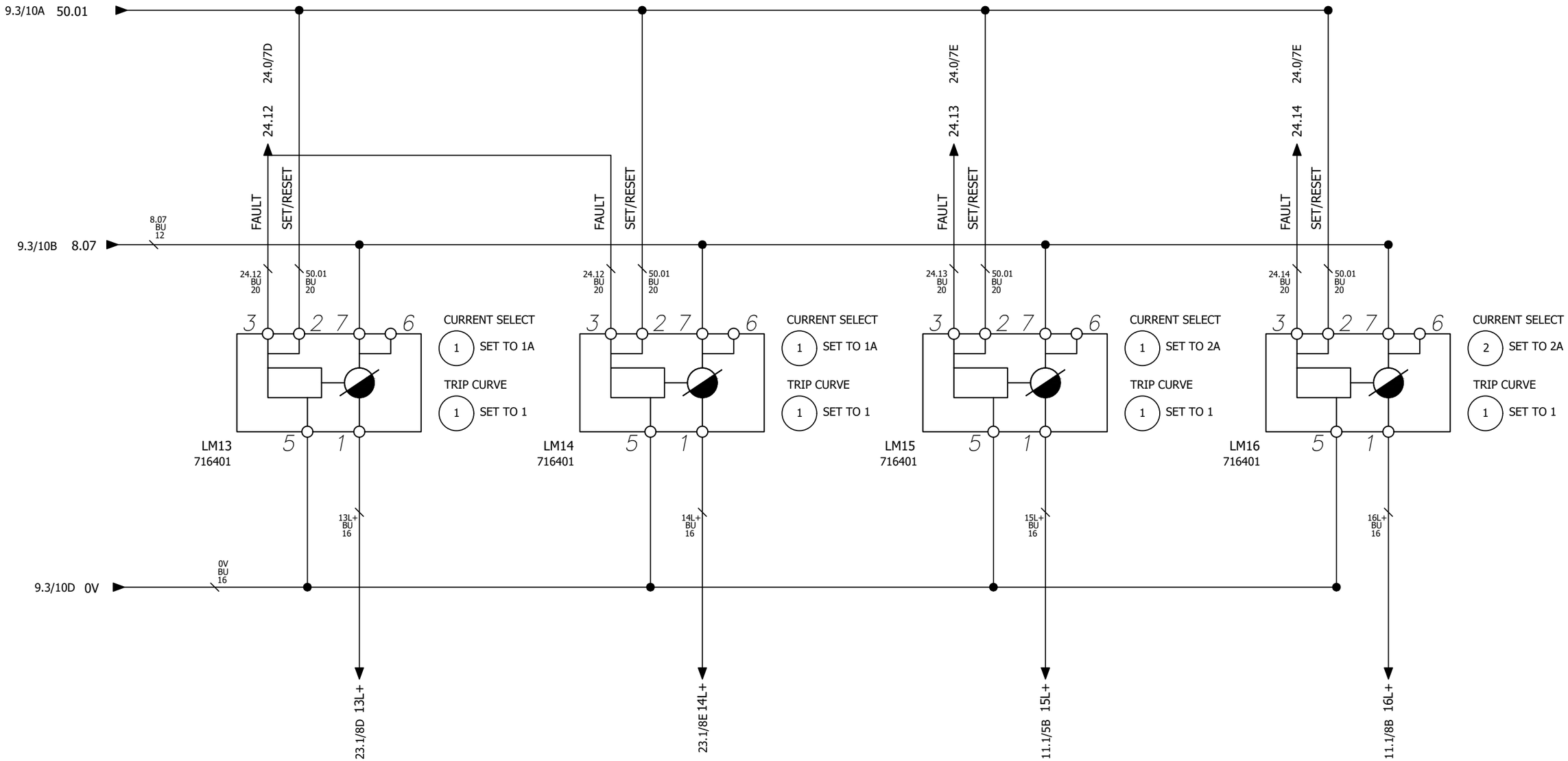
SHEET 9.3 OF 56

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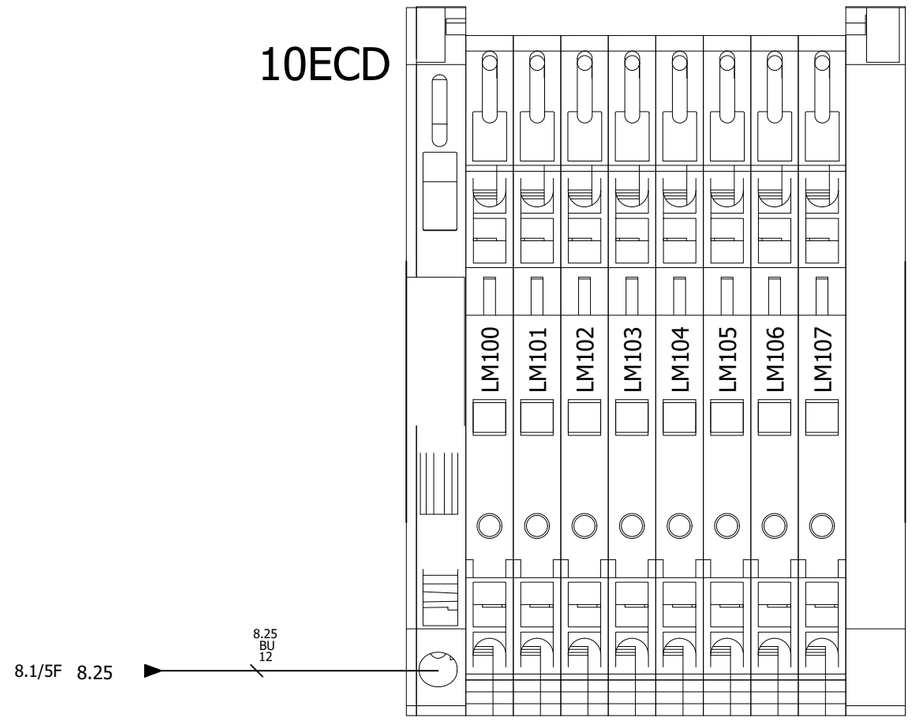


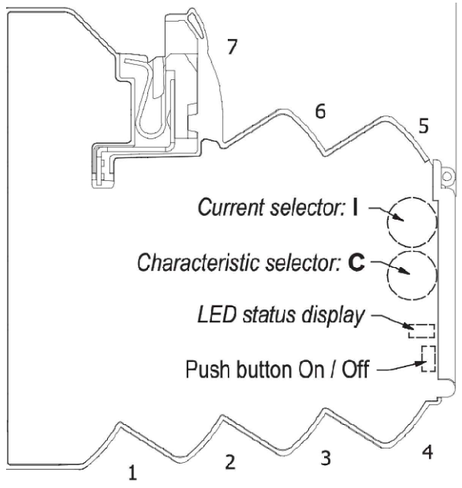

## LOAD MONITORING(LM) PIN ASSIGNMENT

| |
|---|
| PIN 1–OUT+(LOAD) |
| PIN 2–CTRL INPUT (SET/RESET) |
| PIN 3–STATUS OUTPUT |
| PIN 4–NC |
| PIN 5–0V |
| PIN 6–SUPPLY (ALTERNATIVE) |
| PIN 7–SUPPLY |

## TRIP CURVES:

1. CHARACTERISTIC FAST
2. CHARACTERISTIC MEDIUM
3. CHARACTERISTIC SLOW–1
4. CHARACTERISTIC SLOW–2
5. CHARACTERISTIC SLOW–3

## ELECTRONIC CIRCUIT BREAKER (LM)

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PAGE DESCRIPTION: 24V ELECTRONIC LOAD MONITORING MAIN PANEL FOR REMOTE COMPONENTS

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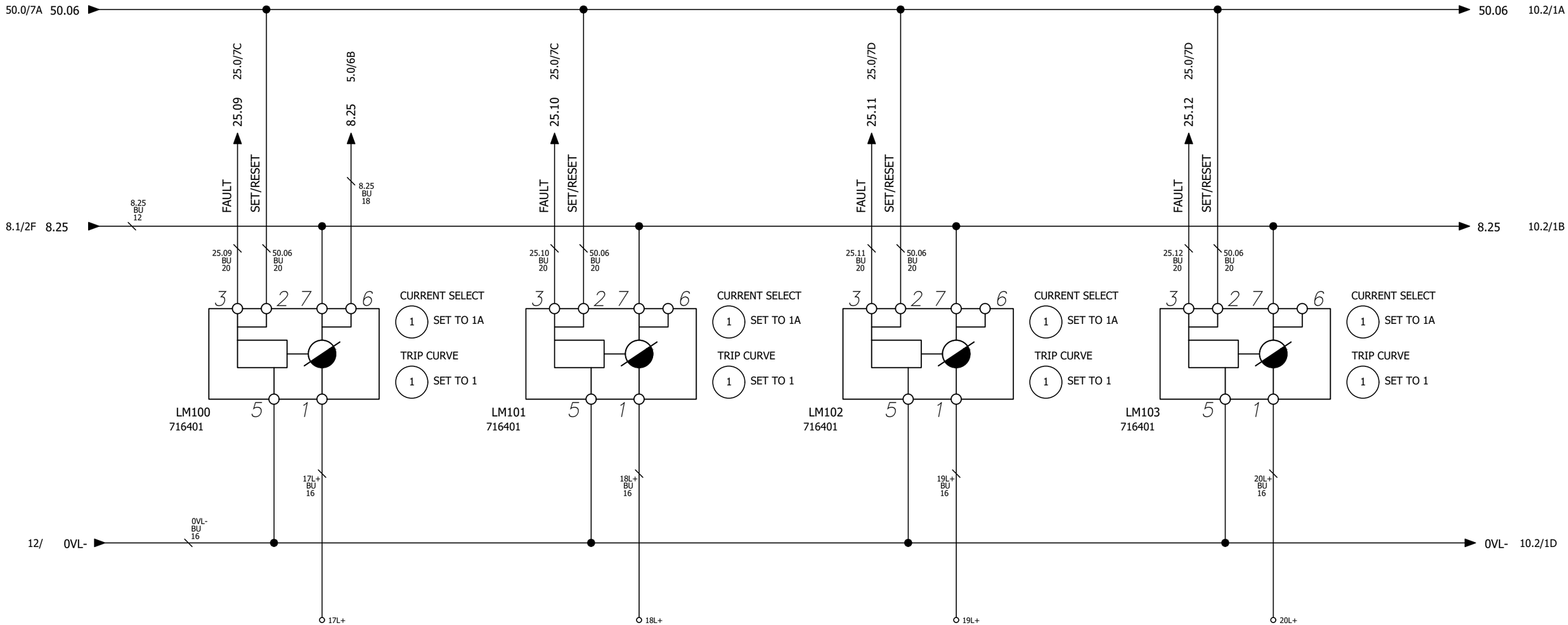


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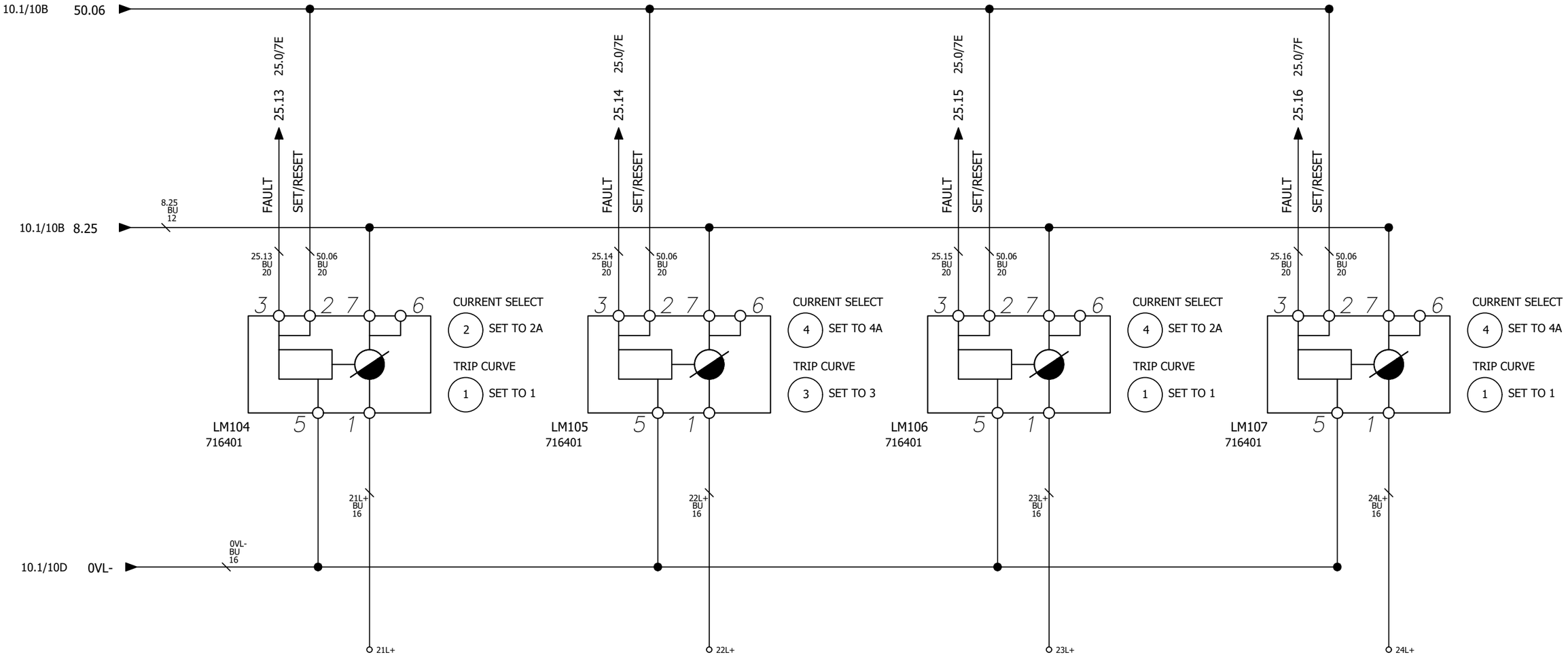


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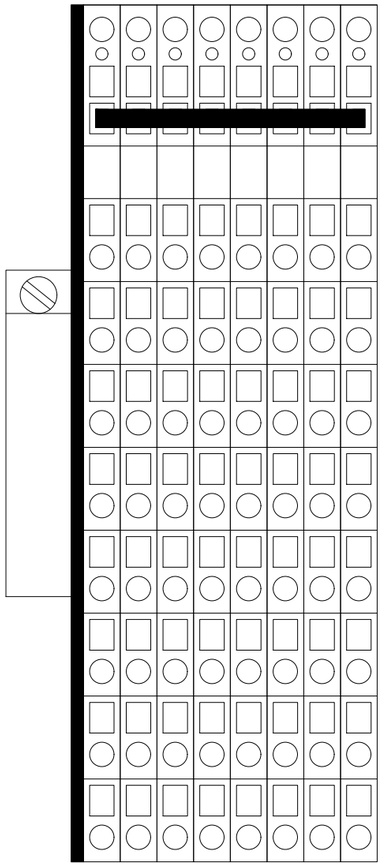

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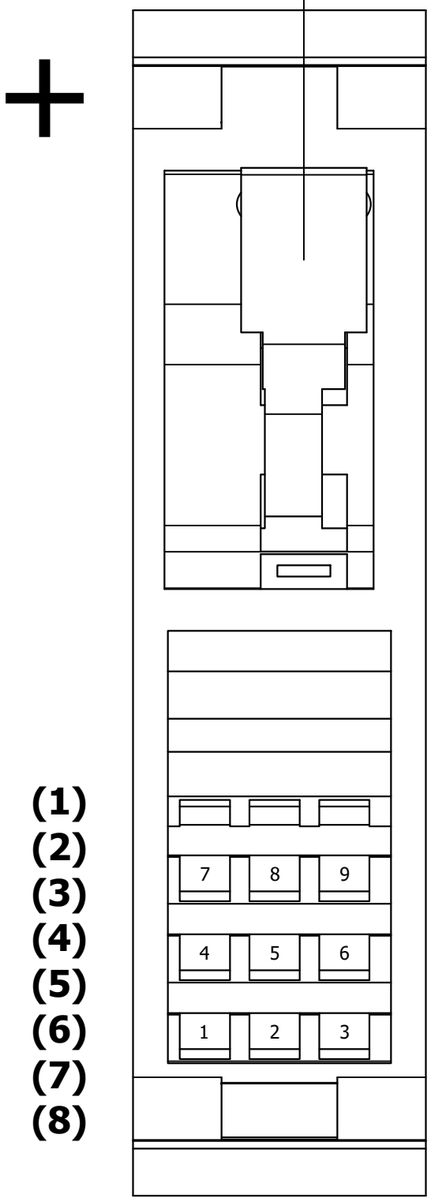
SHEET 11.0 OF 56

9.3/3E 9L+

# 9L+

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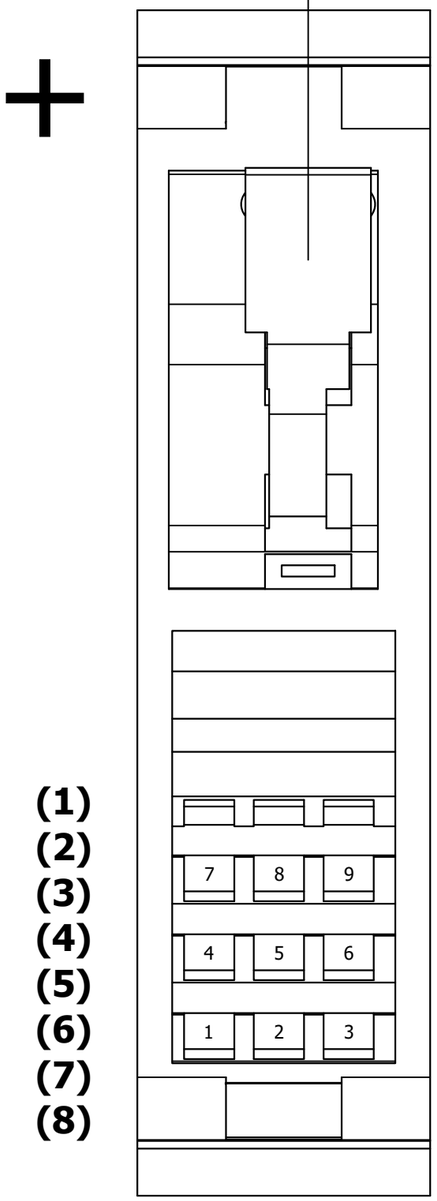


9.3/7E 11L+

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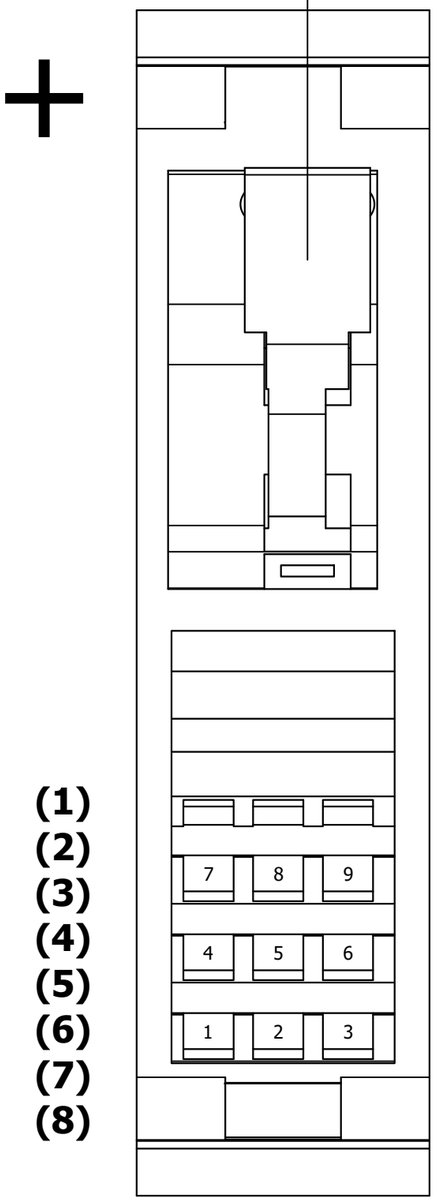


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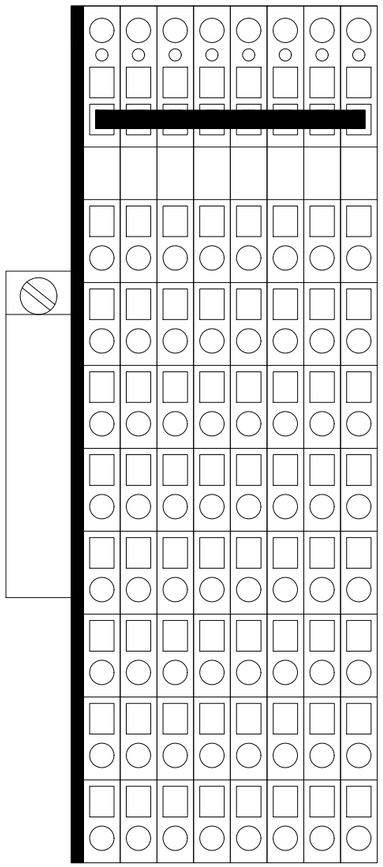

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JOB # C40572

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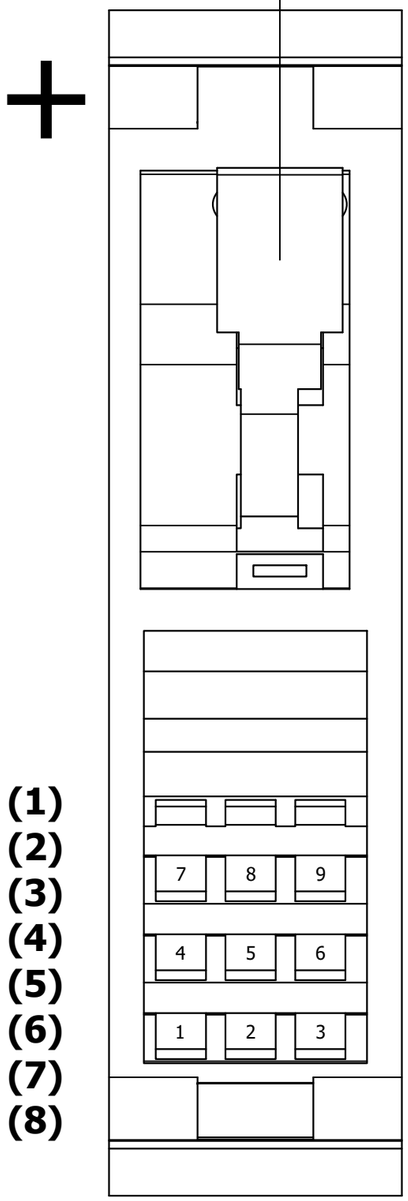
SHEET 12.0 OF 56

10.1/7E 19L+

# 19L+

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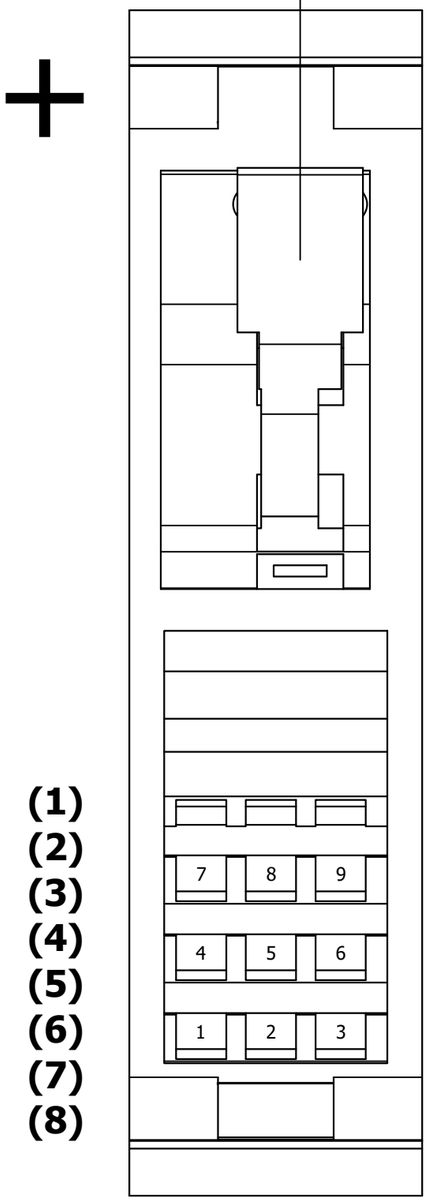
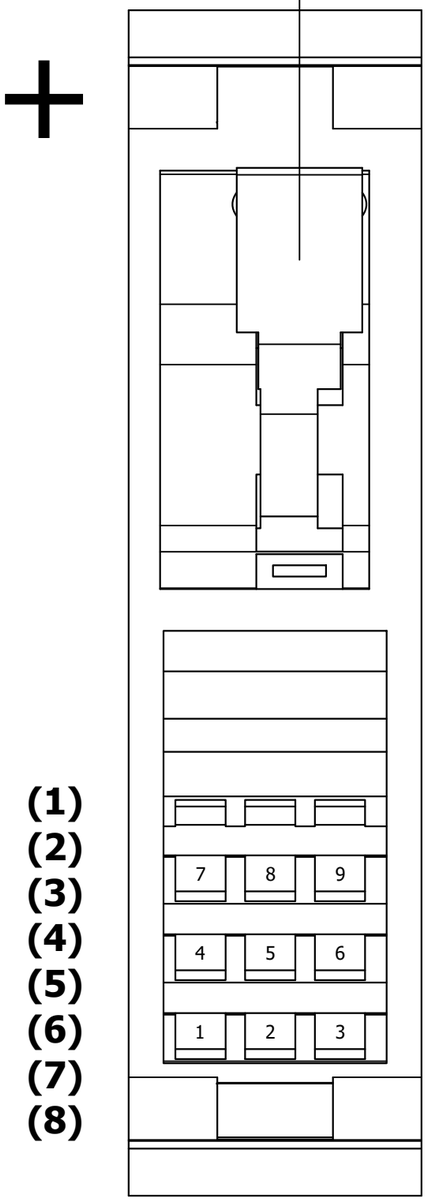



10.2/3E 21L+

# 21L+

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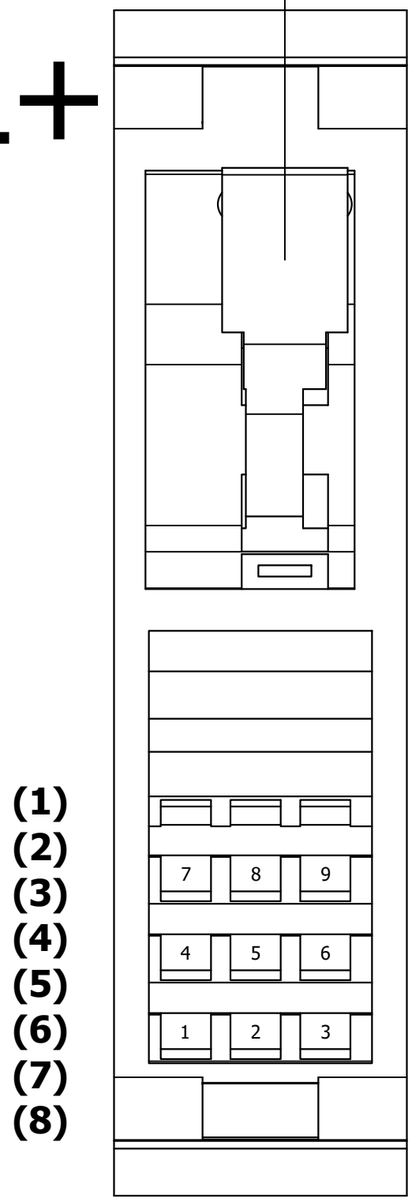
SHEET 12.1 OF 56

10.2/5E 22L+

# 22L+

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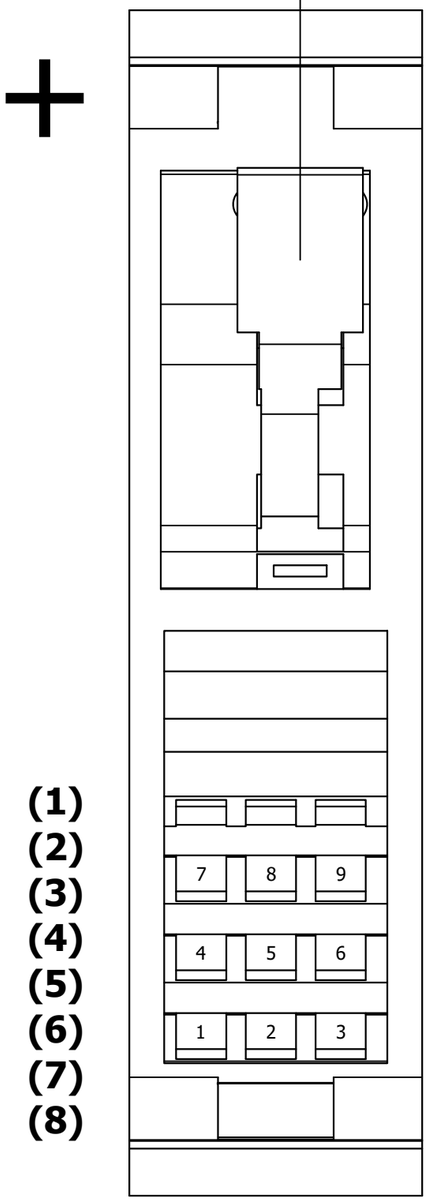


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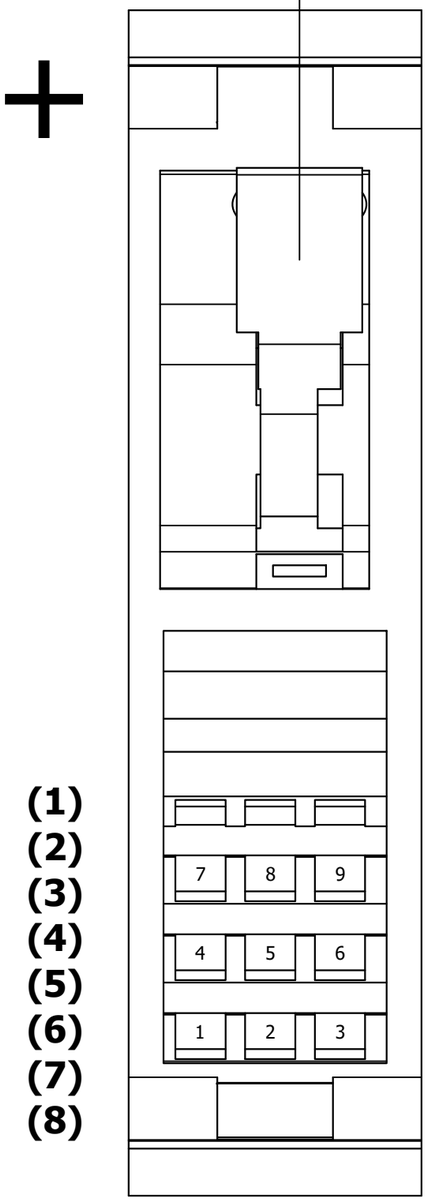


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TOTAL PAGES: 175

SHEET 14.0 OF 56

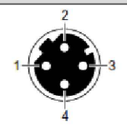
# 15VFD1

### X4233: Ethernet fieldbus

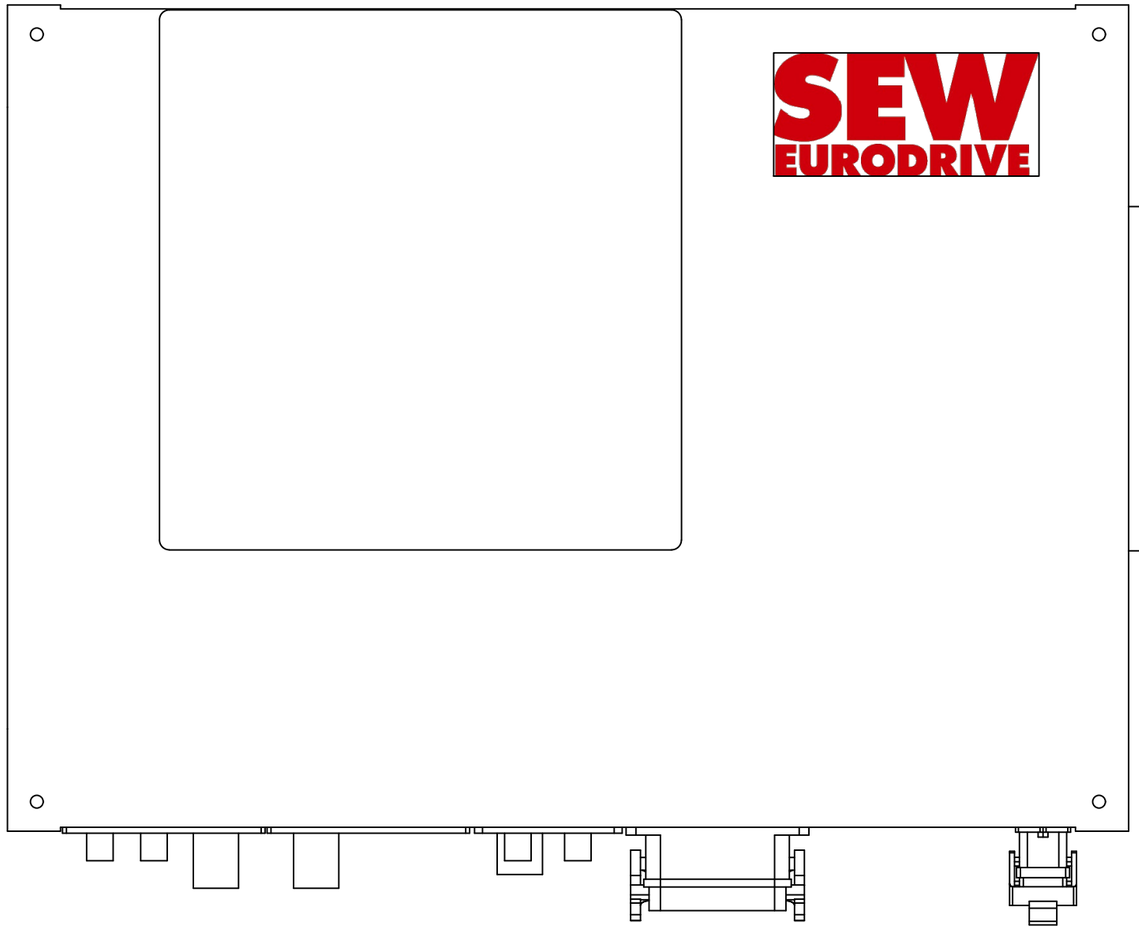
| Function |
|---|
| Ethernet fieldbus interface, 4-pin |

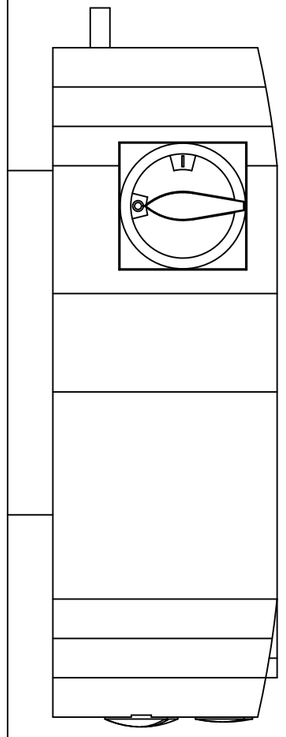
| Connection type |
|---|
| M12, 4-pin, female, D-coded |

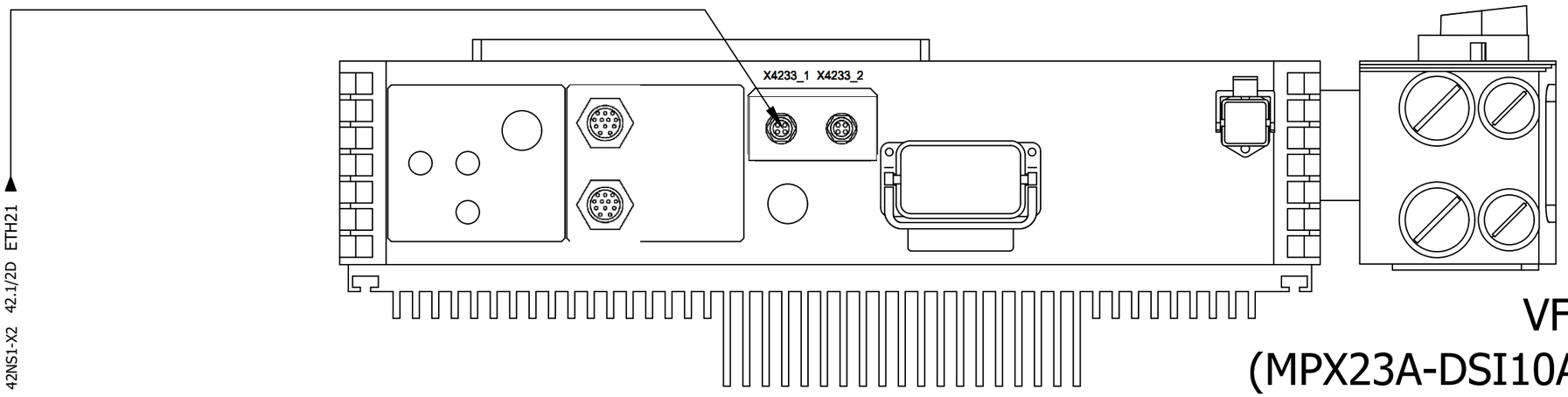
**Wiring diagram**




| No. | Name | Function |
|---|---|---|
| 1 | TX+ | Sending cable (+) |
| 2 | RX+ | Receiving cable (+) |
| 3 | TX- | Sending cable (-) |
| 4 | RX- | Receiving cable (-) |





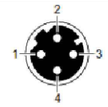



### X4224: Ethernet engineering

| Function |
|---|
| Ethernet engineering interface, 4-pin |

| Connection type |
|---|
| M12, 4-pin, female, D-coded |

**Connection diagram**




| No. | Name | Function |
|---|---|---|
| 1 | TX+ | Sending cable (+) |
| 2 | RX+ | Receiving cable (+) |
| 3 | TX- | Sending cable (-) |
| 4 | RX- | Receiving cable (-) |

## VFD 15kW
## (MPX23A-DSI10A-MPI-0460-5E3-CA-C)

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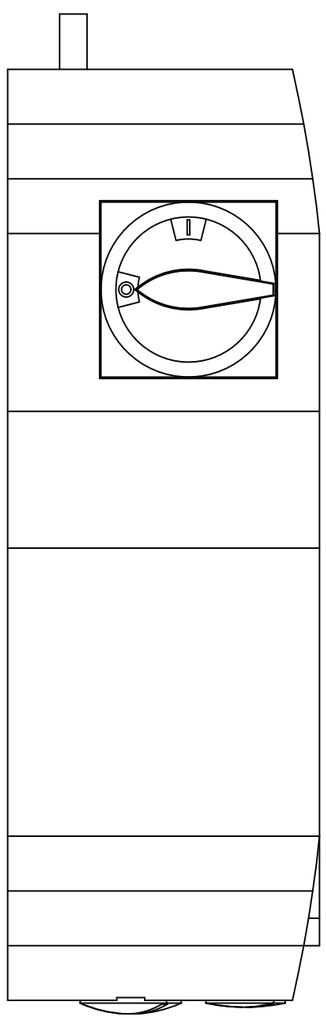
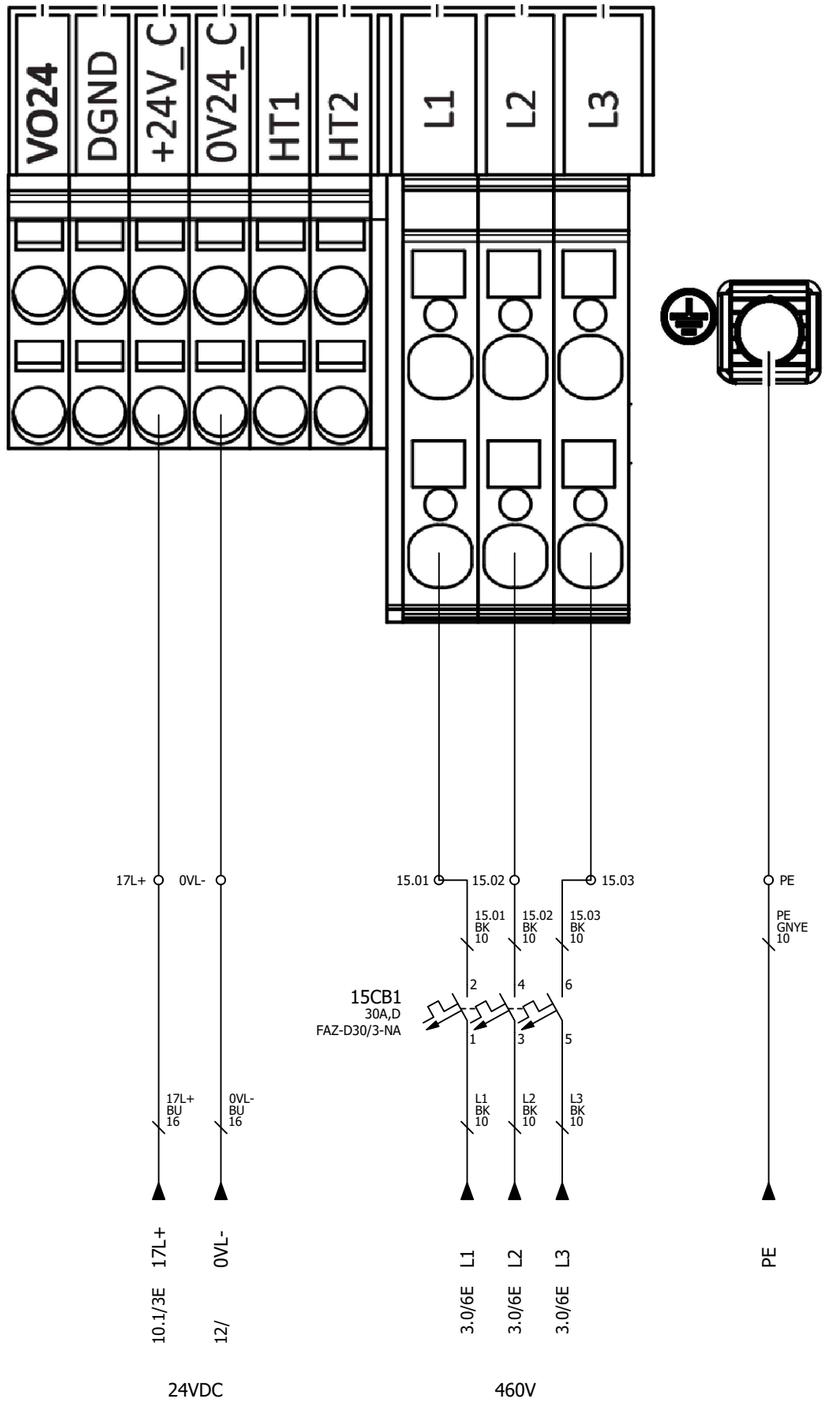
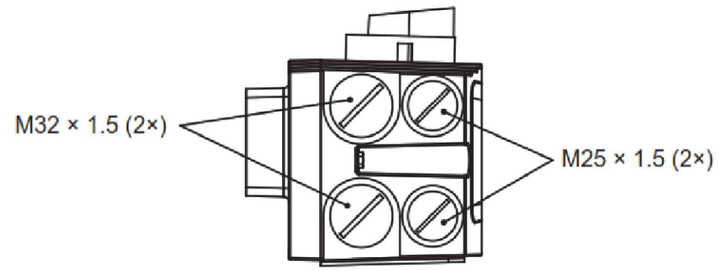
SHEET 15.0 OF 56






| Interface box | PZM2xA.. |
|---|---|
| Line terminals L1 – L3 | 57 A at 16 mm² |
| 24 V terminals | 41 A at 6 mm² |




| Name | Function |
|---|---|
| VO24 | DC 24 V output |
| DGND | Not assigned |
| +24V_C | DC 24 V input |
| 0V24_C | 0V24 reference potential – input |
| HT1 | Auxiliary terminal for additional voltage levels (without internal function) |
| HT2 | Auxiliary terminal for additional voltage levels (without internal function) |
| L1 | Line connection, phase 1 |
| L2 | Line connection, phase 2 |
| L3 | Line connection, phase 3 |
| PE | Protective earth connection |

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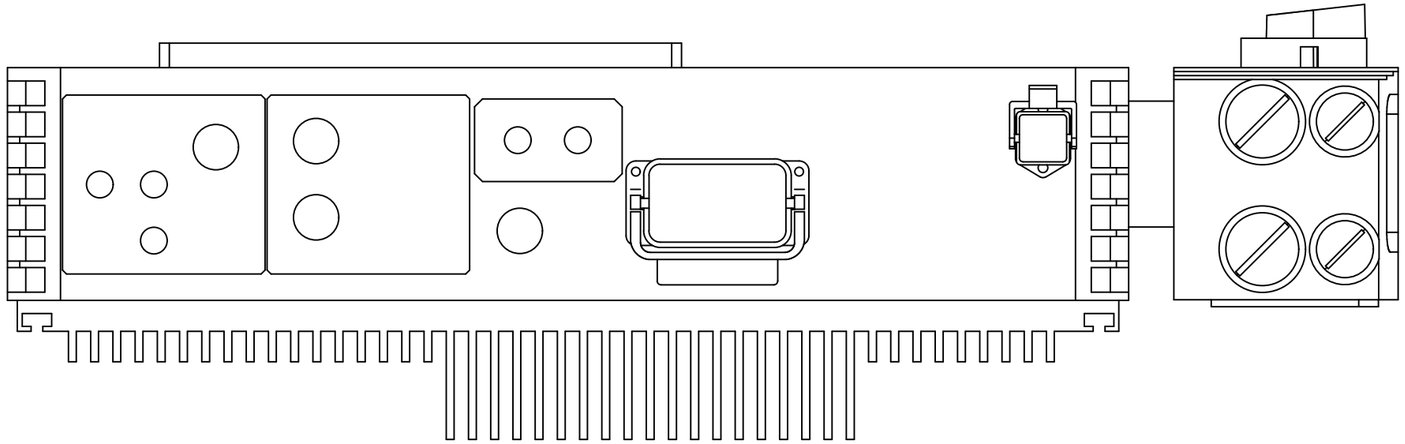
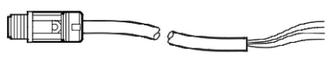
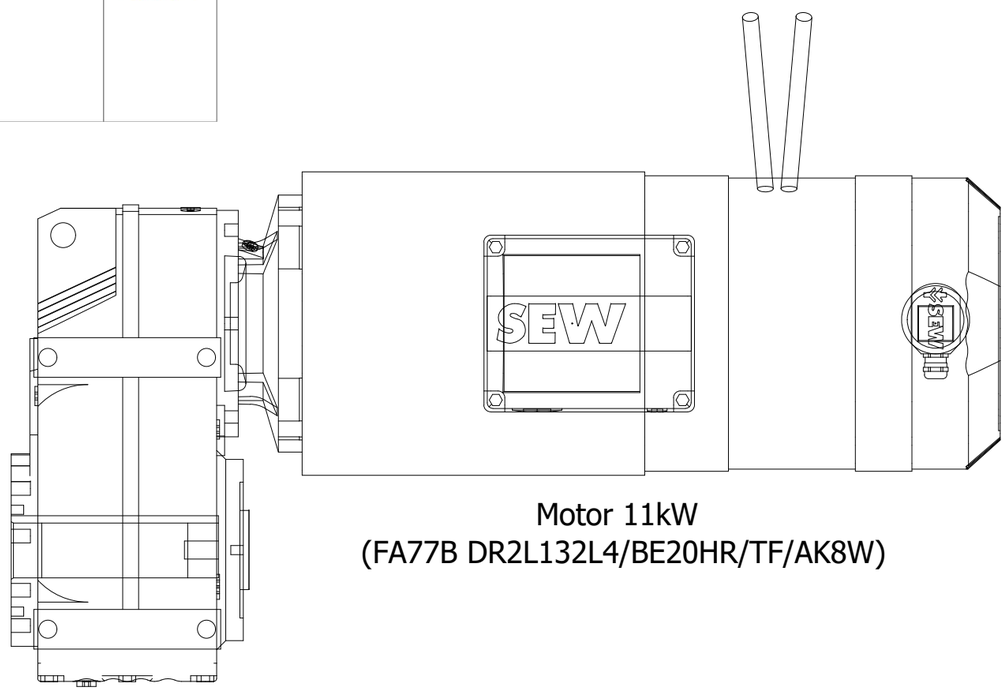


| | | | | |
|---|---|---|---|---|
|  M23, coding ring: brown, male — Open, core end sleeves | CE/UL: 28126009 | LEONI LEHC® 005272 | Variable | 1.5 mm² / AC 500 V |




Motor 11kW
(FA77B DR2L132L4/BE20HR/TF/AK8W)

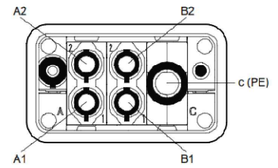
*Cable cross section 16 mm²*

| Connection cable | Conformity/ part number | Cable type | Length/installation type | Cable cross section/operating voltage |
|---|---|---|---|---|
| Han® 10 B, male — open | CE/UL: 28178599 | HELUKABEL TOPSERV 109 Li9YC11Y-HF | Variable | 16 mm² / AC 500 V |

| Function |
|---|
| Connection of motor phases |

| Connection type |
|---|
| Han-Modular® 10 B, female |

**Connection diagram**




| [a] Han® C module, male | | |
|---|---|---|
| **Contact** | **Function** | |
| 1 | U | Motor connection, phase U |
| 2 | V | Motor connection, phase V |

| [b] Han® EE module, male | | |
|---|---|---|
| **Contact** | **Function** | |
| 1 | W | Motor connection, phase W |
| 2 | Res. | Reserved |

| [c] Articulated frame | | |
|---|---|---|
| **Contact** | **Function** | |
| – | PE | Protective earth connection |

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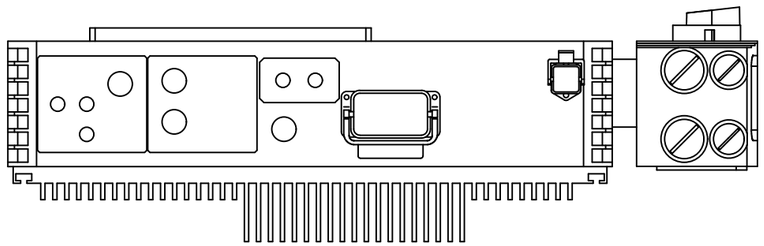
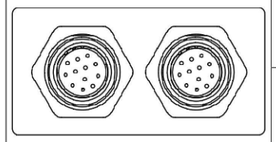
SHEET 15.2 OF 56



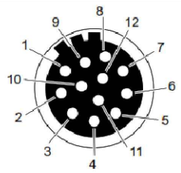
| Connections | | | Encoder type |
|---|---|---|---|
| [1] [2]  | [1] | X3222 | Multi distance encoder:<br>HIPERFACE®, SSI, Sin/Cos, HTL |
| | [2] | X3011 | Motor encoder:<br>HIPERFACE®, sin/cos, TTL, HTL, RS422 |

**X3222: Multi-distance encoder**

| Function |
|---|
| Connection for HIPERFACE®, SSI, sin/cos, TTL, HTL and RS422 encoders |

| Connection type |
|---|
| M23, insulator, P part, 12-pin, female, +20°-coded |

**Wiring diagram**




| No. | Name | Function |
|---|---|---|
| 1 | CLK (C) | Clock line (Signal track C (K0)) |
| 2 | /CLK (/C) | Negated clock line (Negated signal track C (/K0)) |
| 3 | A | Signal track A (K1) |
| 4 | /A | Negated signal track A (/K1) |
| 5 | B | Signal track B (K2) |
| 6 | /B | Negated signal track B (/K2) |
| 7 | Data- | Data line (-) |
| 8 | Data+ | Data line (+) |
| 9 | GND | Reference potential |
| 10 | +24V | DC 24 V output<br>Total current load of DC 24 V encoder supply ≤ 400 mA |
| 11 | GND | Reference potential |
| 12 | +12V | DC 12 V output<br>Total current load of DC 12 V encoder supply ≤ 650 mA |

| Part number: 18121942<br>Cable design: (6X2X0.25) | | |
|---|---|---|
| <br>M23, 12-pin, 20°-coded ↔ Open with conductor end sleeves | Variable length | HIPERFACE®/SSI encoder (24 V) |

| Part number | Signal name | Color coding |
|---|---|---|
| 18121942 | CLK | Brown |
| | /CLK | White |
| | A | Red |
| | /A | Blue |
| | B | Yellow |
| | /B | Green |
| | Data- | Purple |
| | Data+ | Black |
| | GND | Gray/Pink + Pink |
| | +24V | Red/Blue + Gray |

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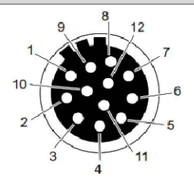
SHEET 15.3 OF 56

**X3222: Multi-distance encoder**

| Function |
|---|
| Connection for HIPERFACE®, SSI, sin/cos, TTL, HTL and RS422 encoders |

| Connection type |
|---|
| M23, insulator, P part, 12-pin, female, +20°-coded |

**Wiring diagram**



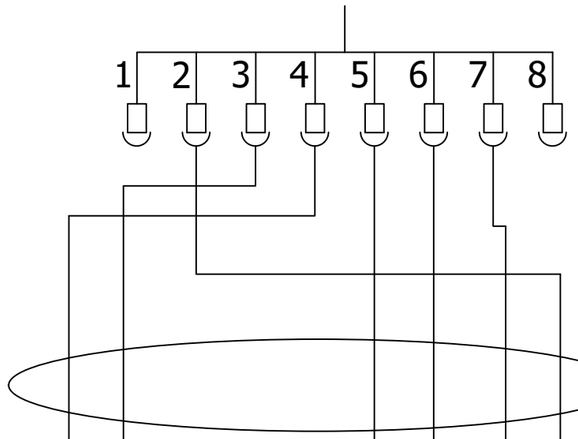

| No. | Name | Function |
|---|---|---|
| 1 | CLK (C) | Clock line (Signal track C (K0)) |
| 2 | /CLK (/C) | Negated clock line (Negated signal track C (/K0)) |
| 3 | A | Signal track A (K1) |
| 4 | /A | Negated signal track A (/K1) |
| 5 | B | Signal track B (K2) |
| 6 | /B | Negated signal track B (/K2) |
| 7 | Data- | Data line (-) |
| 8 | Data+ | Data line (+) |
| 9 | GND | Reference potential |
| 10 | +24V | DC 24 V output<br>Total current load of DC 24 V encoder supply ≤ 400 mA |
| 11 | GND | Reference potential |
| 12 | +12V | DC 12 V output<br>Total current load of DC 12 V encoder supply ≤ 650 mA |

PORT # X3222 (male connector
M23, 12-pin, +20°-coded)

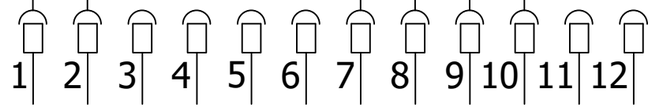
| SICK DL50-P2228 | WIRE COLOR | SIGNAL | SEW X3222 | SEW SIGNAL |
|---|---|---|---|---|
| PIN 4 | YELLOW | RX+ / CLK+ | PIN 1 | CLK |
| PIN 3 | GREEN | RX− / CLK− | PIN 2 | /CLK |
| PIN 6 | PINK | TX+ / DATA+ | PIN 8 | DATA+ |
| PIN 5 | GRAY | TX− / DATA− | PIN 7 | DATA− |
| PIN 2 | BROWN | L+ | PIN 10 | +24 VDC |
| PIN 7 | BLUE | M / 0V | PIN 9 | GND |
| PIN 1 | WHITE | Q1 | NOT REQUIRED FOR SSI | |
| PIN 8 | RED | Q2/MF | NOT REQUIRED FOR SSI | |

SSI ENCODER
(male connector M12,
8-pin, A-coded)

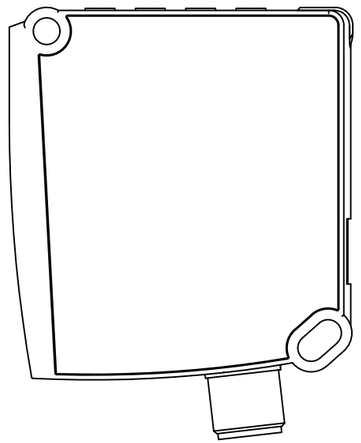
15DE1




15VFD1

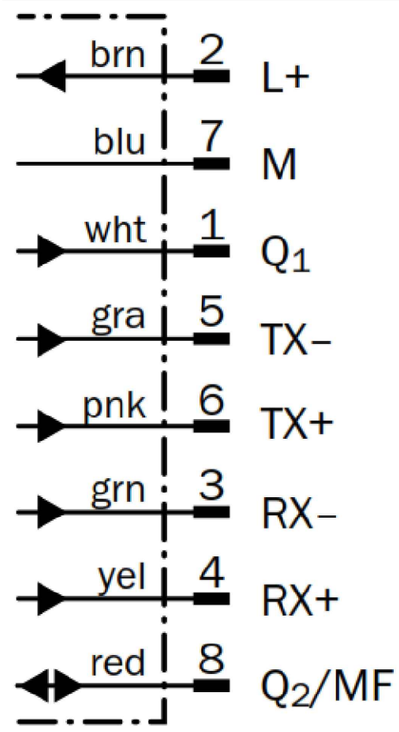



PORT # X3222 (male connector
M23, 12-pin, +20°-coded)



# DL50-P2228

(Male connector M12, 8-pin, A-coded)

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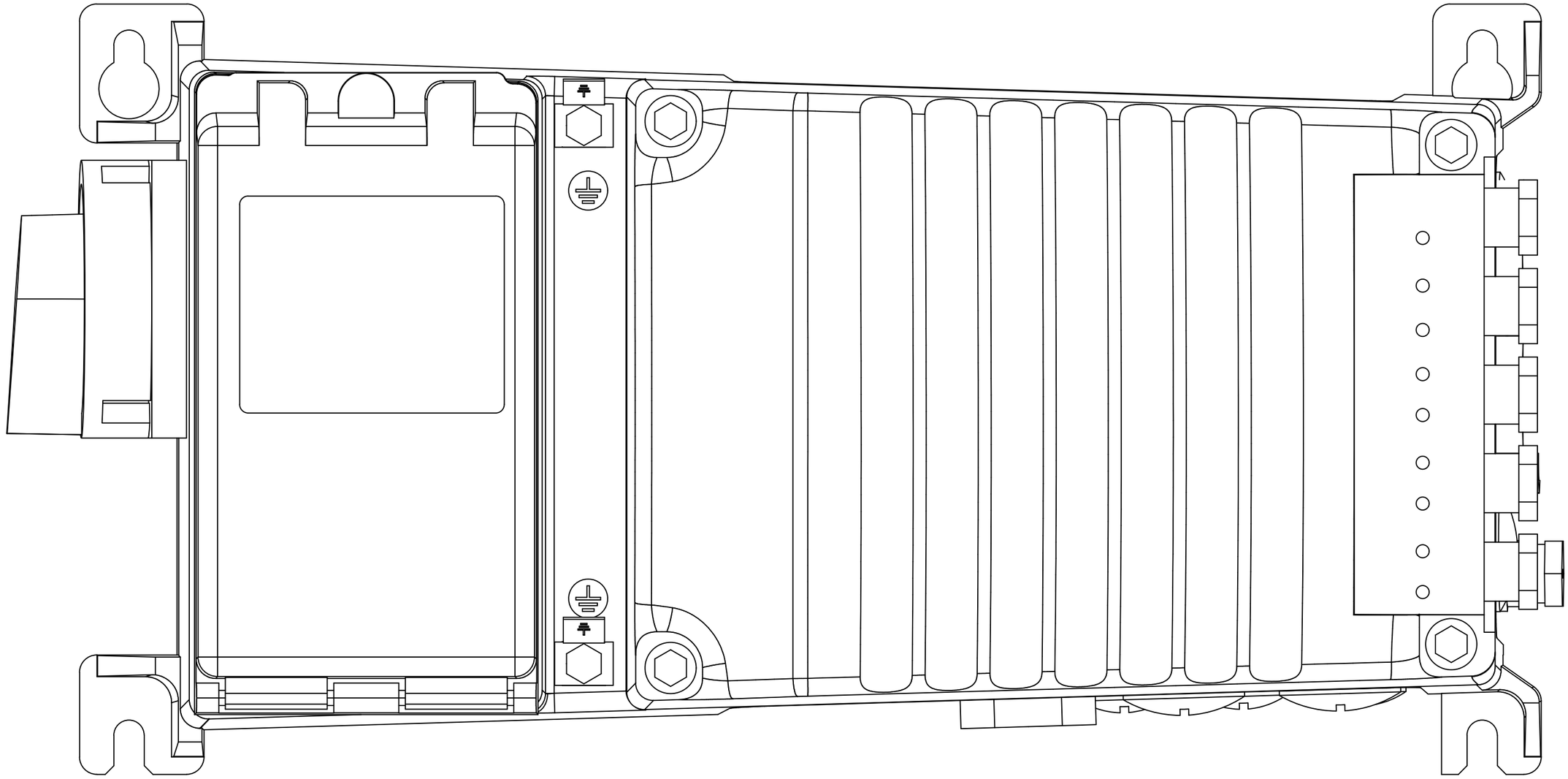
SHEET 15.4 OF 56



# 16VFD1

VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

VTD MOBILE EXTRACTOR CONVEYOR AREA VFD

ALL .75 VFD REFERENCES FOUND IN THIS SECTION

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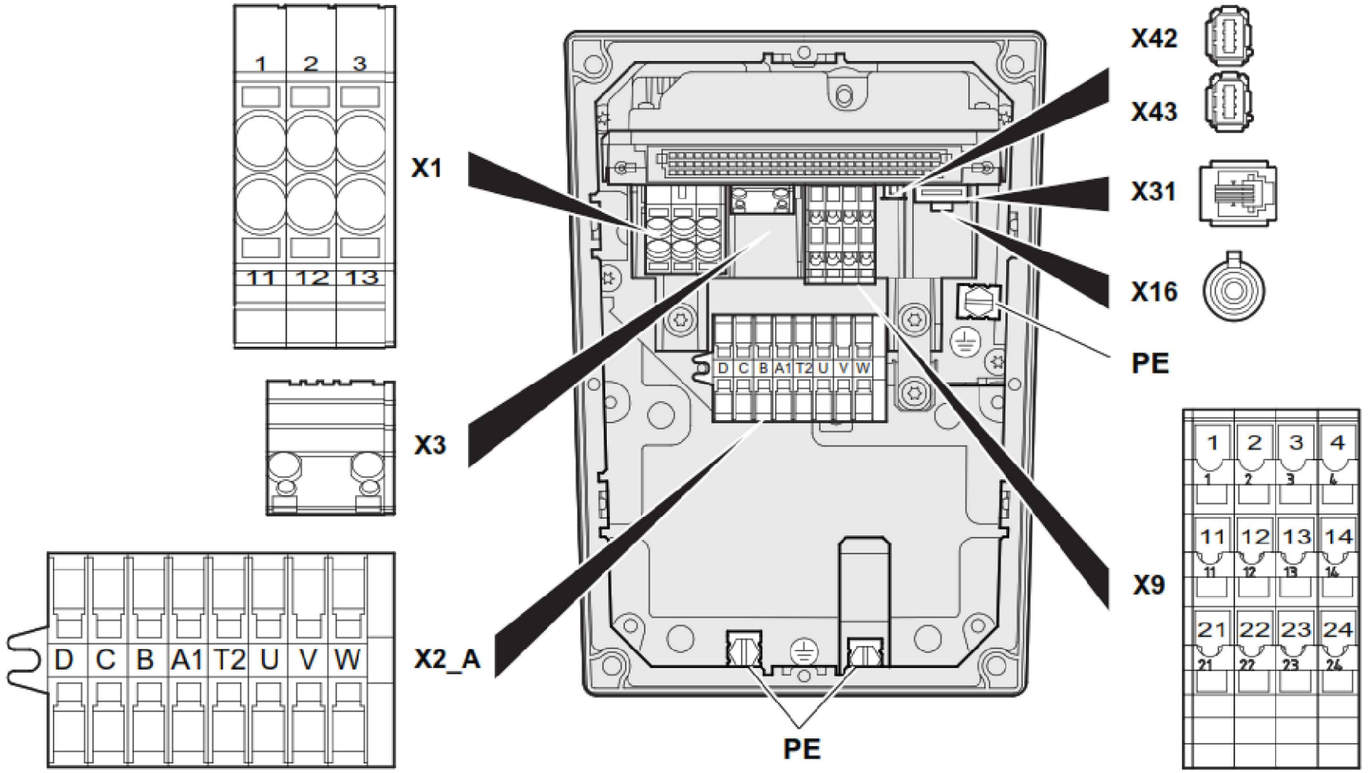
SHEET 16.0 OF 56

| X2_A terminals for motor and brake | Brake control /B for AC brakes 100 – 525 V. |
|---|---|
| D | **Connection of brake 14**<br>• Brake coil: White<br>• Hybrid cable: Violet (D) |
| C | **Connection of brake 13**<br>• Brake coil: Red<br>• Hybrid cable: Pink (C) |
| B | **Connection of brake 15**<br>• Brake coil: Blue<br>• Hybrid cable: Orange (B) |
| A | Reserved<br>• Hybrid cable: Yellow (A) |
| U | **Motor connection, phase U** |
| V | **Motor connection, phase V** |
| W | **Motor connection, phase W** |

| Assignment | | | | |
|---|---|---|---|---|
| **Terminal** | **No.** | **Name** | **Marking** | **Function** |
| **X1**<br>**line terminals** | 1 | L1 | Brown | Line connection, phase L1 – IN |
| | 2 | L2 | Black | Line connection, phase L2 – IN |
| | 3 | L3 | Gray | Line connection, phase L3 – IN |
| | 11 | L1 | Brown | Line connection, phase L1 – OUT |
| | 12 | L2 | Black | Line connection, phase L2 – OUT |
| | 13 | L3 | Gray | Line connection, phase L3 – OUT |
| ⏚ | – | PE | – | PE connection |
| **X3**<br>**braking resistor terminals** | 1 | BW | – | Braking resistor connection |
| | 2 | BW | – | Braking resistor connection |

| Assignment | | | | |
|---|---|---|---|---|
| **Terminal** | **No.** | **Name** | **Marking** | **Function** |
| **X9**<br>**control terminals** | 1 | F_STO_P1 | Yellow | Input STO+ |
| | 2 | F_STO_P1 | Yellow | Input STO+ (to loop through) |
| | 3 | 0V24_out | – | 0V24 reference potential for DC 24 V auxiliary output |
| | 4 | 24V_out | – | DC 24 V auxiliary output (**only for internal voltage supply of for non-safety-related enable signal via STO input**) |
| | 11 | F_STO_M | Yellow | Input STO_ground |
| | 12 | F_STO_M | Yellow | Input STO_ground (to loop through) |
| | 13 | 24V_in | – | DC 24 V input |
| | 14 | 24V_in | – | DC 24 V input (to loop through) |
| | 21 | F_STO_P2 | Yellow | Input STO+ |
| | 22 | F_STO_P2 | Yellow | Input STO+ (to loop through) |
| | 23 | 0V24_in | – | 0V24 reference potential |
| | 24 | 0V24_in | – | 0V24 reference potential (to loop through) |
| **X42**<br>**EtherCAT®/SBus$^{PLUS}$ IN** | 1 | TX+ | – | Transmit line + |
| | 2 | TX- | – | Transmit line - |
| | 3 | RX+ | – | Receive line + |
| | 6 | RX- | – | Receive line - |
| **X43**<br>**EtherCAT®/SBus$^{PLUS}$ OUT** | 1 | TX+ | – | Transmit line + |
| | 2 | TX- | – | Transmit line - |
| | 3 | RX+ | – | Receive line + |
| | 6 | RX- | – | Receive line - |
| **X31**<br>**engineering interface** | – | – | – | Engineering interface with DC 24 V supply voltage for use in conjunction with suitable SEW engineering options |
| **X16**<br>**MOVILINK® DDI interface** | 1 | DDI | – | MOVILINK® DDI supply/communication |
| | 2 | DDI_GND | – | MOVILINK® DDI reference potential |

## 5.5.1 Design MMF1., MMF31




27021626766467339

16VFD1

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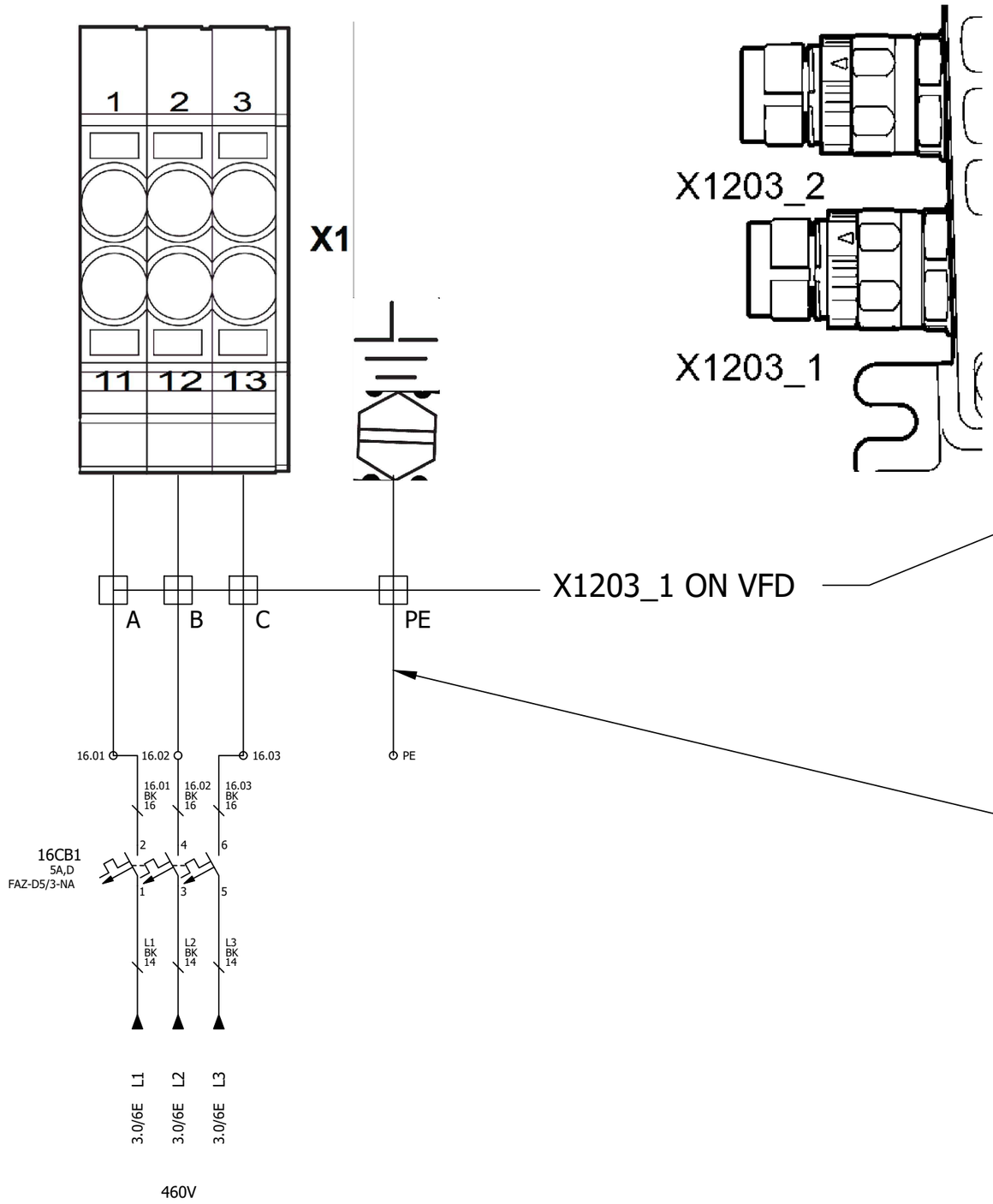
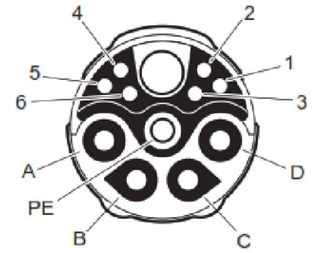
SHEET 16.1 OF 56

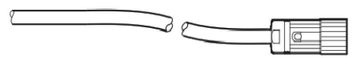



**Connection diagram**




**Assignment**

| Contact | Function | |
|---|---|---|
| A | L1 | Line connection, phase L1 |
| B | L2 | Line connection, phase L2 |
| C | L3 | Line connection, phase L3 |
| D | Res. | Reserved |
| PE | PE | Protective earth connection |
| 1 | Res. | Reserved |
| 2 | Res. | Reserved |
| 3 | Res. | Reserved |
| 4 | Res. | Reserved |
| 5 | Res. | Reserved |
| 6 | Res. | Reserved |

| | | | | |
|---|---|---|---|---|
|  Open M23, coding ring: black, male | UL: 18153283 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

# 16VFD1

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PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA DRIVE POWER

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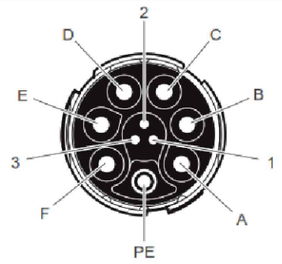
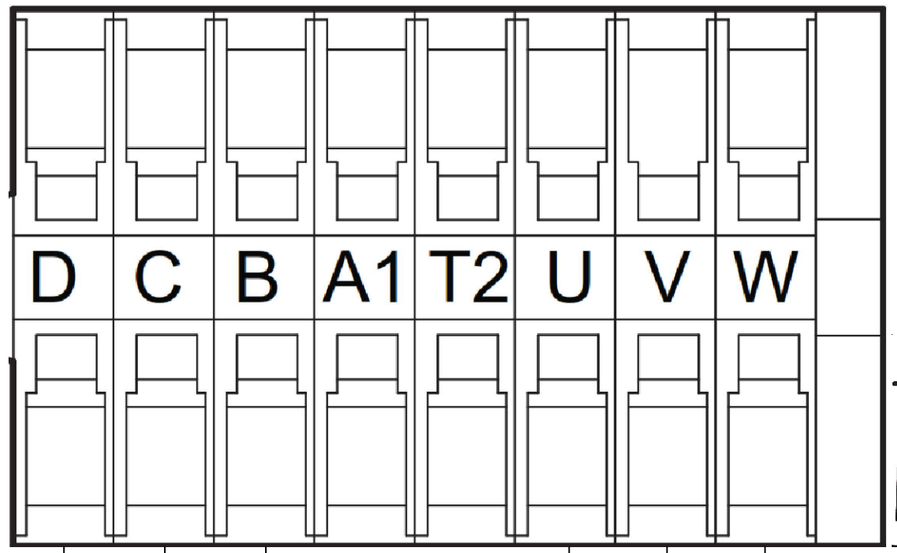
SHEET 16.2 OF 56

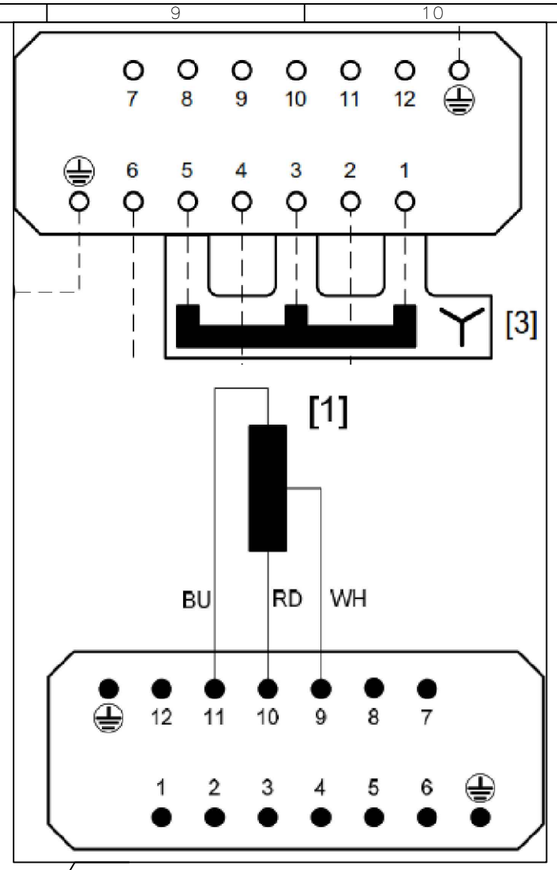
## Connection diagram




## Assignment

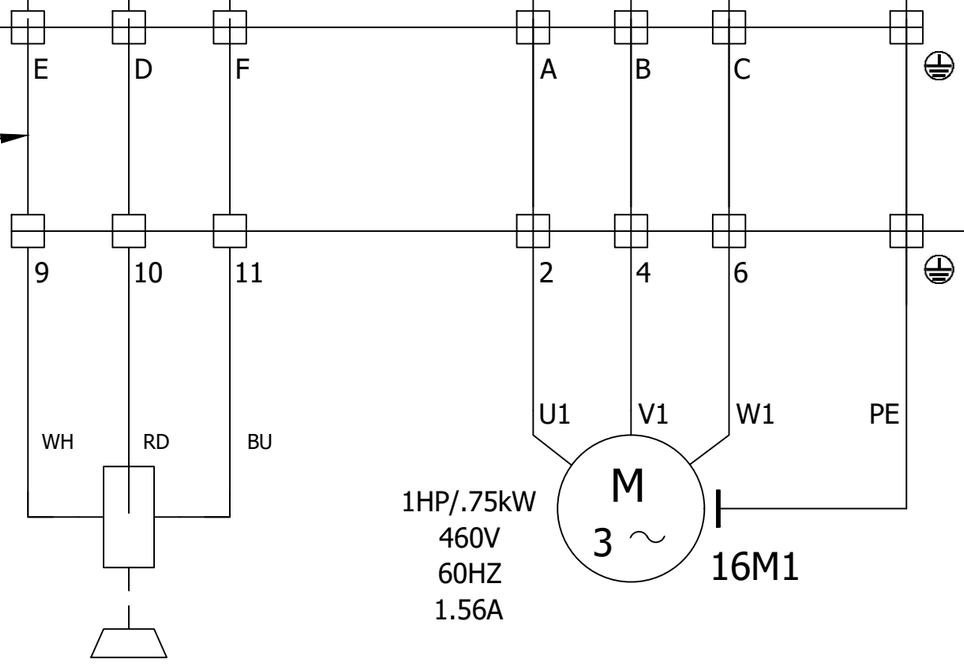
| Contact | Function | |
|---|---|---|
| A | U | Motor connection, phase U |
| B | V | Motor connection, phase V |
| C | W | Motor connection, phase W |
| D | 13 | Brake connection 13 |
| E | 14 | Brake connection 14 |
| F | 15 | Brake connection 15 |
| PE | PE | Protective earth connection |
| 1 | Temp+ | Temperature sensor+ connection[1)] |
| 2 | res. | Reserved |
| 3 | Temp - | Connection [1)] |

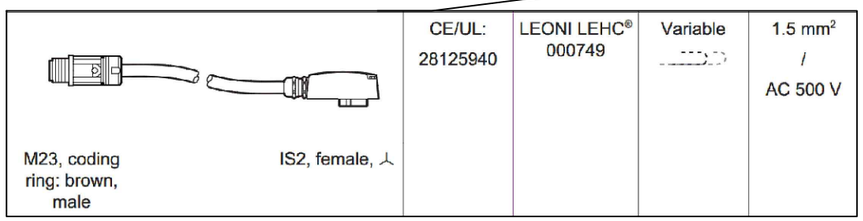



ON MOTOR

X2041 ON VFD

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PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA MOTOR POWER

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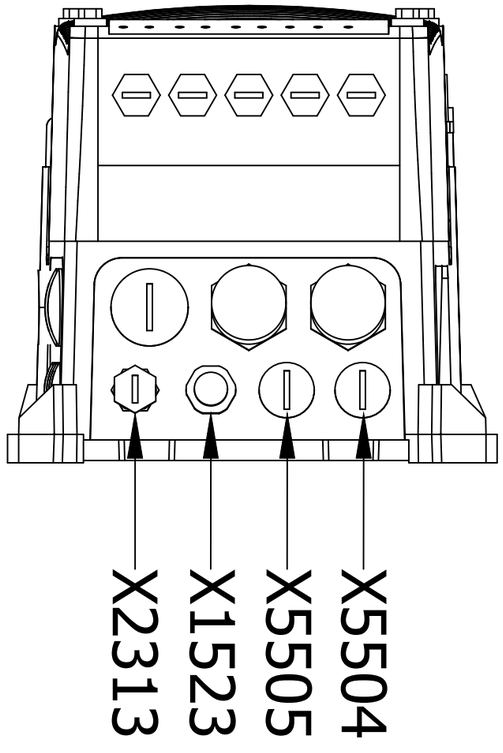
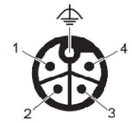



# X1523

| Function |
|---|
| Input DC 24 V backup voltage |

| Connection type |
|---|
| M12, 5-pin, male, L-coded, color: light gray |

**Connection diagram**

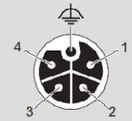


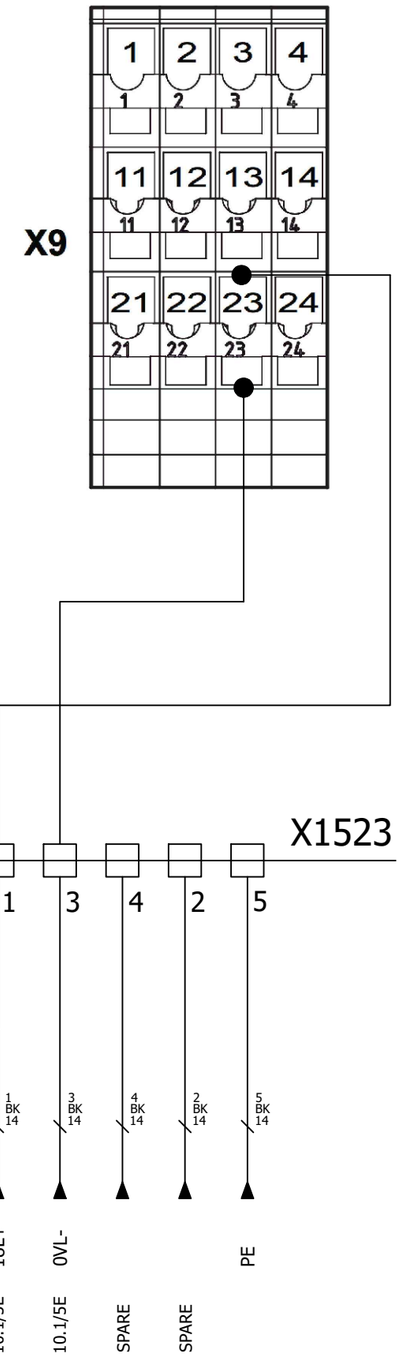

**Assignment**

| Contact | Function | |
|---|---|---|
| 1 | +24V/L1 | DC 24 V input/L1<br>(for backup mode) |
| 2 | 0V24/N2 | 0V24 reference potential/N2<br>(for DC 24 V /BES brake rectifier) |
| 3 | 0V24/N1 | 0V24 reference potential/N1<br>(for backup mode) |
| 4 | +24V/L2 | DC 24 V connection/L2<br>(for DC 24 V /BES brake rectifier) |
| ⏚ | FE | Functional earth |

| Part numbers |
|---|
| 28117786 |

**Assembly**

| Open cable end | | | Description | Prefabricated connector | |
|---|---|---|---|---|---|
| | | | |  | |
| Core color/ core cross section | Identification | Assembly | | Signal | Contact |
| Black 2.5 mm² | 1 | Not prefabricated | DC 24 V output/L1<br>(for backup voltage/supply) | +24V/L1 | 1 |
| Black 2.5 mm² | 2 | Not prefabricated | 0V24 reference potential/N2<br>(for DC 24 V /BES brake rectifier) | 0V24/N2 | 2 |
| Black 2.5 mm² | 3 | Not prefabricated | 0V24 reference potential/N1<br>(for backup voltage/supply) | 0V24/N1 | 3 |
| Black 2.5 mm² | 4 | Not prefabricated | DC 24 V output/L2<br>(for DC 24 V /BES brake rectifier) | +24V/L2 | 4 |
| Black 2.5 mm² | 5 | Not prefabricated | Functional earth | FE | ⏚ |

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PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD 24V DC POWER

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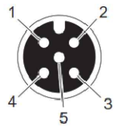
# X5504: STO (3 cores)

| Function |
|---|
| Connection for safe torque off (STO, 3 cores) |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: yellow |

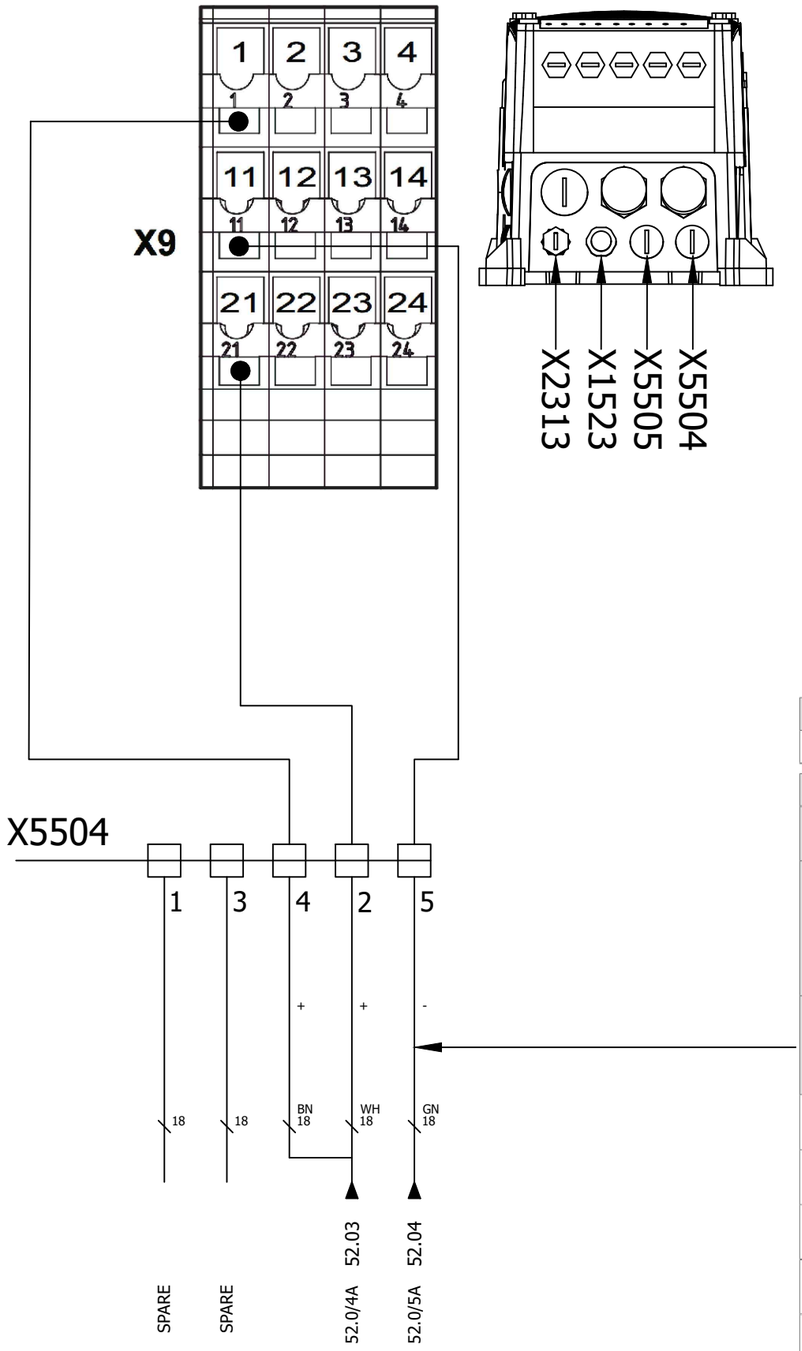
**Connection diagram**




**Assignment**

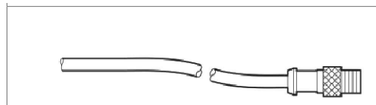
| Contact | Signal | Description |
|---|---|---|
| 1 | 24V_OUT | DC 24 V auxiliary output (permitted only for operation with STO jumper plug) |
| 2 | F_STO_P2 | F_STO_P2 connection |
| 3 | 0V24_OUT | 0V24 reference potential for DC 24 V auxiliary output (permitted only for operation with STO jumper plug) |
| 4 | F_STO_P1 | F_STO_P1 connection |
| 5 | F_STO_M | F_STO_M connection |

Do not use the 24 V output (pins 1 and 3) for safety-related applications.




**Plug connectors**

| Designation | Coding ring/ color | Function |
|---|---|---|
| X5504 | Yellow | STO (3-core connection)[1] |
| X5505 | Yellow | STO (3-core connection)[1] |
| X1523 | Light gray | DC 24 V backup voltage – input[2] |
| X2313 | Light gray | DC 24 V backup voltage – output |
| X5004_1 | Black | Digital inputs[3] |
| X5004_2 | Black | Digital inputs[3] |
| X5004_3 | Black | Digital inputs[3] |

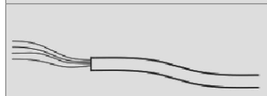



| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110943 | LiYCY | | |

**Part numbers**

28110978, 28110943

**Assembly**

| Open cable end | | | Description | Prefabricated plug connectors | |
|---|---|---|---|---|---|
|  | | | | | |
| Core color/ core cross section | Identi-fication | Assembly | | Signal | Contact |
| [1] | – | Not pre-fabricated | DC 24 V auxiliary output | 24V_OUT | 1 |
| White 0.75 mm² | – | Not pre-fabricated | F_STO_P2 connection | F_STO_P2 | 2 |
| [1] | – | Not pre-fabricated | 0V24 reference potential for DC 24 V auxiliary output | 0V24_OUT | 3 |
| Brown 0.75 mm² | – | Not pre-fabricated | F_STO_P1 connection | F_STO_P1 | 4 |
| Green 0.75 mm² | – | Not pre-fabricated | F_STO_M connection | F_STO_M | 5 |

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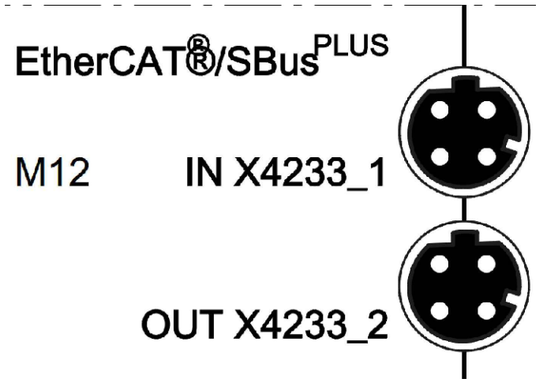
SHEET 16.5 OF 56




**X4233_1: EtherCAT®/SBus^PLUS interface, IN**

The following table shows information about this connection:

| Function |
|---|
| EtherCAT®/SBus^PLUS interface, IN |

| Connection type |
|---|
| M12, 4-pin, female, D-coded, color: black |

| Connection diagram |
|---|
| |



| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

**X4233_2: EtherCAT®/SBus^PLUS interface, OUT**

The following table shows information about this connection:

| Function |
|---|
| EtherCAT®/SBus^PLUS interface, OUT |

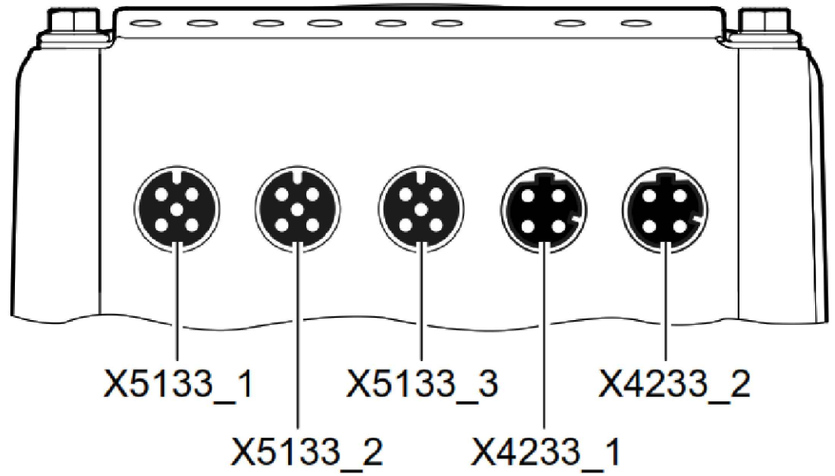
| Connection type |
|---|
| M12, 4-pin, female, D-coded, color: black |

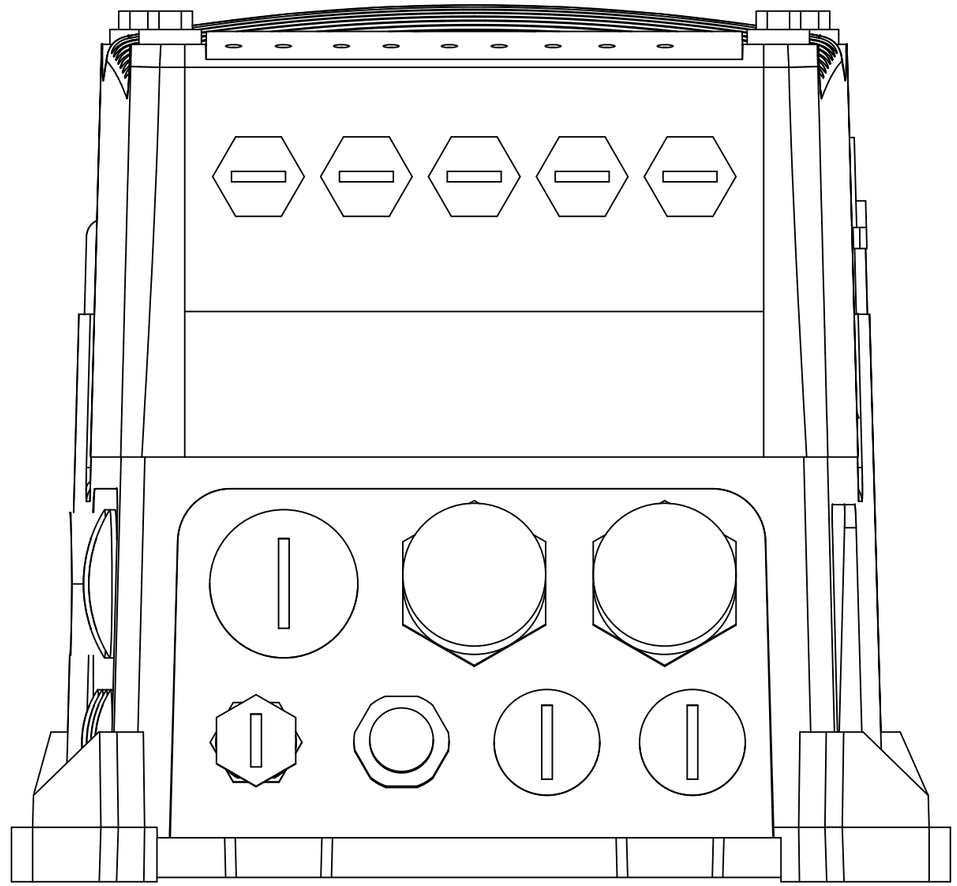
| Connection diagram |
|---|
| |



| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |



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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD EtherCAT INTERFACE

| REV | DESCRIPTION | DATE |
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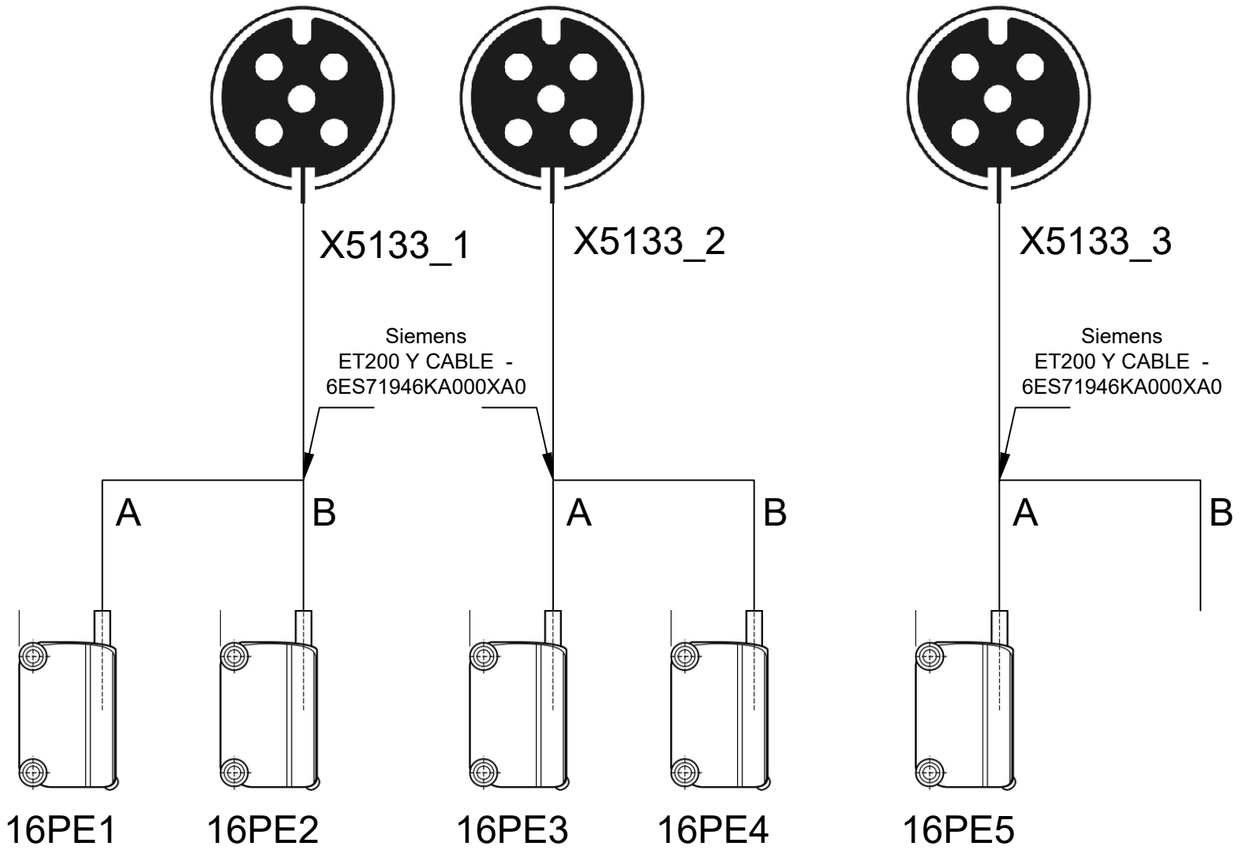
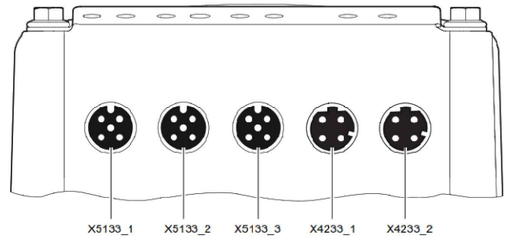
SHEET 16.6 OF 56




**Plug connector positions at the DSI.. electronics cover**

The following figure shows an example of the plug connector positions:




*18014420217585035*

| Plug connector | Function |
|---|---|
| X5133_1[1)] | Digital inputs DI01 and DI02 |
| X5133_2[1)] | Digital inputs DI03 and DI04 |
| X5133_3[1)] | Digital inputs/outputs DIO01 and DIO02 |

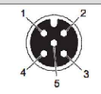
**X5133_1: Digital inputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**




**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor supply |
| 2 | DI02 | Sensor input DI02 |
| 3 | 0V24 | 0V24 reference potential for sensors |
| 4 | DI01 | Sensor input DI01 |
| 5 | FE | Functional earth |

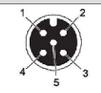
**X5133_2: Digital inputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**




**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor supply |
| 2 | DI04 | Sensor input DI04 |
| 3 | 0V24 | 0V24 reference potential for sensors |
| 4 | DI03 | Sensor input DI03 |
| 5 | FE | Functional earth |

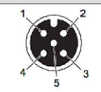
**X5133_3: Digital inputs/outputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs/outputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**




**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor/actuator supply |
| 2 | DIO02 | Sensor input DIO02/actuator output DIO02 |
| 3 | 0V24 | 0V24 reference potential for sensors/actuators |
| 4 | DIO01 | Sensor input DIO01/actuator output DIO01 |
| 5 | FE | Functional earth |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

| REV | DESCRIPTION | DATE |
|---|---|---|
| | | |
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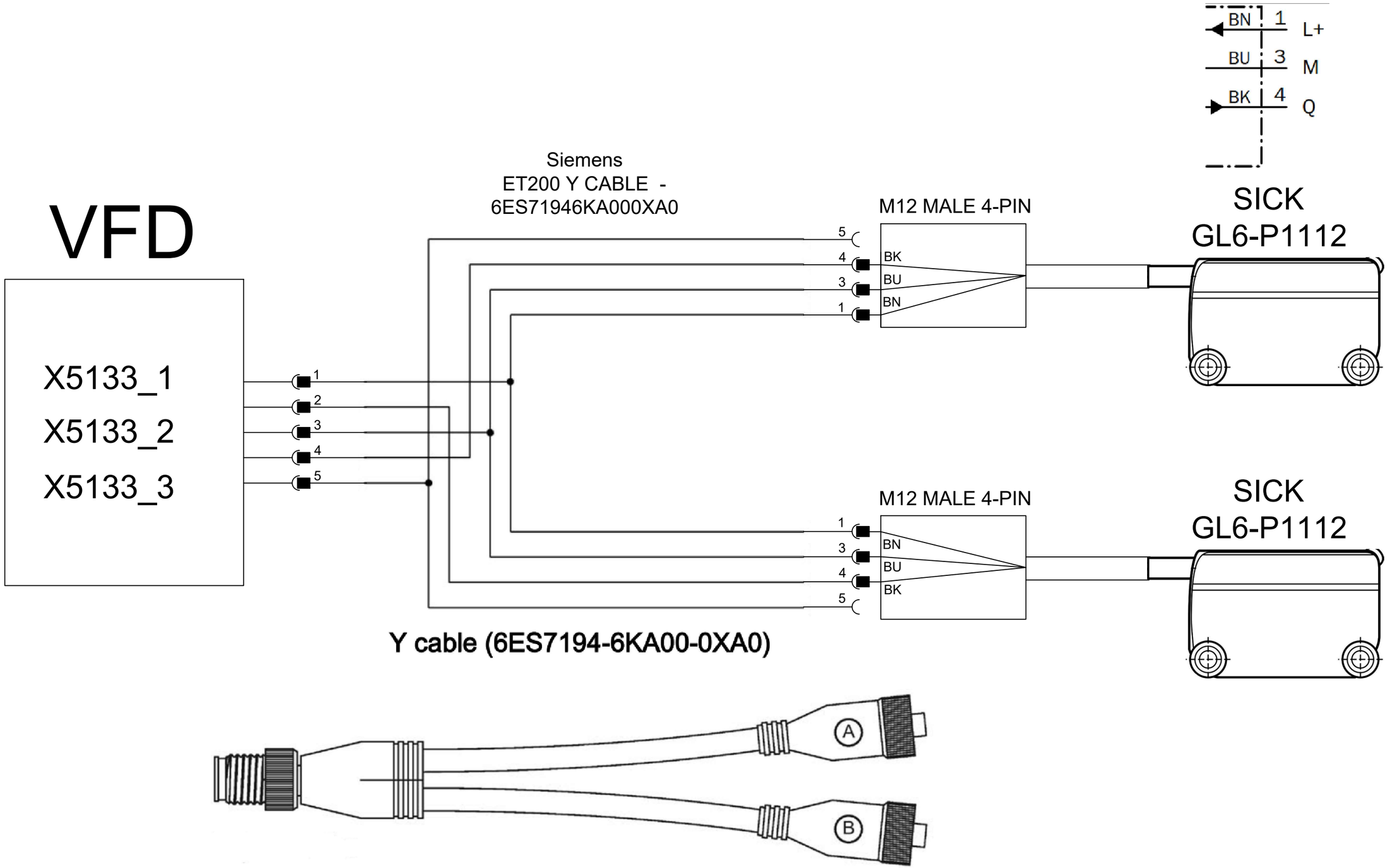
TOTAL PAGES: 175

SHEET 16.7 OF 56




Y cable (6ES7194-6KA00-0XA0)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

| REV | DESCRIPTION | DATE |
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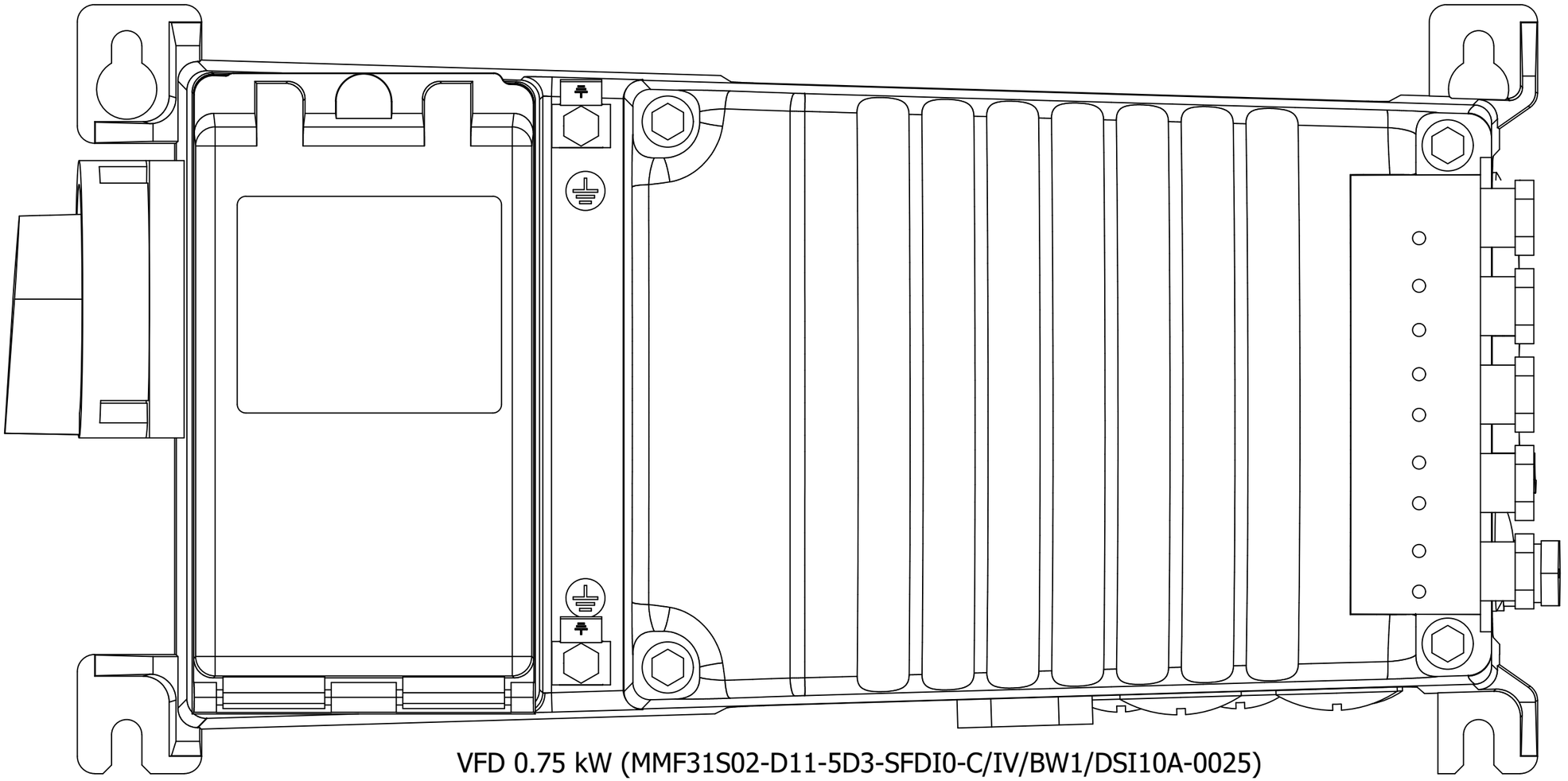
TOTAL PAGES: 175

SHEET 16.8 OF 56



VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

BUFFER CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 3 VFD's

| 17VFD1 | 17VFD2 | 17VFD3 |
|---|---|---|
| CONVEYOR WITH STOPPER LEVEL 1 | CONVEYOR WITH STOPPER LEVEL 2 | CONVEYOR WITH STOPPER LEVEL 3 |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1

| REV | DESCRIPTION | DATE |
|---|---|---|
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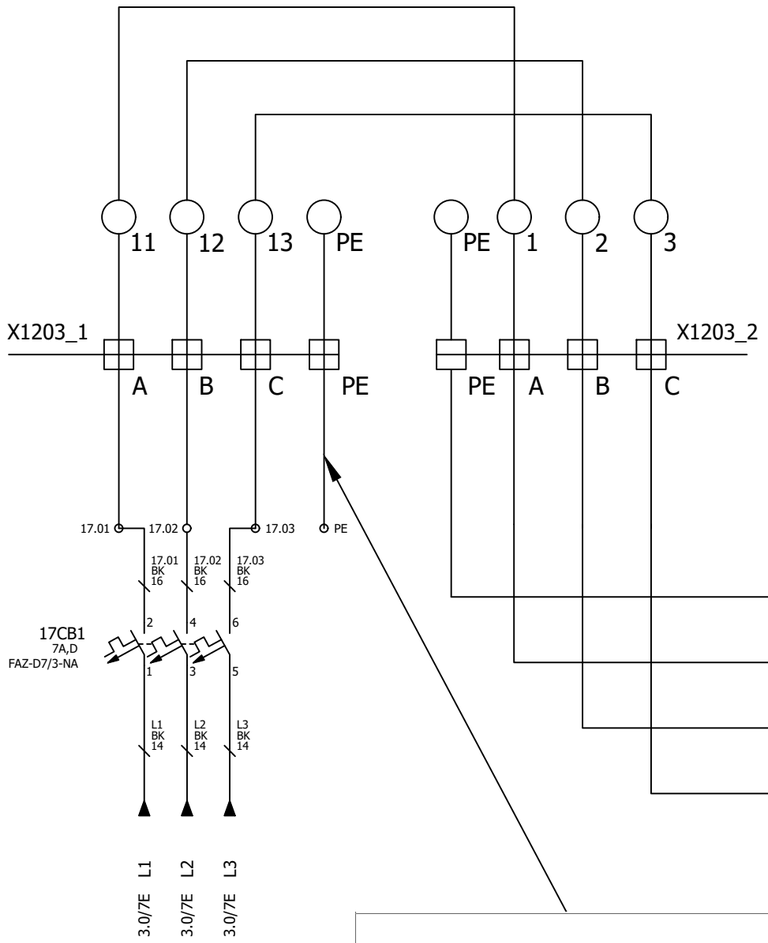
SHEET 17.0 OF 56

# CONVEYOR WITH STOPPER LEVEL 1

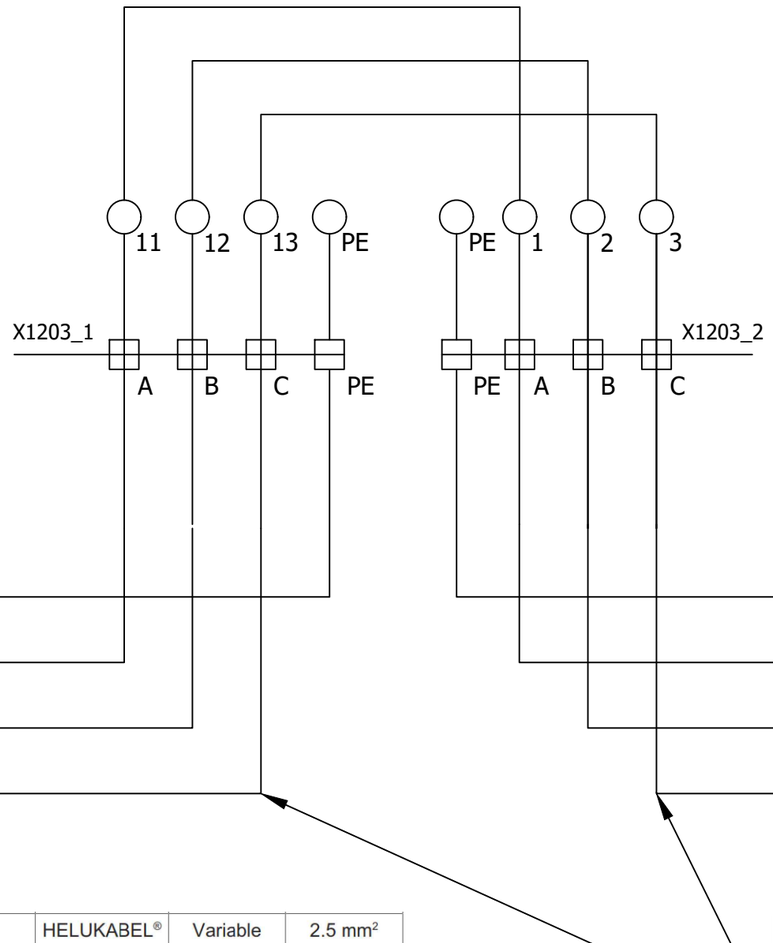
## 17VFD1




| | | | | |
|---|---|---|---|---|
| Open — M23, coding ring: black, male | UL: 18153283 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

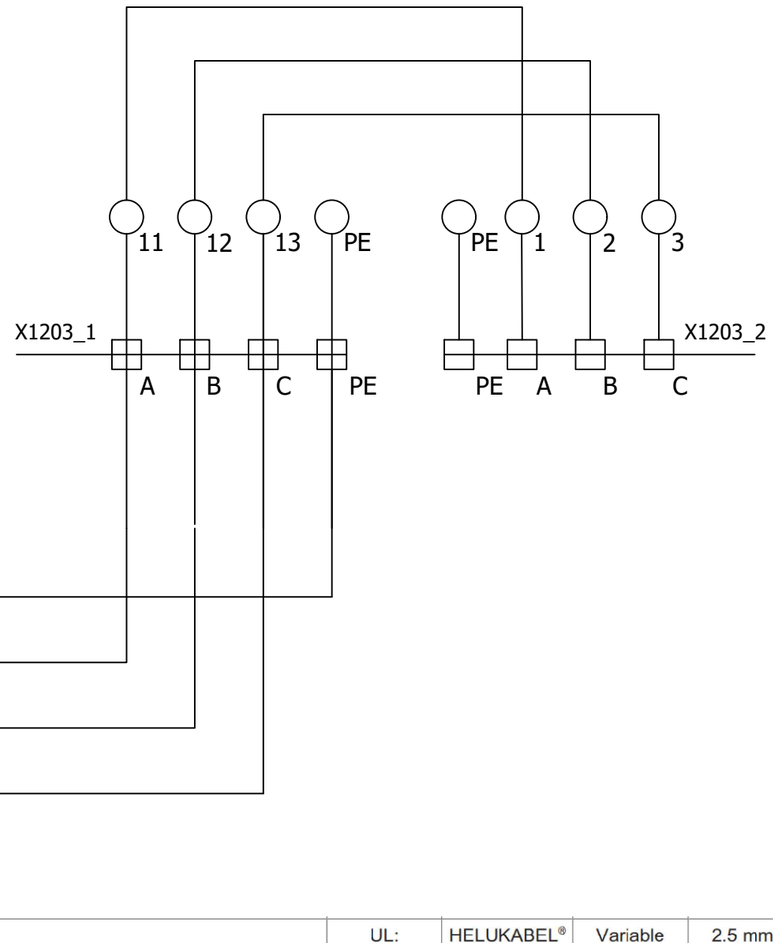
# CONVEYOR WITH STOPPER LEVEL 2

## 17VFD2




# CONVEYOR WITH STOPPER LEVEL 3

## 17VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: black, male — M23, coding ring: black, male | UL: 18153267 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 POWER

| REV | DESCRIPTION | DATE |
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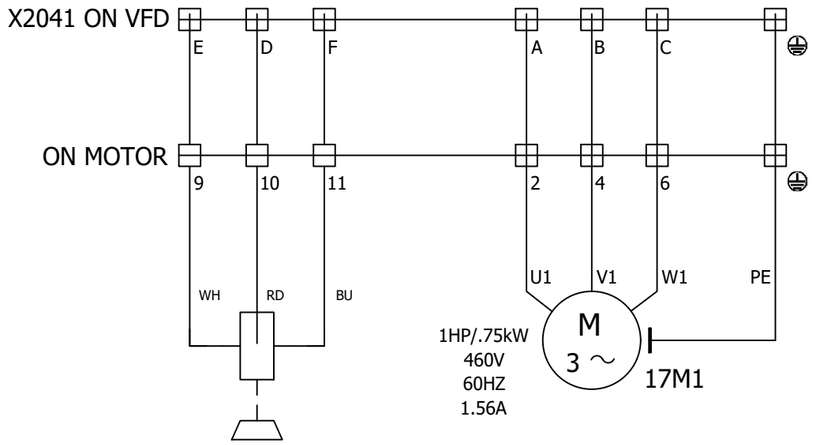
SHEET 17.1 OF 56

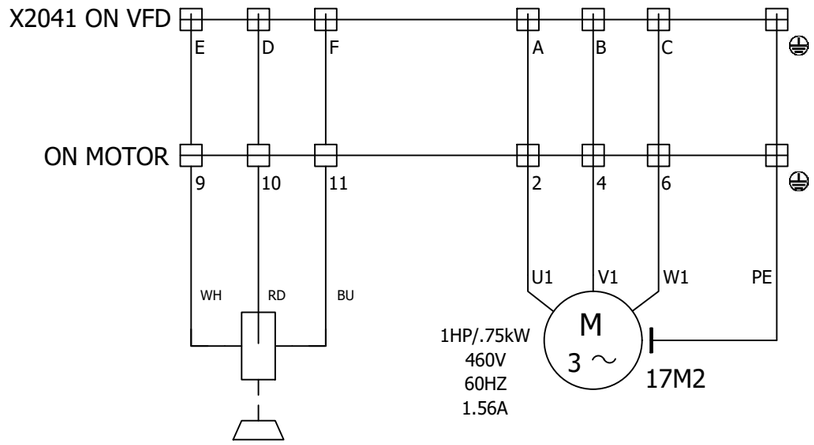
# CONVEYOR WITH STOPPER LEVEL 1

## 17VFD1



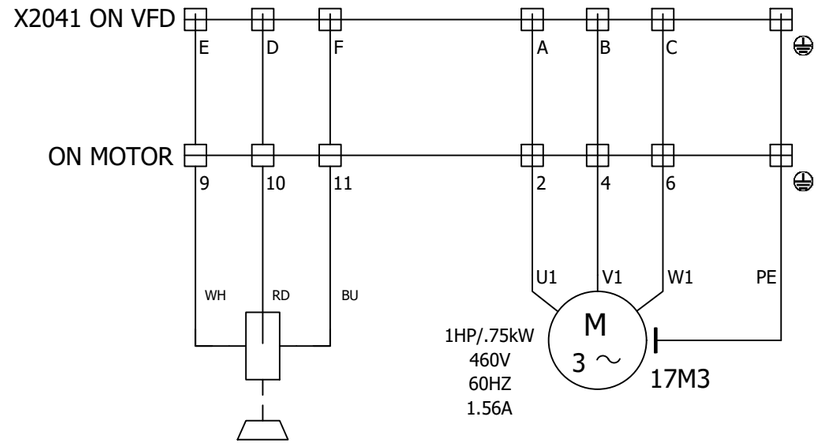

# CONVEYOR WITH STOPPER LEVEL 2

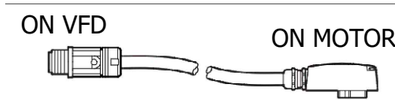
## 17VFD2




# CONVEYOR WITH STOPPER LEVEL 3

## 17VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: brown, male | IS2, female, ⅄ | CE/UL: 28125940 | LEONI LEHC® 000749 | Variable | 1.5 mm² / AC 500 V |

## Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

| REV | DESCRIPTION | DATE |
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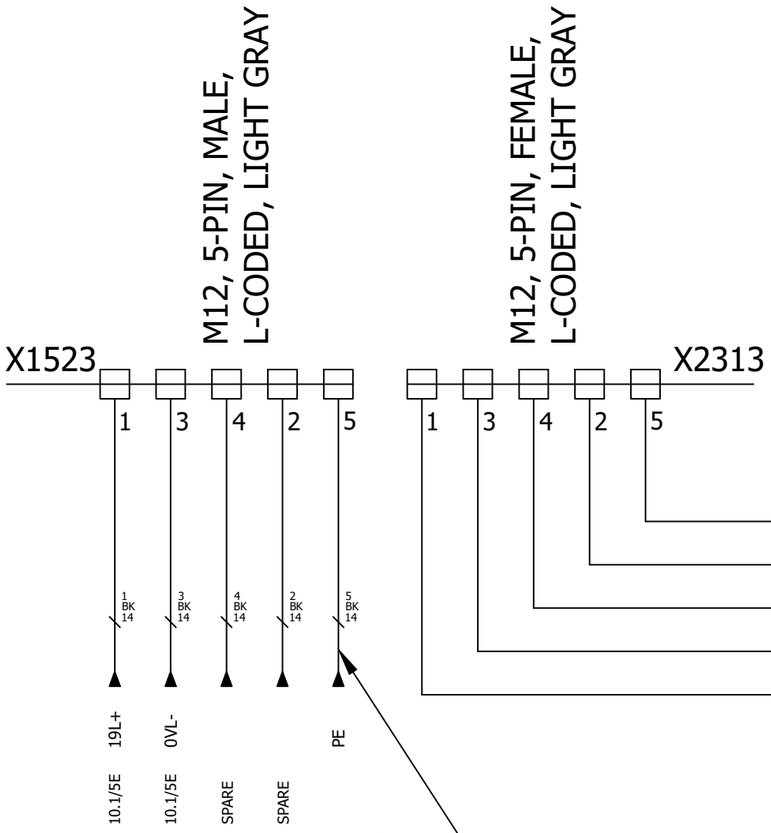
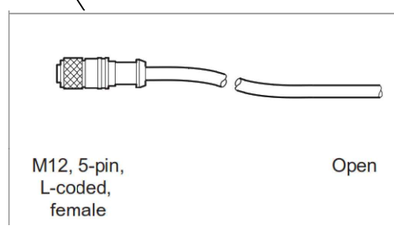
SHEET 17.2 OF 56

# CONVEYOR WITH STOPPER LEVEL 1

## 17VFD1



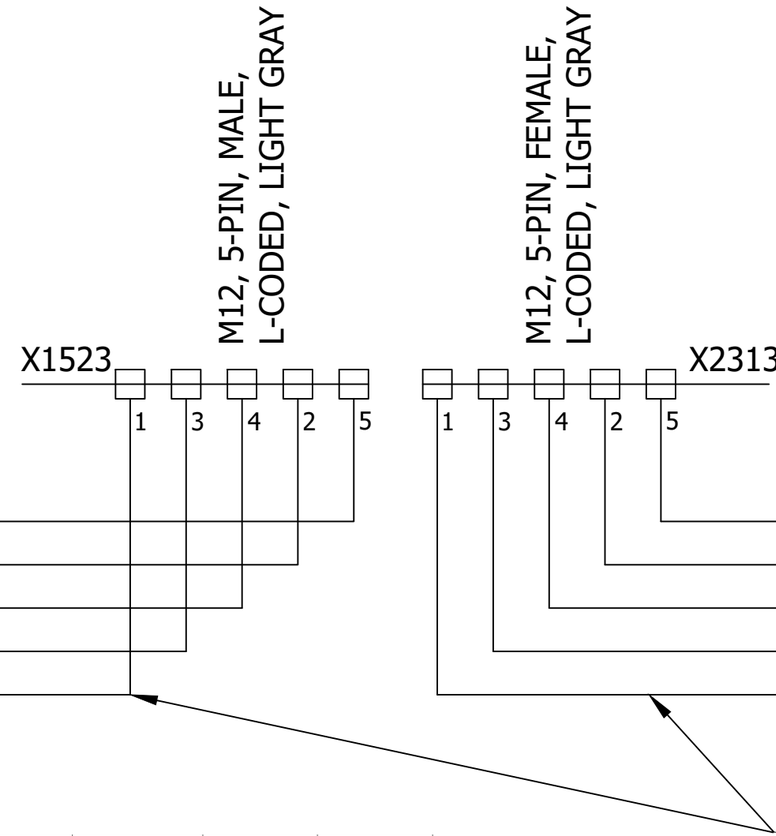



M12, 5-pin, L-coded, female — Open

| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28117786 | JZ-500 | ✕ | |

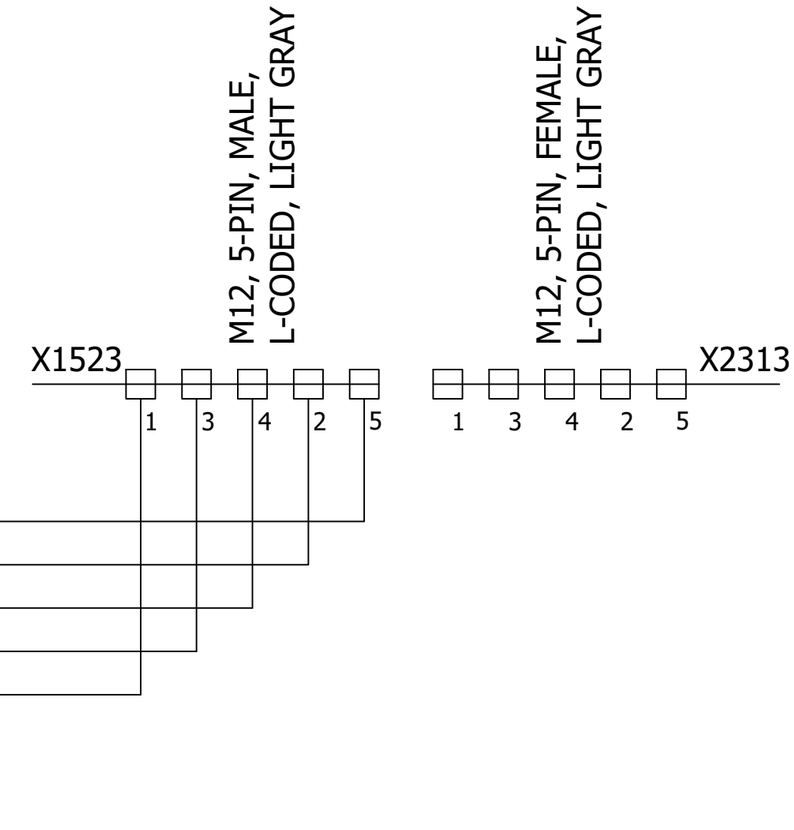
# CONVEYOR WITH STOPPER LEVEL 2

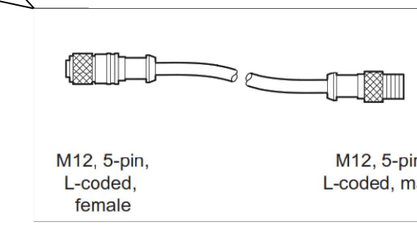
## 17VFD2




# CONVEYOR WITH STOPPER LEVEL 3

## 17VFD3



M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28114345 | JZ-500 | ✕ | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER

| REV | DESCRIPTION | DATE |
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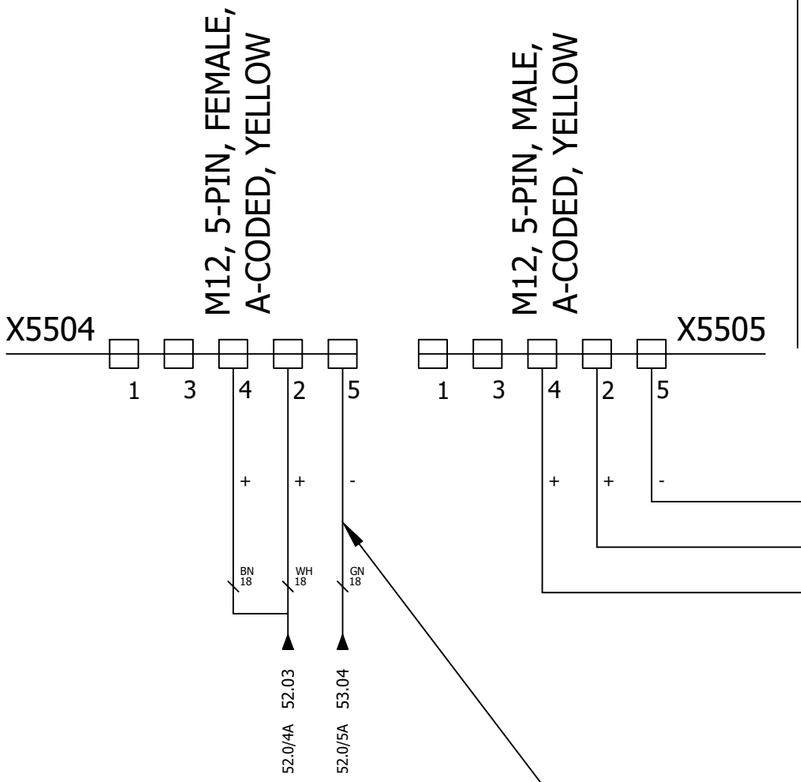
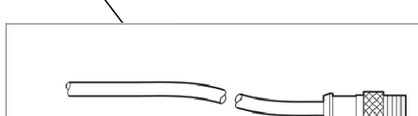
TOTAL PAGES: 175

SHEET 17.3 OF 56

# CONVEYOR WITH STOPPER LEVEL 1

## 17VFD1

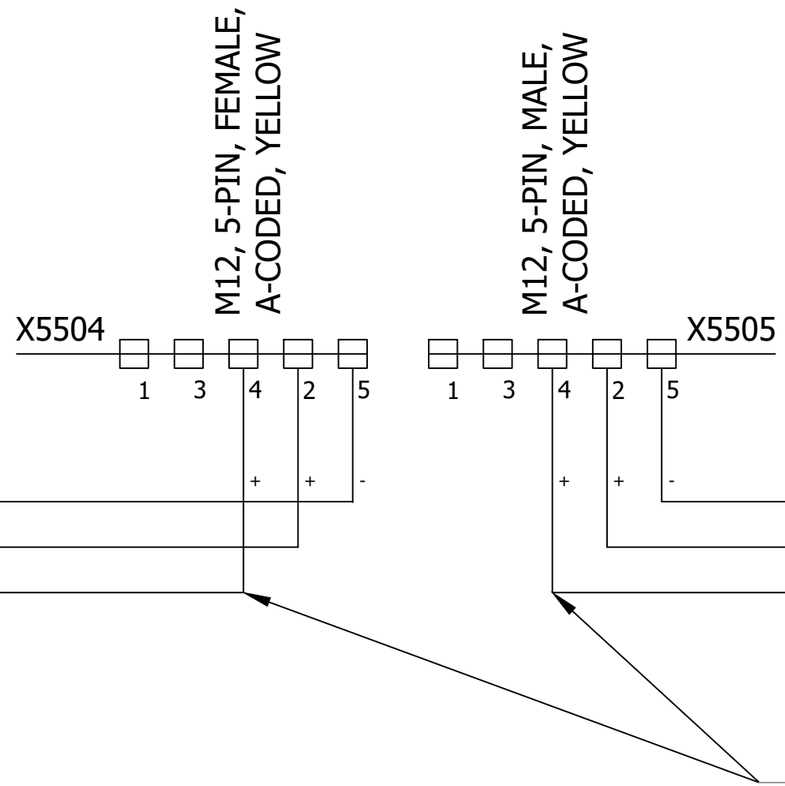


| Open | M12, 5-pin, A-coded, male | CE/UL: 28110943 | HELUKABEL® LiYCY | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|---|---|

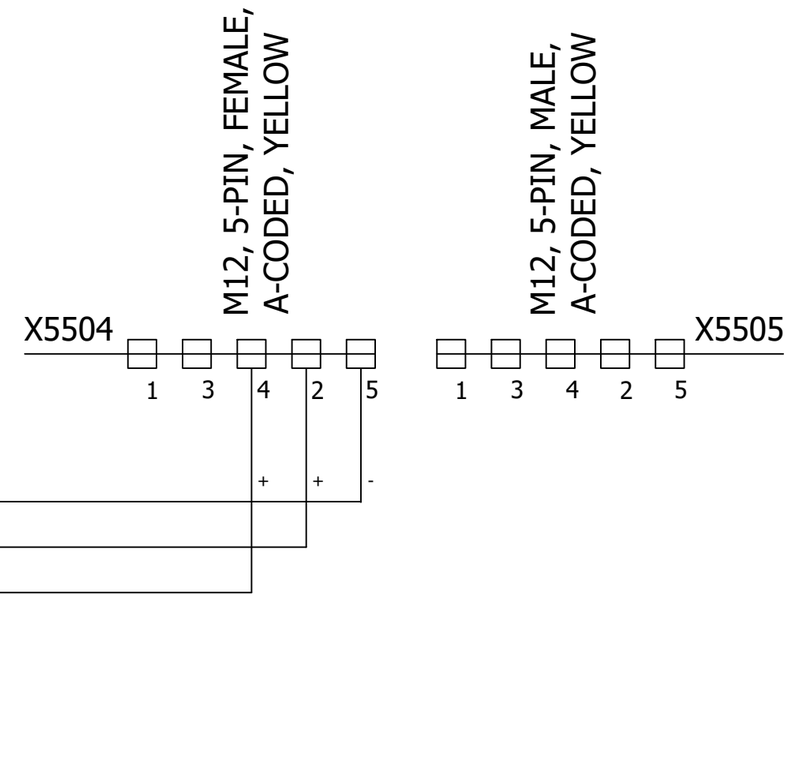
# CONVEYOR WITH STOPPER LEVEL 2

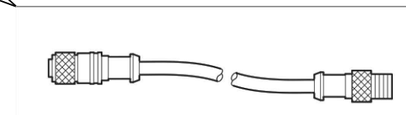
## 17VFD2




# CONVEYOR WITH STOPPER LEVEL 3

## 17VFD3



| M12, 5-pin, A-coded, female | M12, 5-pin, A-coded, male | CE/UL: 28110935 | HELUKABEL® LiYCY | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|---|---|

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD STO

| REV | DESCRIPTION | DATE |
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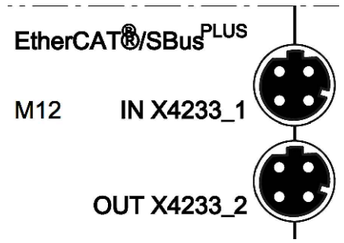
TOTAL PAGES: 175

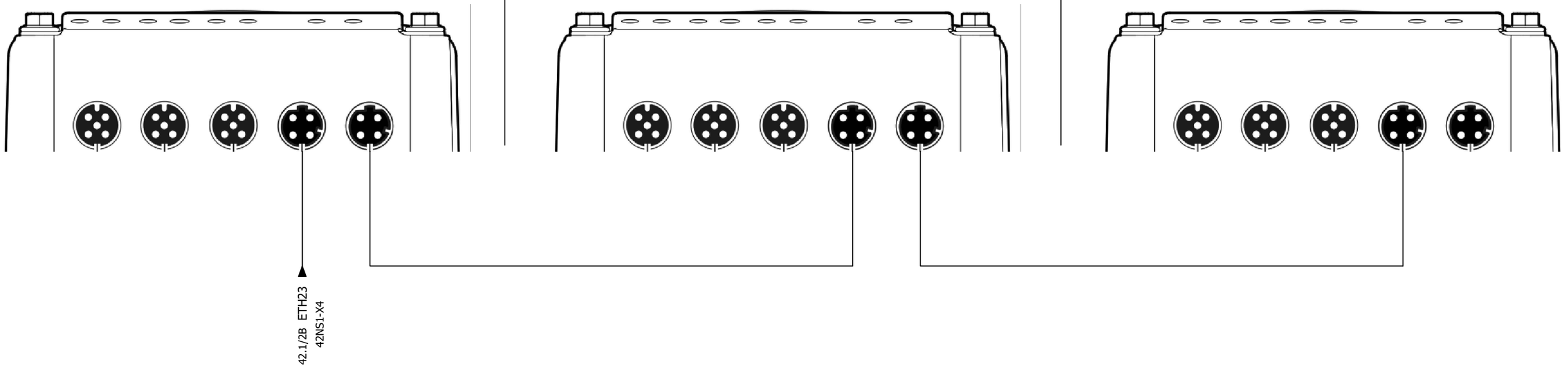
SHEET 17.4 OF 56

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

## 17VFD1




X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK




CONVEYOR
WITH STOPPER
LEVEL 1

## 17VFD2

CONVEYOR
WITH STOPPER
LEVEL 2

## 17VFD3

CONVEYOR
WITH STOPPER
LEVEL 3

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

| REV | DESCRIPTION | DATE |
|---|---|---|
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|---|---|---|
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| DATE | DATE | CURRENT REVISION: |
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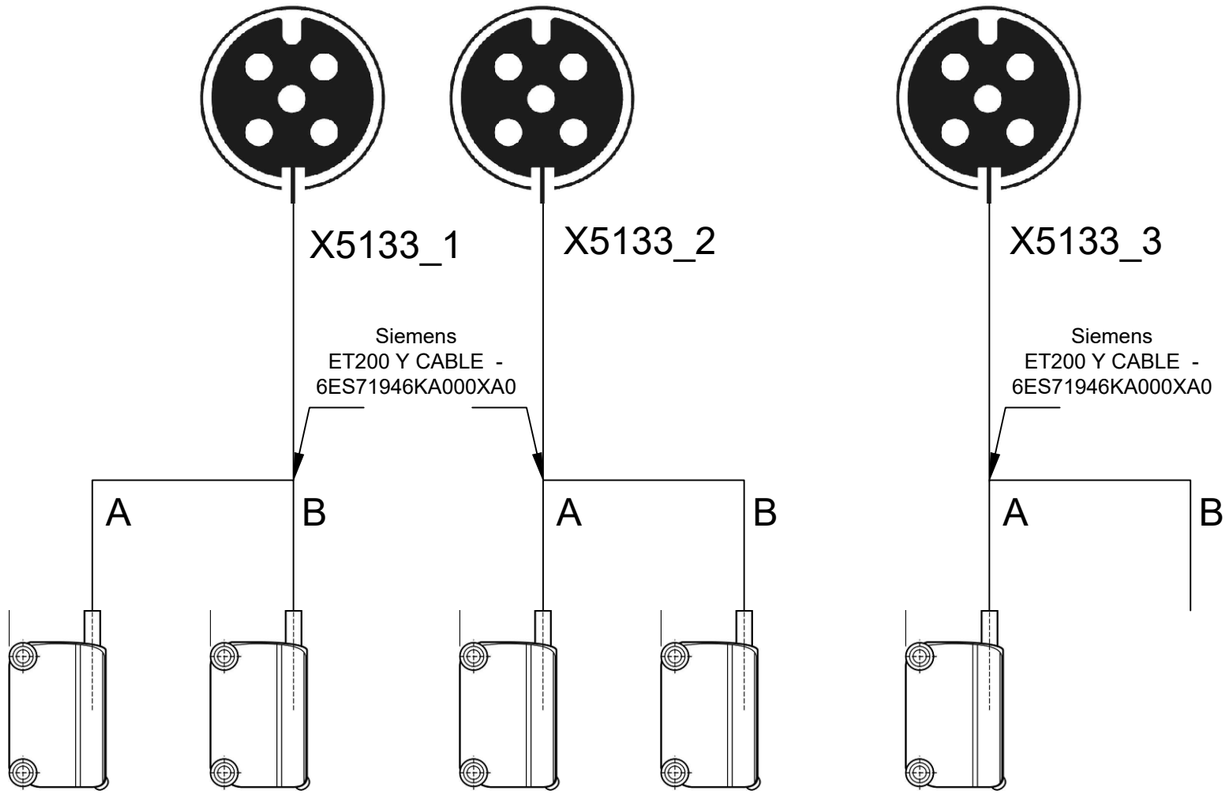
TOTAL PAGES: 175

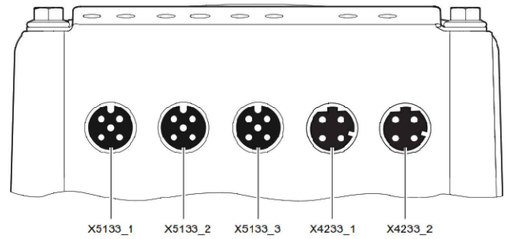
SHEET 17.5 OF 56




| | | | | | |
|---|---|---|---|---|---|
| 17PE1 | 17PE2 | 17PE3 | 17PE4 | 17PE5 | TO 17VFD1 |
| 17PE7 | 17PE8 | 17PE9 | 17PE10 | 17PE11 | TO 17VFD2 |
| 17PE13 | 17PE14 | 17PE15 | 17PE16 | 17PE17 | TO 17VFD3 |

**Plug connector positions at the DSI.. electronics cover**

The following figure shows an example of the plug connector positions:




*18014420217585035*

| Plug connector | Function |
|---|---|
| X5133_1[1)] | Digital inputs DI01 and DI02 |
| X5133_2[1)] | Digital inputs DI03 and DI04 |
| X5133_3[1)] | Digital inputs/outputs DIO01 and DIO02 |

**X5133_1: Digital inputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**

**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor supply |
| 2 | DI02 | Sensor input DI02 |
| 3 | 0V24 | 0V24 reference potential for sensors |
| 4 | DI01 | Sensor input DI01 |
| 5 | FE | Functional earth |

**X5133_2: Digital inputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**

**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor supply |
| 2 | DI04 | Sensor input DI04 |
| 3 | 0V24 | 0V24 reference potential for sensors |
| 4 | DI03 | Sensor input DI03 |
| 5 | FE | Functional earth |

**X5133_3: Digital inputs/outputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs/outputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**

**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor/actuator supply |
| 2 | DIO02 | Sensor input DIO02/actuator output DIO02 |
| 3 | 0V24 | 0V24 reference potential for sensors/actuators |
| 4 | DIO01 | Sensor input DIO01/actuator output DIO01 |
| 5 | FE | Functional earth |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

| REV | DESCRIPTION | DATE |
|---|---|---|
| | | |
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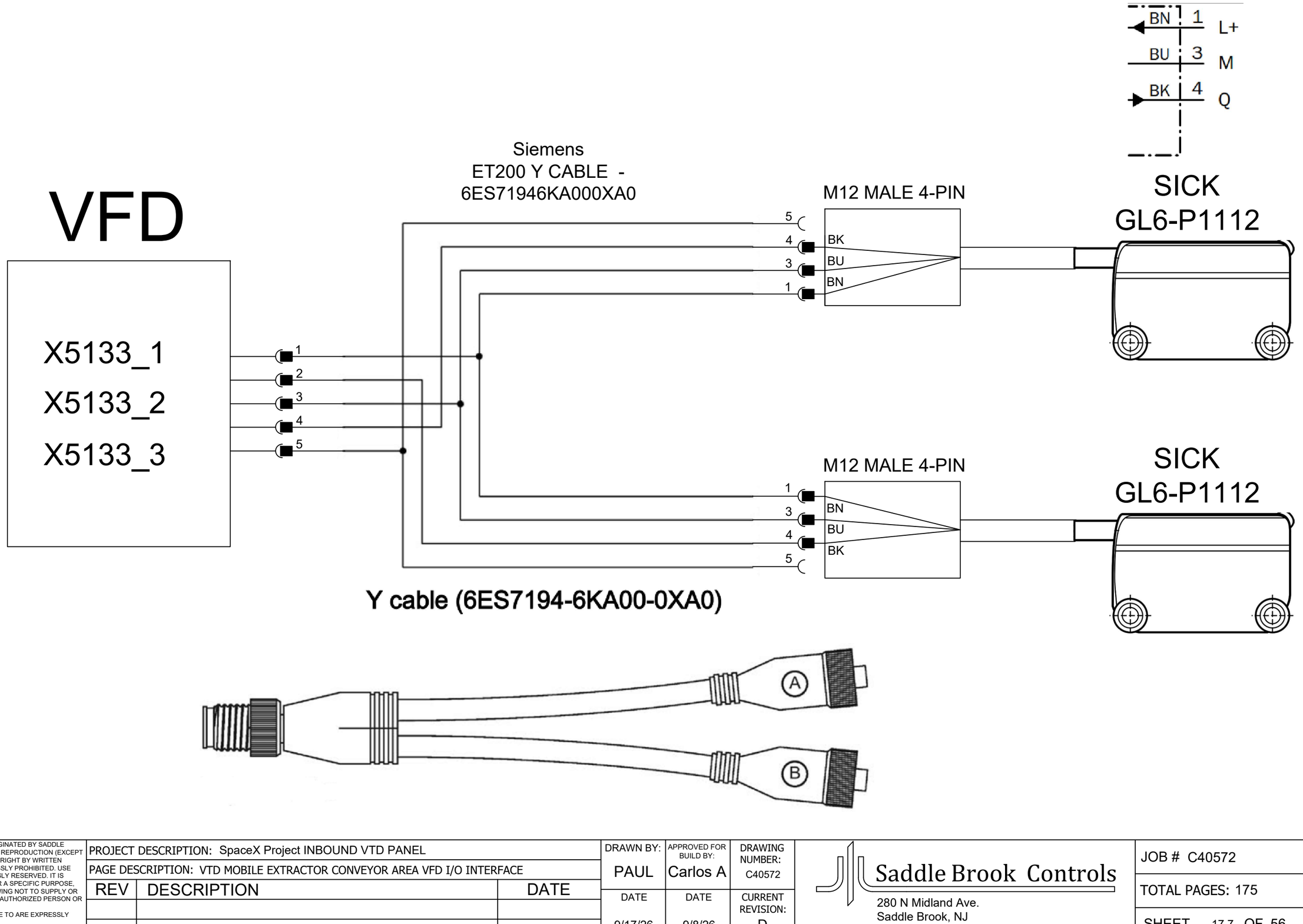
TOTAL PAGES: 175

SHEET 17.6 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

| REV | DESCRIPTION | DATE |
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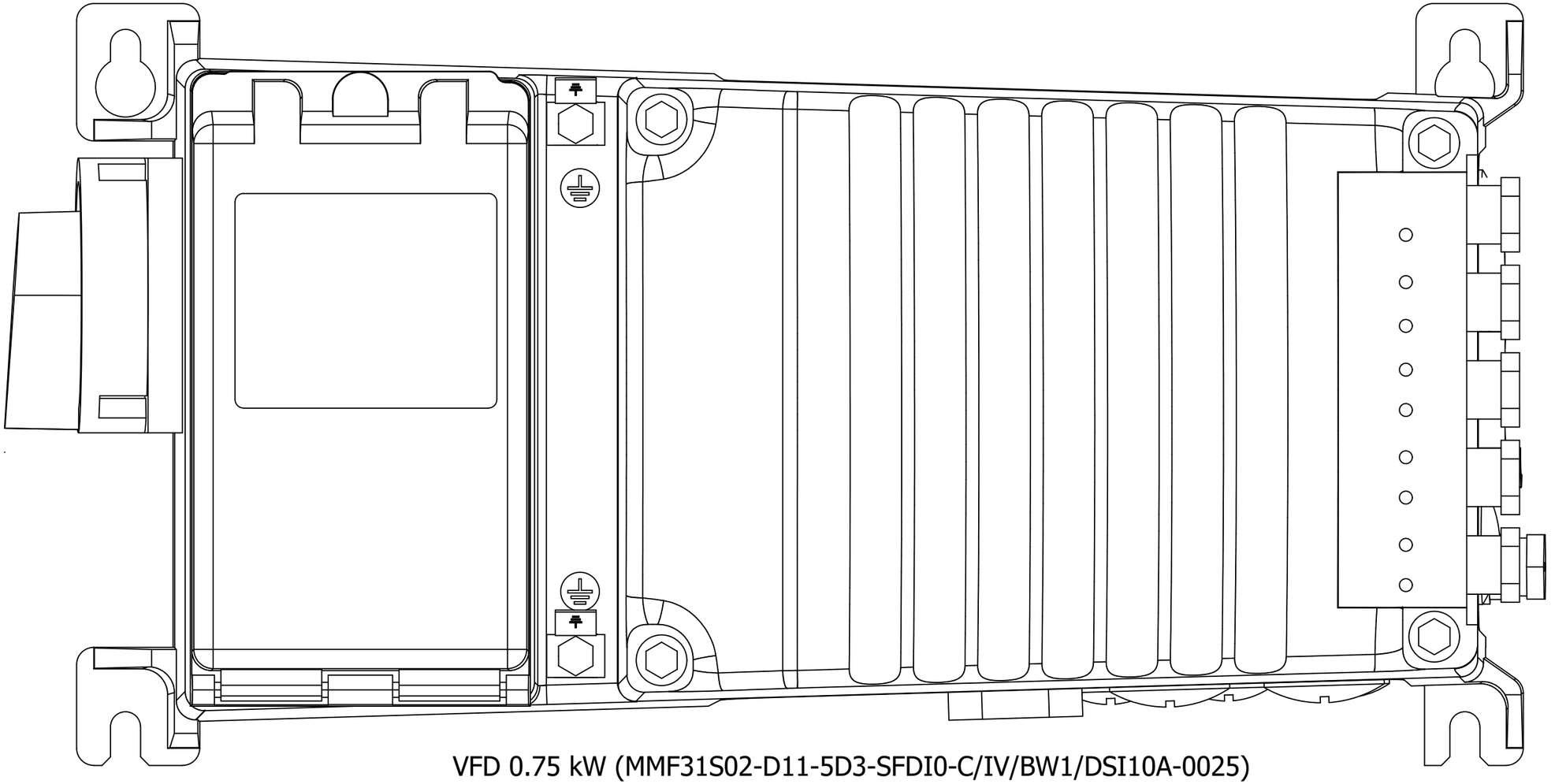
SHEET 17.7 OF 56



VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

BUFFER CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

## 18VFD1
CONVEYOR
WITH STOPPER
LEVEL 4

## 18VFD2
CONVEYOR
WITH STOPPER
LEVEL 5

## 18VFD3
CONVEYOR
WITH STOPPER
LEVEL 6

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2

| REV | DESCRIPTION | DATE |
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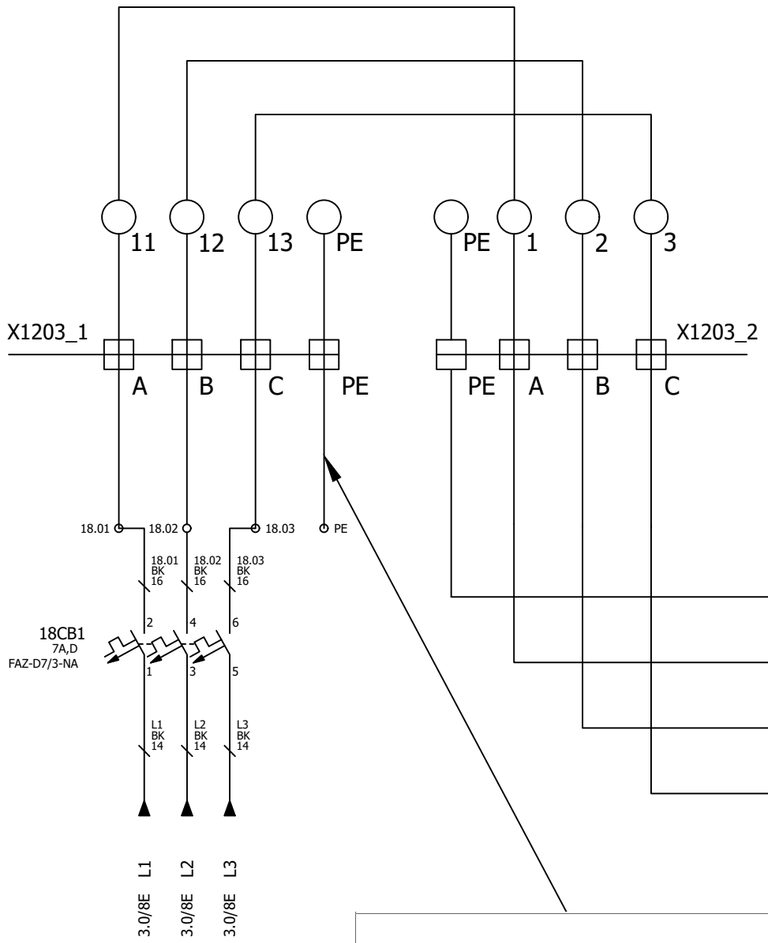
SHEET 18.0 OF 56

# CONVEYOR WITH STOPPER LEVEL 4

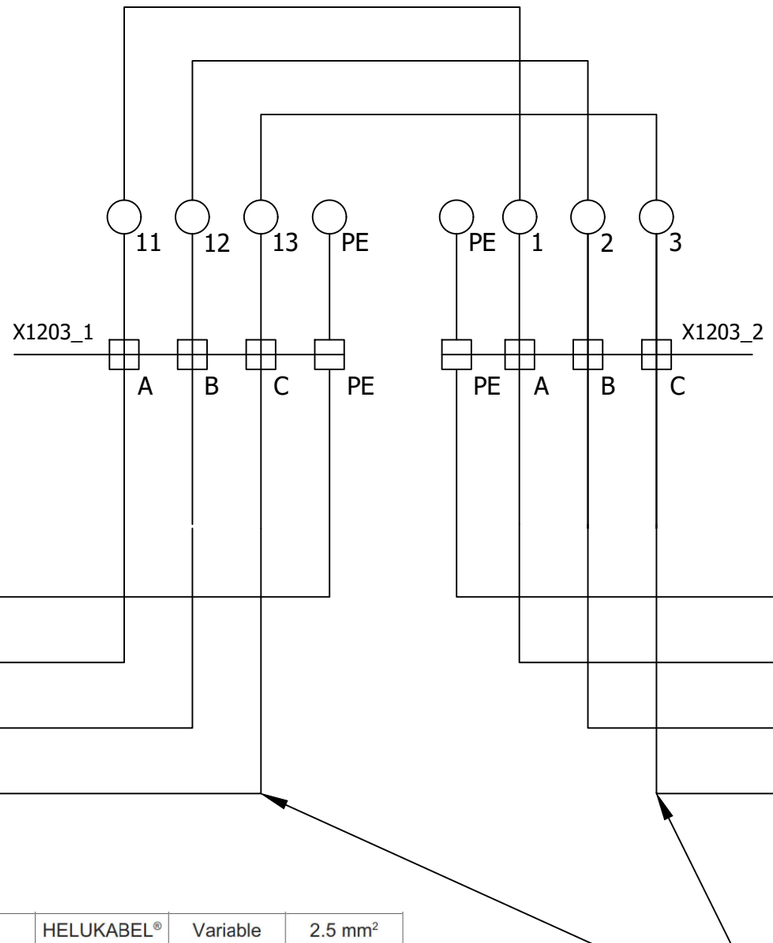
## 18VFD1




| | | | | |
|---|---|---|---|---|
| Open — M23, coding ring: black, male | UL: 18153283 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

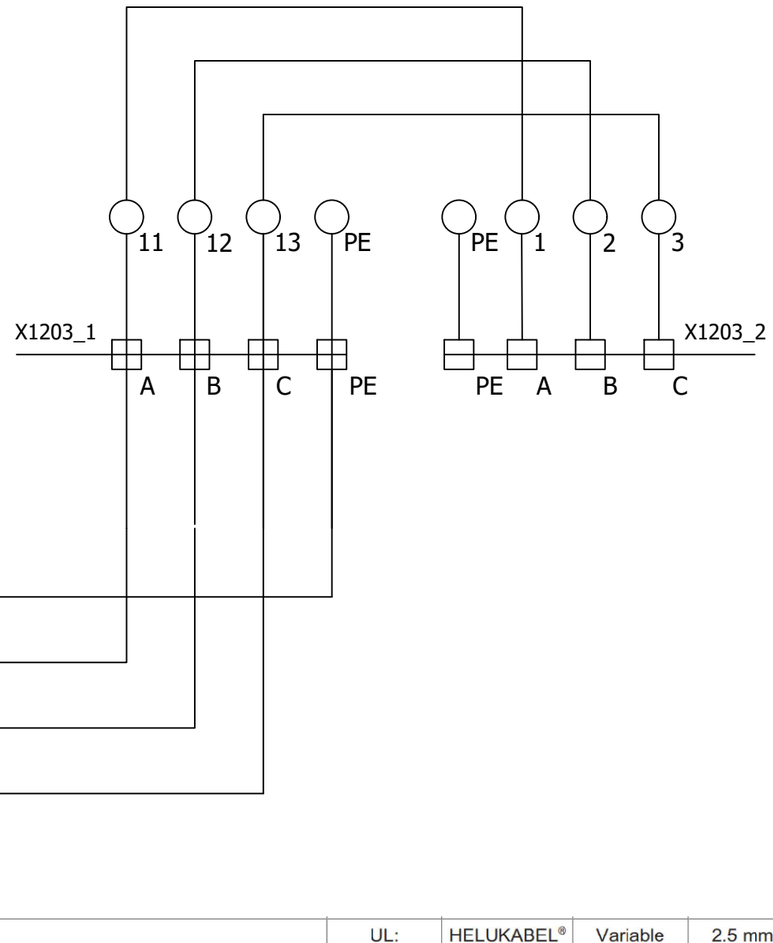
# CONVEYOR WITH STOPPER LEVEL 5

## 18VFD2




# CONVEYOR WITH STOPPER LEVEL 6

## 18VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: black, male — M23, coding ring: black, male | UL: 18153267 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 POWER

| REV | DESCRIPTION | DATE |
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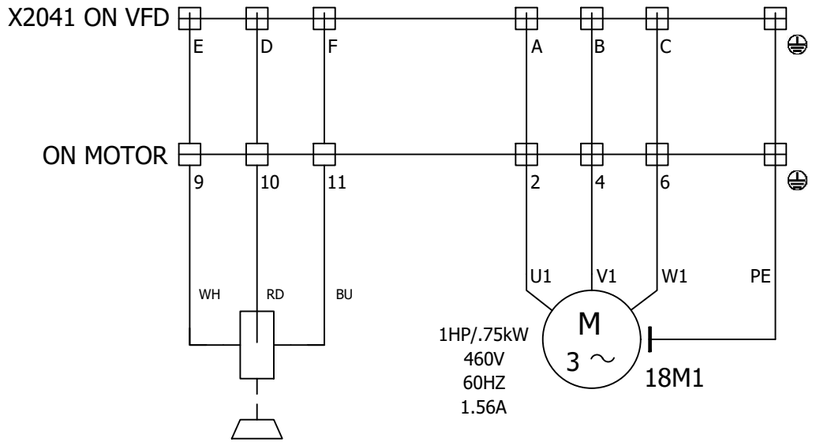
TOTAL PAGES: 175

SHEET 18.1 OF 56

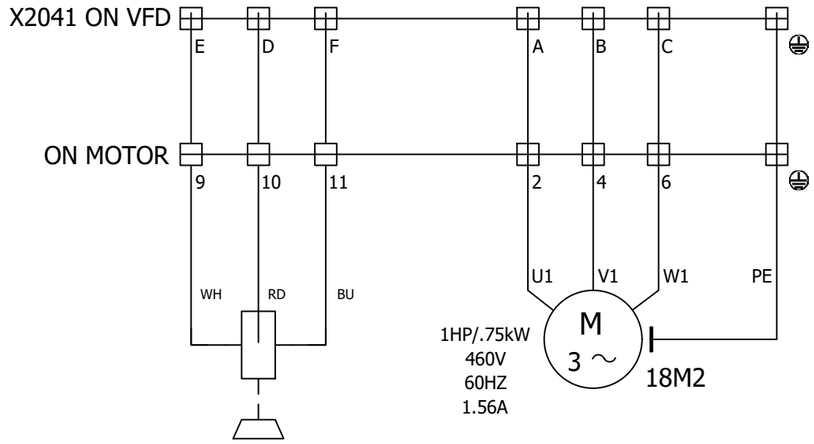
# CONVEYOR WITH STOPPER LEVEL 4

## 18VFD1



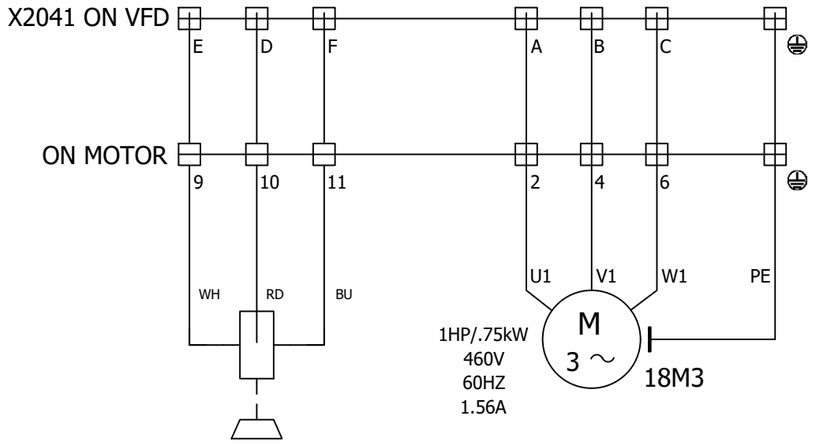

# CONVEYOR WITH STOPPER LEVEL 5

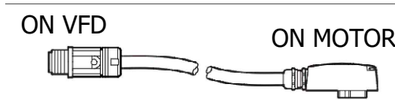
## 18VFD2




# CONVEYOR WITH STOPPER LEVEL 6

## 18VFD3




| CE/UL: | LEONI LEHC® | Variable | 1.5 mm² |
|---|---|---|---|
| 28125940 | 000749 | | / AC 500 V |

## Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 MOTOR POWER

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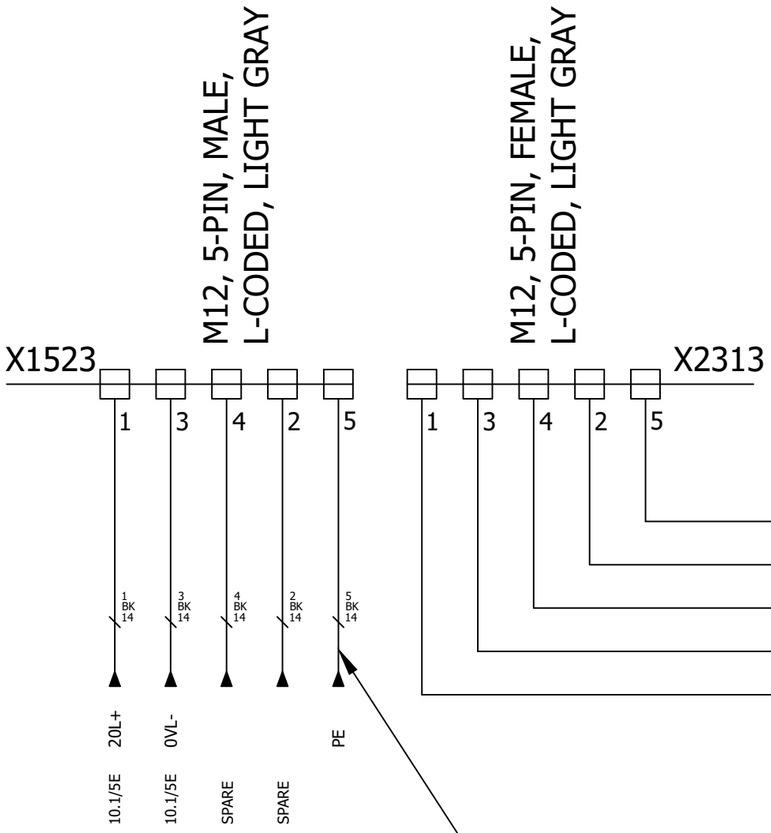
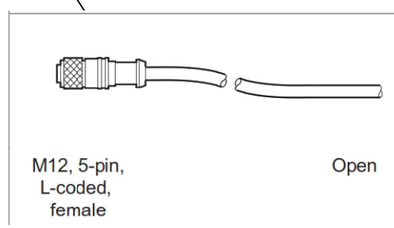
TOTAL PAGES: 175

SHEET 18.2 OF 56

# CONVEYOR WITH STOPPER LEVEL 4

## 18VFD1



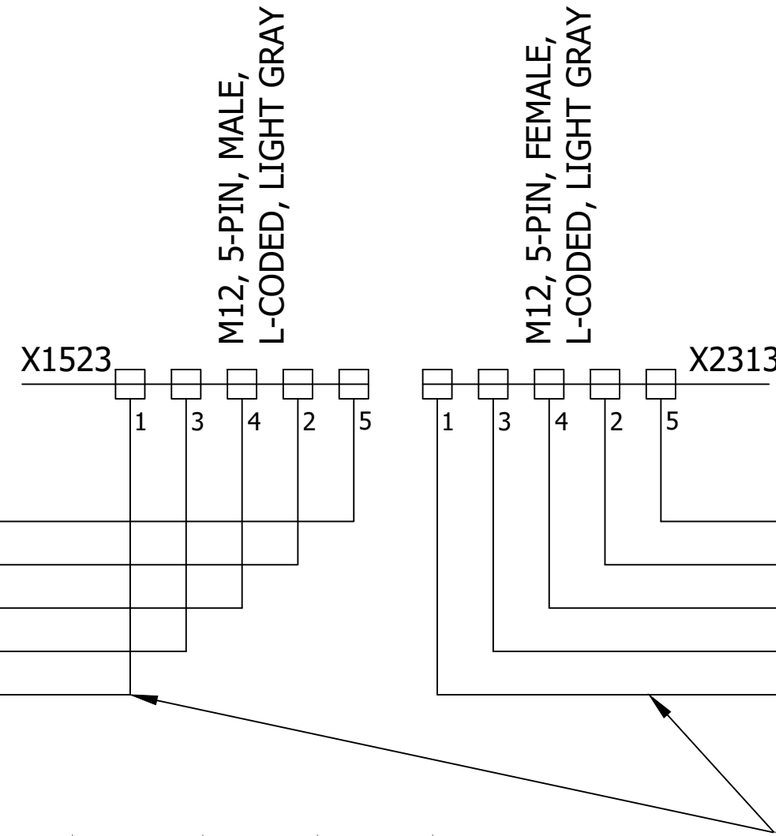



M12, 5-pin, L-coded, female — Open

| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28117786 | JZ-500 | ✕ | |

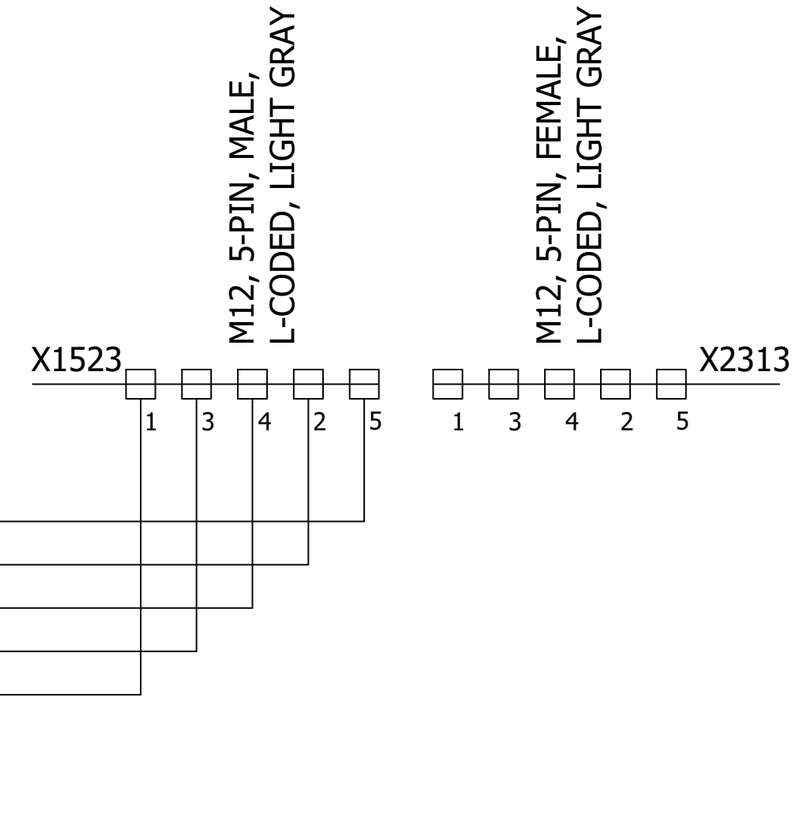
# CONVEYOR WITH STOPPER LEVEL 5

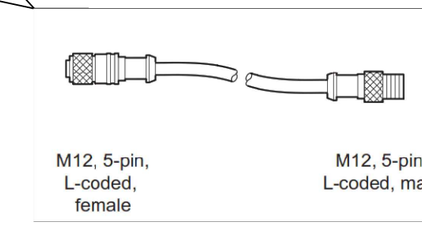
## 18VFD2




# CONVEYOR WITH STOPPER LEVEL 6

## 18VFD3



M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28114345 | JZ-500 | ✕ | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 VFD 24VDC POWER

| REV | DESCRIPTION | DATE |
|---|---|---|
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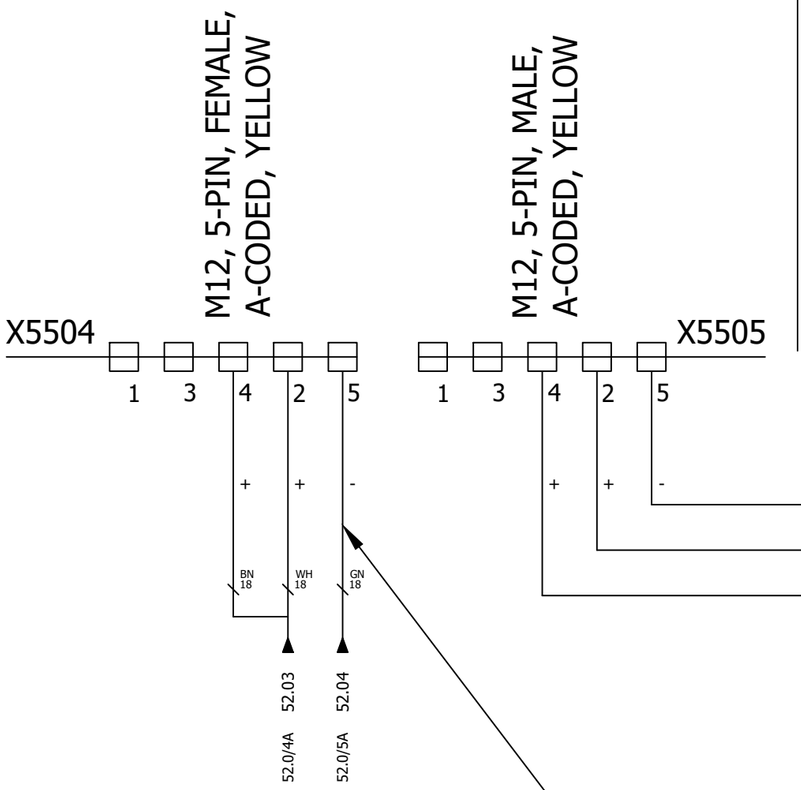
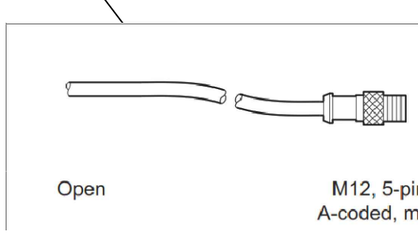
TOTAL PAGES: 175

SHEET 18.3 OF 56

# CONVEYOR WITH STOPPER LEVEL 4

## 18VFD1

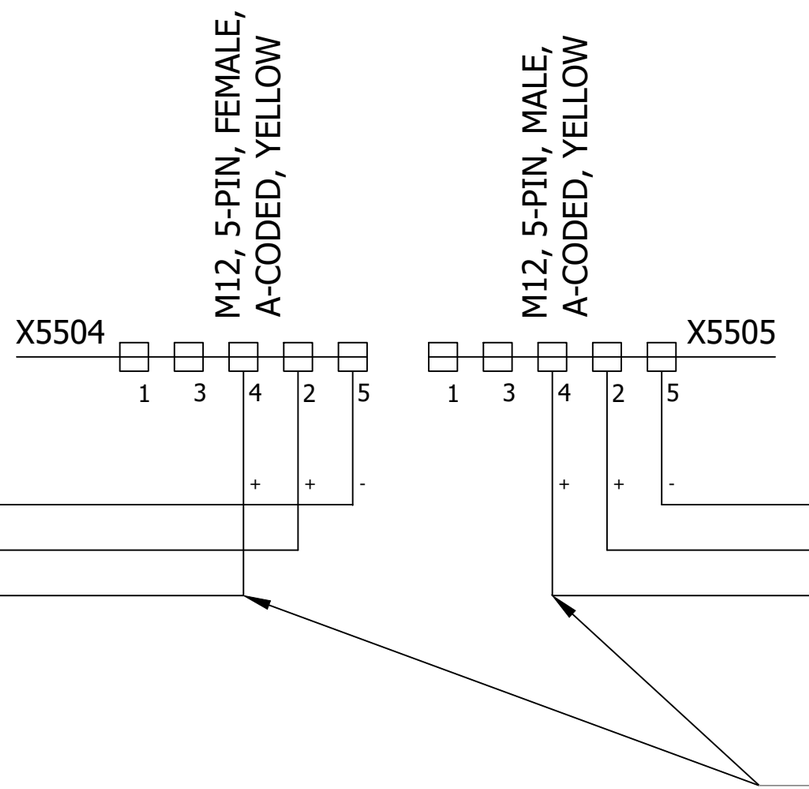


| | | | | |
|---|---|---|---|---|
| Open / M12, 5-pin, A-coded, male | CE/UL: 28110943 | HELUKABEL® LiYCY | Variable | 3 × 0.75 mm² / DC 60 V |

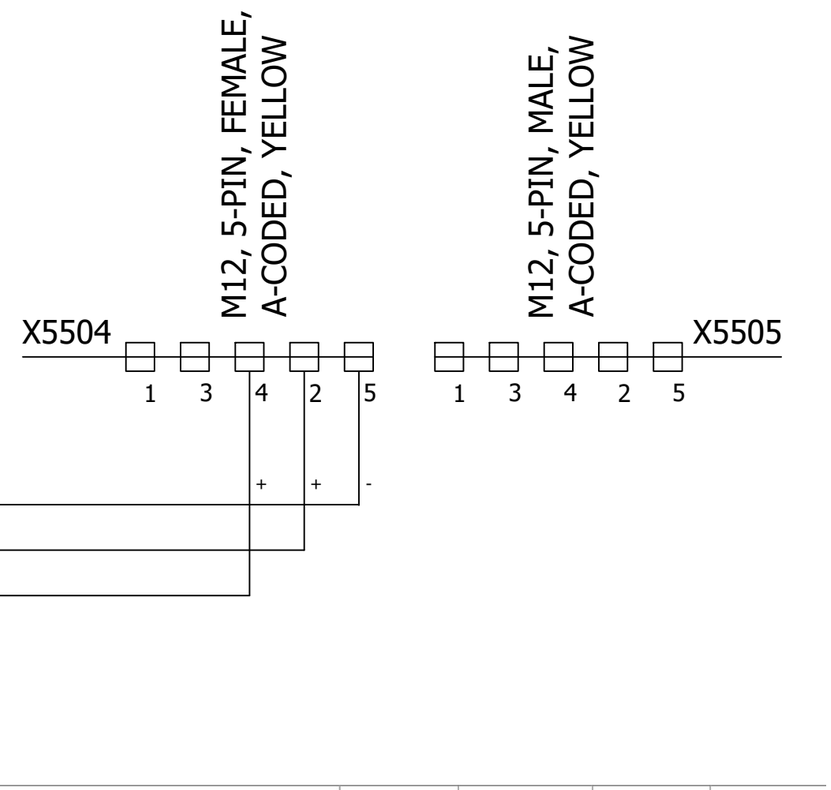
# CONVEYOR WITH STOPPER LEVEL 5

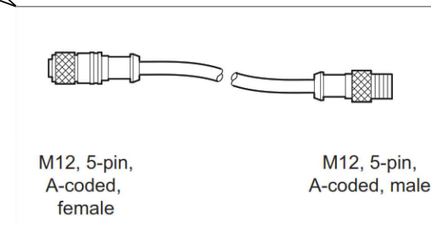
## 18VFD2




# CONVEYOR WITH STOPPER LEVEL 6

## 18VFD3



| | | | | |
|---|---|---|---|---|
| M12, 5-pin, A-coded, female / M12, 5-pin, A-coded, male | CE/UL: 28110935 | HELUKABEL® LiYCY | Variable | 3 × 0.75 mm² / DC 60 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 VFD STO

| REV | DESCRIPTION | DATE |
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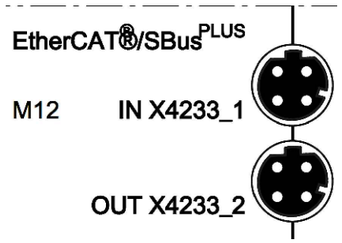
TOTAL PAGES: 175

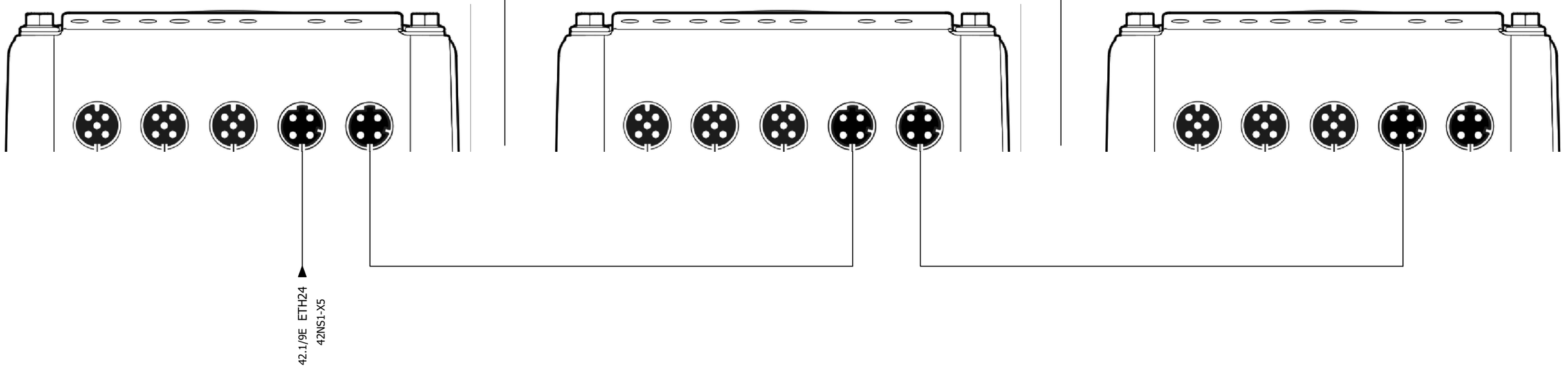
SHEET 18.4 OF 56

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

## 18VFD1




X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK




CONVEYOR
WITH STOPPER
LEVEL 4

## 18VFD2

CONVEYOR
WITH STOPPER
LEVEL 5

## 18VFD3

CONVEYOR
WITH STOPPER
LEVEL 6

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: BUFFER CONVEYORS AREA VFD's SECTION 2 VFD EtherCAT INTERFACE

| REV | DESCRIPTION | DATE |
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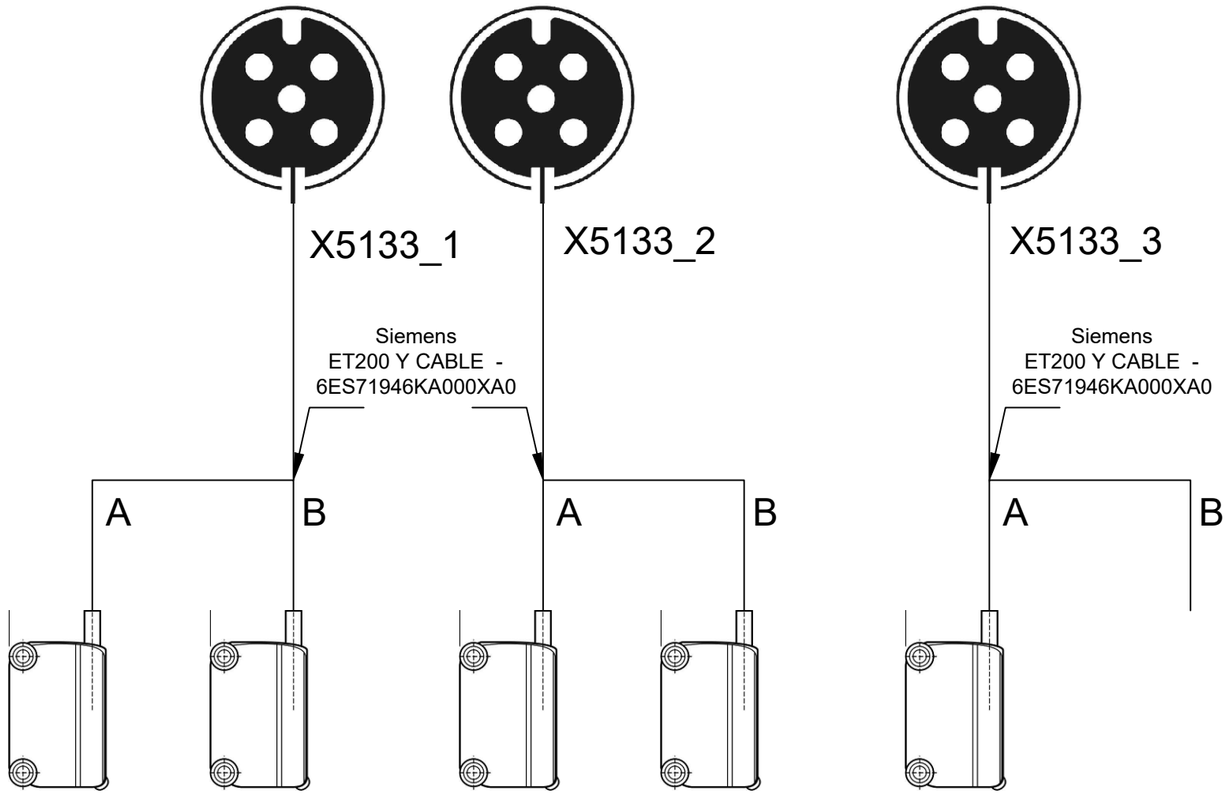
JOB # C40572

TOTAL PAGES: 175

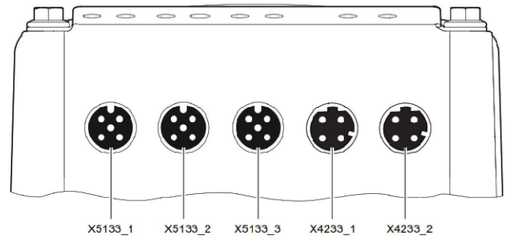
SHEET 18.5 OF 56




| | | | | | |
|---|---|---|---|---|---|
| 18PE1 | 18PE2 | 18PE3 | 18PE4 | 18PE5 | TO 18VFD1 |
| 18PE7 | 18PE8 | 18PE9 | 18PE10 | 18PE11 | TO 18VFD2 |
| 18PE13 | 18PE14 | 18PE15 | 18PE16 | 18PE17 | TO 18VFD3 |

**Plug connector positions at the DSI.. electronics cover**

The following figure shows an example of the plug connector positions:




*18014420217585035*

| Plug connector | Function |
|---|---|
| X5133_1[1)] | Digital inputs DI01 and DI02 |
| X5133_2[1)] | Digital inputs DI03 and DI04 |
| X5133_3[1)] | Digital inputs/outputs DIO01 and DIO02 |

**X5133_1: Digital inputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**

**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor supply |
| 2 | DI02 | Sensor input DI02 |
| 3 | 0V24 | 0V24 reference potential for sensors |
| 4 | DI01 | Sensor input DI01 |
| 5 | FE | Functional earth |

**X5133_2: Digital inputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**

**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor supply |
| 2 | DI04 | Sensor input DI04 |
| 3 | 0V24 | 0V24 reference potential for sensors |
| 4 | DI03 | Sensor input DI03 |
| 5 | FE | Functional earth |

**X5133_3: Digital inputs/outputs**

The following table shows information about this connection:

| Function |
|---|
| Digital inputs/outputs |

| Connection type |
|---|
| M12, 5-pin, female, A-coded, color: black |

**Connection diagram**

**Assignment**

| Contact | Signal | Description |
|---|---|---|
| 1 | +24V | DC 24 V sensor/actuator supply |
| 2 | DIO02 | Sensor input DIO02/actuator output DIO02 |
| 3 | 0V24 | 0V24 reference potential for sensors/actuators |
| 4 | DIO01 | Sensor input DIO01/actuator output DIO01 |
| 5 | FE | Functional earth |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

| REV | DESCRIPTION | DATE |
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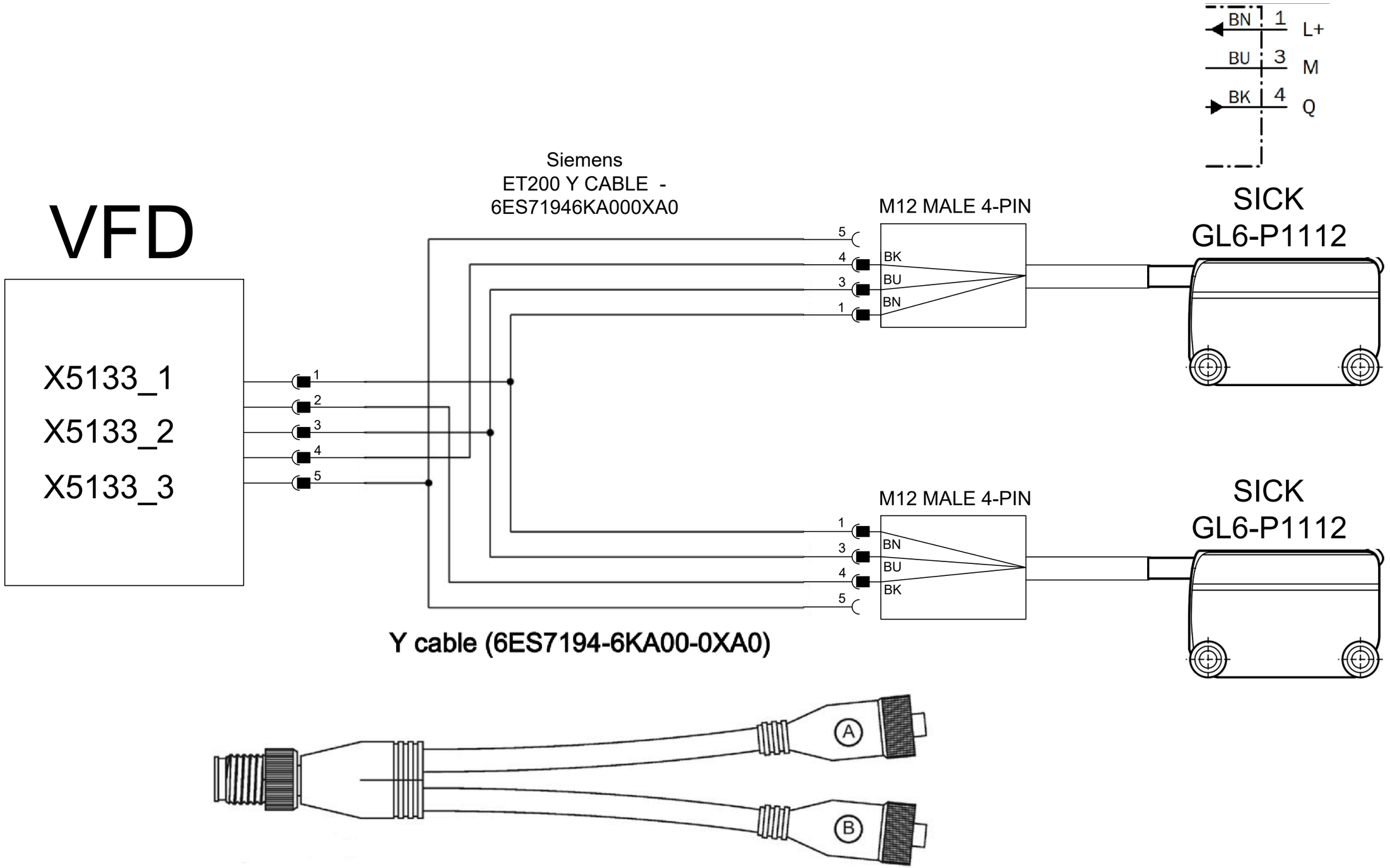


Y cable (6ES7194-6KA00-0XA0)




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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: VTD MOBILE EXTRACTOR CONVEYOR AREA VFD I/O INTERFACE

| REV | DESCRIPTION | DATE |
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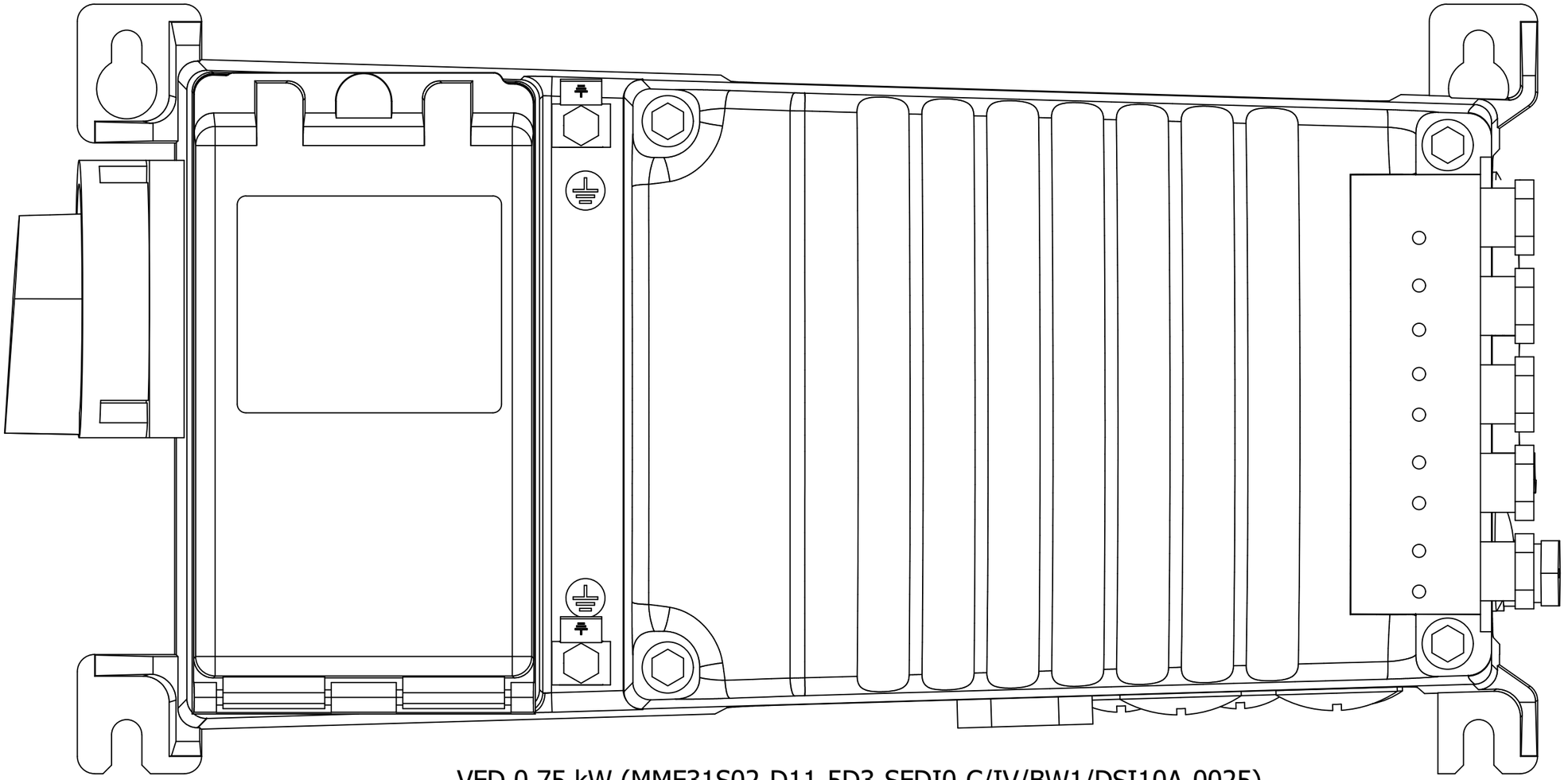

VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

CAROUSEL CONVEYORS AREA VFD's
SECTION 3 TOTAL OF 4 VFD's

| 19VFD1 | 19VFD2 | 19VFD3 | 19VFD4 |
|---|---|---|---|
| CAROUSEL CONVEYOR POPUP 1 | CAROUSEL CONVEYOR POPUP 2 | CAROUSEL CONVEYOR TURNTABLE+ROLLER | CAROUSEL CONVEYOR TURNTABLE |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3

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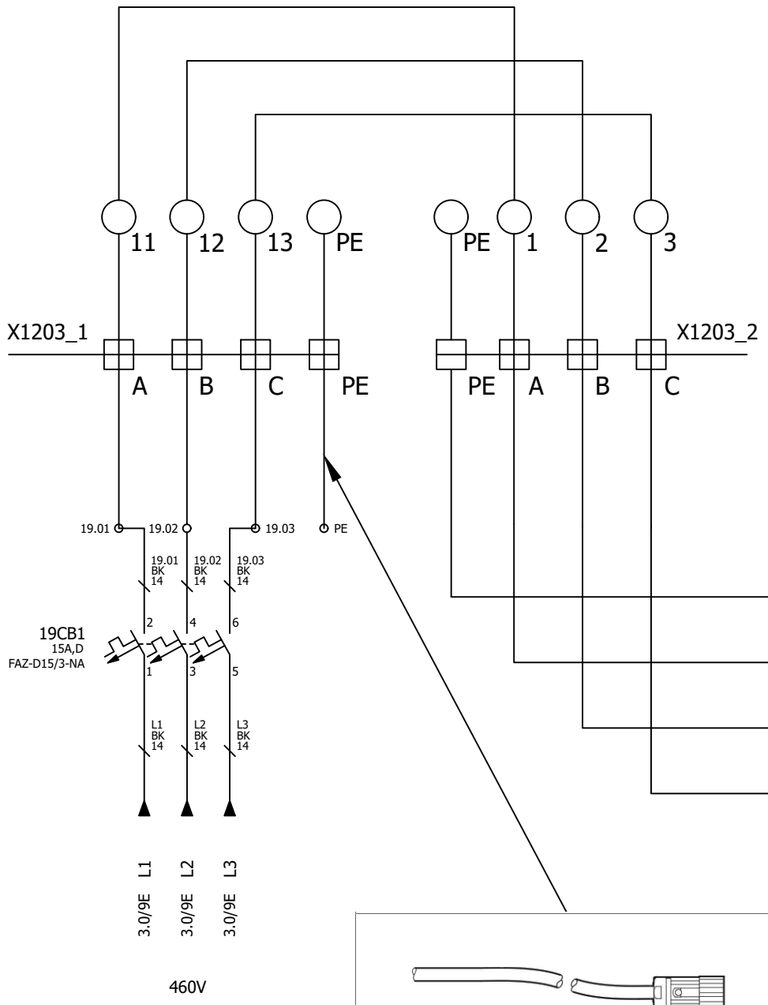
# CAROUSEL CONVEYOR POPUP 1

## 19VFD1

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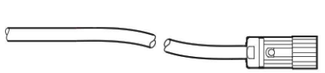
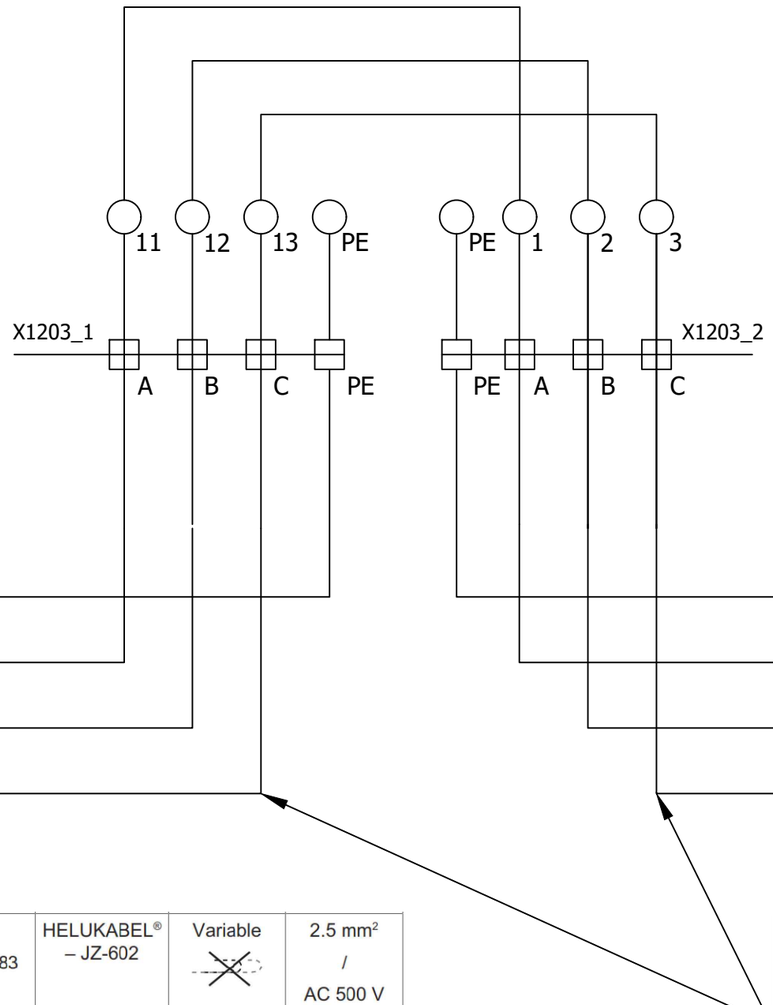
|  Open — M23, coding ring: black, male | UL: 18153283 | HELUKABEL® – JZ-602 | Variable  | 2.5 mm² / AC 500 V |

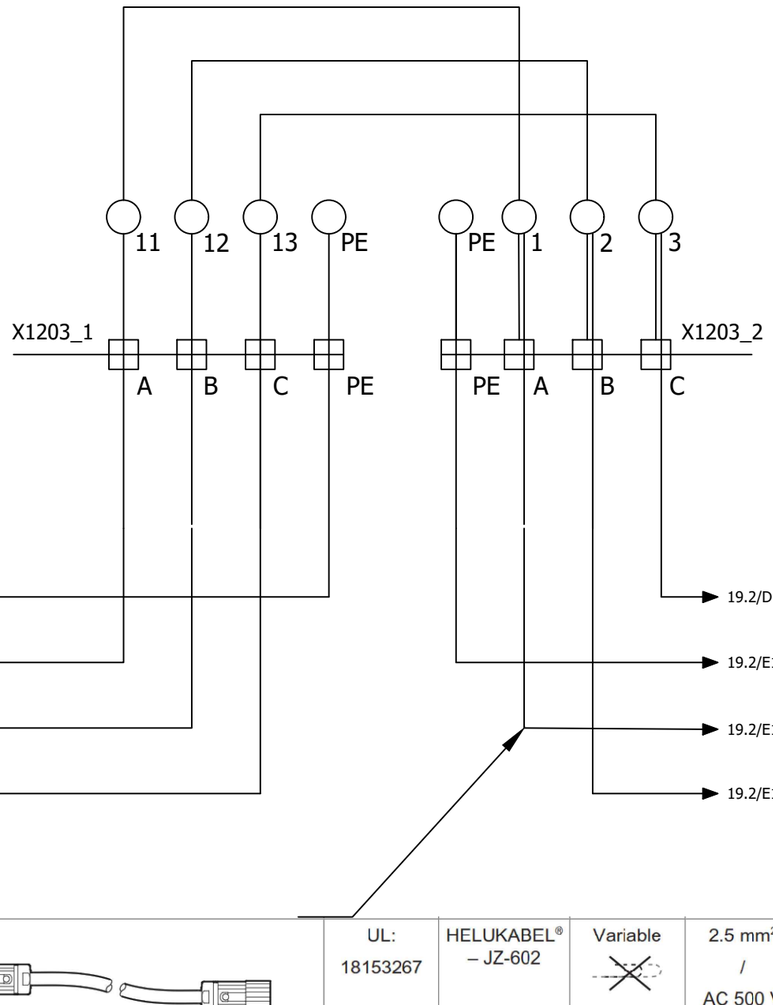
# CAROUSEL CONVEYOR POPUP 2

## 19VFD2




# CAROUSEL ROLLER CONVEYOR

## 19VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: black, male — M23, coding ring: black, male | UL: 18153267 | HELUKABEL® – JZ-602 | Variable  | 2.5 mm² / AC 500 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 POWER

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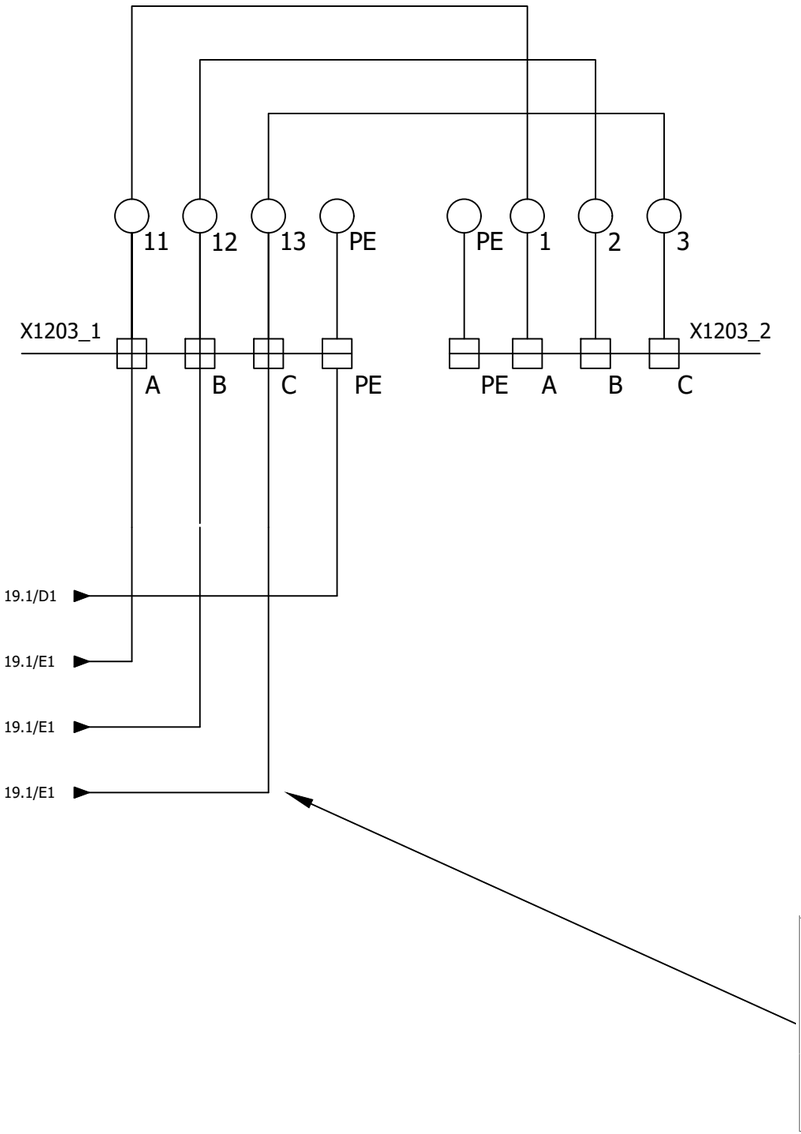
SHEET 19.1 OF 56

# CAROUSEL
# TURNTABLE
# CONVEYOR

# 19VFD4

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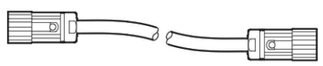
|  M23, coding ring: black, male — M23, coding ring: black, male | UL: 18153267 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 POWER

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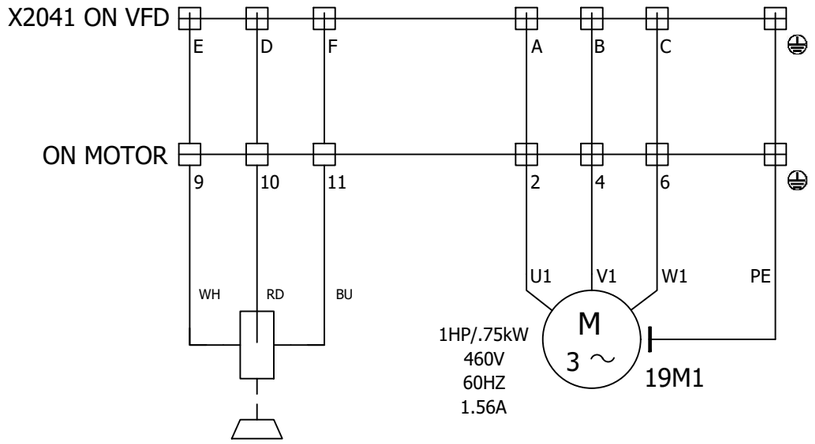
SHEET 19.2 OF 56

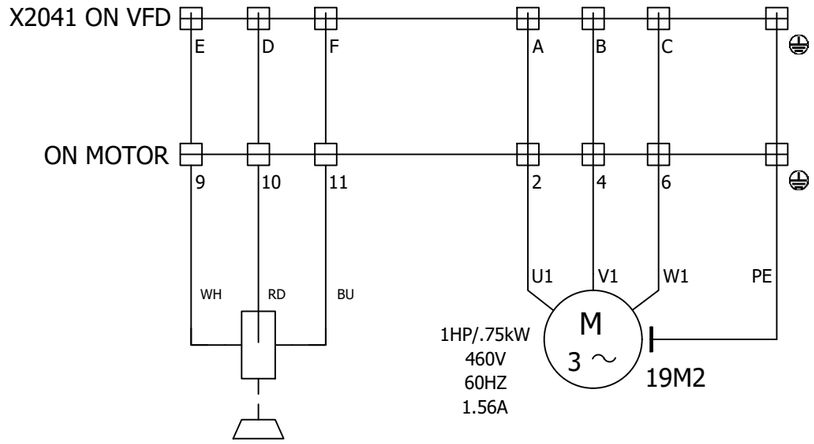
# CAROUSEL CONVEYOR POPUP 1

## 19VFD1



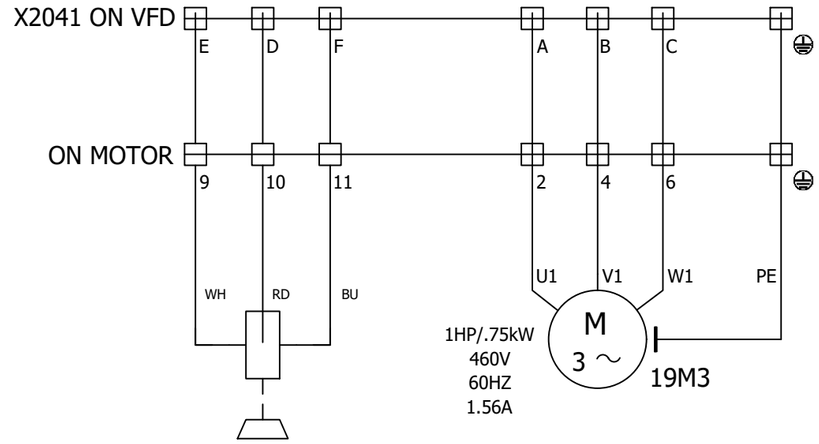

# CAROUSEL CONVEYOR POPUP 2

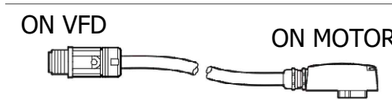
## 19VFD2




# CAROUSEL ROLLER CONVEYOR

## 19VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: brown, male | IS2, female, ⅄ | CE/UL: 28125940 | LEONI LEHC® 000749 | Variable | 1.5 mm² / AC 500 V |

## Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 MOTOR POWER

| REV | DESCRIPTION | DATE |
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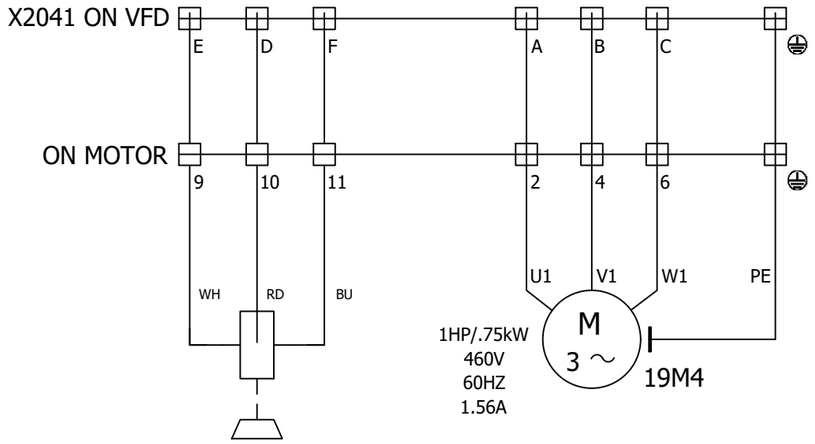
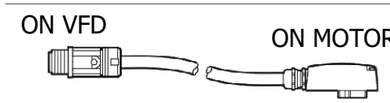
SHEET 19.3 OF 56

# CAROUSEL
# TURNTABLE
# CONVEYOR

# 19VFD4

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|---|---|---|---|---|
| M23, coding ring: brown, male | IS2, female, ⅄ | CE/UL: 28125940 | LEONI LEHC® 000749 | Variable | 1.5 mm² / AC 500 V |

## Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 MOTOR POWER

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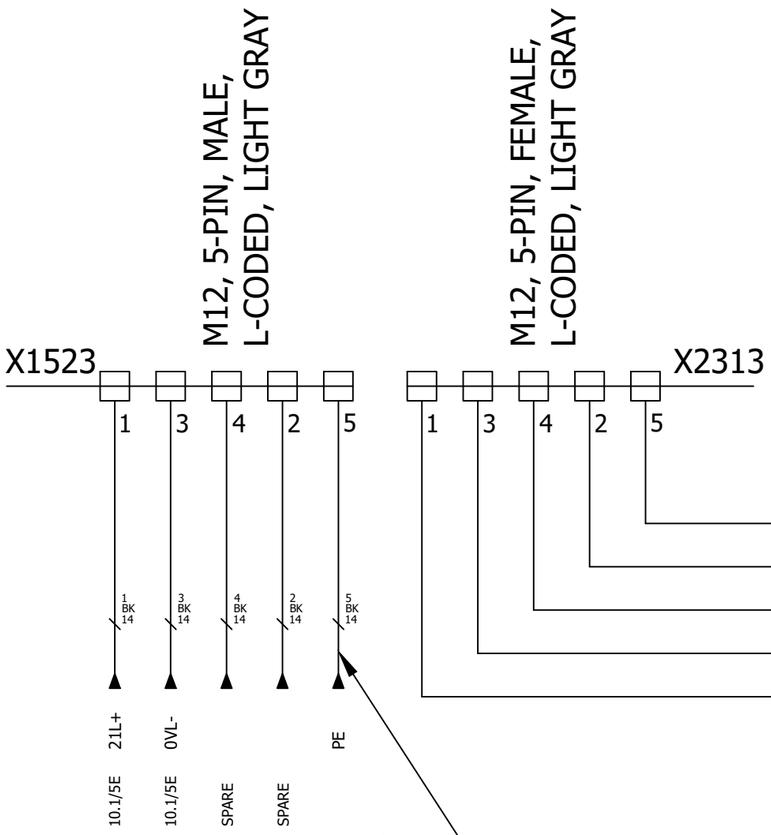
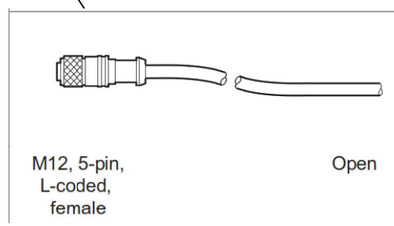
SHEET 19.4 OF 56

## CAROUSEL CONVEYOR POPUP 1

# 19VFD1



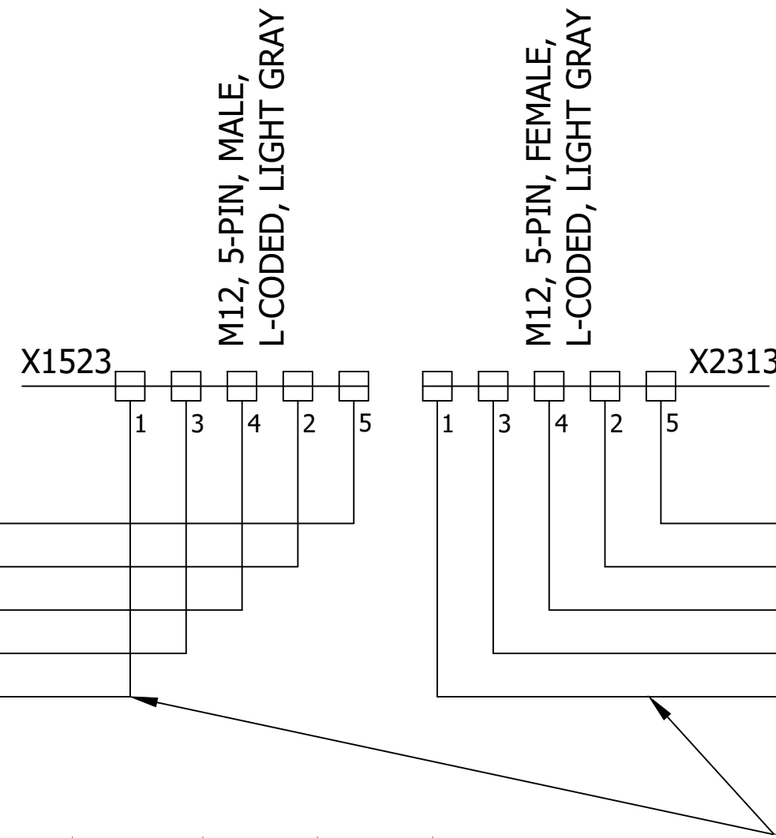



M12, 5-pin, L-coded, female — Open

| CE/UL: | HELUKABEL® JZ-500 | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28117786 | | | |

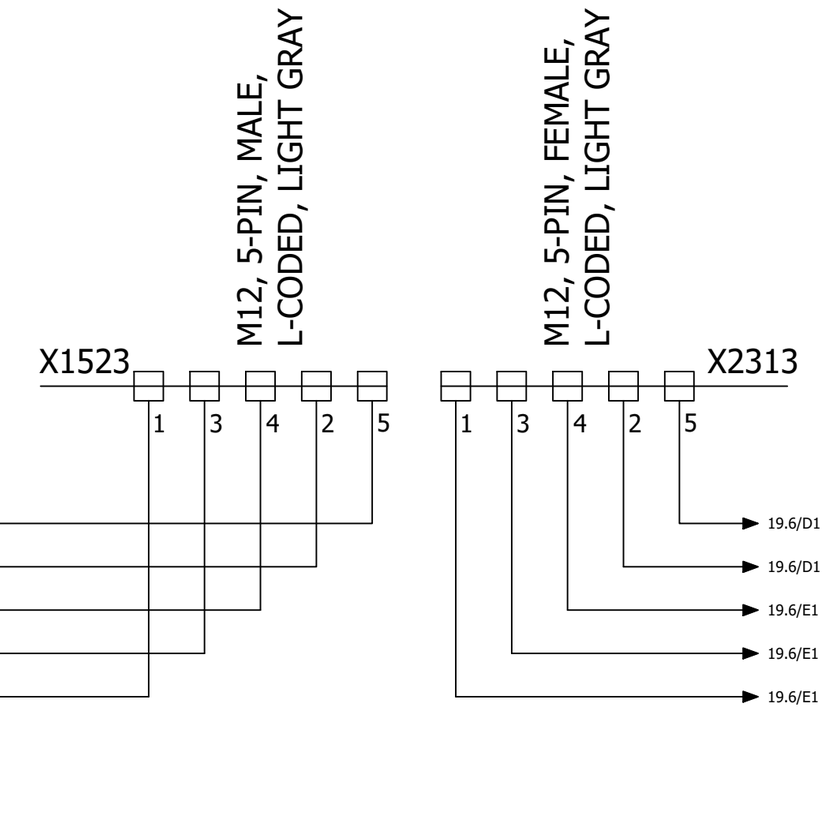
## CAROUSEL CONVEYOR POPUP 2

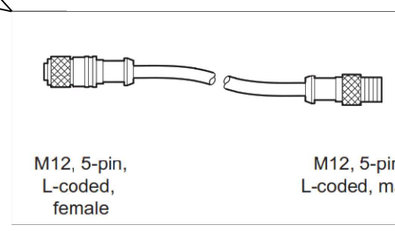
# 19VFD2




## CAROUSEL ROLLER CONVEYOR

# 19VFD3



M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

| CE/UL: | HELUKABEL® JZ-500 | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28114345 | | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD 24VDC POWER

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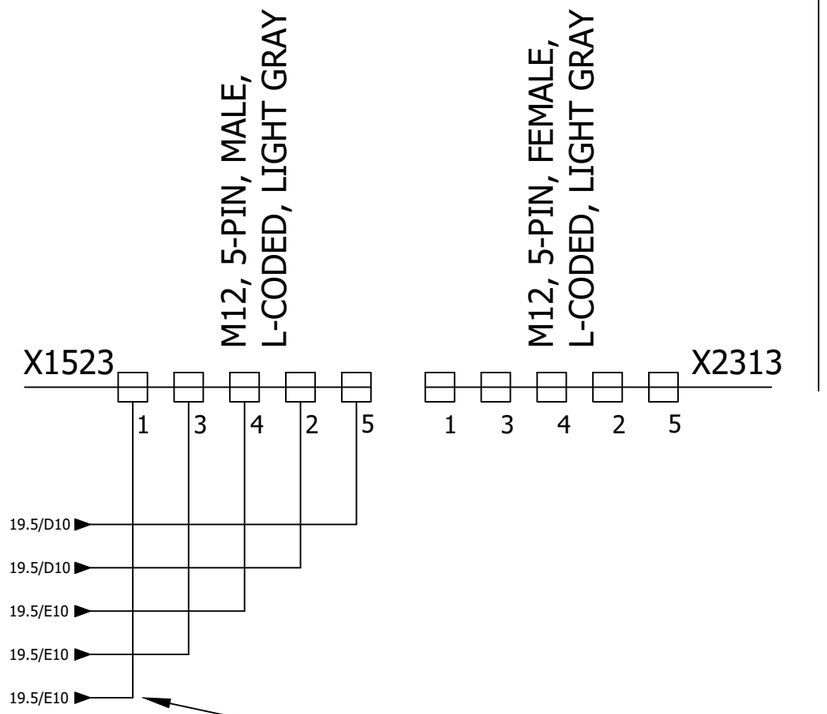
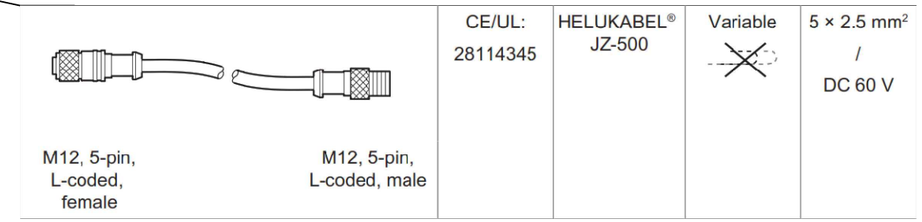
SHEET 19.5 OF 56

# CAROUSEL TURNTABLE CONVEYOR

# 19VFD4




| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28114345 | JZ-500 | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD 24VDC POWER

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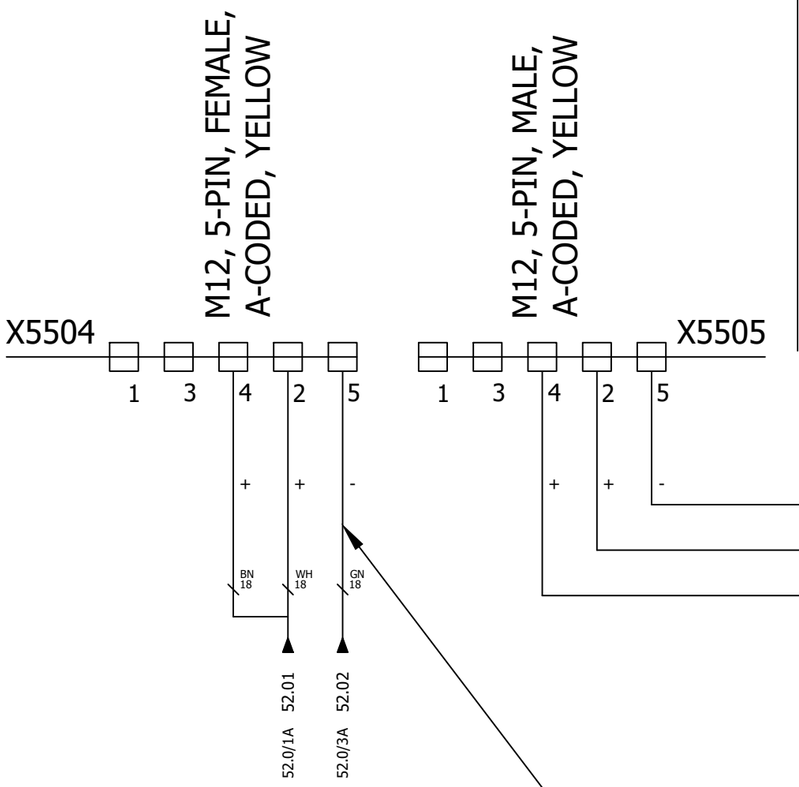
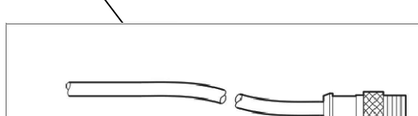
SHEET 19.6 OF 56

# CAROUSEL CONVEYOR POPUP 1

## 19VFD1

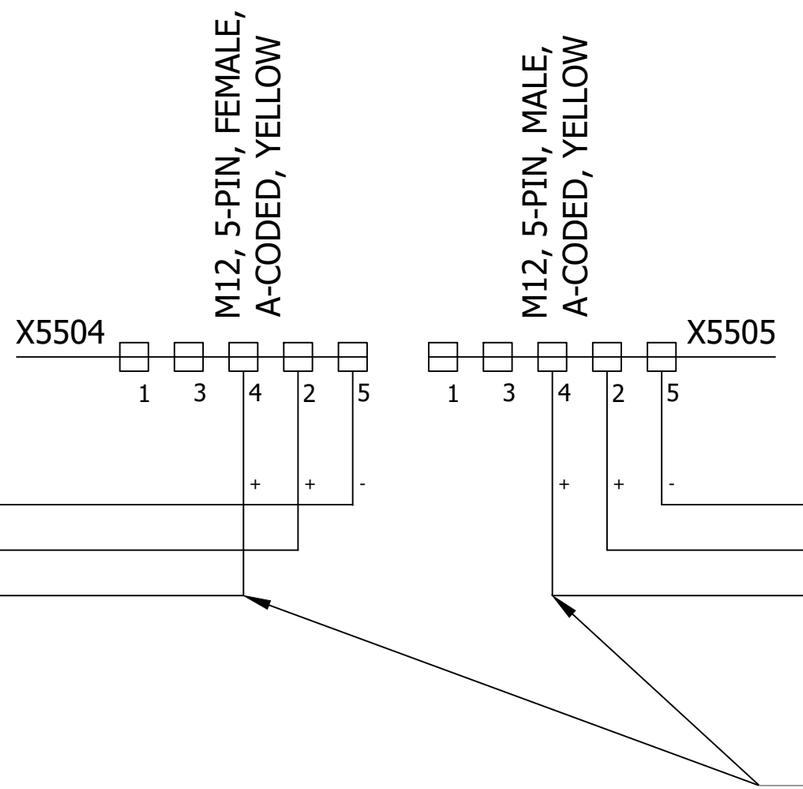



| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110943 | LiYCY | | |

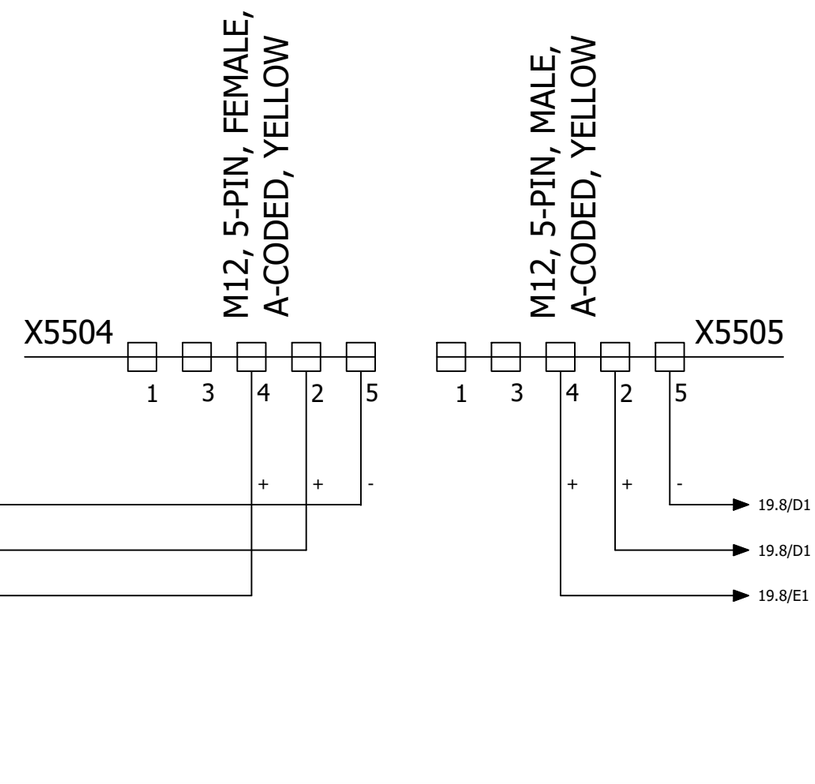
# CAROUSEL CONVEYOR POPUP 2

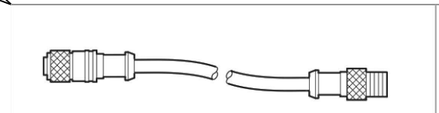
## 19VFD2




# CAROUSEL ROLLER CONVEYOR

## 19VFD3




| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110935 | LiYCY | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD STO

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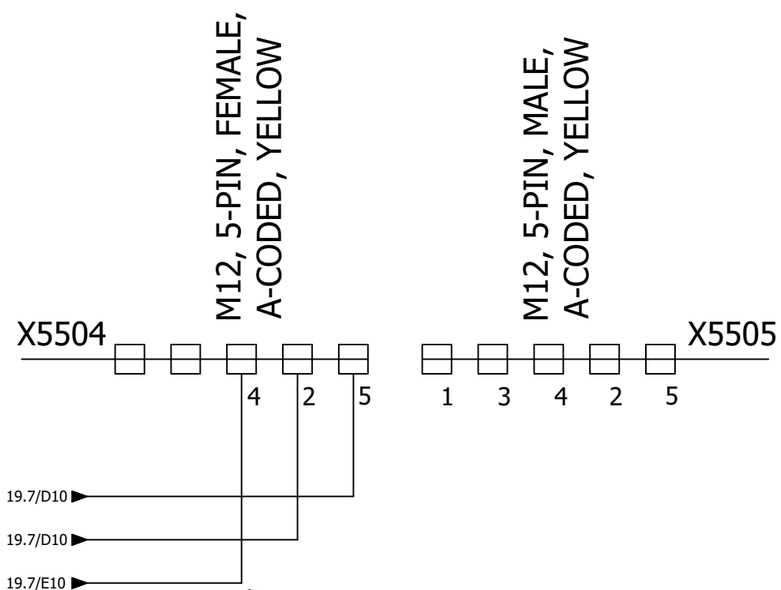
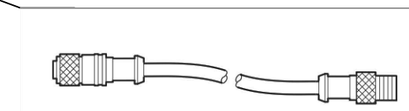
SHEET 19.7 OF 56

# CAROUSEL
# TURNTABLE
# CONVEYOR

# 19VFD4



M12, 5-pin, A-coded, female

M12, 5-pin, A-coded, male

| | | | | |
|---|---|---|---|---|
| CE/UL: 28110935 | HELUKABEL® LiYCY | Variable | 3 × 0.75 mm² / DC 60 V | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD STO

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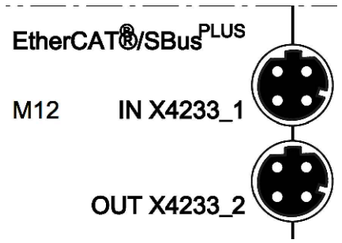
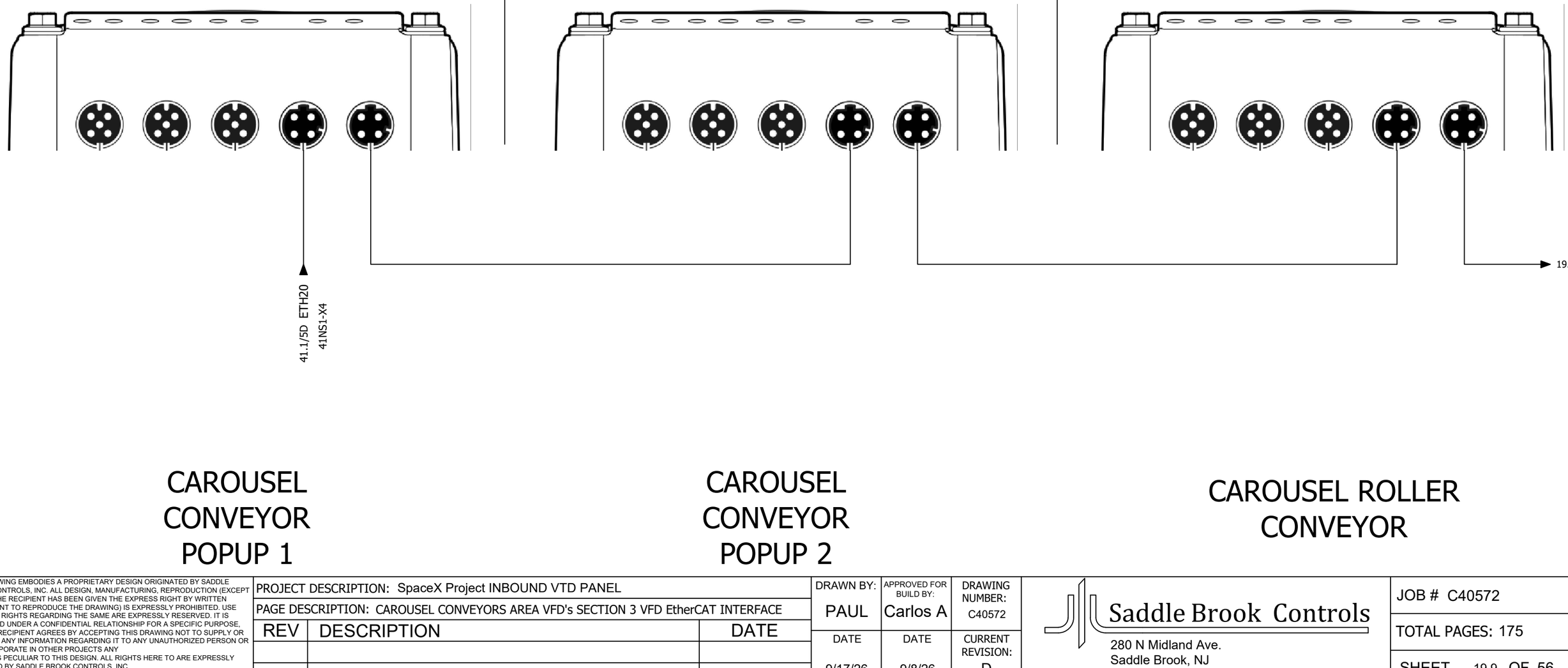
SHEET 19.8 OF 56

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

# 19VFD1




X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK




# 19VFD2

# 19VFD3

## CAROUSEL CONVEYOR POPUP 1

## CAROUSEL CONVEYOR POPUP 2

## CAROUSEL ROLLER CONVEYOR

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD EtherCAT INTERFACE

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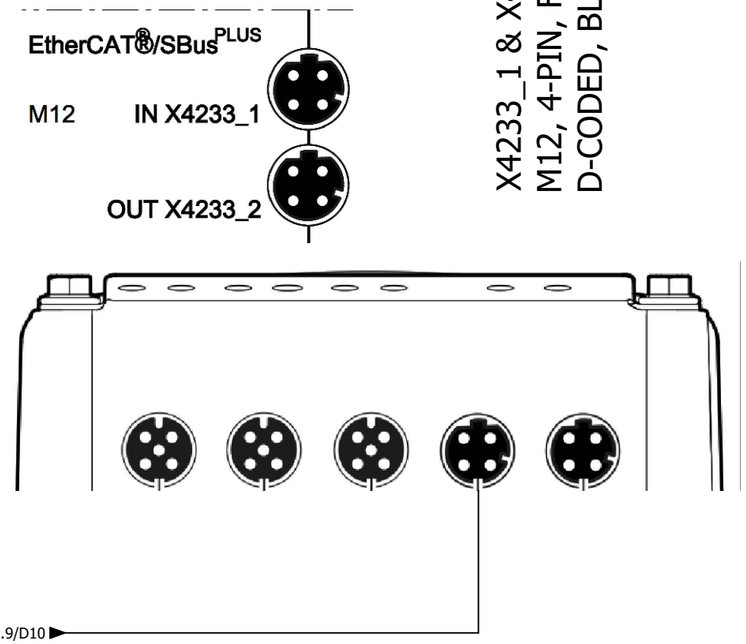
TOTAL PAGES: 175

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

# 19VFD4




# CAROUSEL TURNTABLE CONVEYOR

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 3 VFD EtherCAT INTERFACE

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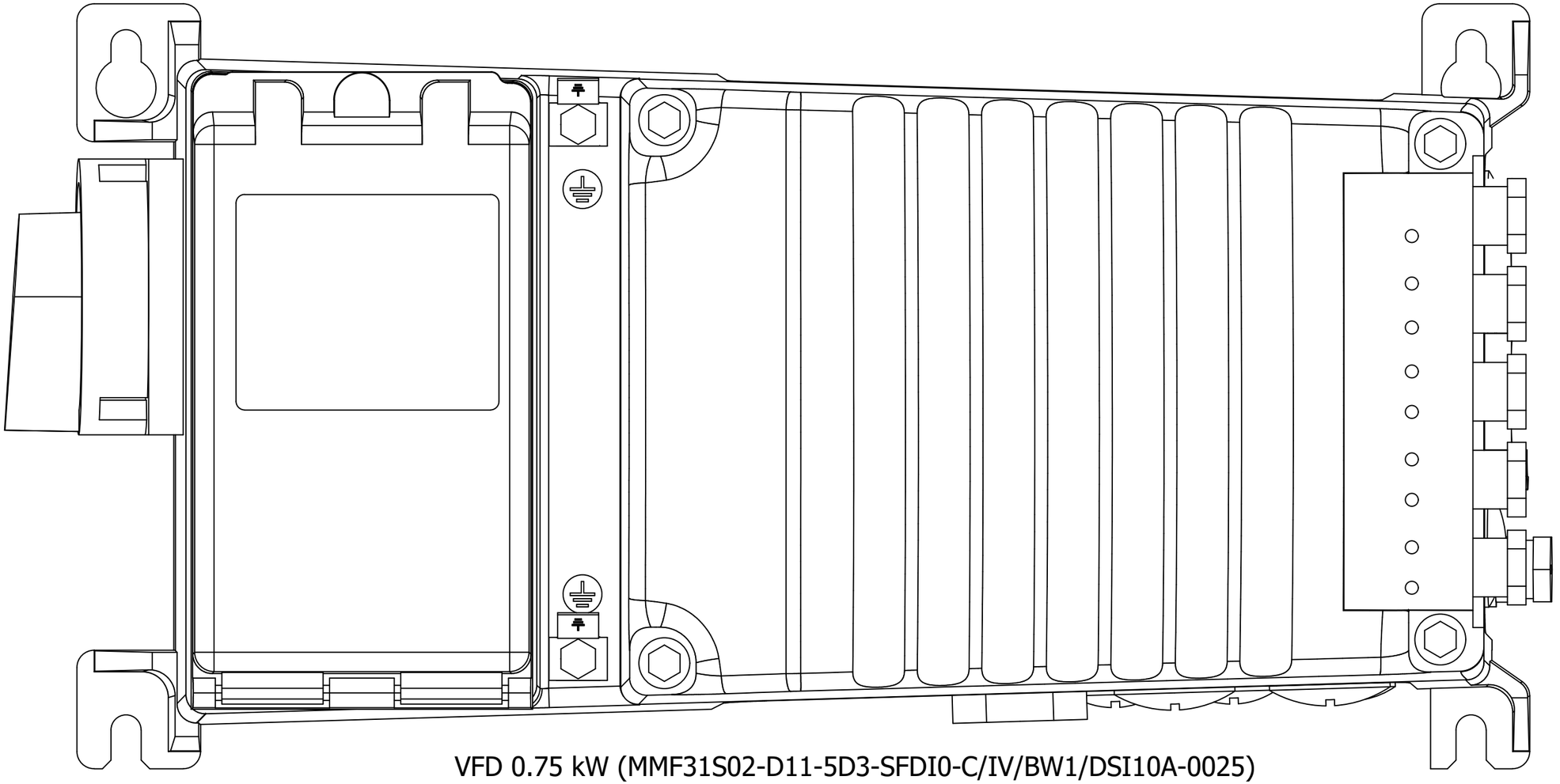

VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

CAROUSEL CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 5 VFD's

| 20VFD1 | 20VFD2 | 20VFD3 | 20VFD4 | 20VFD5 |
|---|---|---|---|---|
| ROLLER CONVEYOR 1 | ROLLER CONVEYOR 2 | ROLLER CONVEYOR 3 | ROLLER CONVEYOR 4 | ROLLER CONVEYOR 5 |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1

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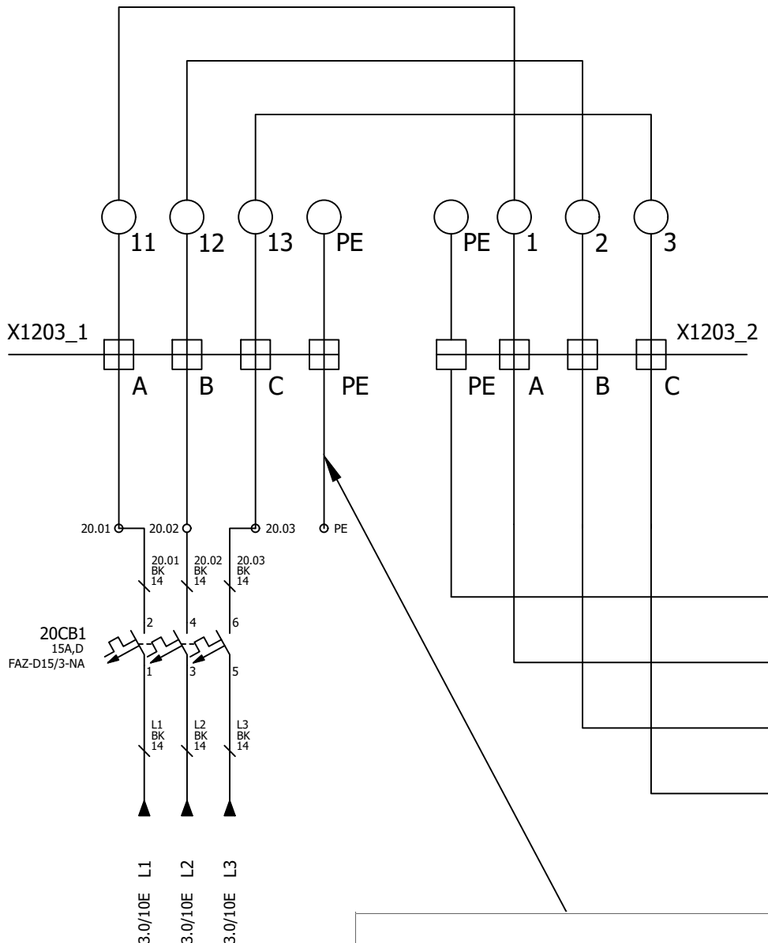
SHEET 20.0 OF 56

# ROLLER CONVEYOR 1

## 20VFD1

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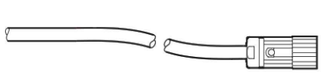
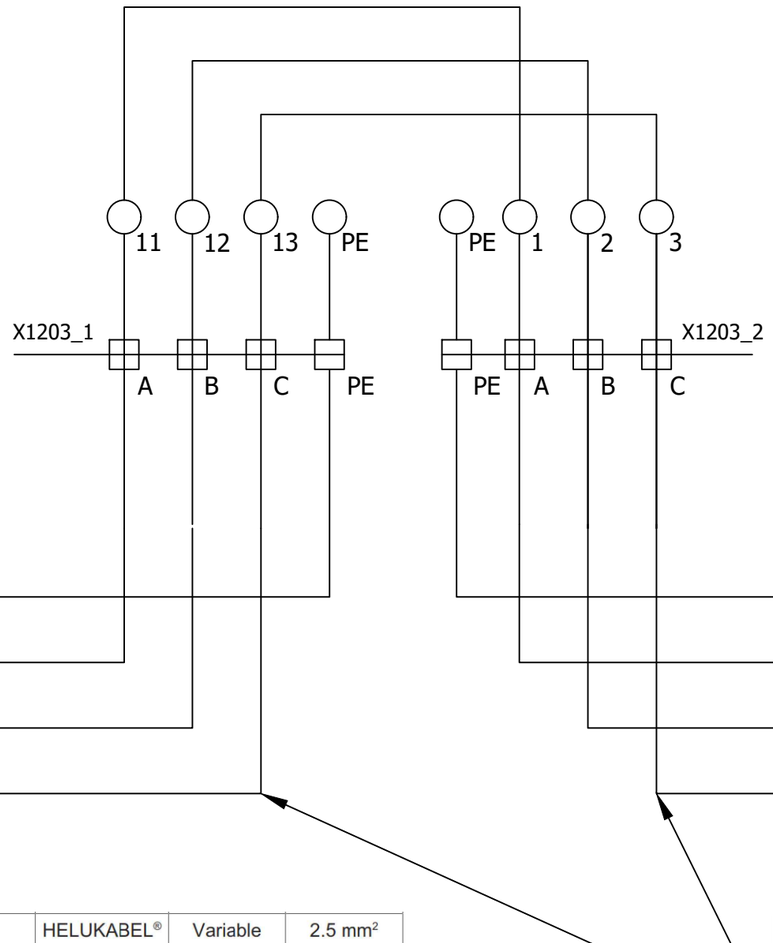
|  Open — M23, coding ring: black, male | UL: 18153283 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

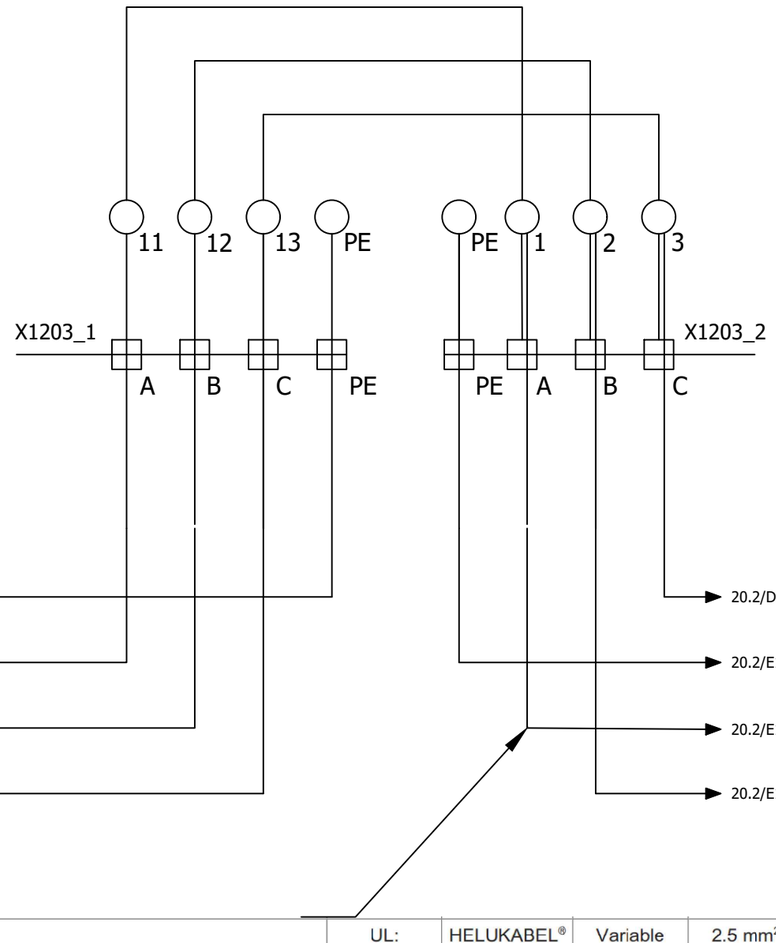
# ROLLER CONVEYOR 2

## 20VFD2




# ROLLER CONVEYOR 3

## 20VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: black, male — M23, coding ring: black, male | UL: 18153267 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER

| REV | DESCRIPTION | DATE |
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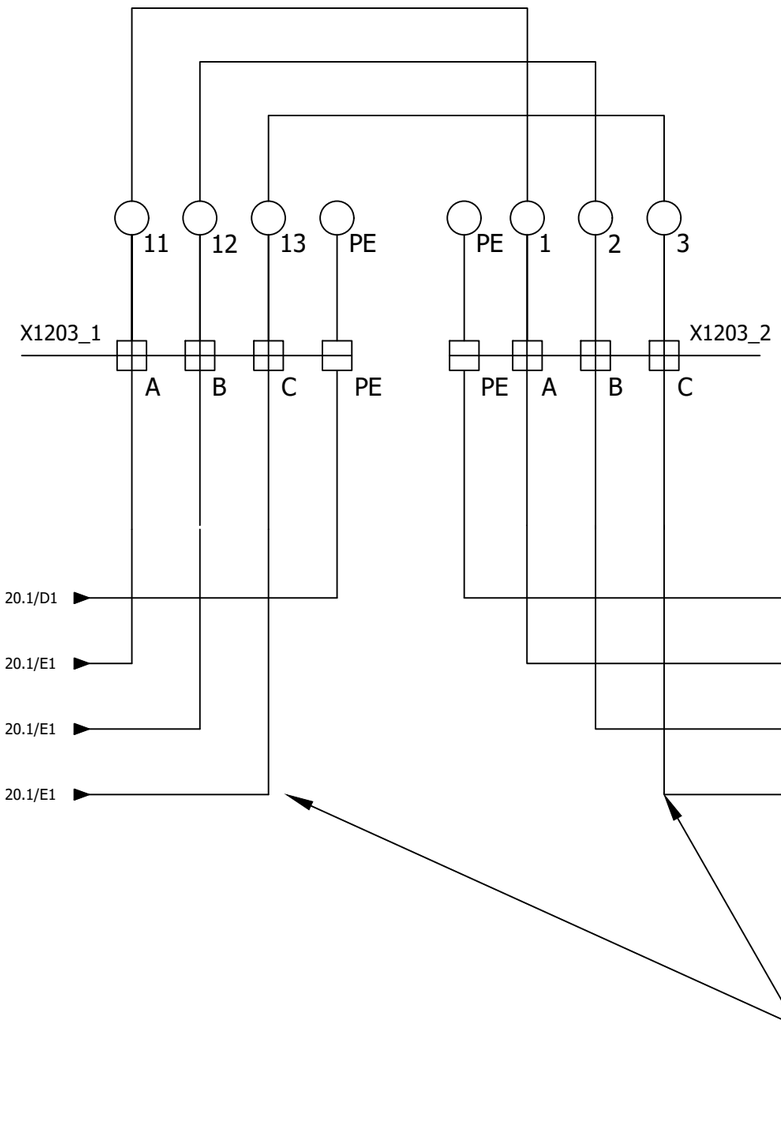
TOTAL PAGES: 175

SHEET 20.1 OF 56

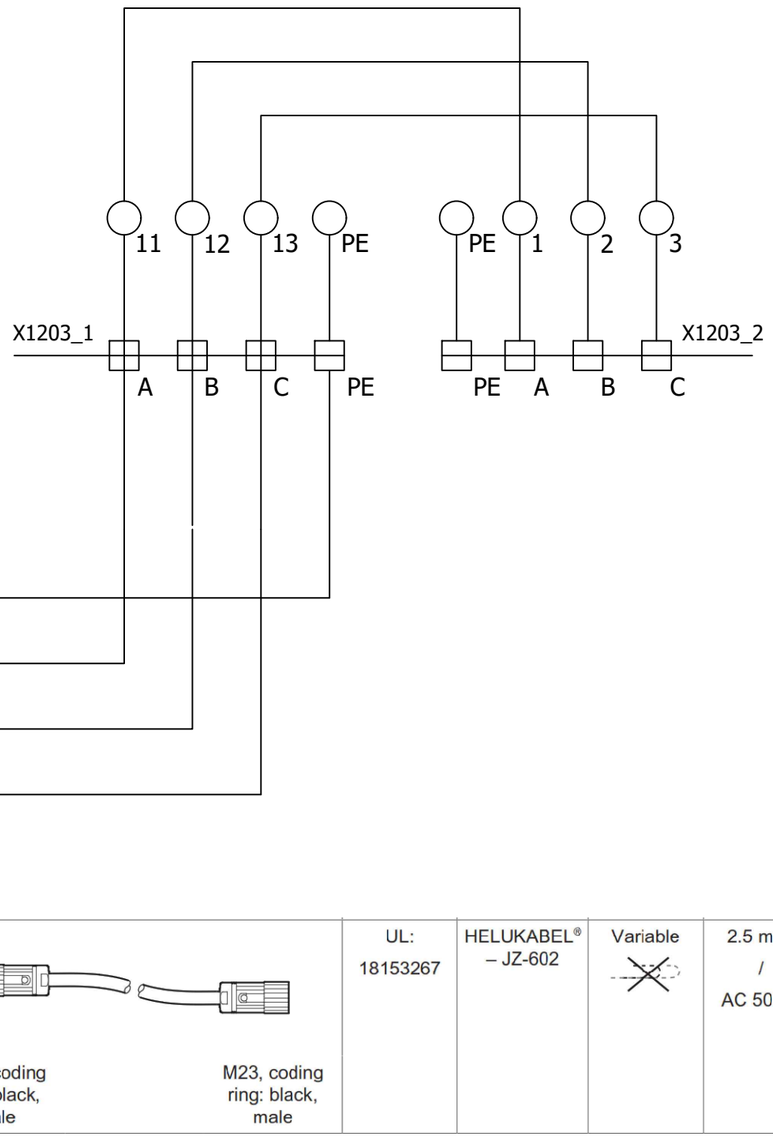
# ROLLER CONVEYOR 4

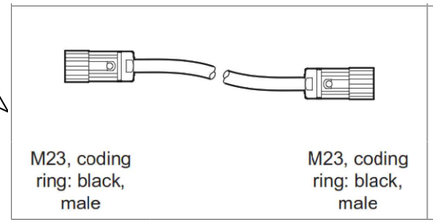
## 20VFD4




# ROLLER CONVEYOR 5

## 20VFD5




| UL: | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |
|---|---|---|---|
| 18153267 | | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER

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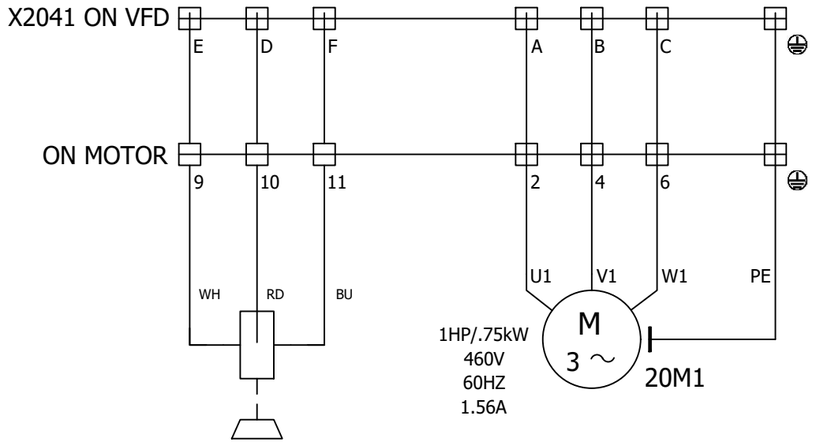
TOTAL PAGES: 175

SHEET 20.2 OF 56

# ROLLER CONVEYOR 1

## 20VFD1




# ROLLER CONVEYOR 2

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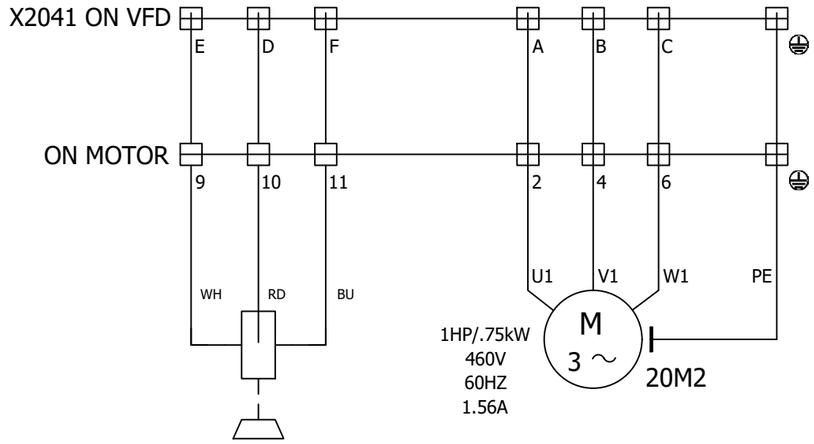
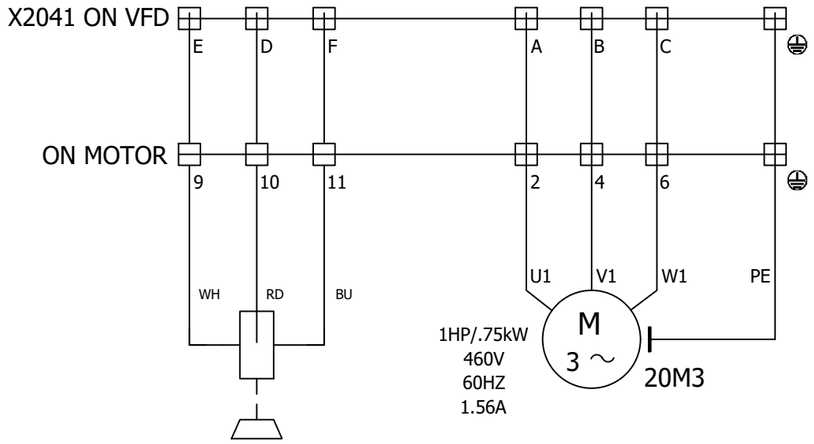
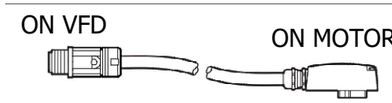



# ROLLER CONVEYOR 3

## 20VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: brown, male | IS2, female, ⅄ | CE/UL: 28125940 | LEONI LEHC® 000749 | Variable | 1.5 mm² / AC 500 V |

## Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

| REV | DESCRIPTION | DATE |
|---|---|---|
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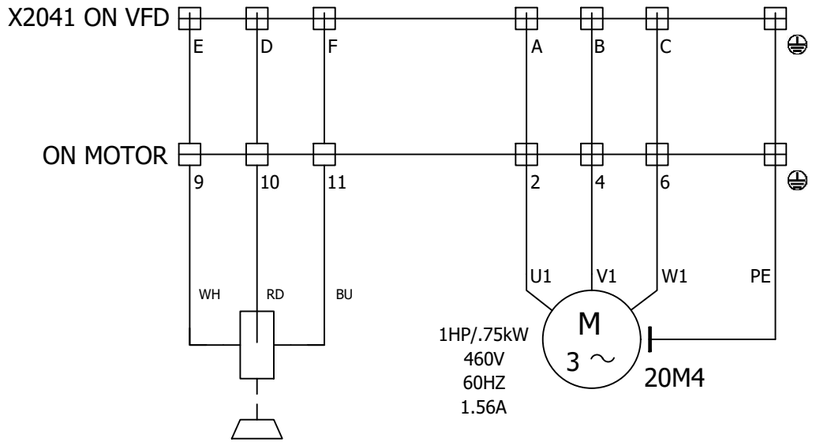
TOTAL PAGES: 175

SHEET 20.3 OF 56

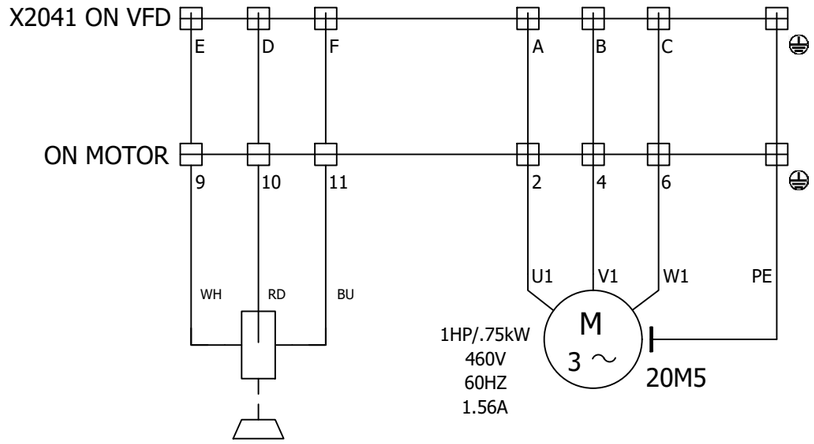
# ROLLER CONVEYOR 4

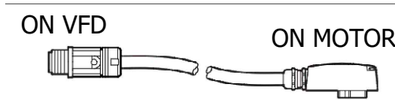
## 20VFD4




# ROLLER CONVEYOR 5

## 20VFD5




| | | | | |
|---|---|---|---|---|
| M23, coding ring: brown, male | IS2, female, ⅄ | CE/UL: 28125940 | LEONI LEHC® 000749 | Variable | 1.5 mm² / AC 500 V |

## Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 MOTOR POWER

| REV | DESCRIPTION | DATE |
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|---|---|---|
| PAUL | Carlos A | C40572 |
| DATE 9/17/26 | DATE 9/8/26 | CURRENT REVISION: D |

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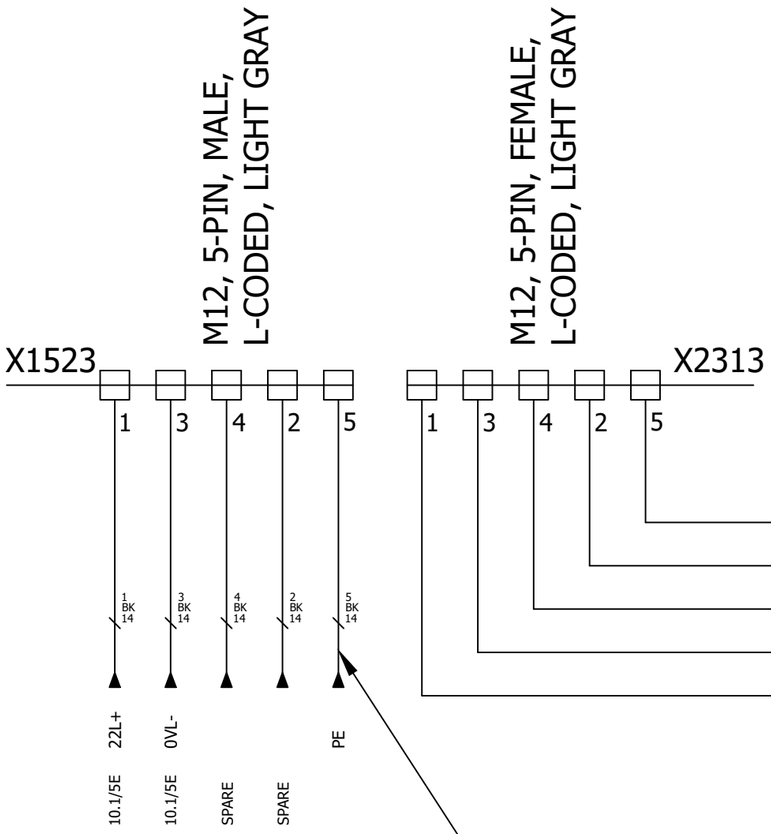
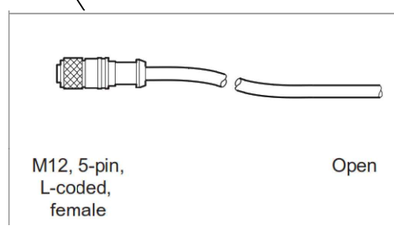
TOTAL PAGES: 175

SHEET 20.4 OF 56

## ROLLER CONVEYOR 1

# 20VFD1



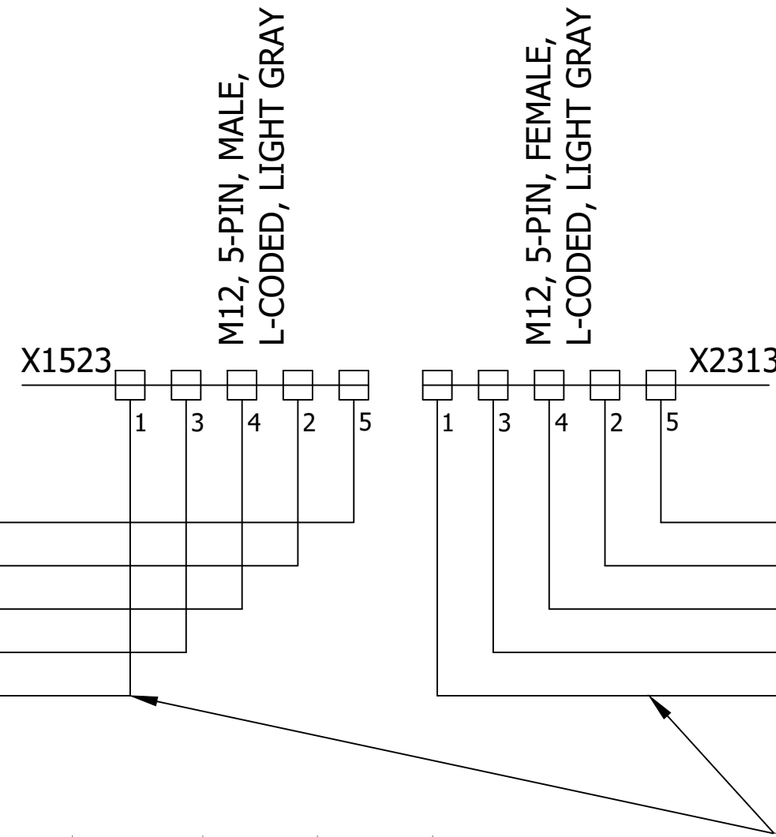



M12, 5-pin, L-coded, female — Open

| CE/UL: | HELUKABEL® JZ-500 | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28117786 | | | |

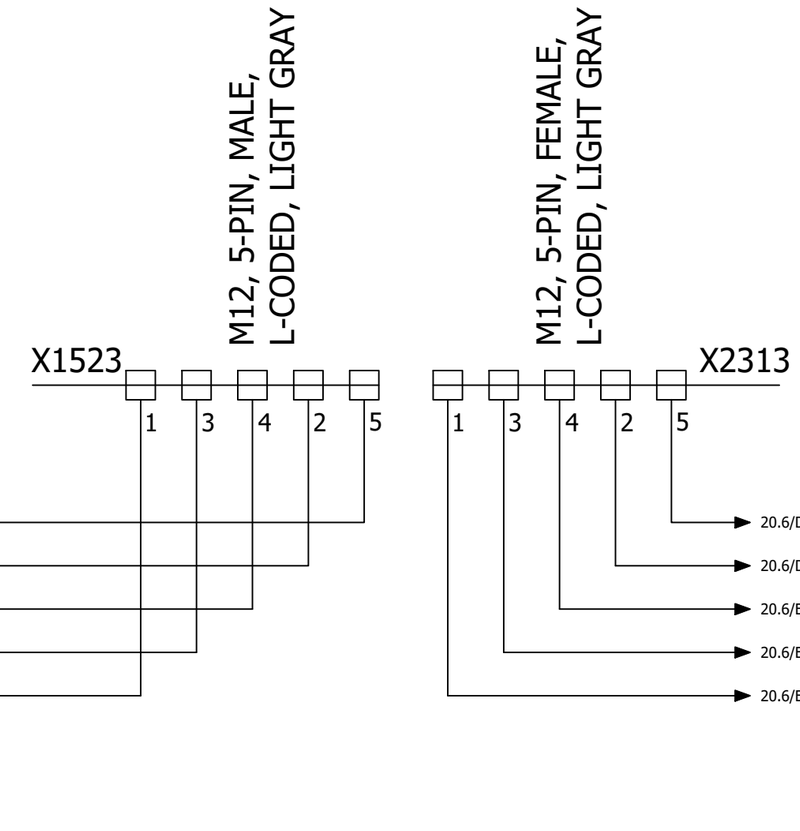
## ROLLER CONVEYOR 2

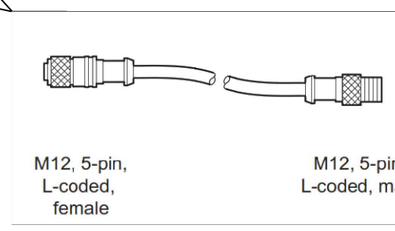
# 20VFD2




## ROLLER CONVEYOR 3

# 20VFD3



M12, 5-pin, L-coded, female — M12, 5-pin, L-coded, male

| CE/UL: | HELUKABEL® JZ-500 | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28114345 | | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER

| REV | DESCRIPTION | DATE |
|---|---|---|
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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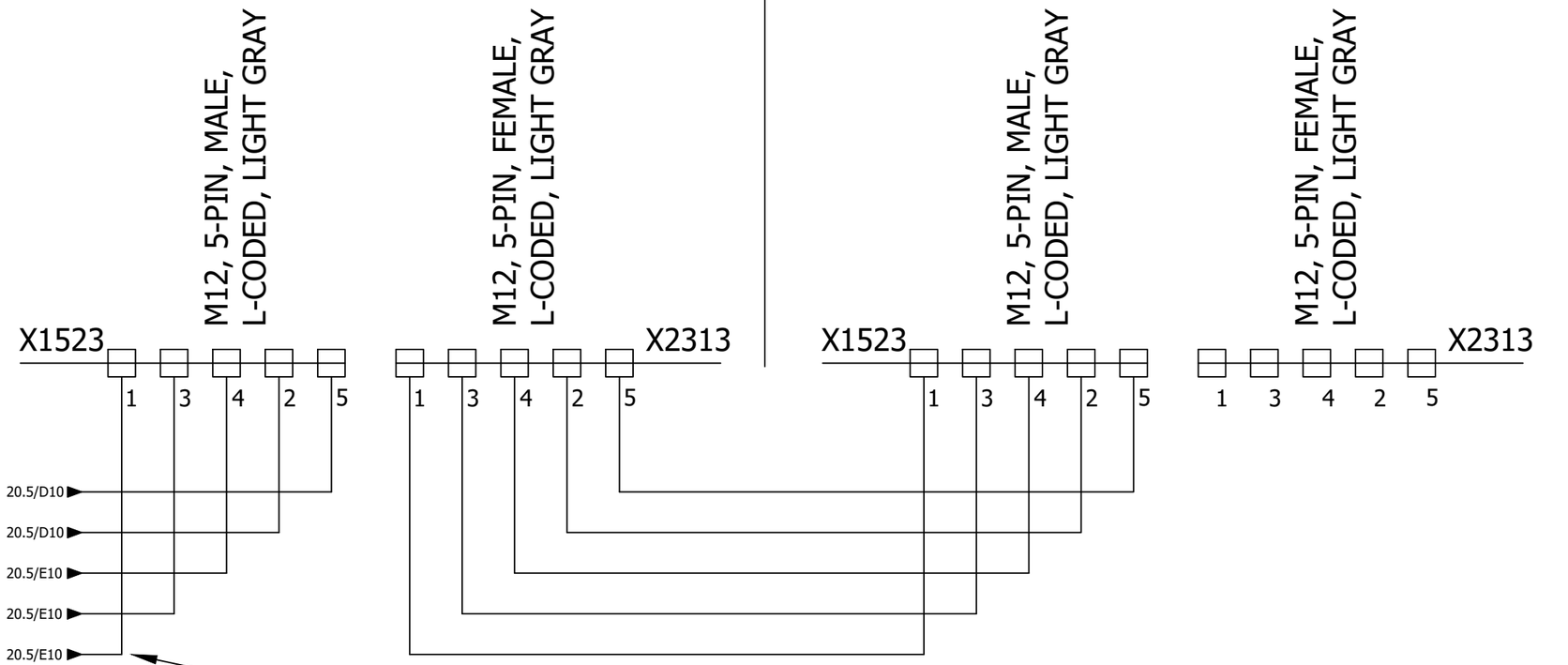
TOTAL PAGES: 175

SHEET 20.5 OF 56

## CAROUSEL TURNTABLE CONVEYOR

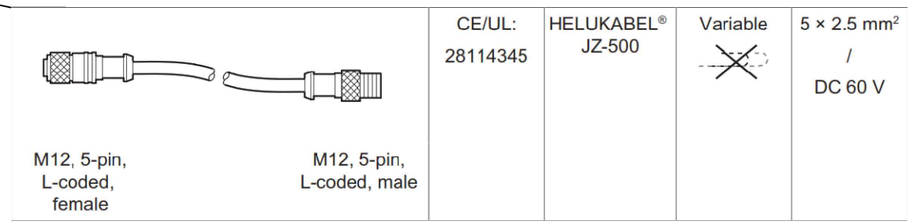
# 20VFD4




## ROLLER CONVEYOR 5

# 20VFD5

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD 24VDC POWER

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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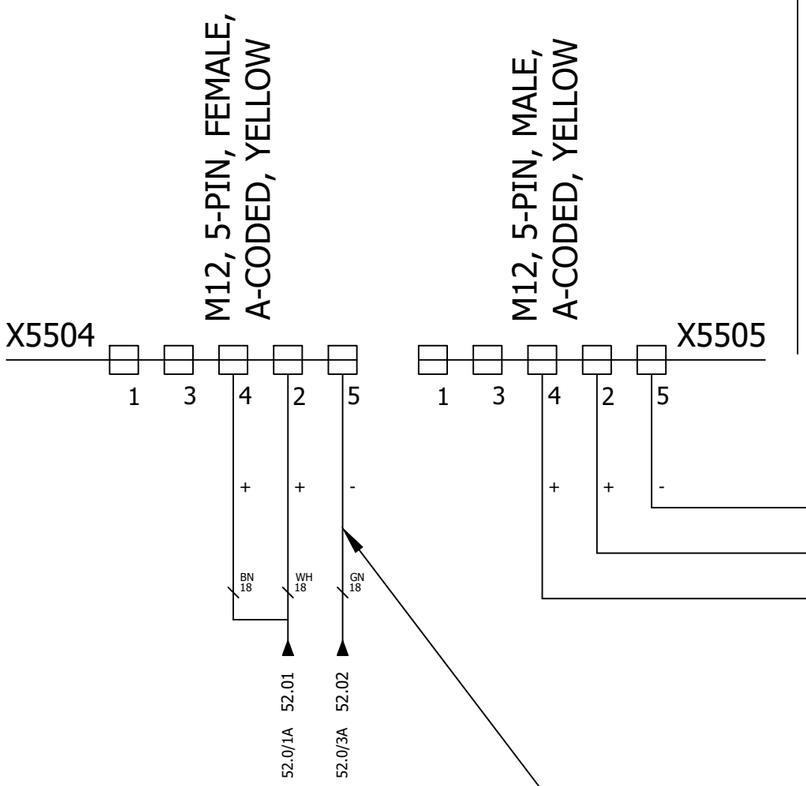
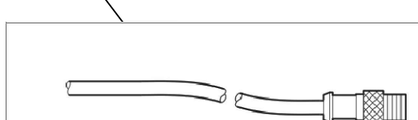
TOTAL PAGES: 175

SHEET 20.6 OF 56

# ROLLER CONVEYOR 1

## 20VFD1



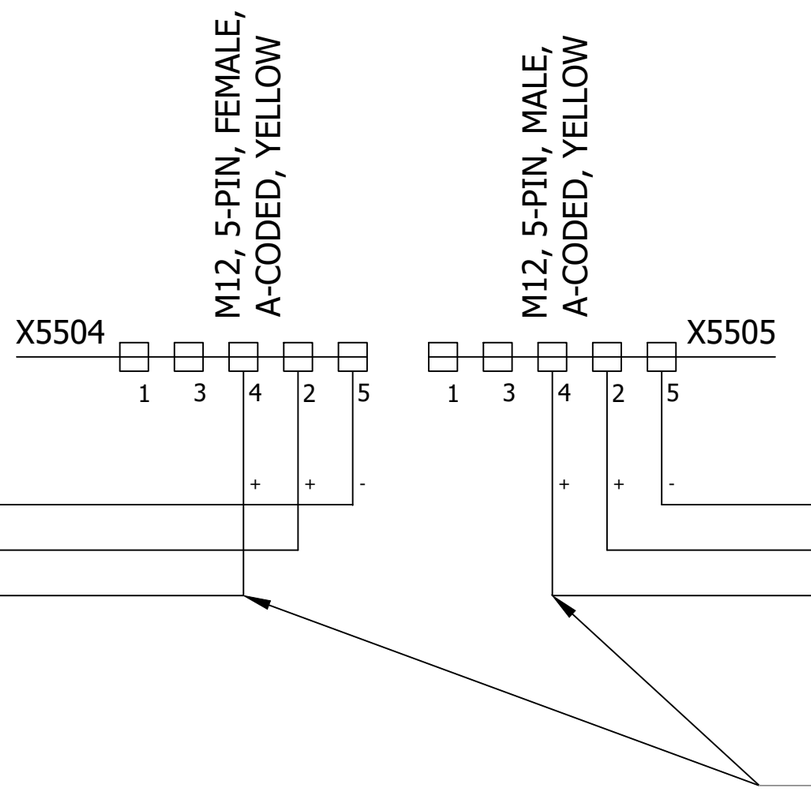



Open — M12, 5-pin, A-coded, male

| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110943 | LiYCY | | |

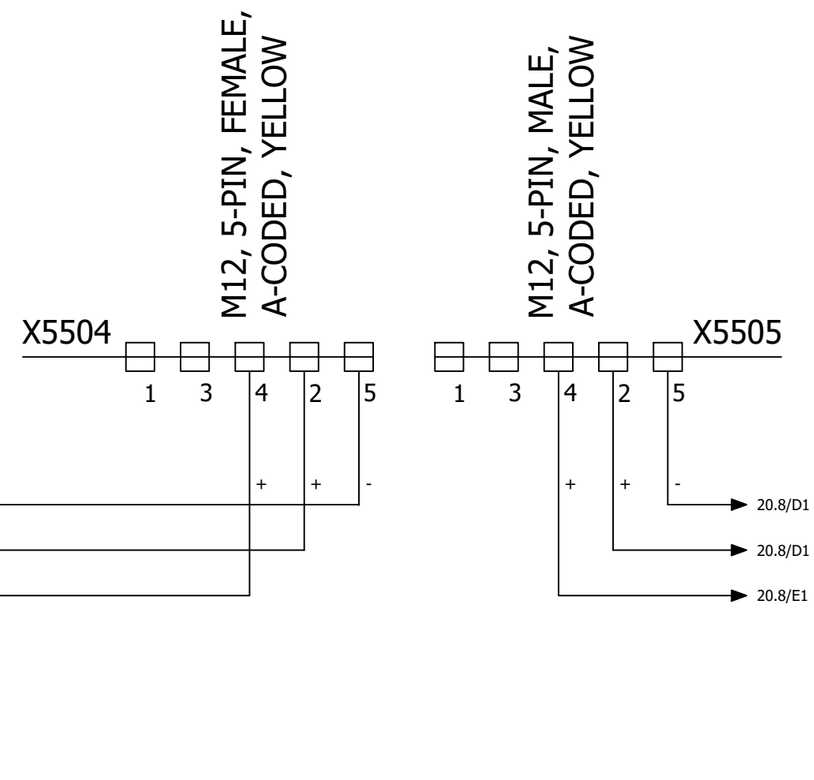
# ROLLER CONVEYOR 2

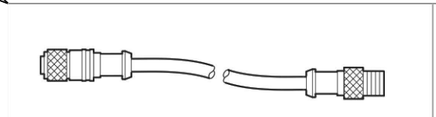
## 20VFD2




# ROLLER CONVEYOR 3

## 20VFD3



M12, 5-pin, A-coded, female — M12, 5-pin, A-coded, male

| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110935 | LiYCY | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD STO

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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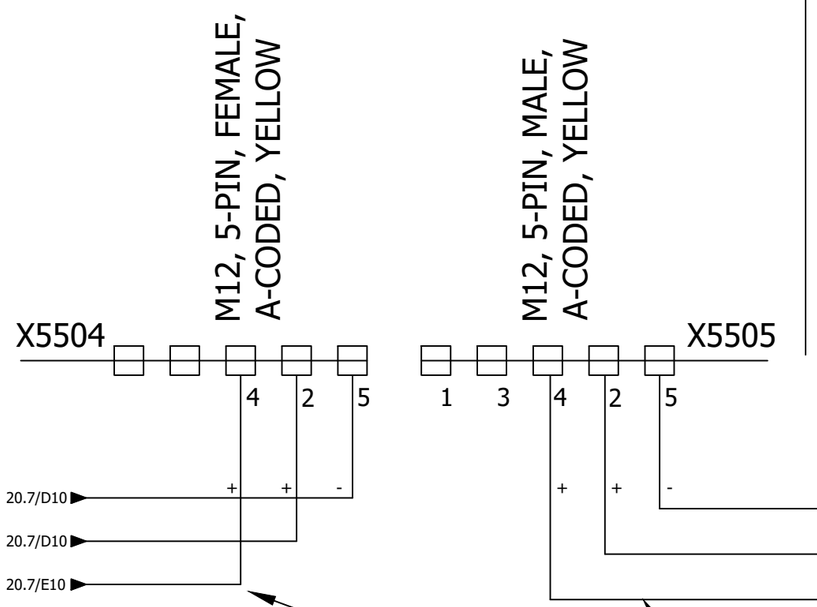
TOTAL PAGES: 175

SHEET 20.7 OF 56

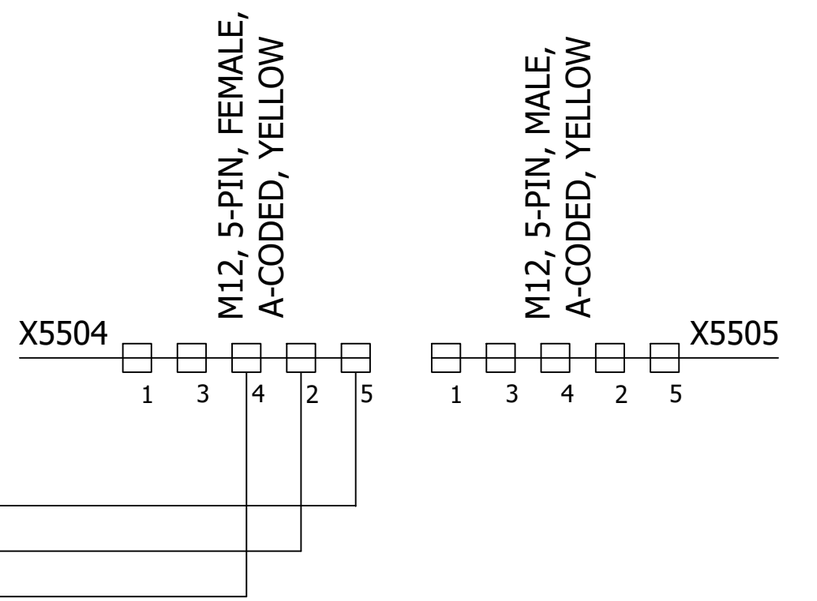
## ROLLER CONVEYOR 1

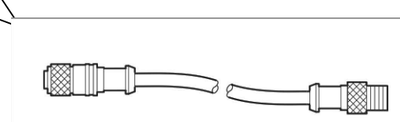
# 20VFD4




## ROLLER CONVEYOR 5

# 20VFD5



M12, 5-pin, A-coded, female

M12, 5-pin, A-coded, male

| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110935 | LiYCY | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD STO

| REV | DESCRIPTION | DATE |
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TOTAL PAGES: 175

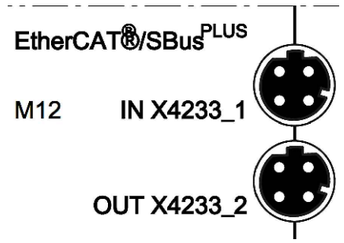
SHEET 20.8 OF 56

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

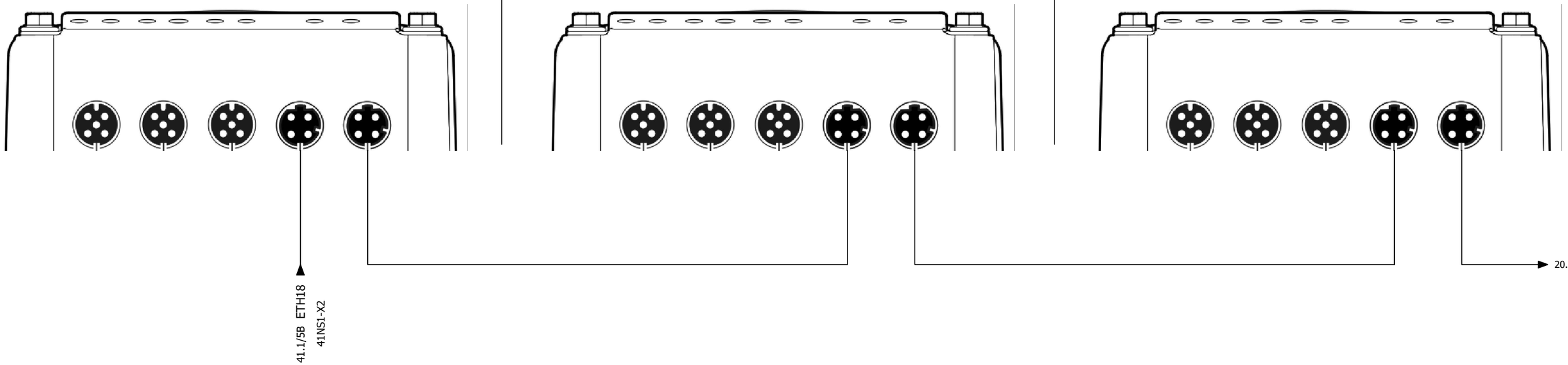
# 20VFD1

# 20VFD2

# 20VFD3




X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK




# ROLLER CONVEYOR 1

# ROLLER CONVEYOR 2

# ROLLER CONVEYOR 3

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

| REV | DESCRIPTION | DATE |
|---|---|---|
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
|---|---|---|
| PAUL | Carlos A | C40572 |
| DATE | DATE | CURRENT REVISION: |
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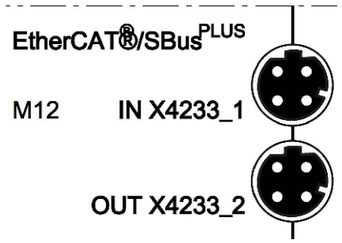
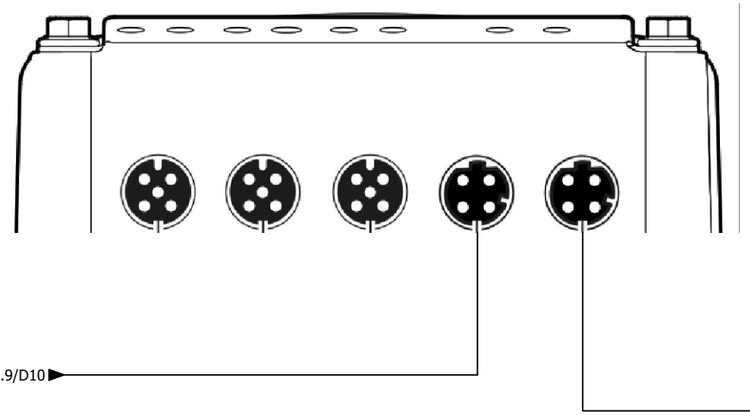
TOTAL PAGES: 175

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

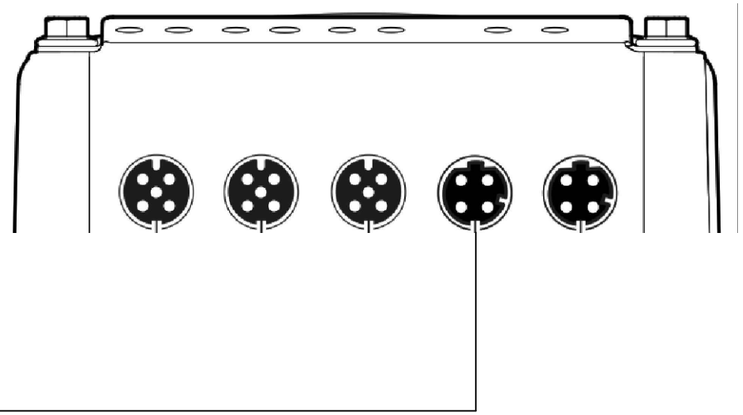
## 20VFD4




X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK




## 20VFD5



ROLLER
CONVEYOR 1

ROLLER
CONVEYOR 5

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

| REV | DESCRIPTION | DATE |
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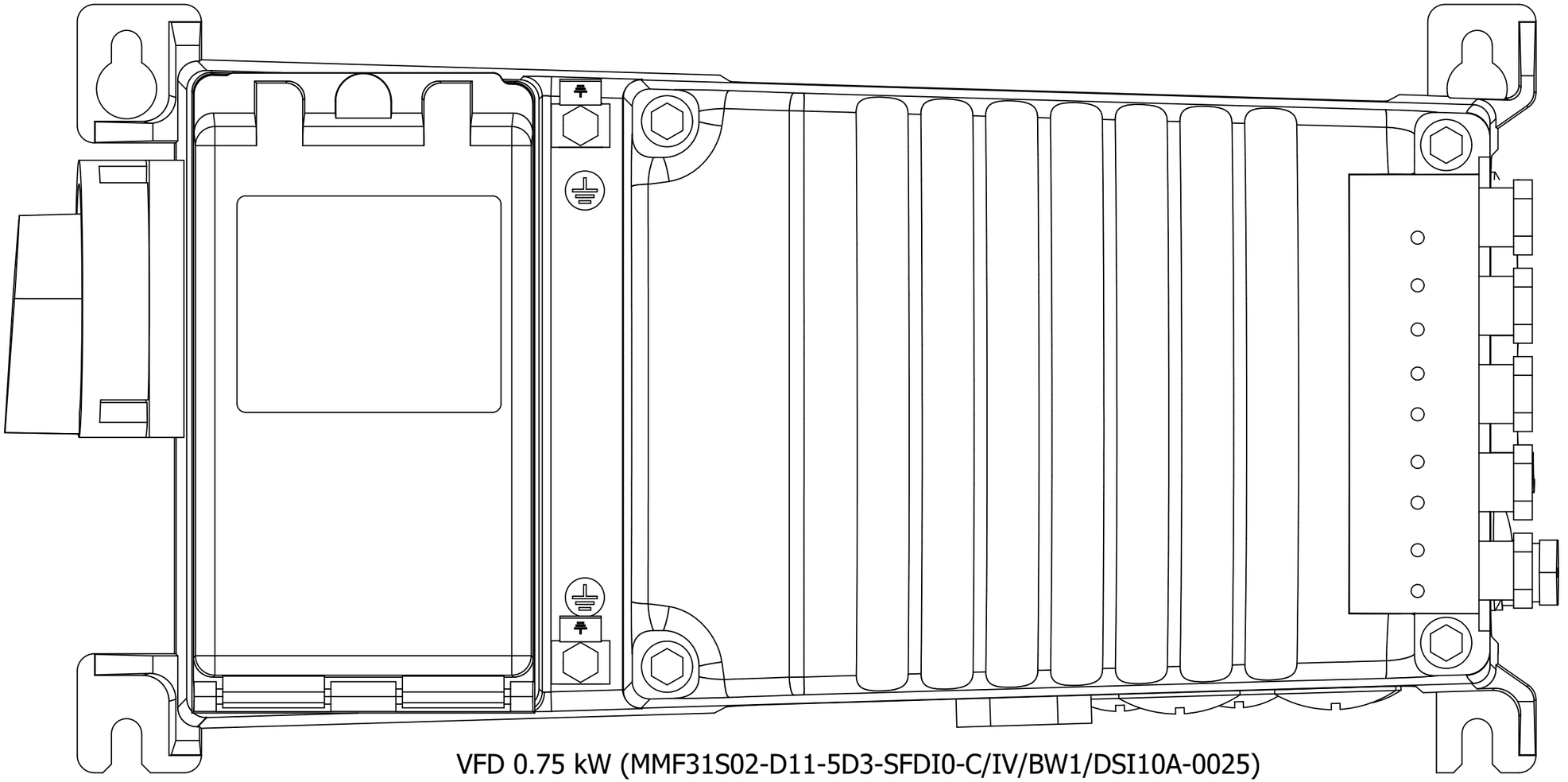
TOTAL PAGES: 175



VFD 0.75 kW (MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025)

CAROUSEL CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

| 21VFD1 | 21VFD2 | 21VFD3 |
|---|---|---|
| CHAIN CONVEYOR 1 | Transfer Conveyor 1 | Transfer Conveyor 2 |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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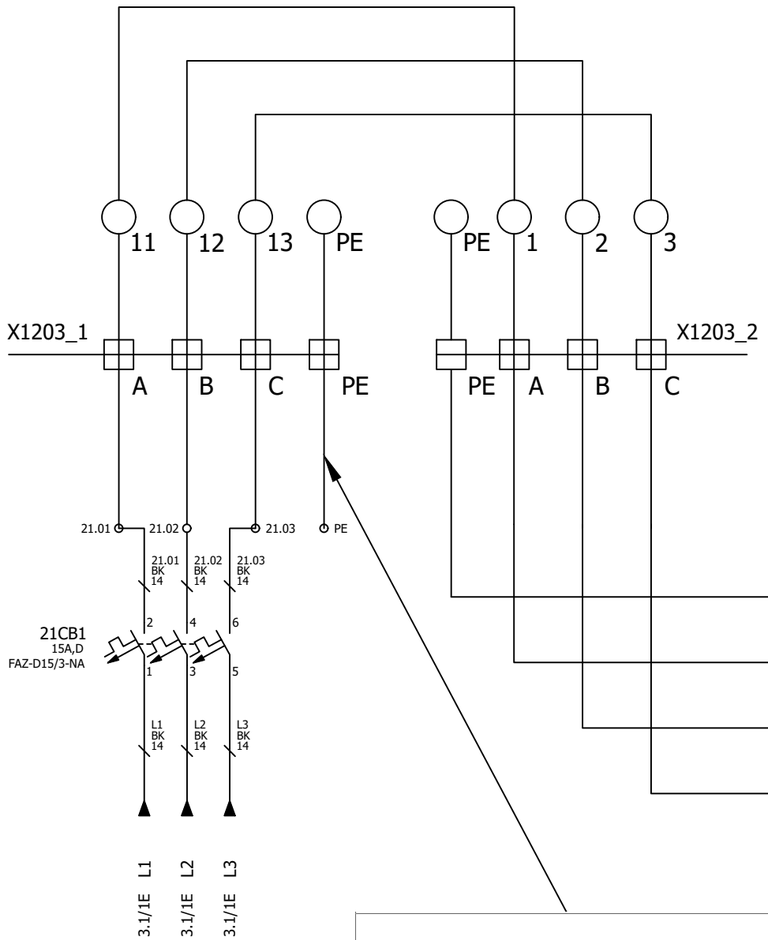
JOB # C40572

TOTAL PAGES: 175

SHEET 21.0 OF 56

# CHAIN CONVEYOR 1

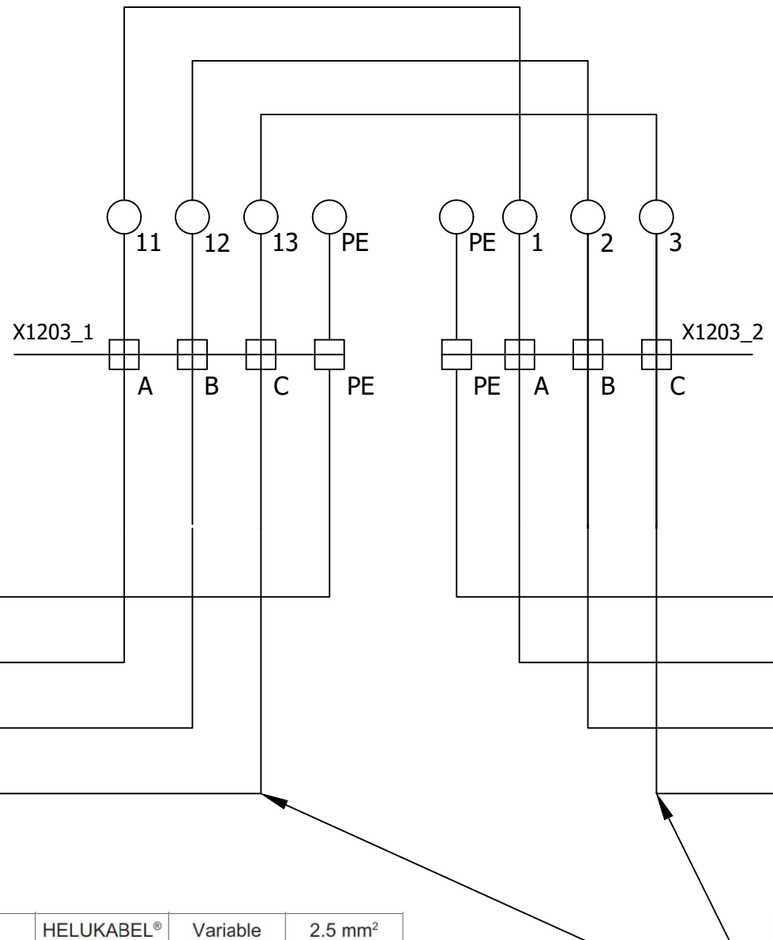
## 21VFD1




| | | | | |
|---|---|---|---|---|
| Open — M23, coding ring: black, male | UL: 18153283 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

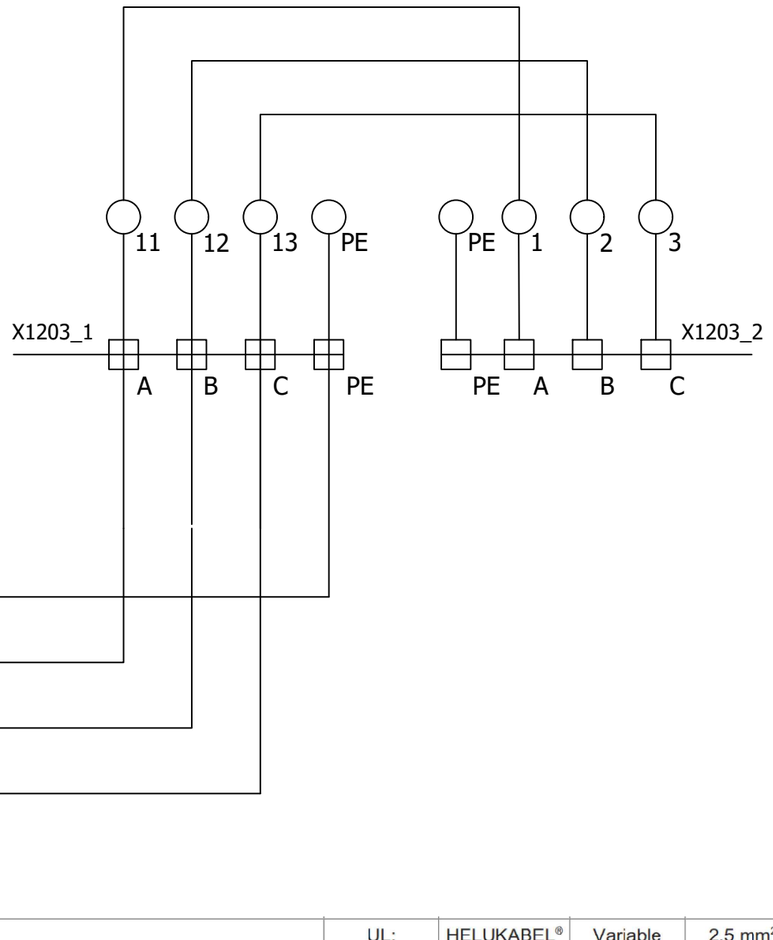
# Transfer Conveyor 1

## 21VFD2




# Transfer Conveyor 2

## 21VFD3




| | | | | |
|---|---|---|---|---|
| M23, coding ring: black, male — M23, coding ring: black, male | UL: 18153267 | HELUKABEL® – JZ-602 | Variable | 2.5 mm² / AC 500 V |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 POWER

| REV | DESCRIPTION | DATE |
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| DATE 9/17/26 | DATE 9/8/26 | CURRENT REVISION: D |

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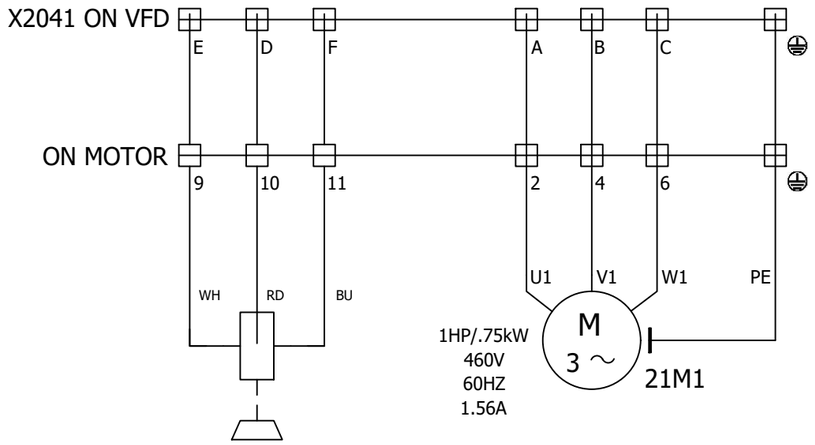
TOTAL PAGES: 175

SHEET 21.1 OF 56

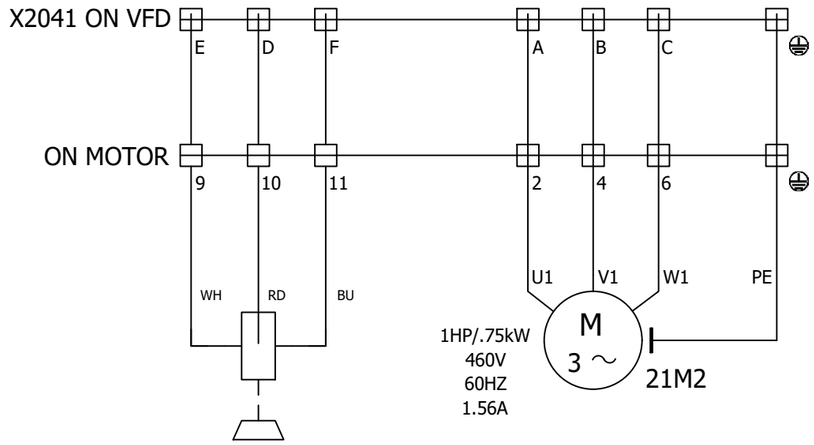
## CHAIN CONVEYOR 1

# 21VFD1



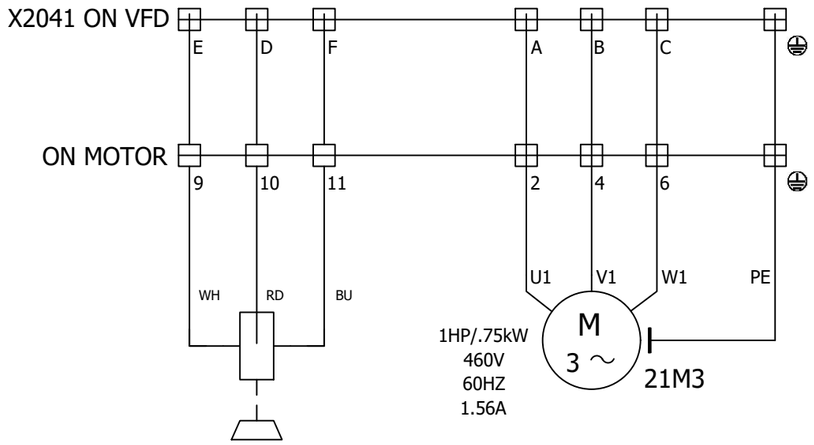

## Transfer Conveyor 1

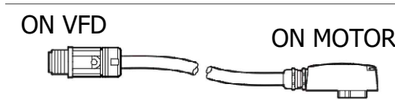
# 21VFD2




## Transfer Conveyor 2

# 21VFD3




| CE/UL: | LEONI LEHC® | Variable | 1.5 mm² |
|---|---|---|---|
| 28125940 | 000749 | | / AC 500 V |

# Motor 0.75kW (DRN80M4/BE1HR/ISU/TF)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2 MOTOR POWER

| REV | DESCRIPTION | DATE |
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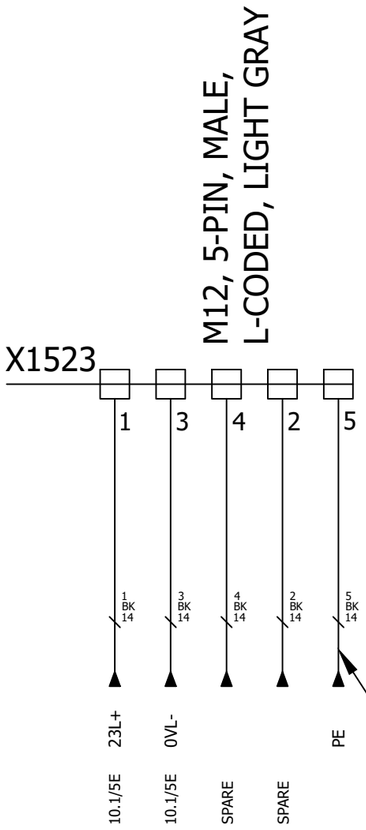
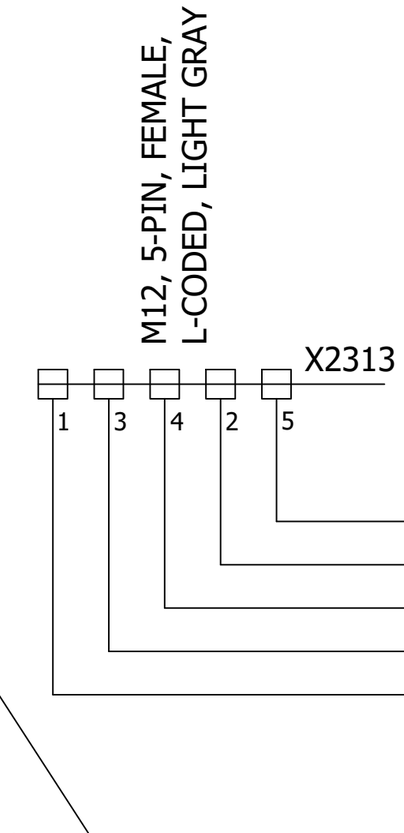
TOTAL PAGES: 175

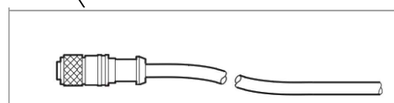
SHEET 21.2 OF 56

## CHAIN CONVEYOR 1

# 21VFD1



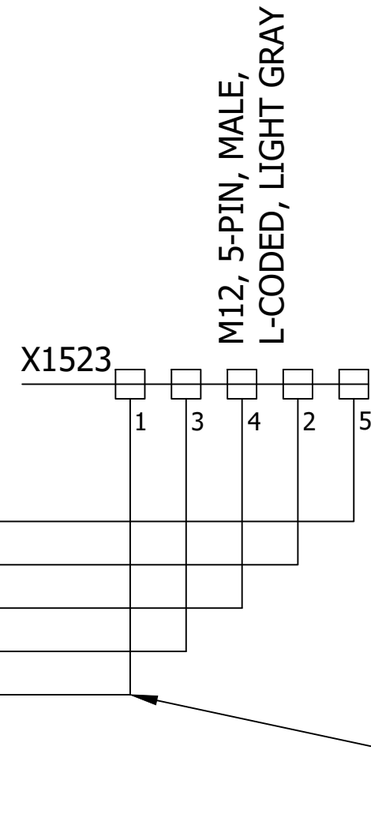



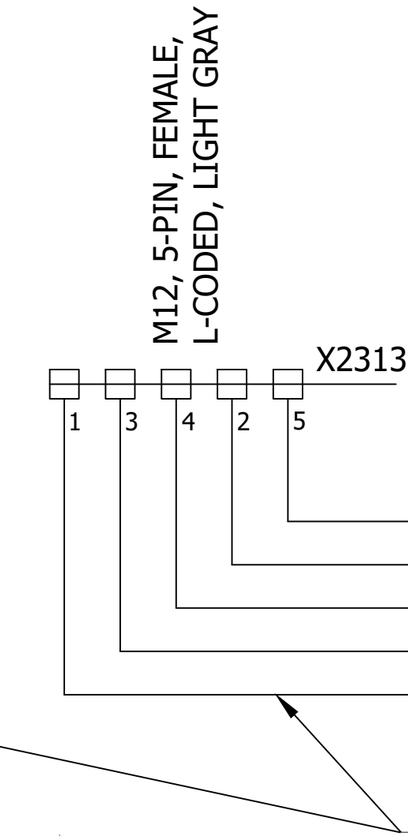




| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28117786 | JZ-500 | | |

## Transfer Conveyor 1

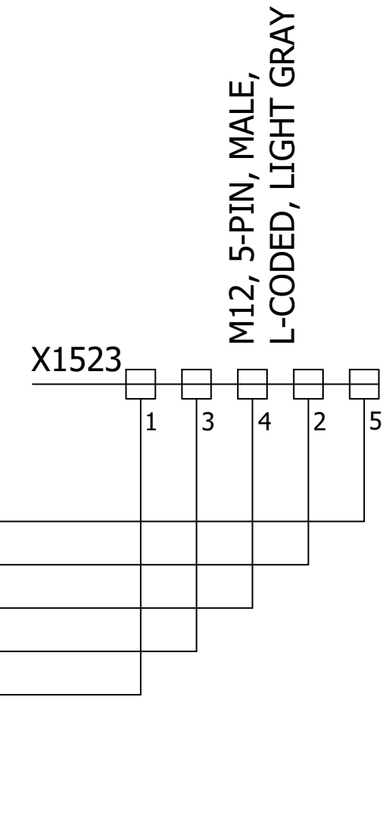
# 21VFD2

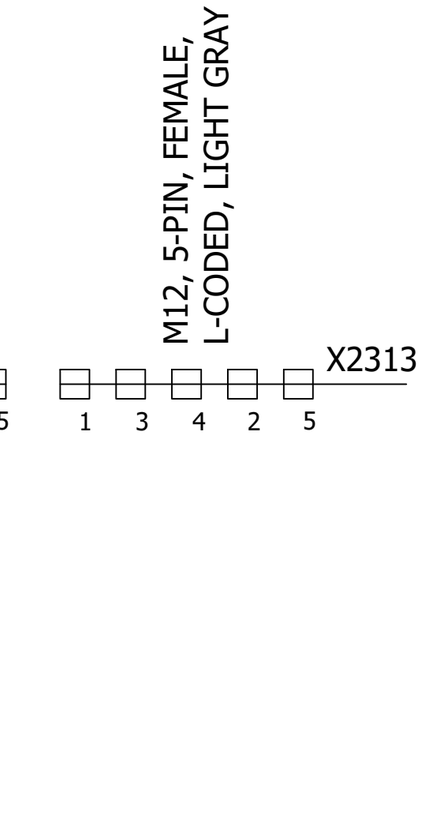


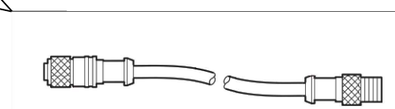




## Transfer Conveyor 2

# 21VFD3




| CE/UL: | HELUKABEL® | Variable | 5 × 2.5 mm² / DC 60 V |
|---|---|---|---|
| 28114345 | JZ-500 | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2 VFD 24VDC POWER

| REV | DESCRIPTION | DATE |
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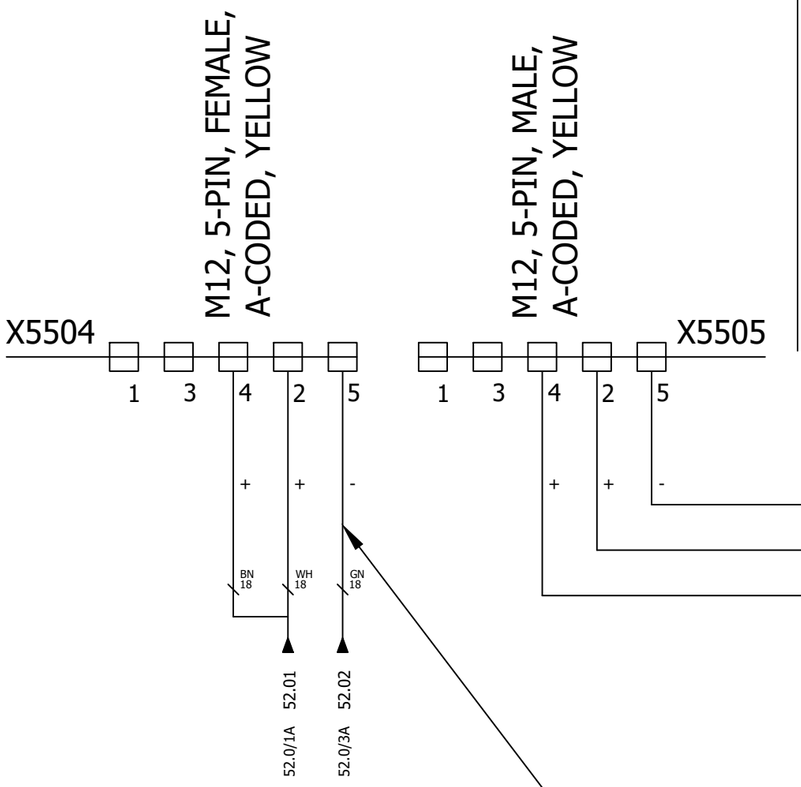
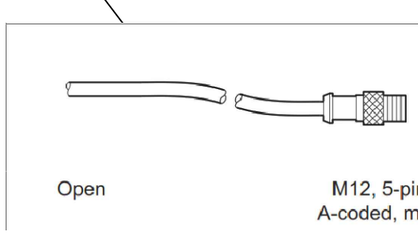
TOTAL PAGES: 175

## CHAIN CONVEYOR 1

# 21VFD1

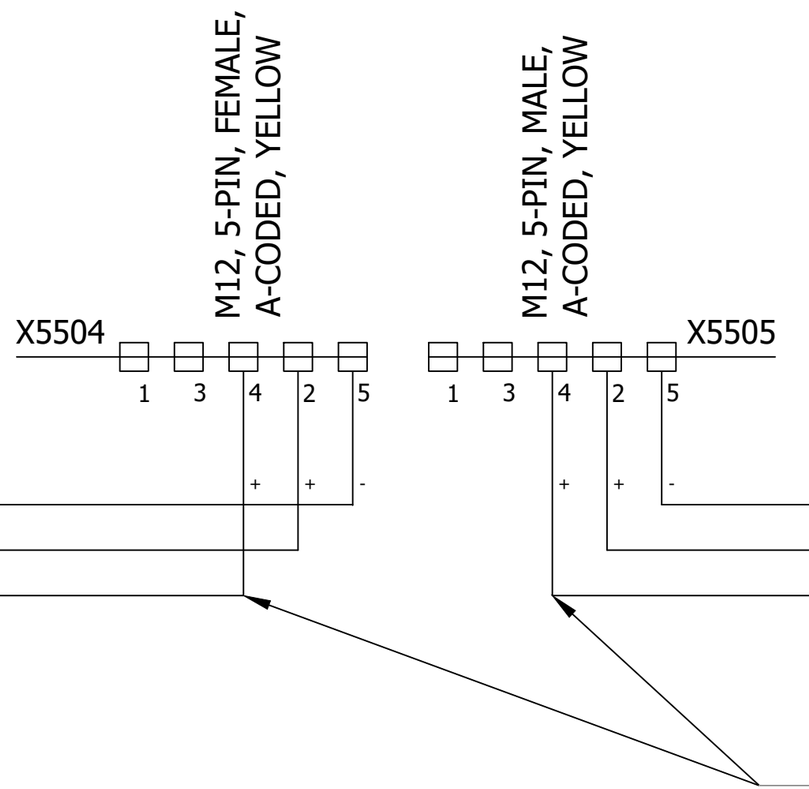



| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110943 | LiYCY | | |

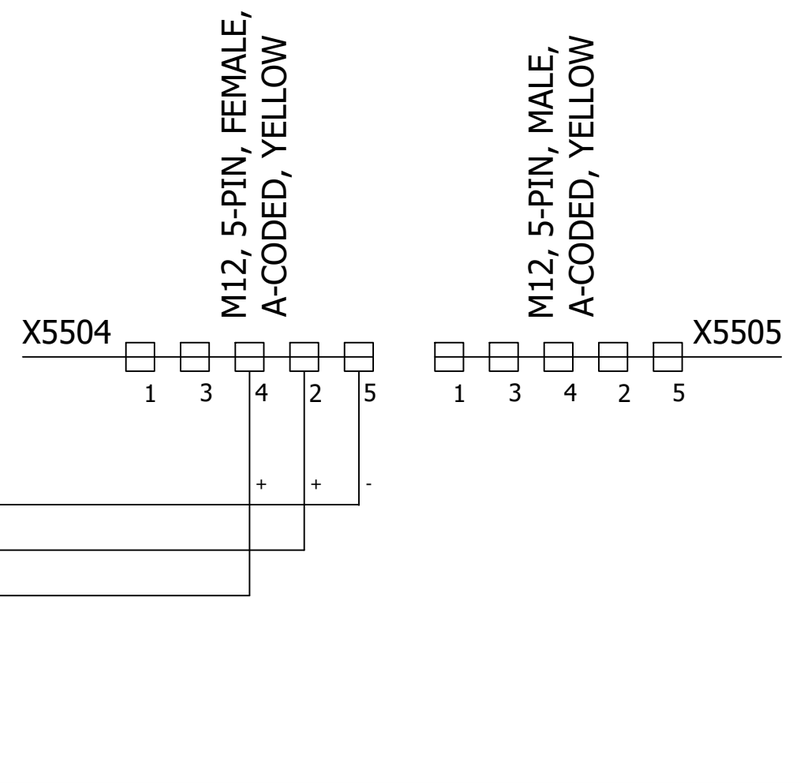
## Transfer Conveyor 1

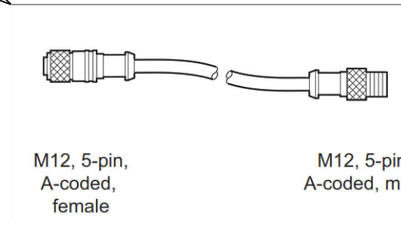
# 21VFD2




## Transfer Conveyor 2

# 21VFD3




| CE/UL: | HELUKABEL® | Variable | 3 × 0.75 mm² / DC 60 V |
|---|---|---|---|
| 28110935 | LiYCY | | |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 2 VFD STO

| REV | DESCRIPTION | DATE |
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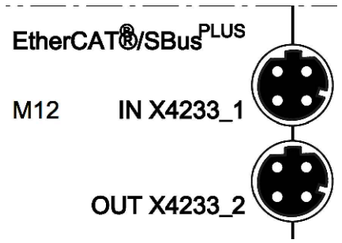
TOTAL PAGES: 175

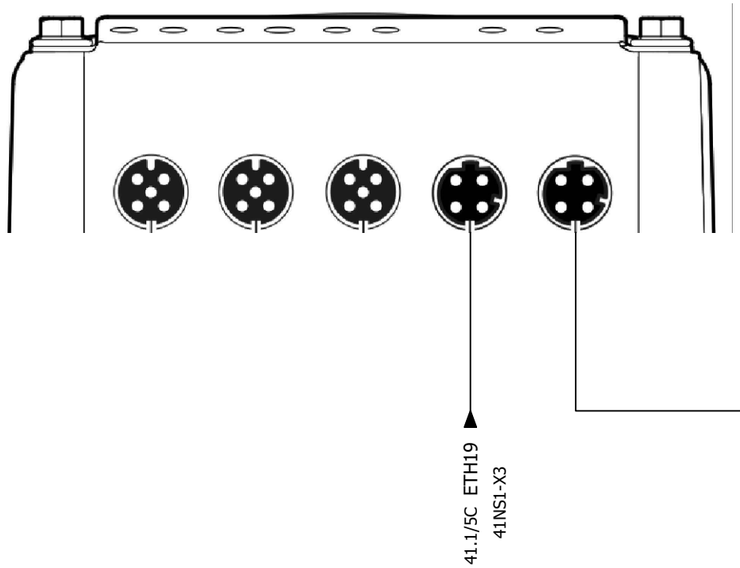
SHEET 21.4 OF 56

| Assignment | | |
|---|---|---|
| **Contact** | **Signal** | **Description** |
| 1 | TX+ | Transmit line (+) |
| 2 | RX+ | Receive line (+) |
| 3 | TX- | Transmit line (-) |
| 4 | RX- | Receive line (-) |

# 21VFD1

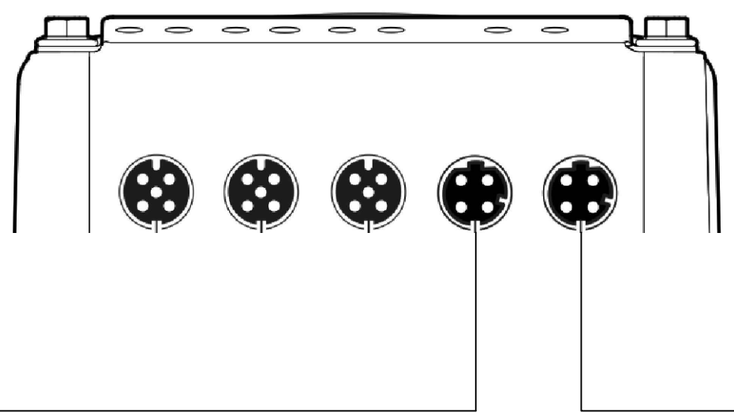



X4233_1 & X4233_2
M12, 4-PIN, FEMALE,
D-CODED, BLACK



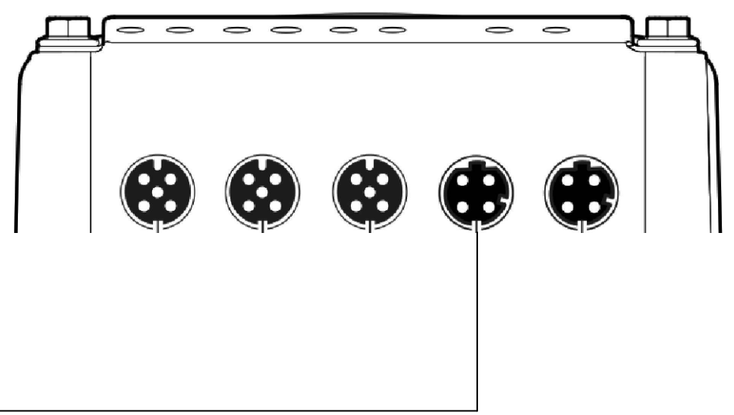

CHAIN
CONVEYOR 1

# 21VFD2



Transfer
Conveyor 1

# 21VFD3



Transfer
Conveyor 2

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: CAROUSEL CONVEYORS AREA VFD's SECTION 1 VFD EtherCAT INTERFACE

| REV | DESCRIPTION | DATE |
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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SPARE

| REV | DESCRIPTION | DATE |
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| PAUL | Carlos A | C40572 |
| DATE | DATE | CURRENT REVISION: |
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JOB # C40572

TOTAL PAGES: 175

SHEET 22.0 OF 56

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EL2809
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EL1918
EL2912
EL9410
EL1322
EL1322
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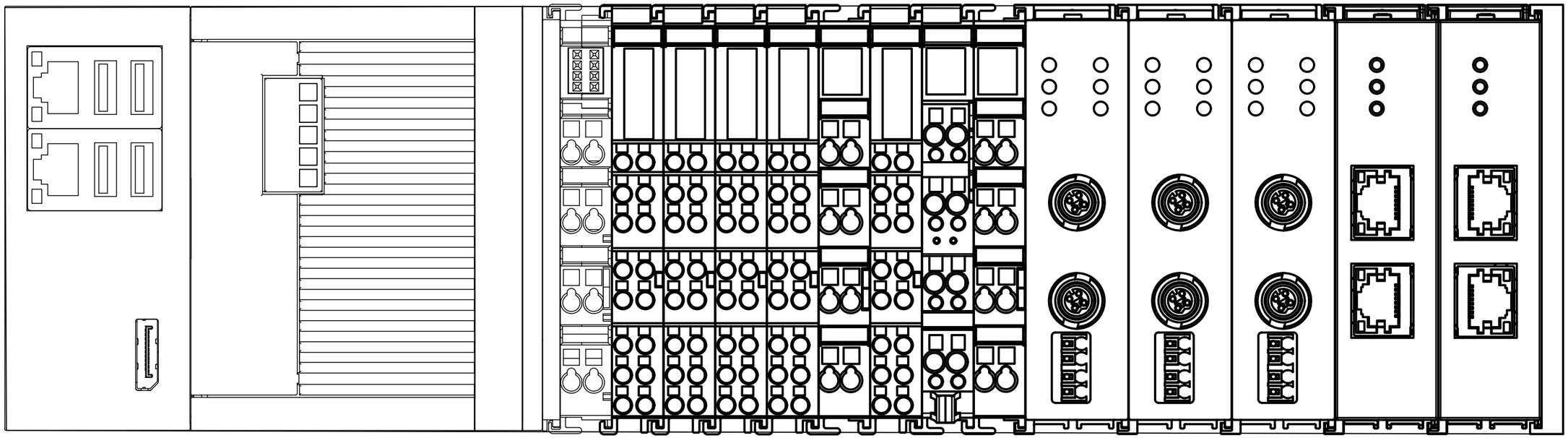
EL6633
EK1122



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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: PLC OVERVIEW RACK

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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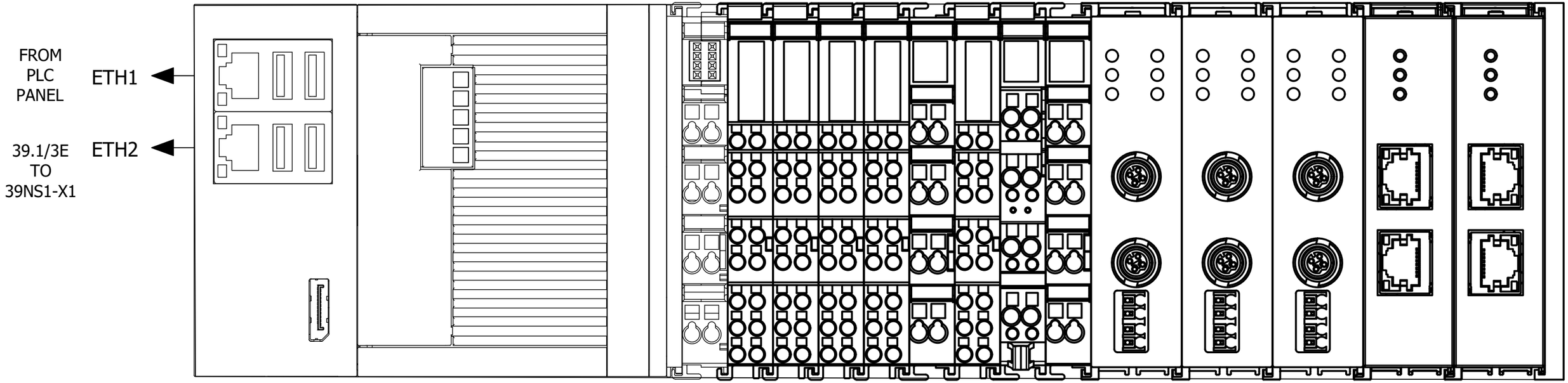
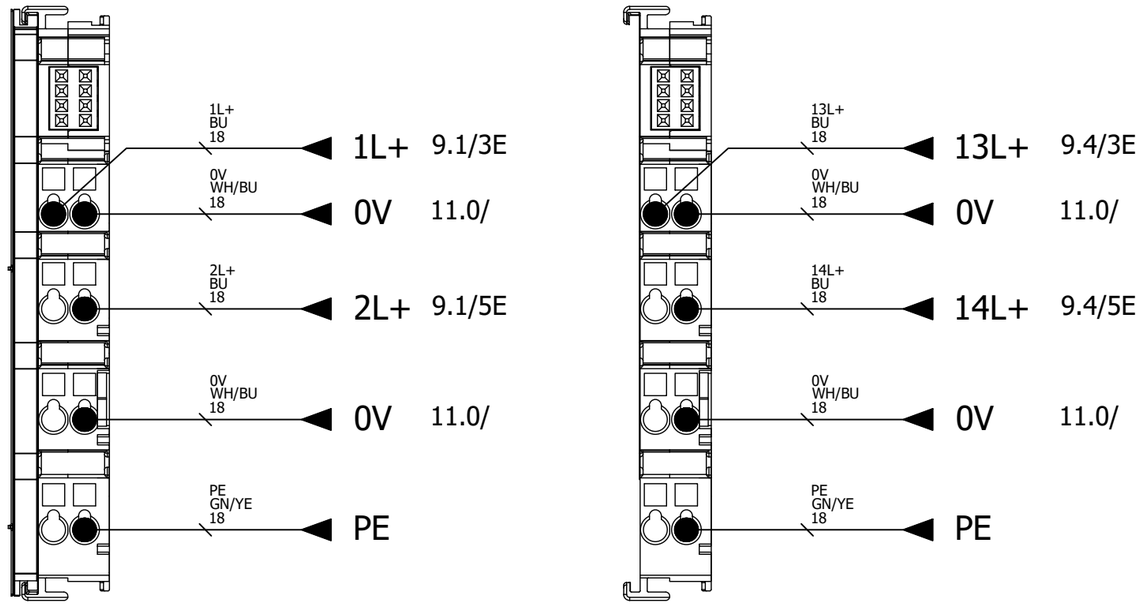
TOTAL PAGES: 175

SHEET 23.0 OF 56







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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: PLC 24VDC

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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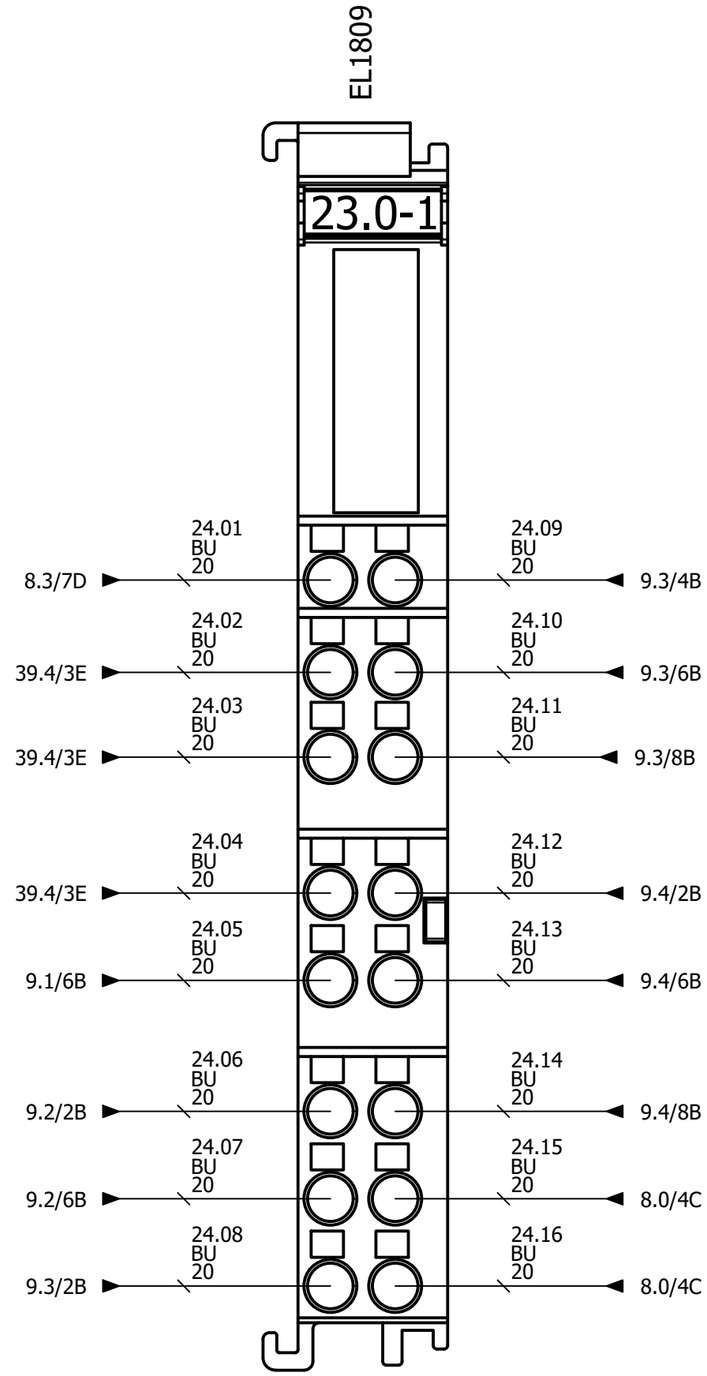
JOB # C40572

TOTAL PAGES: 175

SHEET 23.1 OF 56

# 24VDC INPUTS




4.4/2E
11.0/

LM11
11L+ 0V

REDUNDANCY OK 8RS1
HANDHELD HMI KEY-SWITCH 1 NO
HANDHELD HMI KEY-SWITCH 2 NO
HANDHELD HMI CONNECTED
LM3-4 FAULT
LM5-6 FAULT
LM7-8 FAULT
LM9 FAULT

/9B /9B

8.3/7D
39.4/3E
39.4/3E
39.4/3E
9.1/6B
9.2/2B
9.2/6B
9.3/2B

9.3/4B
9.3/6B
9.3/8B
9.4/2B
9.4/6B
9.4/8B
8.0/4C
8.0/4C

/2F /2F

0V 11L+

LM10 FAULT
LM11 FAULT
LM12 FAULT
LM13-14 FAULT
LM15 FAULT
LM16 FAULT
8PS1 OUT 100%
8PS1 DC OK

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DC INPUTS CARD 23.0-1

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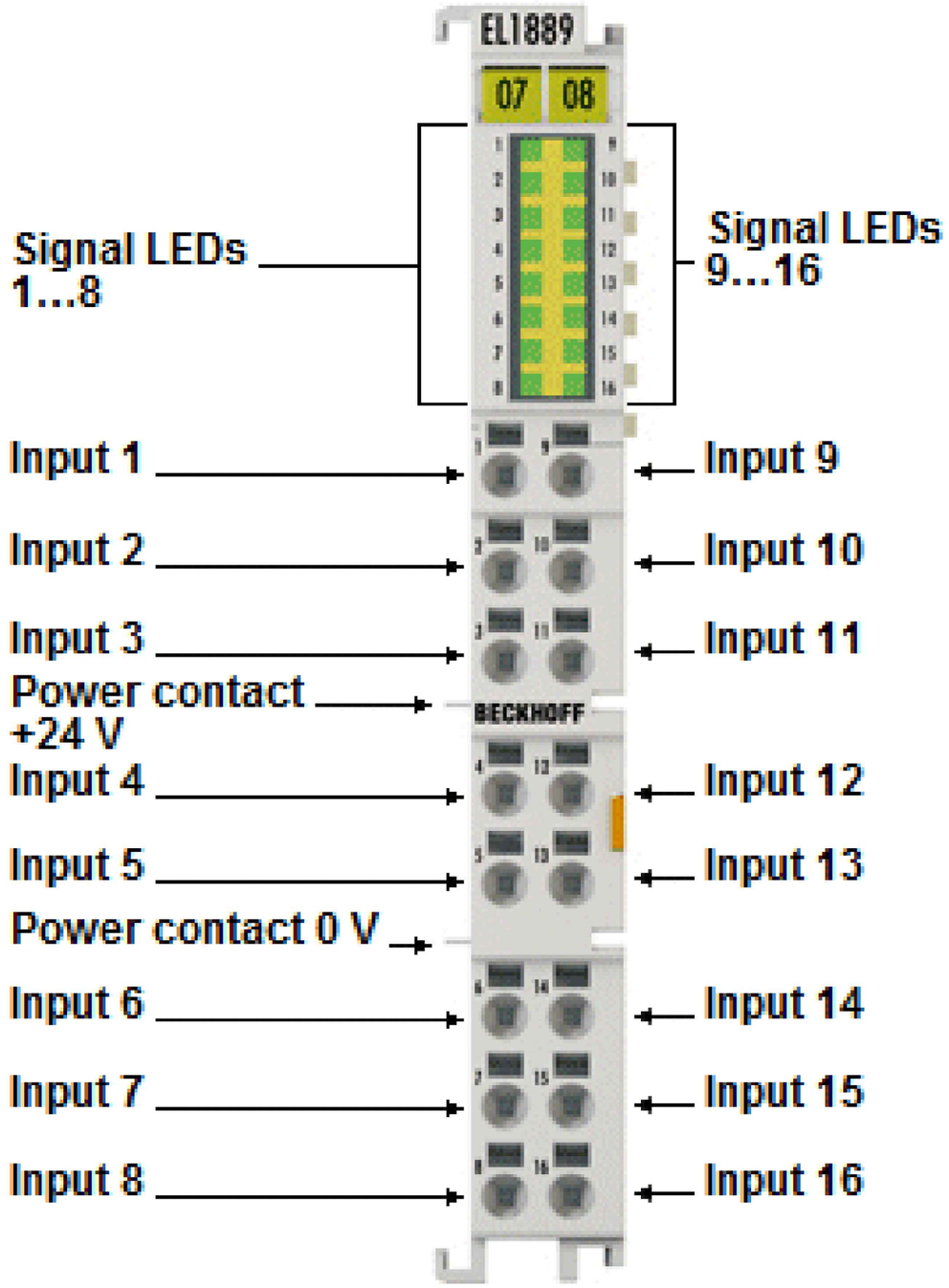
TOTAL PAGES: 175

SHEET 24.0 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DC INPUTS CARD 23.0-1

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JOB # C40572

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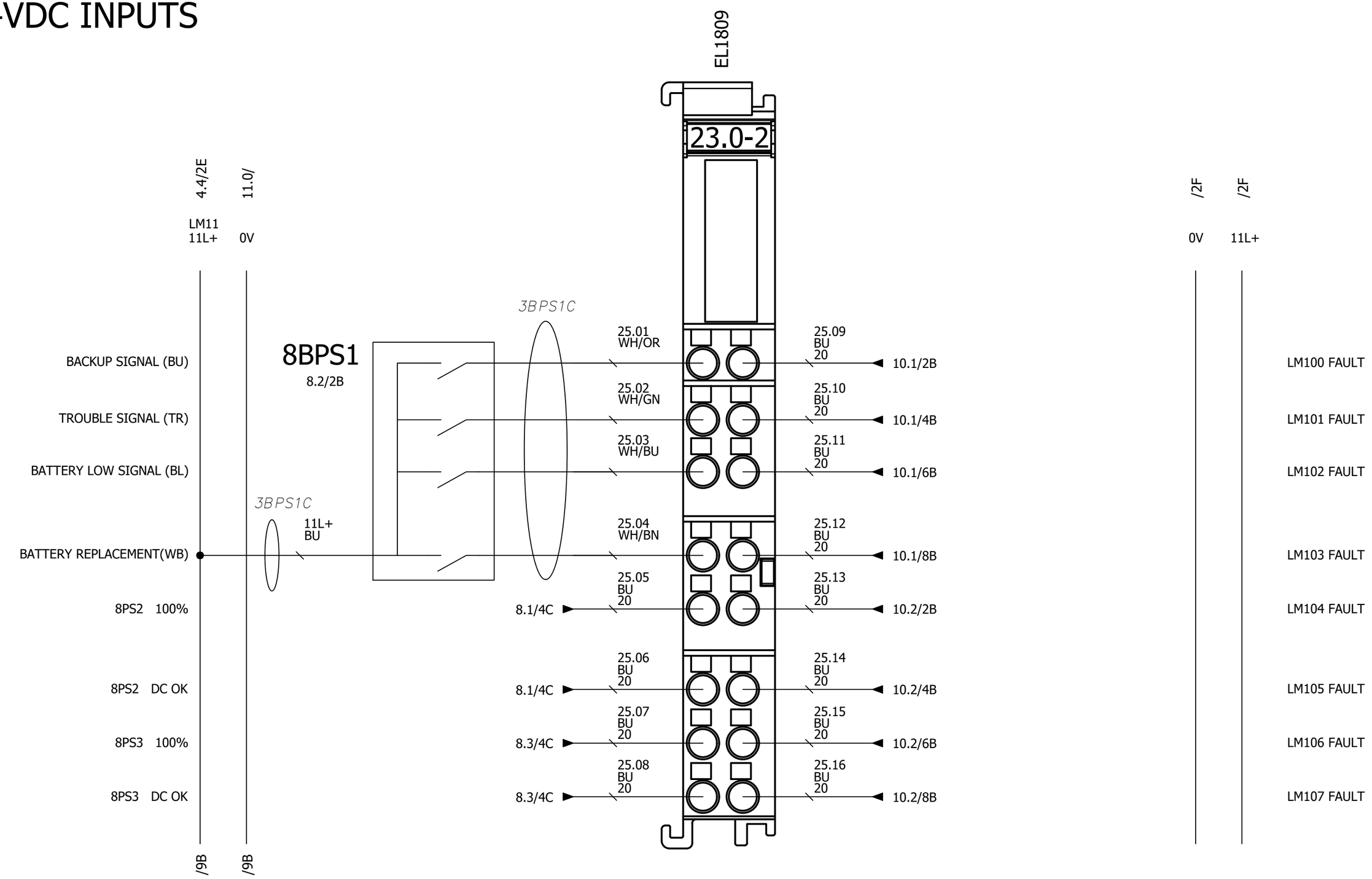
SHEET 24.0 OF 56

# 24VDC INPUTS

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DC INPUTS CARD 23.0-2

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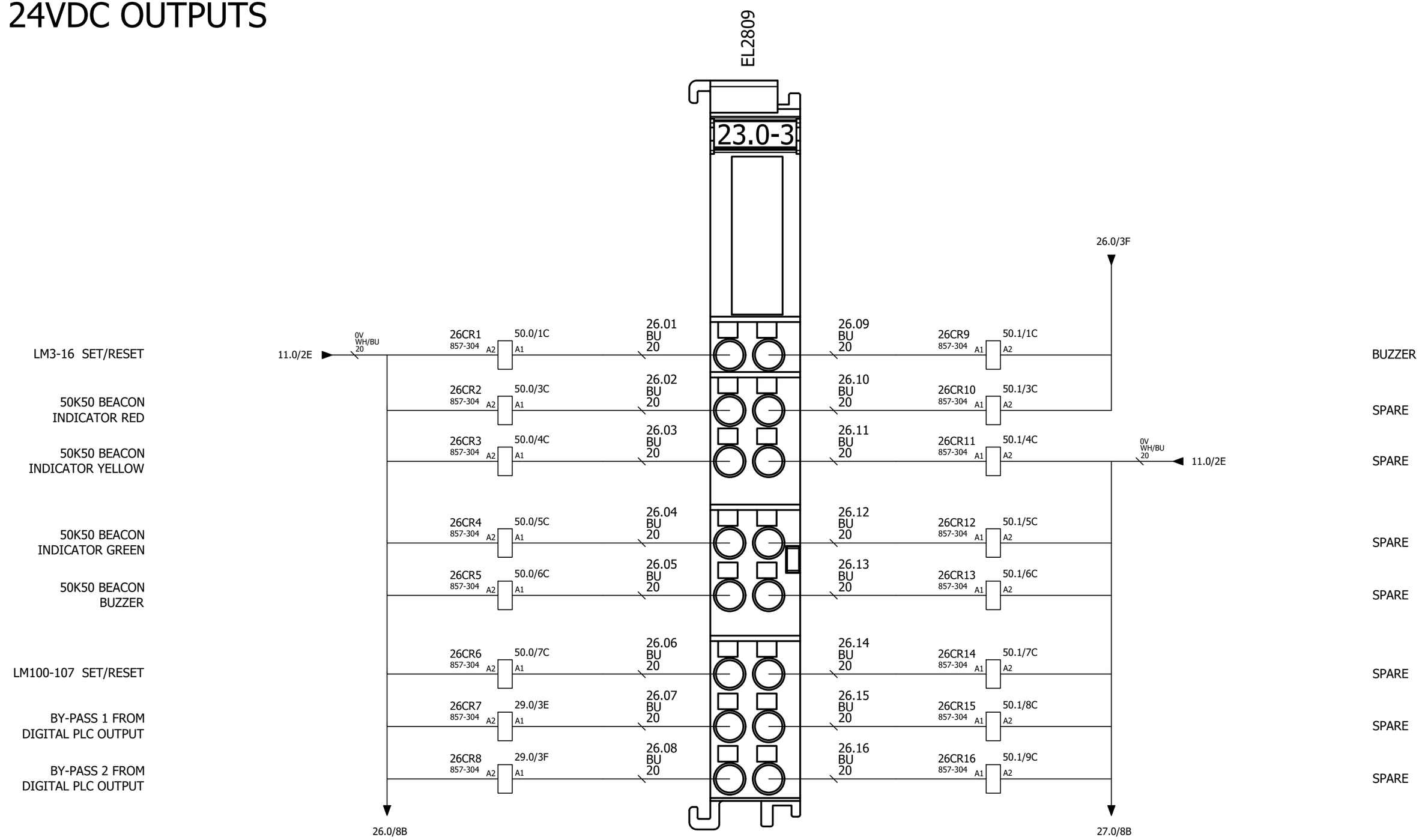
SHEET 25.0 OF 56

# 24VDC OUTPUTS

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DC OUTPUTS CARD 23.0-3

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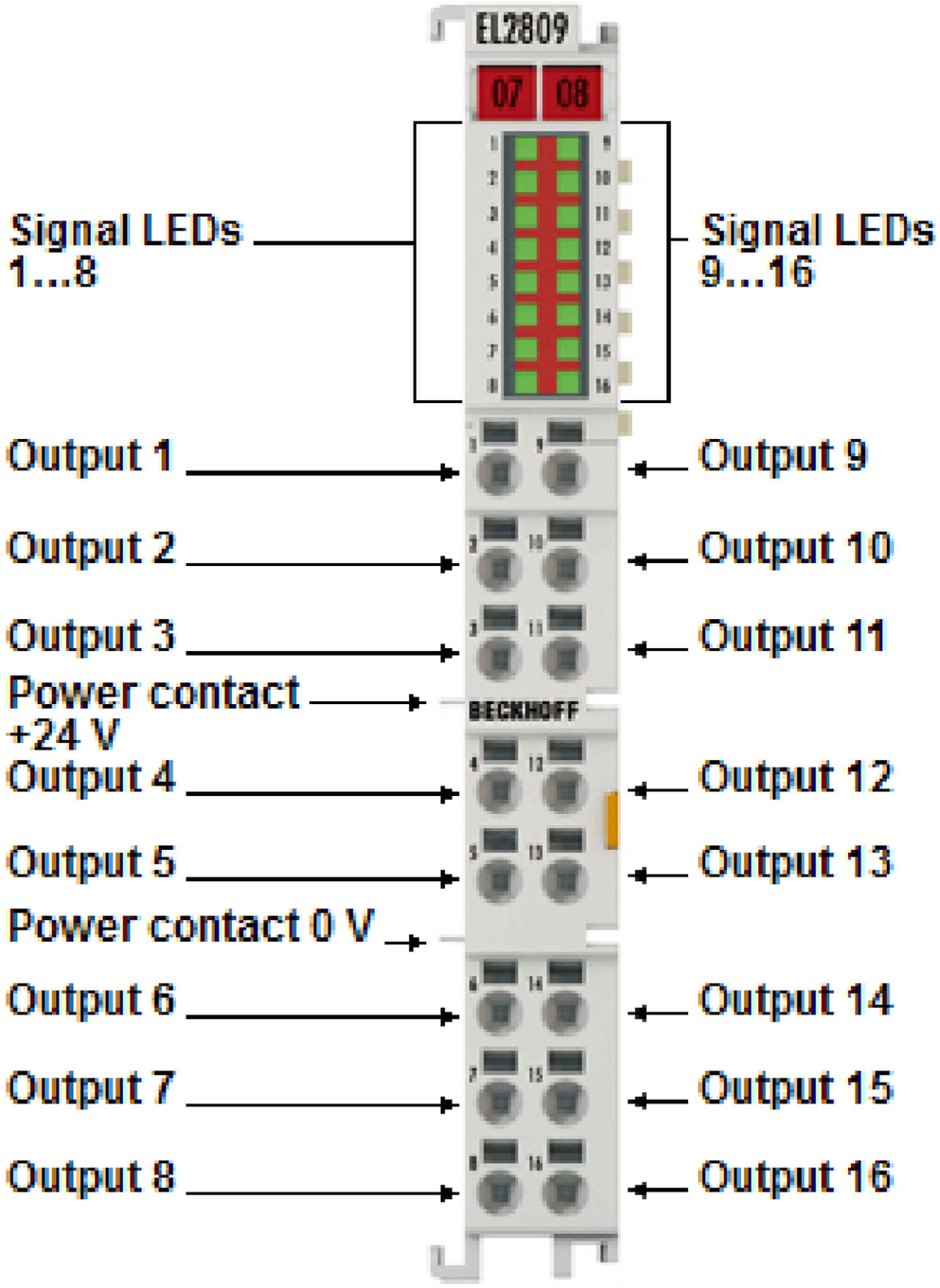
TOTAL PAGES: 175

SHEET 26.0 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DC OUTPUTS CARD 23.0-3

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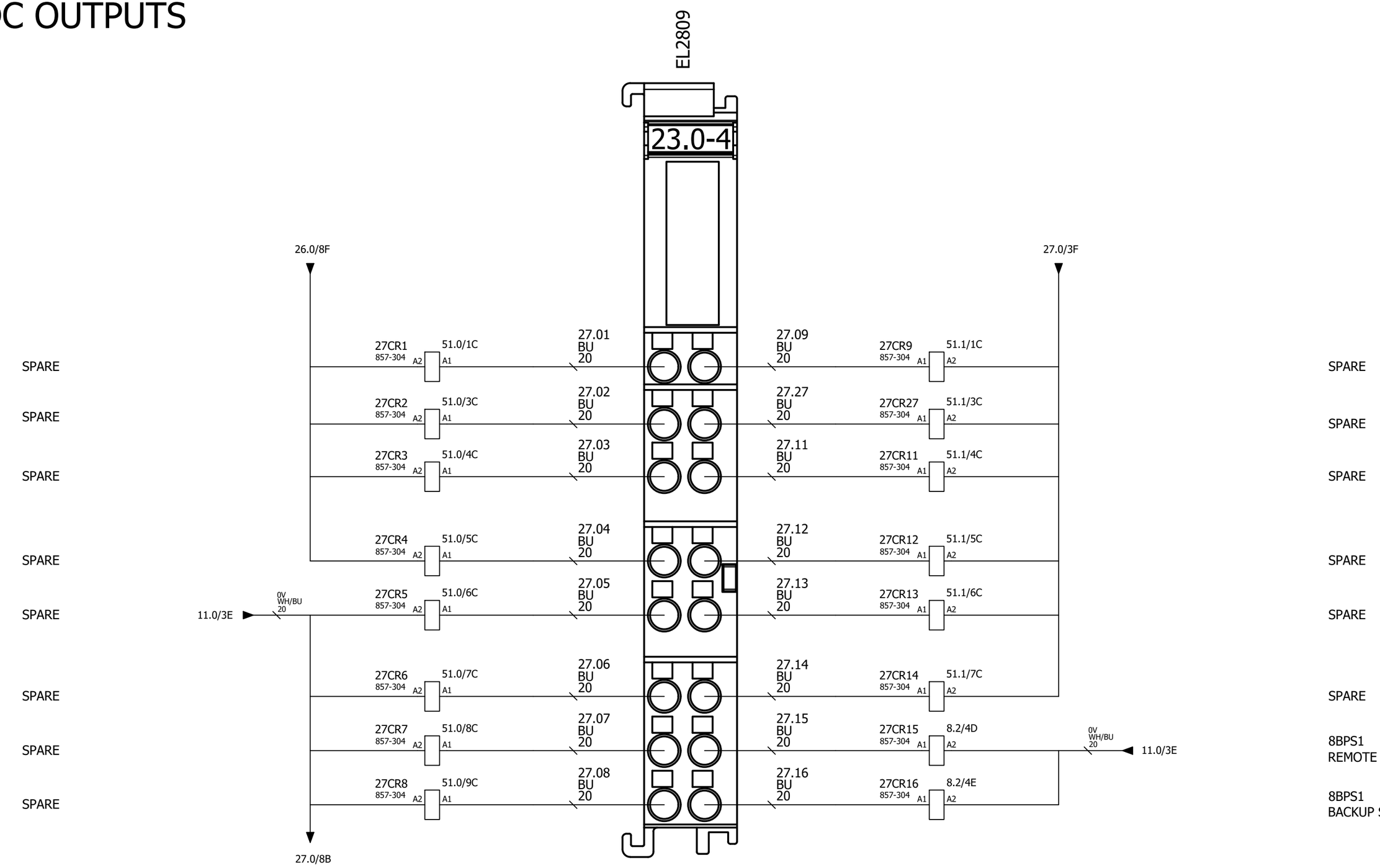
TOTAL PAGES: 175

SHEET 26.1 OF 56

# 24VDC OUTPUTS

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DRIVE LIFTING TRANSFER ROLLER CONVEYOR#3

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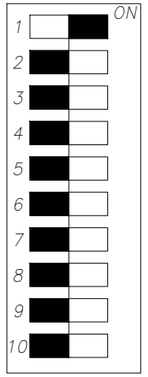
TOTAL PAGES: 175

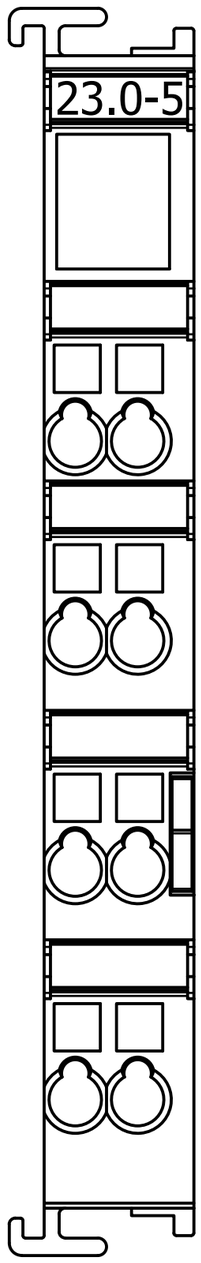
SHEET 27.0 OF 56

# EL 6910 (SAFETY CPU CONTROLLER)

SET
DIPSWITCHES ON
THE MODULE




EL6910

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY CPU CONTROLLER 23.0-5

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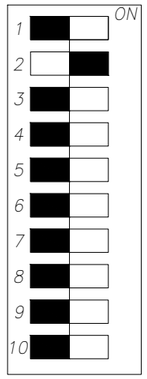
JOB # C40572

TOTAL PAGES: 175

SHEET 28.0 OF 56

# SAFETY INPUTS

SET
DIPSWITCHES ON
THE MODULE

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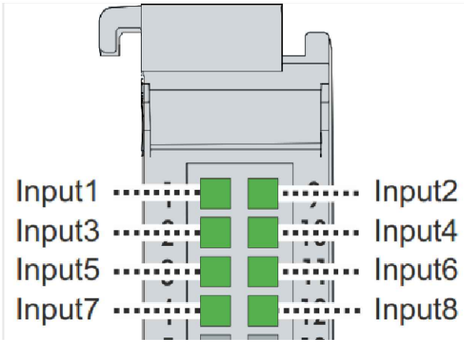


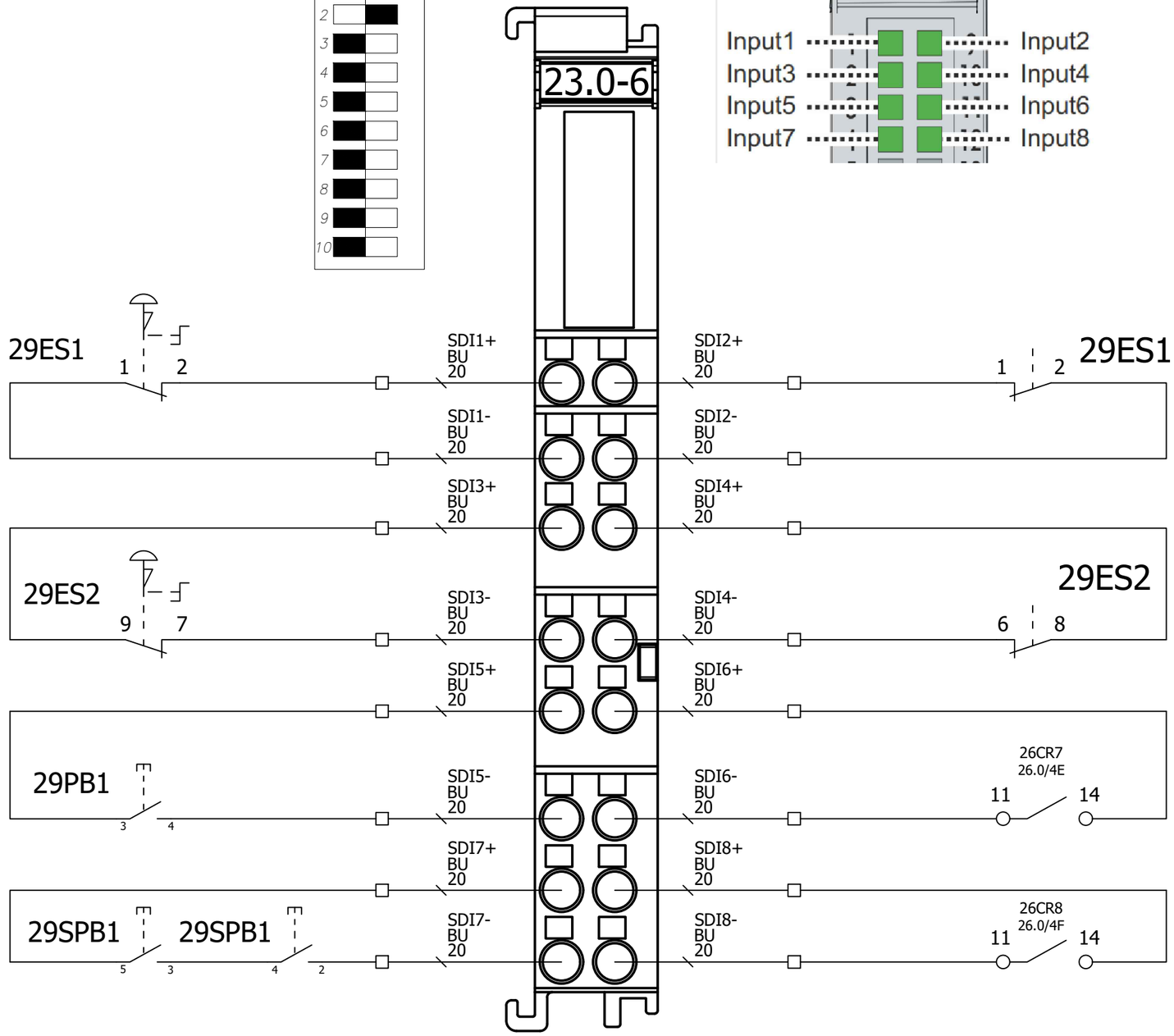


EMERGENCY STOP
ON DOOR

EMERGENCY STOP
ON DOOR

EMERGENCY STOP
ON HMI

EMERGENCY STOP
ON HMI

Emergency Stop
RESET ON DOOR

BY-PASS 1 FROM
DIGITAL PLC OUTPUT

Enabling switch
ON HMI

BY-PASS 2 FROM
DIGITAL PLC OUTPUT

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY INPUTS 23.0-6

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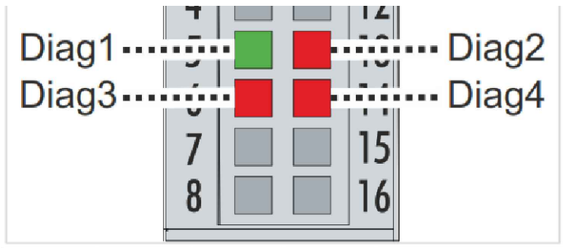
JOB # C40572

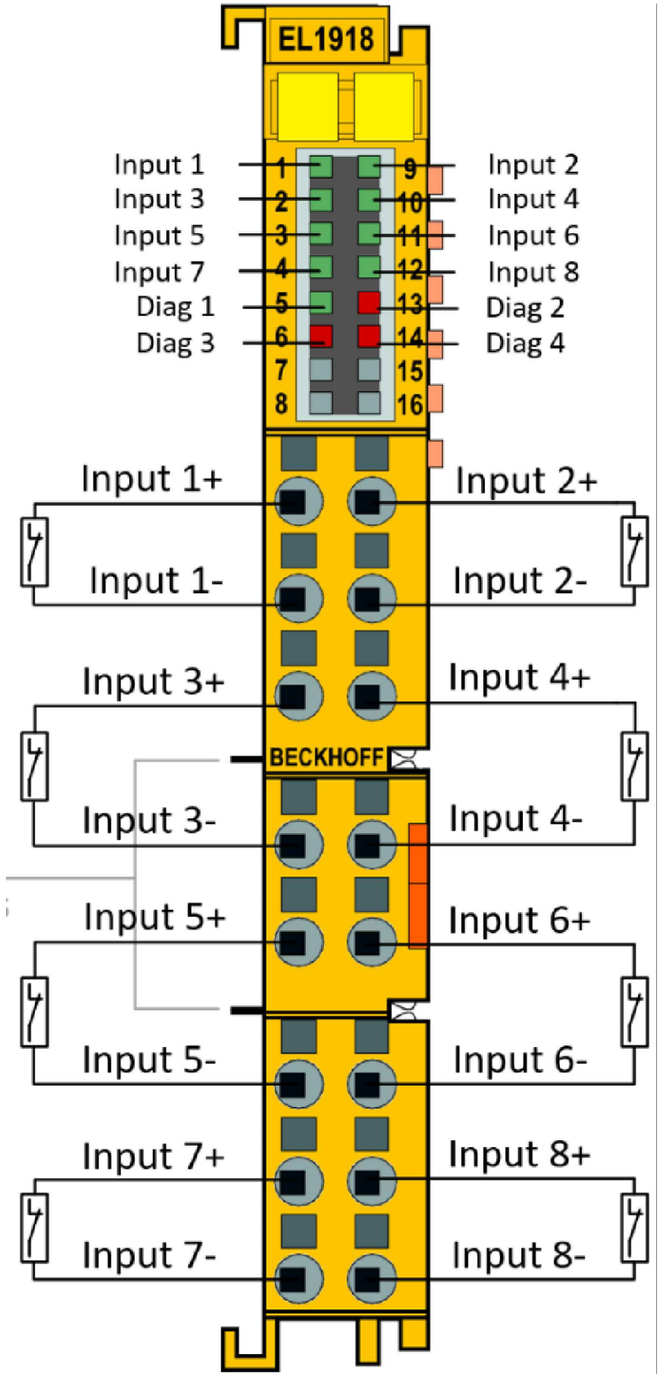
TOTAL PAGES: 175

SHEET 29.0 OF 56

| DIP switch | | | | | | | | | | Address |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 1 |
| OFF | **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 2 |
| **ON** | **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 3 |
| OFF | OFF | **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 4 |
| **ON** | OFF | **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 5 |
| OFF | **ON** | **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 6 |
| **ON** | **ON** | **ON** | OFF | OFF | OFF | OFF | OFF | OFF | OFF | 7 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **ON** | **ON** | **ON** | **ON** | **ON** | **ON** | **ON** | **ON** | **ON** | **ON** | 1023 |




| Diag1 | Diag2 | Diag3 | Diag4 | Description |
|---|---|---|---|---|
| off | off | off | off | The device has no power supply. |
| lights up | off | - | - | The ambient conditions, operating voltage and internal tests are within the valid range. |
| lights up | flashes | - | - | Internal tests are outside the valid range.<br>Diag2 outputs a logic error code.<br>For more information, see the chapter Logic error codes [▶ 57]. |
| off | flashes | - | - | Ambient conditions or electrical parameters are outside the valid range.<br>Diag2 outputs an environment error code.<br>For more information, see the chapter Environment error codes [▶ 57]. |
| lights up | lights up | lights up | off | There is a global shutdown on µC1. |
| lights up | lights up | off | lights up | There is a global shutdown on µC2. |
| lights up | off | lights up | off | There is a global fault on µC1. |
| lights up | off | off | lights up | There is a global fault on µC2. |
| lights up | flickers | - | - | There is an error in a safe input or output module. |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY INPUTS 23.0-6

| REV | DESCRIPTION | DATE |
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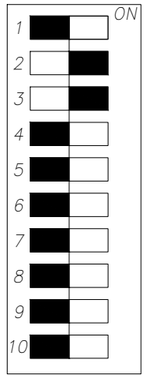
TOTAL PAGES: 175

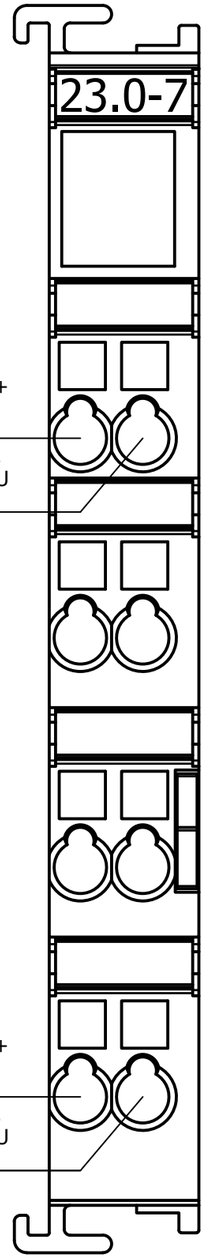
SHEET 29.1 OF 56

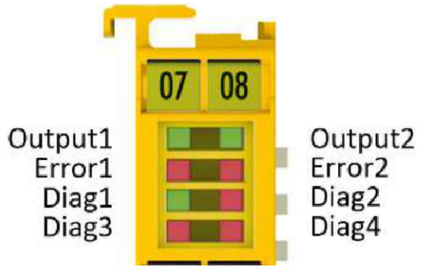
VFD's STO #1

SET
DIPSWITCHES ON
THE MODULE




EL2912

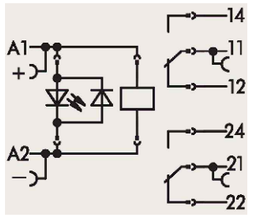


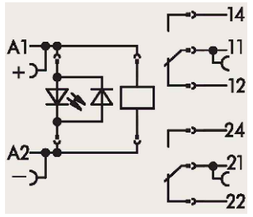

30SCR1
788-384
52.0/1C
A2
A1

SDO1+
BU
20

SDO1-
WH/BU
20




Safety relay SR2M DPDT (2 C/O), with force guided contacts and power indicator




Safety relay SR2M DPDT (2 C/O), with force guided contacts and power indicator

30SCR2
788-384
52.0/4C
A2
A1

SDO2+
BU
20

SDO2-
WH/BU
20

## VFD's STO #1

CAROUSEL CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 5 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 3 TOTAL OF 4 VFD's

VFD's STO #2

## VFD's STO #2

VTD STATIC AREA DRIVE

VTD MOBILE EXTRACTOR CONVEYOR AREA VFD

BUFFER CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 3 VFD's

BUFFER CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SAFETY OUTPUTS 23.0-7

| REV | DESCRIPTION | DATE |
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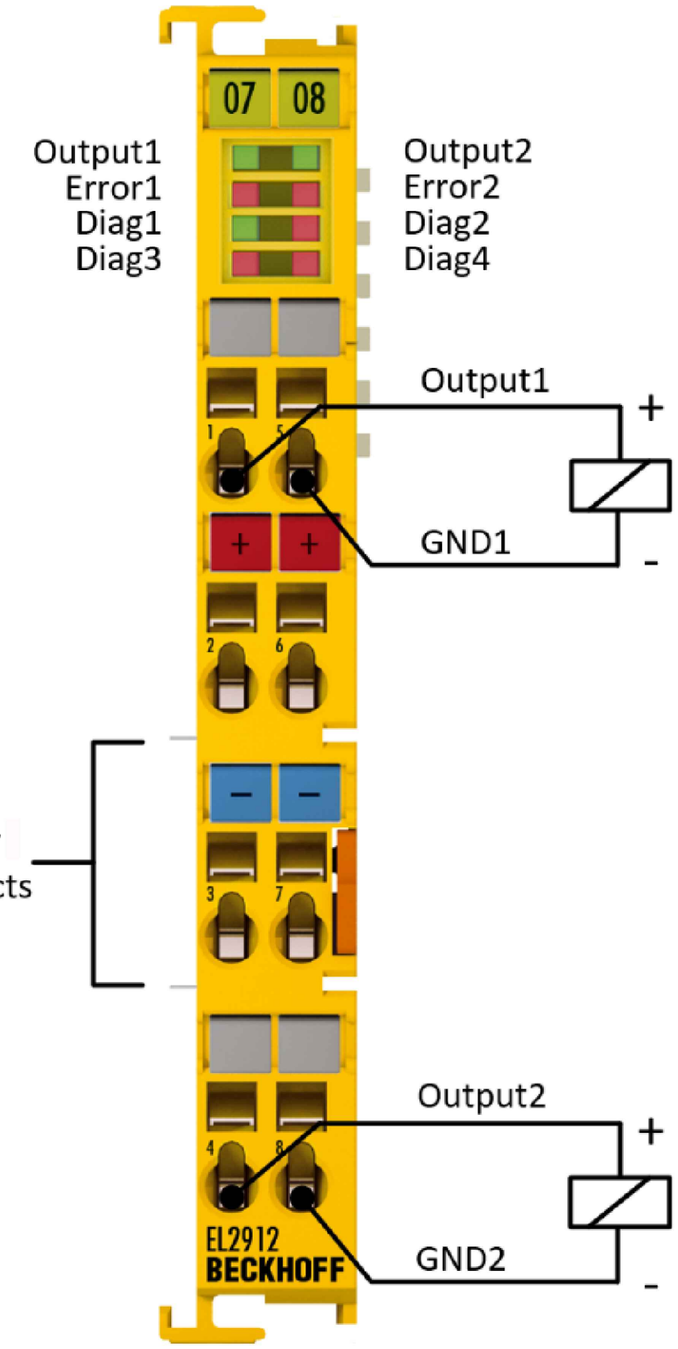
TOTAL PAGES: 175

SHEET 30.0 OF 56

| LED | lit | flashes | off |
|---|---|---|---|
| **Diag 1 (green)** | Environment variables, operating voltage and internal tests are in the valid range<br>• If Diag 2 flashes, a logic error code applies | - | Environment variables, operating voltage and internal tests are outside the valid range<br>• If Diag 2 flashes, an environment error code applies |
| **Diag 2 (red)** | Together with Diag 3 and 4:<br>Global shutdown[1)] has occurred. (see diag history of the TwinSAFE components) | Logic or environment error code according to Diag1 and tables below is output | Together with Diag 3 and 4:<br>Global fault[1)] has occurred. (see diag history of the TwinSAFE components) |
| **Diag 3 (red)** | Global fault or global shutdown on µC1[1)] | - | No global fault or global shutdown on µC1[1)] |
| **Diag 4 (red)** | Global fault or global shutdown on µC2[1)] | - | No global fault or global shutdown on µC2[1)] |

| LED | Color | Description |
|---|---|---|
| Output 1 | green | Status and error display for the respective output<br>LED is lit: output/error is set<br>LED is not lit: output is not set or there is no error |
| Error 1 | red | |
| Output 2 | green | |
| Error 2 | red | |

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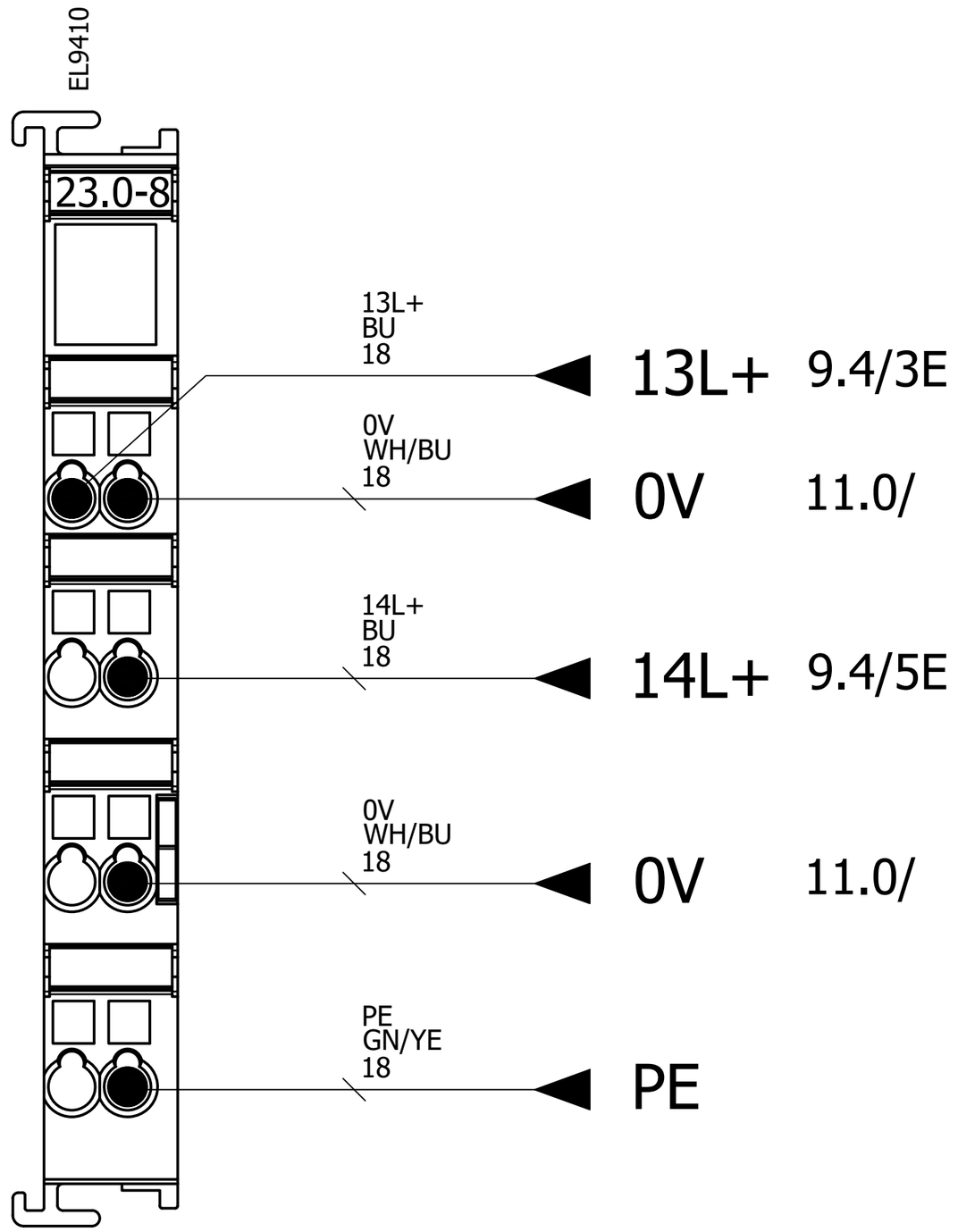


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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: DC SUPPLY FOR NETWORK INTERFACE CARDS 23.0-8

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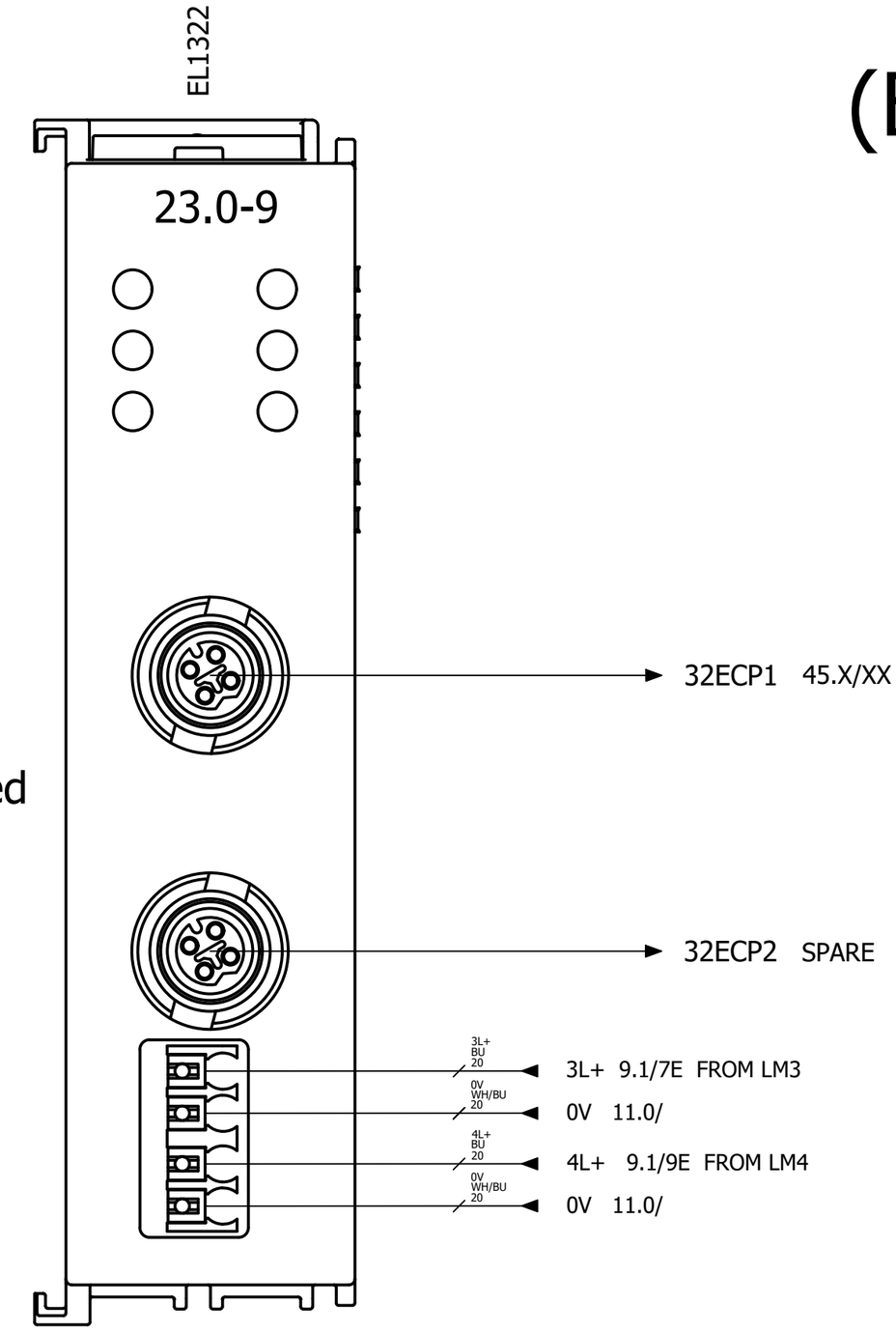
TOTAL PAGES: 175

SHEET 31.0 OF 56

# (EtherCAT P)

| Terminal point | | Description |
|---|---|---|
| Name | No. | |
| $U_S$ 24 V | 1 | +24 V Feed-In $U_S$ (System- and Sensor supply) |
| $U_S$ 0 V | 2 | 0 V of System- and Sensor supply $U_S$ |
| $U_P$ 24 V | 3 | +24 V Feed-In $U_P$ (Peripheral voltage for Actors) |
| $U_P$ 0 V | 4 | 0 V of Peripheral voltage for Actors $U_P$ |

M8 socket, shielded, screw type, EtherCAT P-coded

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE CARD 23.0-9

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TOTAL PAGES: 175

SHEET 32.0 OF 56

| LED | | Display | State | Description |
|---|---|---|---|---|
| Run | green | off | INIT | EtherCAT P module is in state Init |
| | | flashing uniformly | PREOP | EtherCAT P module is in state Pre-Operational |
| | | flashing slowly | SAFEOP | EtherCAT P module is in state Safe-Operational |
| | | on | OP | EtherCAT P module is in state Operational |

### LEDs for fieldbus diagnostics

| LED | | Display | State | Description |
|---|---|---|---|---|
| X1 L/A | green | off | - | No connection on the previous EtherCAT P strand |
| | | on | link | Previous EtherCAT P device connected |
| | | flashing | active | Communication with previous EtherCAT P device |
| X2 L/A | green | off | - | No connection on the subsequent EtherCAT P strand |
| | | on | link | Subsequent EtherCAT P device connected |
| | | flashing | active | Communication with subsequent EtherCAT P device |

### LEDs power supply diagnostics

| LED | | Display | Description |
|---|---|---|---|
| $U_S$ | green | off | System- and Sensor supply $U_S$ not present |
| | | on | System- and Sensor supply $U_S$ present |
| | red | off | System- and Sensor supply $U_S$ >= 19.4 V, no overload/ no case of short circuit |
| | | on | System- and Sensor supply $U_S$ < 19.4 V or overload/ case of short circuit (output current > 3 A) |
| $U_P$ | green | off | Peripheral voltage for Actors $U_P$ not present |
| | | on | Peripheral voltage for Actors $U_P$ present |
| | red | off | Peripheral voltage for Actors $U_P$ >= 19.4 V, no overload/ no case of short circuit |
| | | on | Peripheral voltage for Actors $U_P$ < 19.4 V or overload/ case of short circuit (output current > 3 A) |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE CARD 23.0-9

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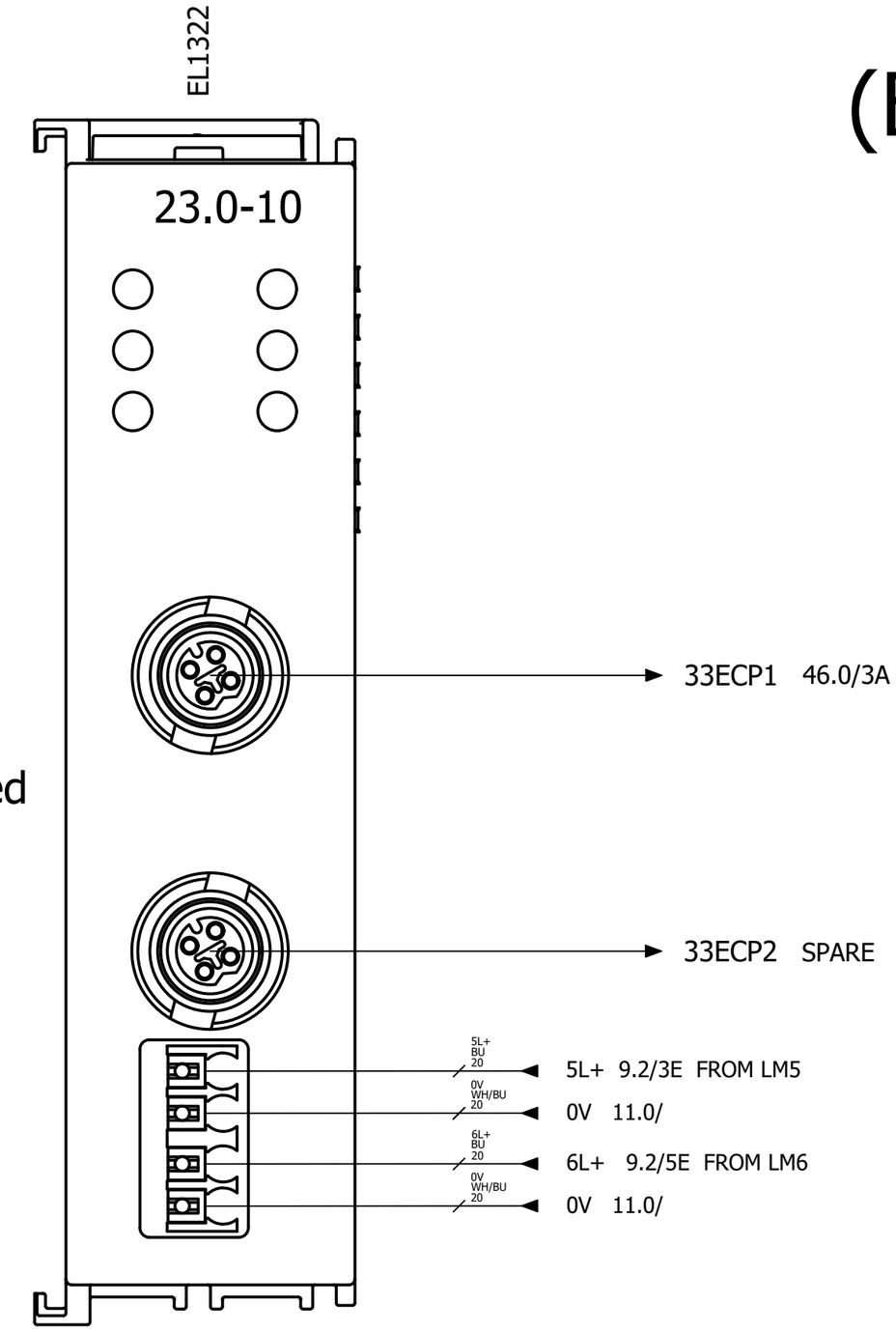
# (EtherCAT P)

| Terminal point | | Description |
|---|---|---|
| Name | No. | |
| $U_S$ 24 V | 1 | +24 V Feed-In $U_S$ (System- and Sensor supply) |
| $U_S$ 0 V | 2 | 0 V of System- and Sensor supply $U_S$ |
| $U_P$ 24 V | 3 | +24 V Feed-In $U_P$ (Peripheral voltage for Actors) |
| $U_P$ 0 V | 4 | 0 V of Peripheral voltage for Actors $U_P$ |

M8 socket, shielded, screw type, EtherCAT P-coded

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PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE CARD 23.0-10

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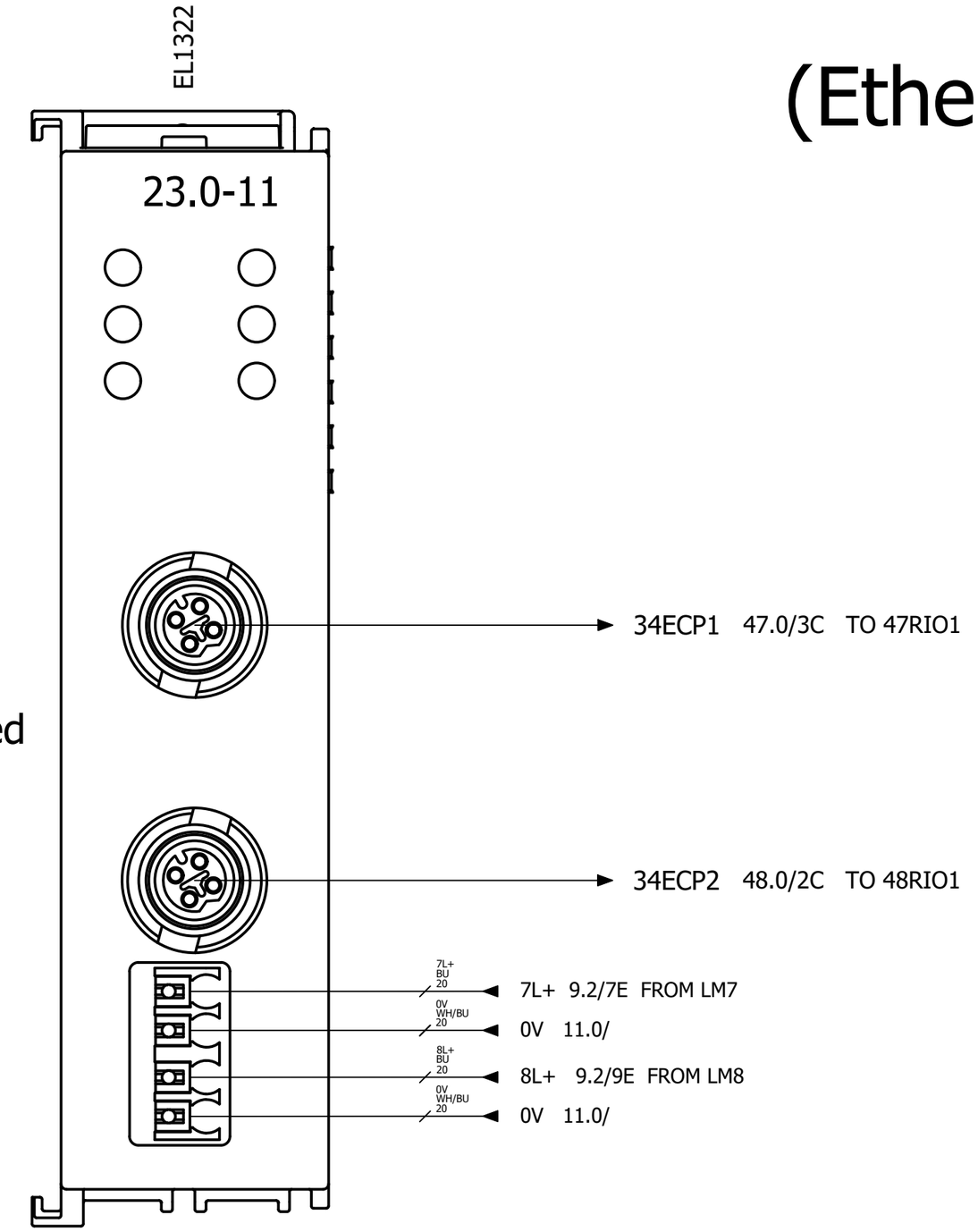
TOTAL PAGES: 175

SHEET 33.0 OF 56

# (EtherCAT P)

| Terminal point | | Description |
|---|---|---|
| Name | No. | |
| $U_S$ 24 V | 1 | +24 V Feed-In $U_S$ (System- and Sensor supply) |
| $U_S$ 0 V | 2 | 0 V of System- and Sensor supply $U_S$ |
| $U_P$ 24 V | 3 | +24 V Feed-In $U_P$ (Peripheral voltage for Actors) |
| $U_P$ 0 V | 4 | 0 V of Peripheral voltage for Actors $U_P$ |

M8 socket, shielded, screw type, EtherCAT P-coded

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PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE CARD 23.0-11

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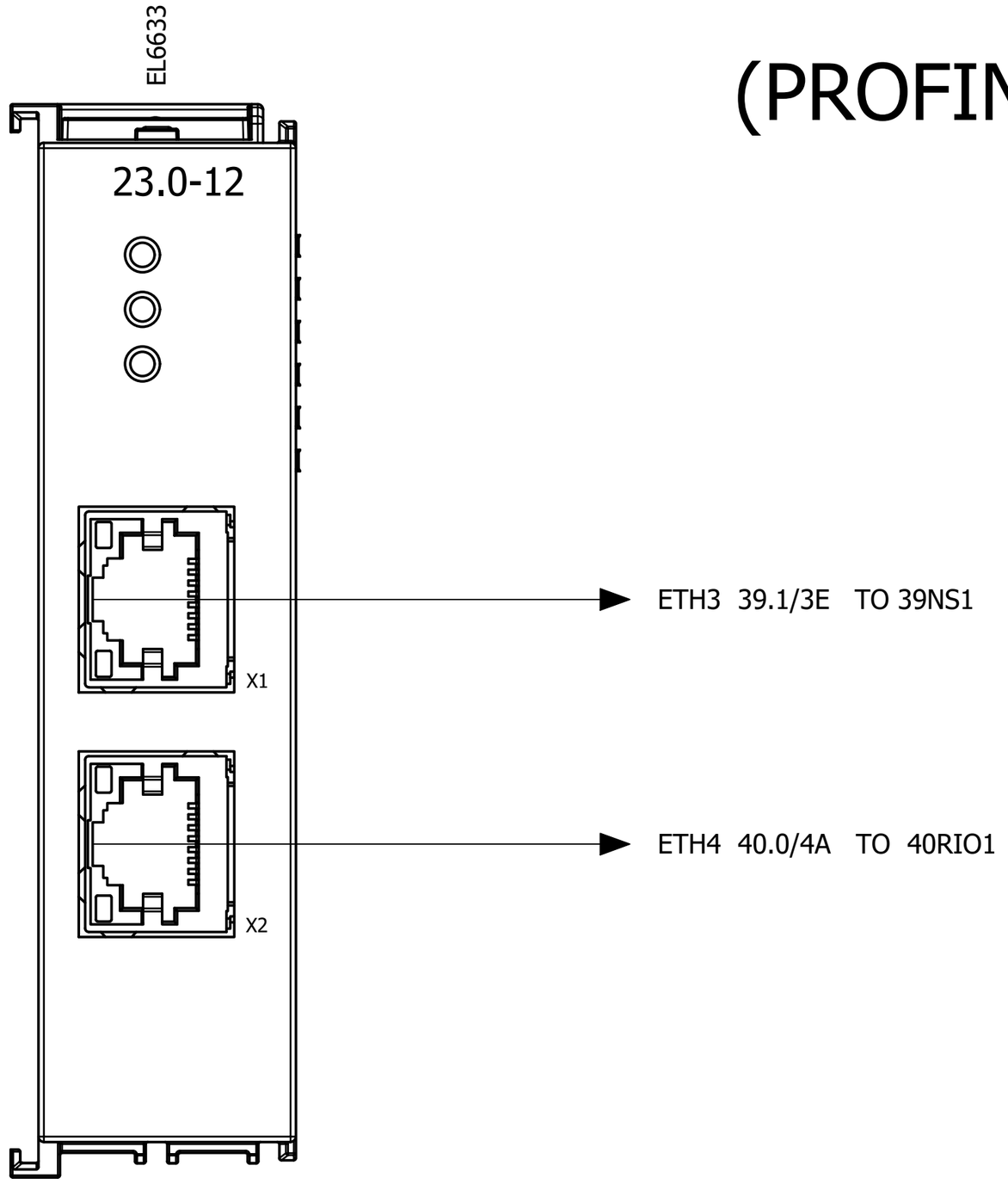
SHEET 34.0 OF 56

# (PROFINET)

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

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TOTAL PAGES: 175

SHEET 35.0 OF 56

| LED | | Display | Description |
|---|---|---|---|
| RUN | green | off | State of the EtherCAT State Machine:<br>**INIT** = initialization of the terminal;<br>**BOOTSTRAP** = function for firmware updates of the terminal |
| | | flashing<br>200 ms | State of the EtherCAT State Machine:<br>**PREOP** = function for mailbox communication and different standard-settings set |
| | | off (1 s)<br>on (200 ms) | State of the EtherCAT State Machine:<br>**SAFEOP** = verification of the sync manager channels and the distributed clocks.<br>Outputs remain in safe state |
| | | on | State of the EtherCAT State Machine:<br>**OP** = normal operating state; mailbox and process data communication is possible |

| Color green | Color red | Meaning |
|---|---|---|
| off | flashing<br>200 ms | Terminal starts |
| flashing<br>200 ms | off | No name |
| 1 s off, 200 ms on | off | No IP address |
| on | off | EL terminal is parameterized |

| Colours green | Colours red | Meaning |
|---|---|---|
| off | flashing<br>500 ms | no AR established with any device |
| off | 1 s off, 200 ms on | At least one device has not established an AR |
| 1 s off, 200 ms on | off | At least one device has signaled an error, e.g. that there is a module difference or that the error bit for an IO-CR is set (Problem Indicator) |
| flashing<br>200 ms | off | At least one device is signaling that its status is 'Stop' (Provider State Stop) |
| on | off | All PROFINET devices are in data exchange mode |

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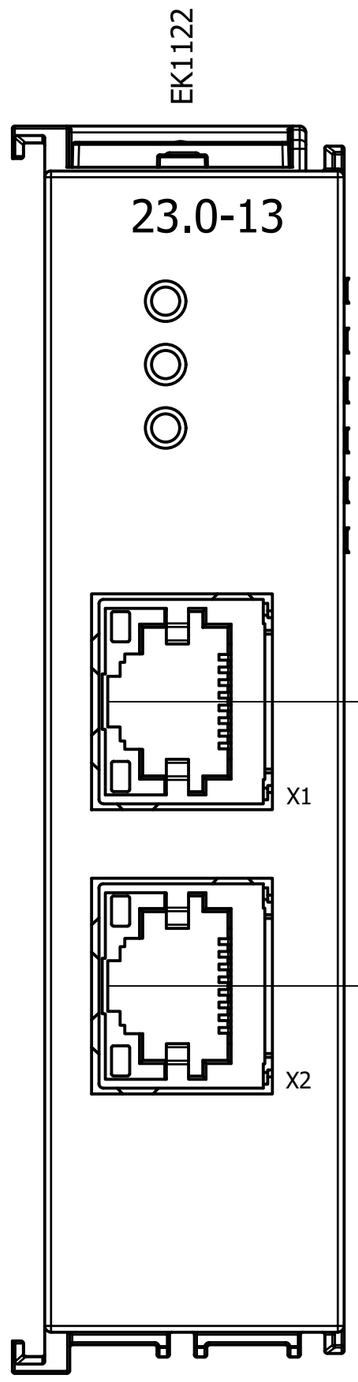
# (EtherCAT)

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PAGE DESCRIPTION: EtherCAT NETWORK INTERFACE CARD 23.0-13

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SHEET 36.0 OF 56

## LEDs for fieldbus diagnostics

| LED | | Display | State | Description |
|---|---|---|---|---|
| LINK / ACT (X1/X2) | green | off | - | no connection on the EtherCAT strand |
| | | on | linked | EtherCAT device connected |
| | | flashing | active | Communication with EtherCAT device |

## LED diagnostics EtherCAT state machine

| LED | Color | Meaning | |
|---|---|---|---|
| RUN | green | This LED indicates the terminal's operating state: | |
| | | off | State of the EtherCAT State Machine: **INIT** = initialization of the terminal |
| | | single flash | State of the EtherCAT State Machine: **PREOP** = function for mailbox communication and different standard-settings set |
| | | flashing | State of the EtherCAT State Machine: **SAFEOP** = verification of the sync manager channels and the distributed clocks.<br>Outputs remain in safe state |
| | | on | State of the EtherCAT State Machine: **OP** = normal operating state; mailbox and process data communication is possible |
| | | flickering | State of the EtherCAT State Machine: **BOOTSTRAP** = function for terminal firmware updates |

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JOB # C40572

TOTAL PAGES: 175

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SPARE

| REV | DESCRIPTION | DATE |
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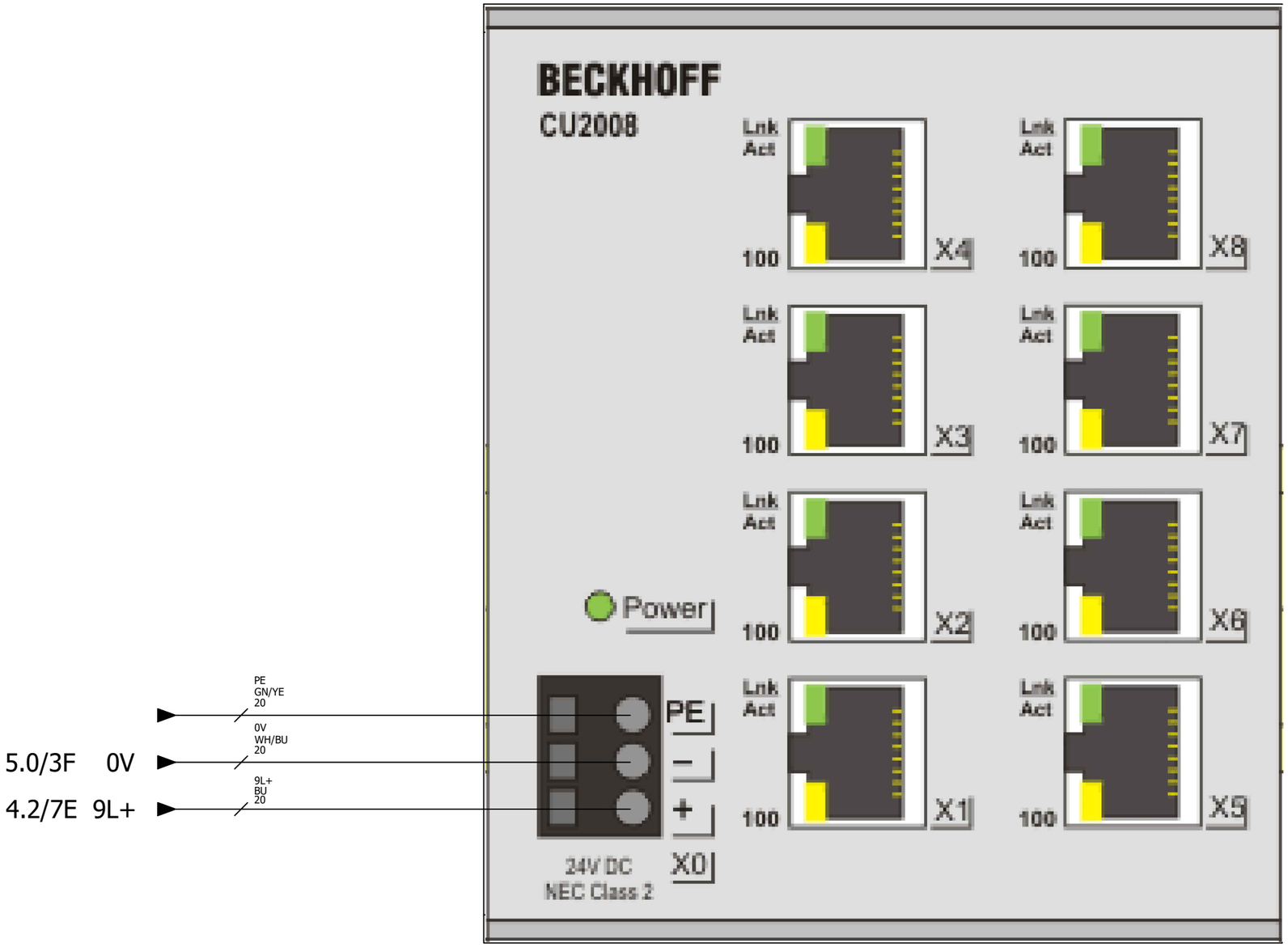
TOTAL PAGES: 175

SHEET 38.0 OF 56

# 39NS1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 39NS1 POWER 24VDC

| REV | DESCRIPTION | DATE |
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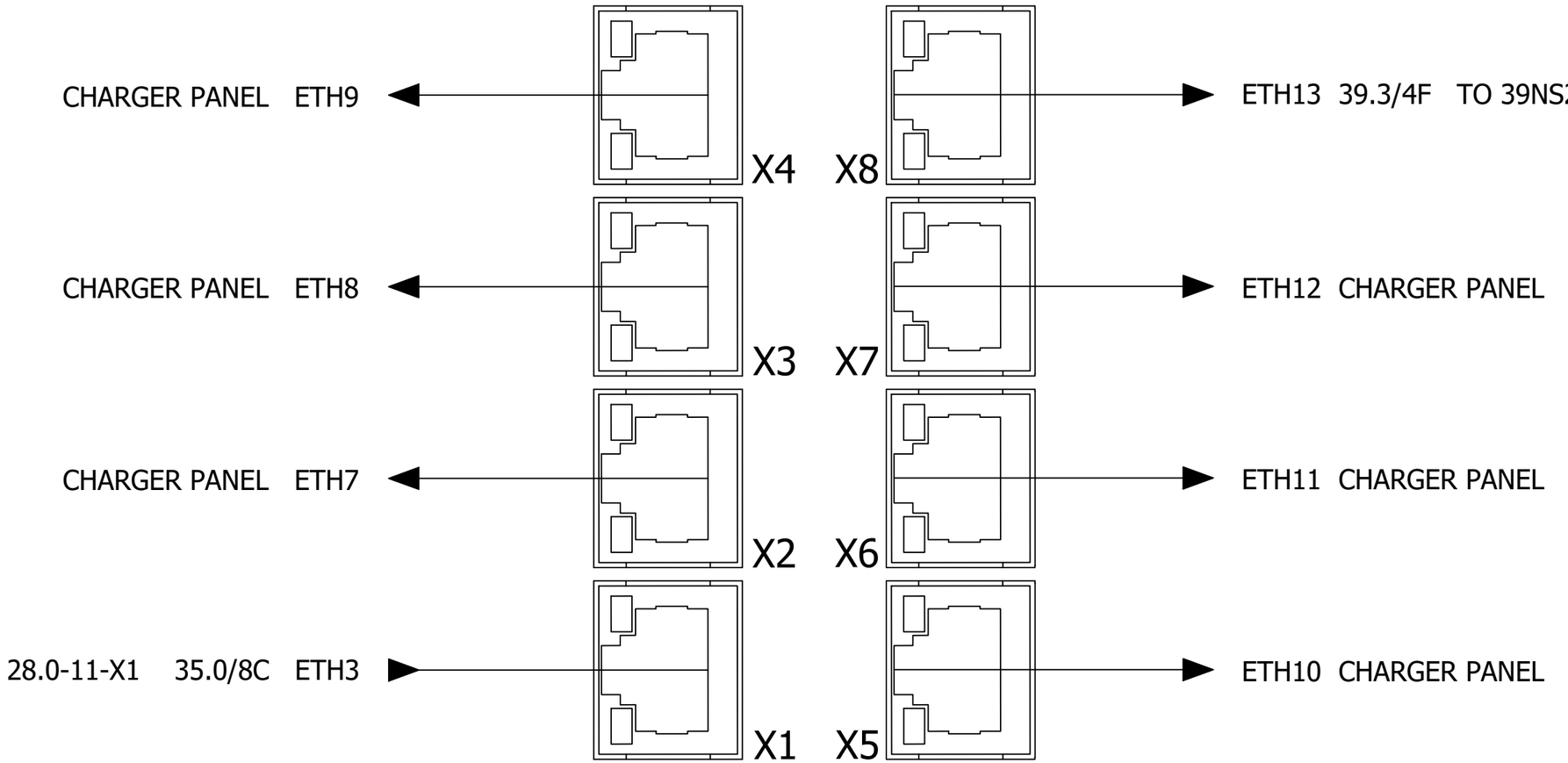
TOTAL PAGES: 175

SHEET 39.0 OF 56

# (PROFINET)

# 39NS1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 39NS1 PROFINET NETWORK INTERFACE

| REV | DESCRIPTION | DATE |
|---|---|---|
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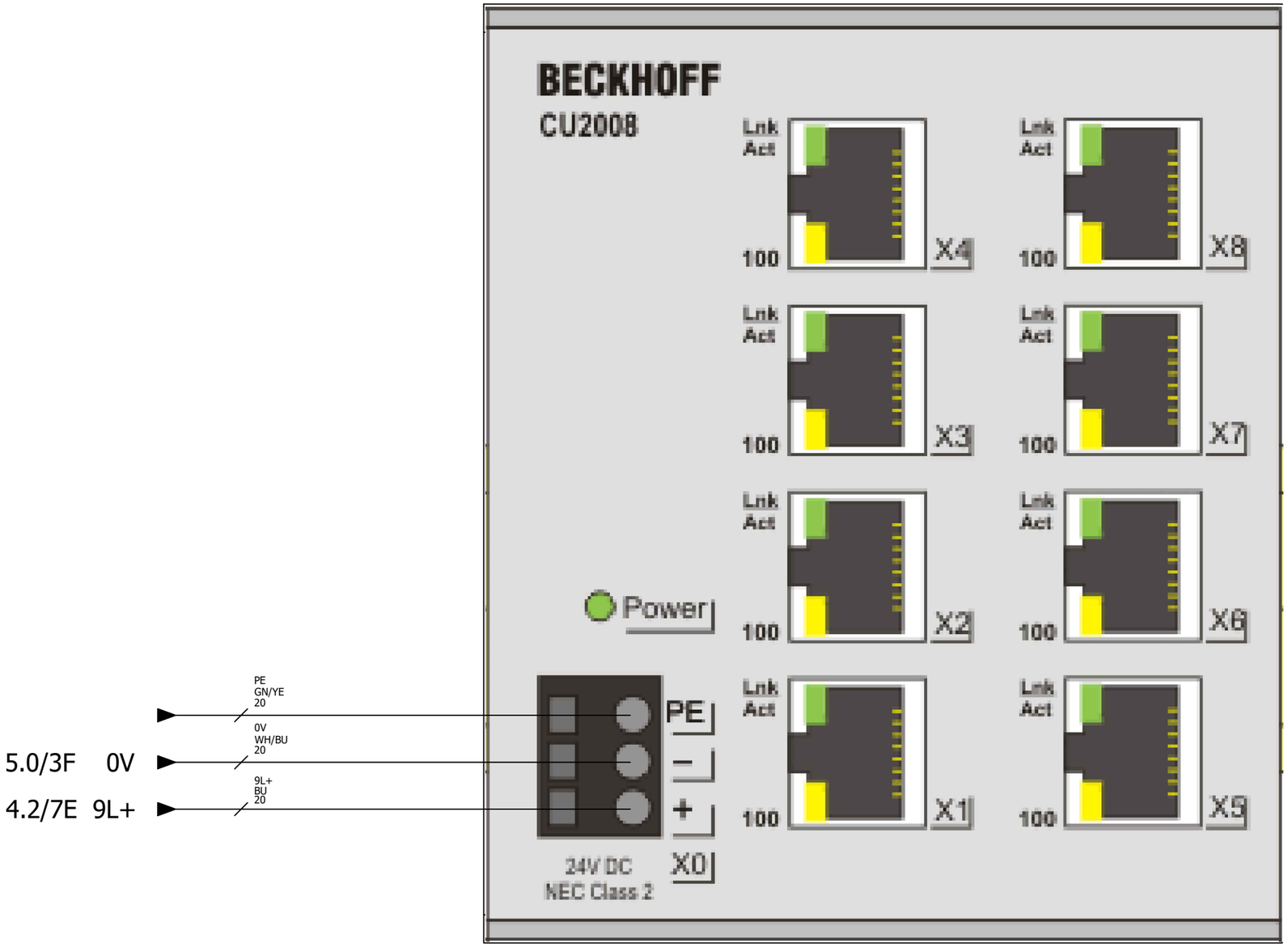
# 39NS2

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PAGE DESCRIPTION: NETWORK SWITCH 39NS2 POWER 24VDC

| REV | DESCRIPTION | DATE |
|---|---|---|
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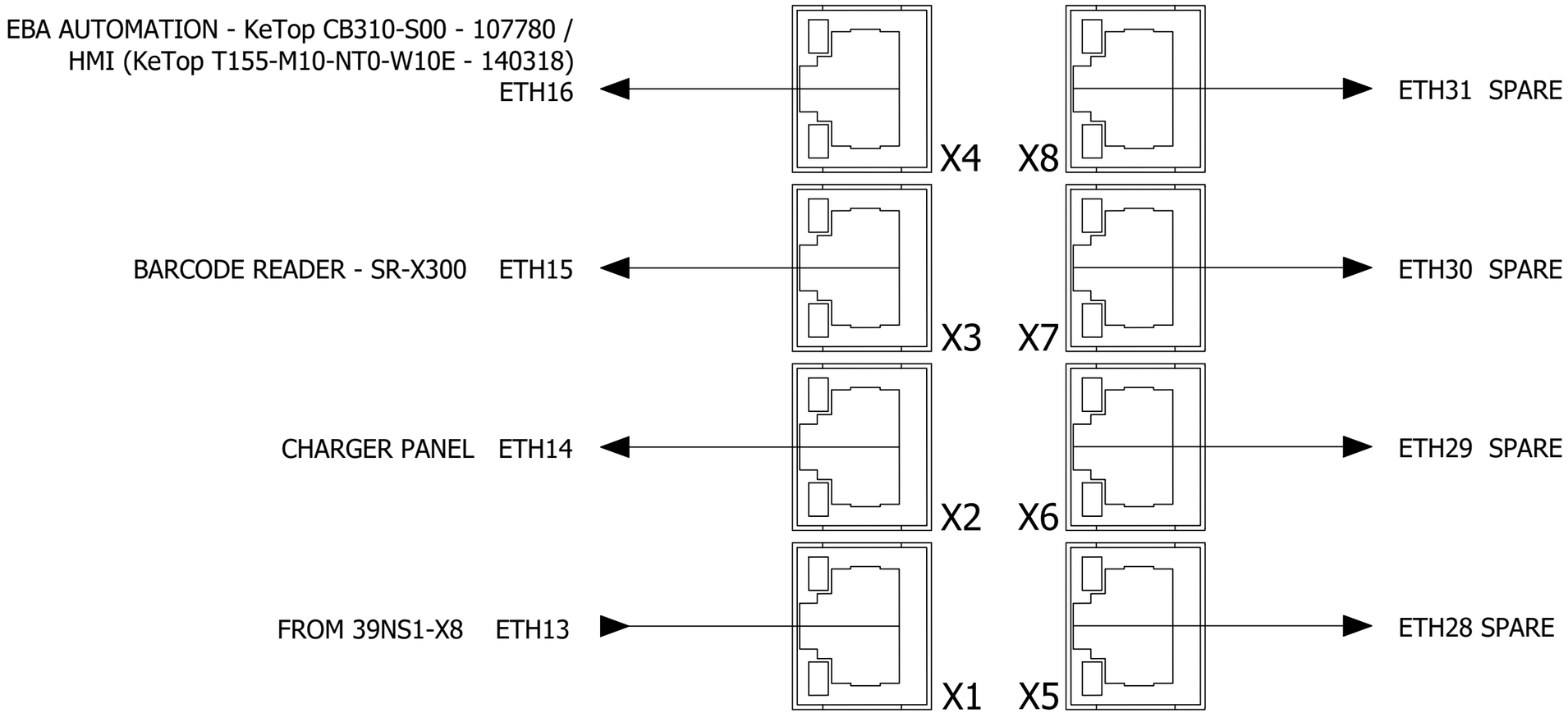
# (PROFINET)

# 39NS2

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PAGE DESCRIPTION: NETWORK SWITCH 39NS2 PROFINET NETWORK INTERFACE

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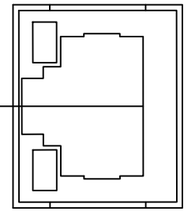
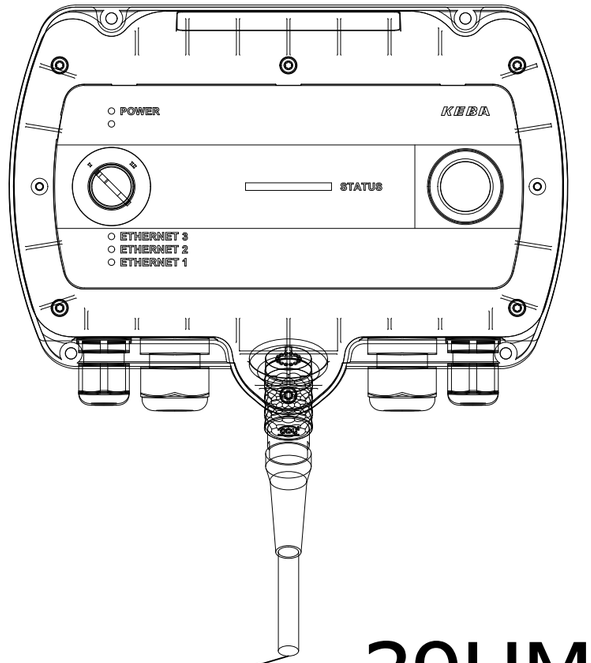
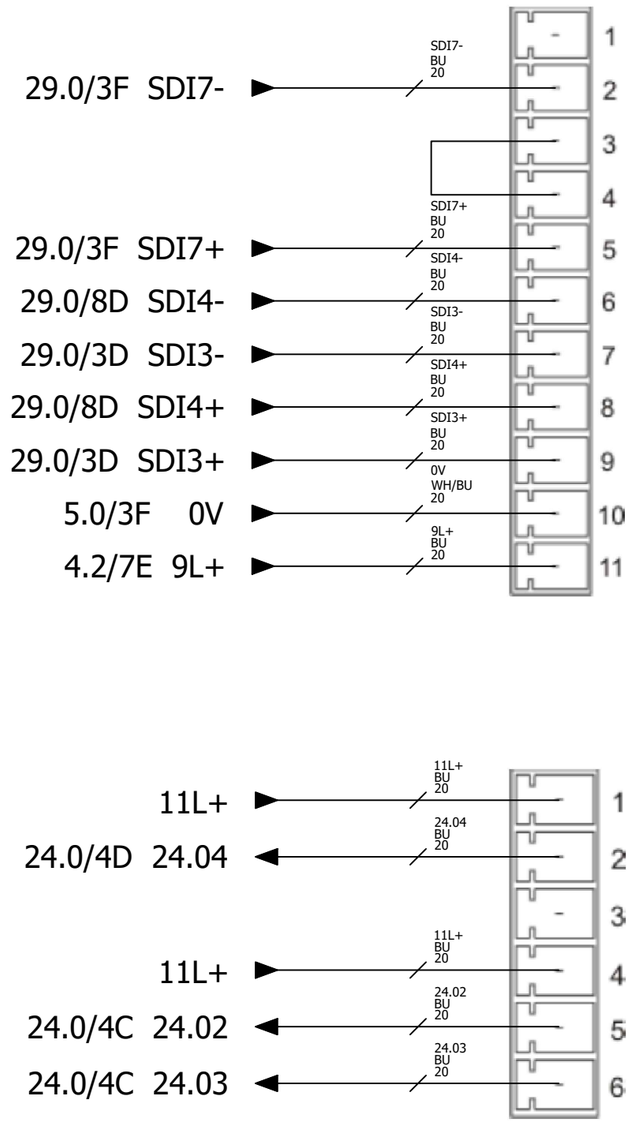
TOTAL PAGES: 175

# 39CBOX

39NS2-X4 ETH16





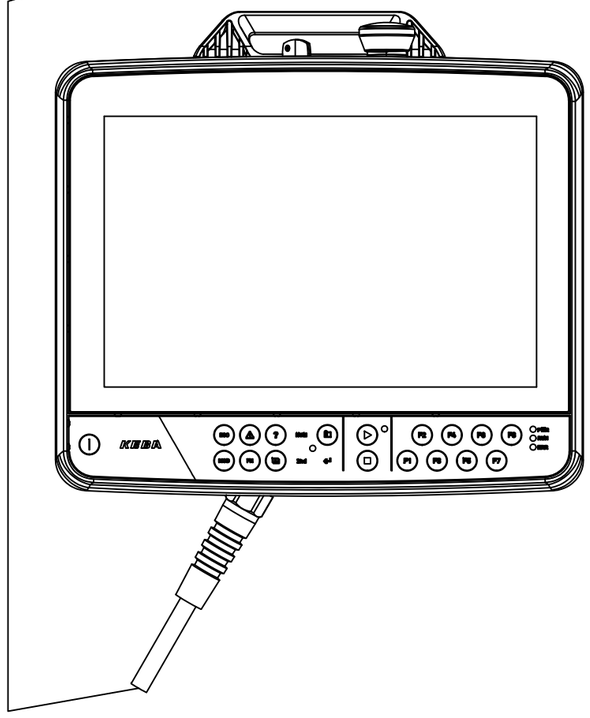




| Pin no. | Signal designation | Description |
|---|---|---|
| 1 | N.C. | not connected |
| 2 | ZT2- | Approval control, circuit 2 negative |
| 3 | ZT1- | Approval control, circuit 1 negative |
| 4 | ZT2+ | Approval control, circuit 2 positive |
| 5 | ZT1+ | Approval control, circuit 1 positive |
| 6 | ES2 | Emergency stop, circuit 2 |
| 7 | ES1 | Emergency stop, circuit 1 |
| 8 | ES2 | Emergency stop, circuit 2 |
| 9 | ES1 | Emergency stop, circuit 1 |
| 10 | - | GND |
| 11 | + | 24 V DC power supply |

| Pin no. | Signal designation | Description |
|---|---|---|
| 1 | DRY | Closed when handheld device is connected |
| 2 | DRY | |
| 3 | KEY3 | Key-operated switch contact (3 pos): Position 3 (right switch position)<br>Illuminated pushbutton: Supply (GND) |
| 4 | KEY2 | Key-operated switch contact (3 pos): COMMON<br>Key-operated switch contact (2 pos): COMMON<br>Illuminated pushbutton: Supply (24 V) |
| 5 | KEY1 | Key-operated switch contact (3 pos): Position 2 (middle switch position)<br>Key-operated switch contact (2 pos): Position 2 (left switch position)<br>Illuminated pushbutton: Circuit illuminated pushbutton |
| 6 | KEY0 | Key-operated switch contact (3 pos): Position 1 (left switch position)<br>Key-operated switch contact (2 pos): Position 1 (right switch position)<br>Illuminated pushbutton: Circuit illuminated pushbutton |

# 39HMI



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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 39NS2 PROFINET NETWORK INTERFACE

| REV | DESCRIPTION | DATE |
|---|---|---|
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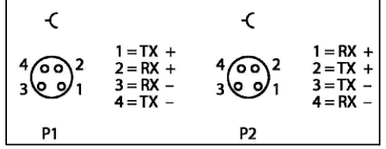
TOTAL PAGES: 175

# P1 & P2 (PROFINET)

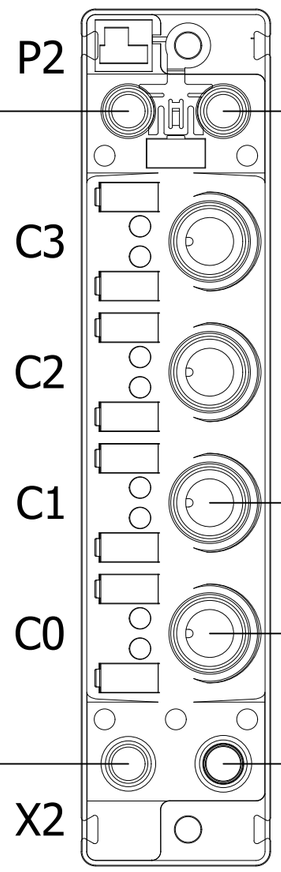
M8 x 1 Ethernet




23.0-12-X2 35.0/8C ETH4

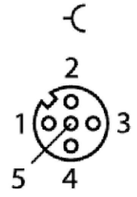
40BCS1 40.1/9C ETH4A

# 40RIO1




# (IO LINK)

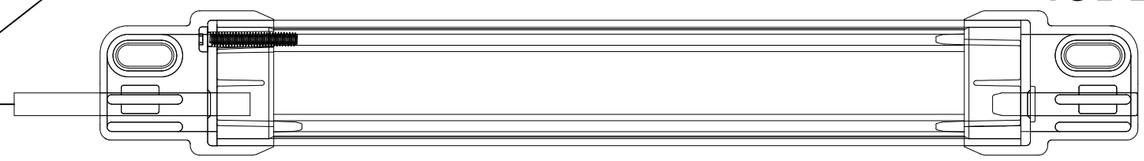
M12 x 1 I/O Port



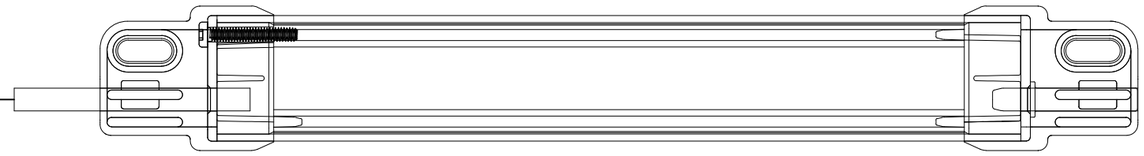

| 4-pin M12 Male | Pin | Wire Color | Description |
|---|---|---|---|
| | 1 | Brown | 18 V DC to 30 V DC |
| | 2 | White | Not used |
| | 3 | Blue | DC common |
| | 4 | Black | IO-Link Communication |



40BL2



40BL1

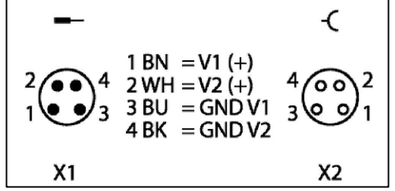


40.1/4E 40PC2

40PC1 XX.X/XX FROM 24L+ & 0VL-

VTD LIGHT
WLS15PX
BANNER IO LINK
LIGHTS

M8 x 1 Voltage Supply

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 40RIO1 NETWORK + I/O + POWER

| REV | DESCRIPTION | DATE |
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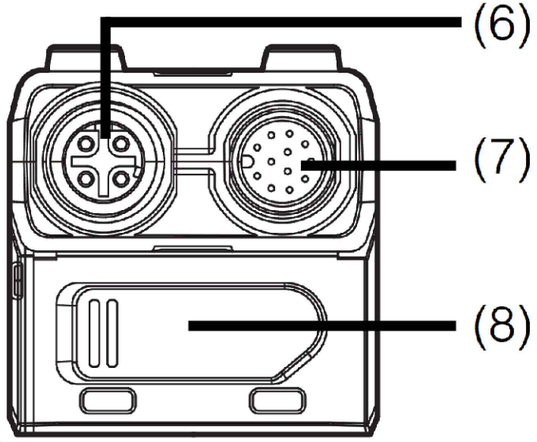
TOTAL PAGES: 175

SHEET 40.0 OF 56

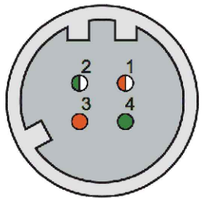
# 40BCS1

KEYENCE
SR-300X

Bottom



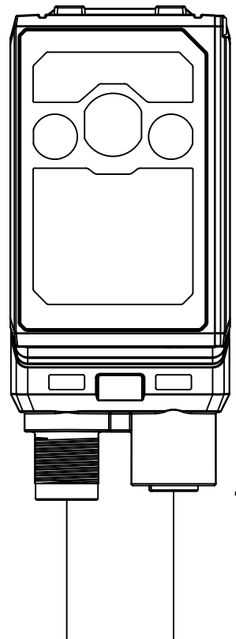

| | | |
|---|---|---|
| (6) | Ethernet port | Port for connecting the Ethernet cable: M12D 4pin (female) |
| (7) | Control port*4 | Port for connecting the control cable: M12A 12pin (male) |
| (8) | USB port | Port for connecting USB memory or a USB cable*3: Type-C |




M12D

| No. | Signal |
|---|---|
| 1 | TX+ |
| 2 | RX+ |
| 3 | TX- |
| 4 | RX- |

Male connector, M12,
12-pin or 17-pin, A-coded



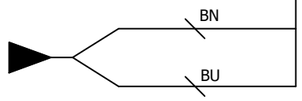
Female connector, M12,
4-pin, D-coded (Ethernet)

M12A

| No. | Color |
|---|---|
| 1 | Brown |
| 2 | Blue |
| 3 | Purple |
| 4 | Yellow |
| 5 | Sky |
| 6 | Pink |
| 7 | Green |
| 8 | White/Blue |
| 9 | Black |
| 10 | White |
| 11 | Grey |
| 12 | Orange |
| SHELL | Braid |

| Wire color | Name | Description | Signal direction | Initial assignment |
|---|---|---|---|---|
| Brown | 24 V | 24 VDC | Input | - |
| Blue | 0 V | Power GND | - | - |
| Orange | RXD (RD) | RS-232C Receive | Input | - |
| Yellow | TXD (SD) | RS-232C Send | Output | - |
| Purple | SGND | RS-232C GND | - | - |
| Green | IN1 | Input signal 1 | Input | Reading start |
| Gray | IN2 | Input signal 2 | Input | - |
| White and blue | IN COM | Input common | - | - |
| Pink | OUT1 | Output signal 1 | Output | Reading OK |
| Light blue | OUT2 | Output signal 2 | Output | Reading ERROR |
| White | OUT3 | Output signal 3 | Output | BUSY |
| Black | OUT COM | Output common | - | - |
| Black tube | FG | Frame ground | - | - |

FROM 40RIO1 40.0/3B C40-2

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FROM 40RIO1-P2 40.0/3B ETH4A

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 40BCS1 - 40BCC1 NETWORK + POWER

| REV | DESCRIPTION | DATE |
|---|---|---|
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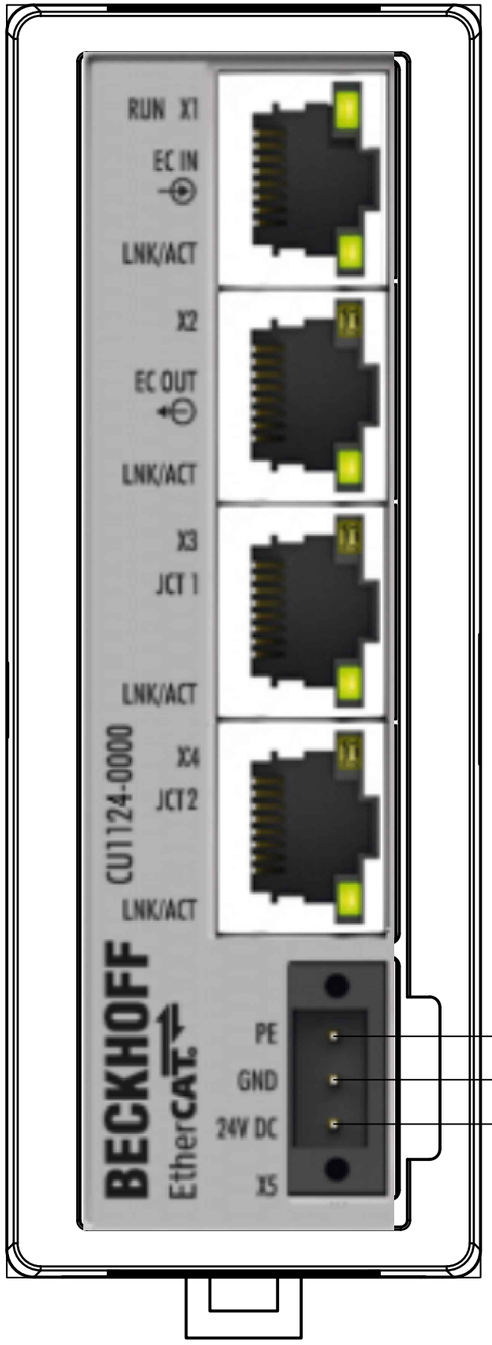
SHEET 40.1 OF 56

# (EtherCAT)




# 41NS1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 41NS1 POWER 24VDC

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|---|---|---|
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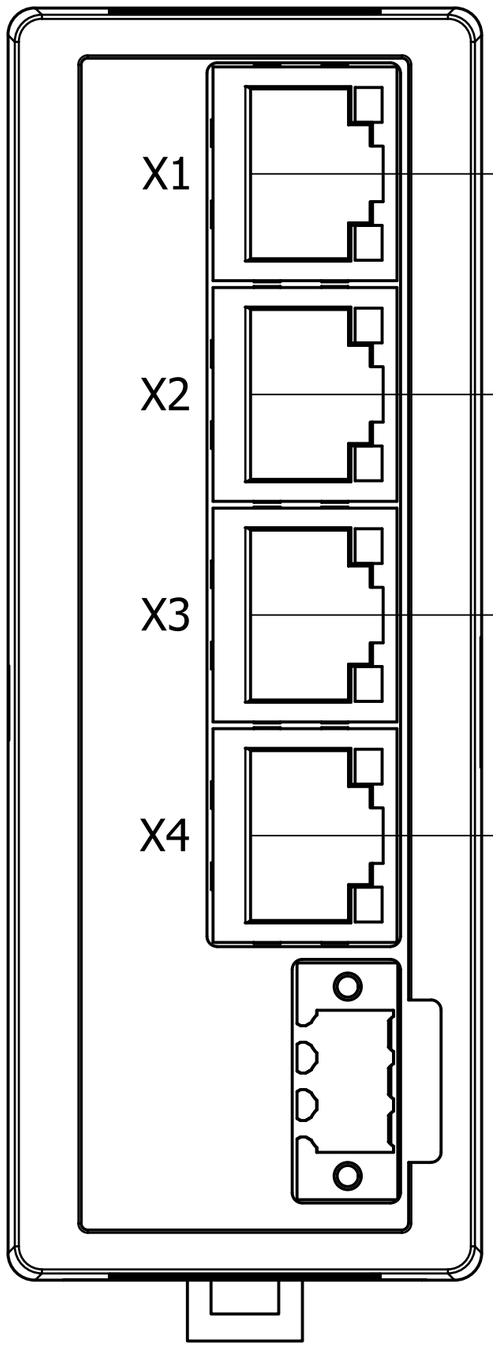
JOB # C40572

TOTAL PAGES: 175

SHEET 41.0 OF 56

# 41NS1

(EtherCAT)




X1 → ETH5 FROM 5.0-13-X1 36.0/8C

X2 → ETH18 20VFD1 CAROUSEL CONVEYORS AREA SECTION #1 20.8/3E

X3 → ETH19 21VFD1 CAROUSEL CONVEYORS AREA SECTION #2 21.5/3E

X4 → ETH20 19VFD1 CAROUSEL CONVEYORS AREA SECTION #3 19.8/3E

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 41NS1 POWER EtherCAT NETWORK INTERFACE

| REV | DESCRIPTION | DATE |
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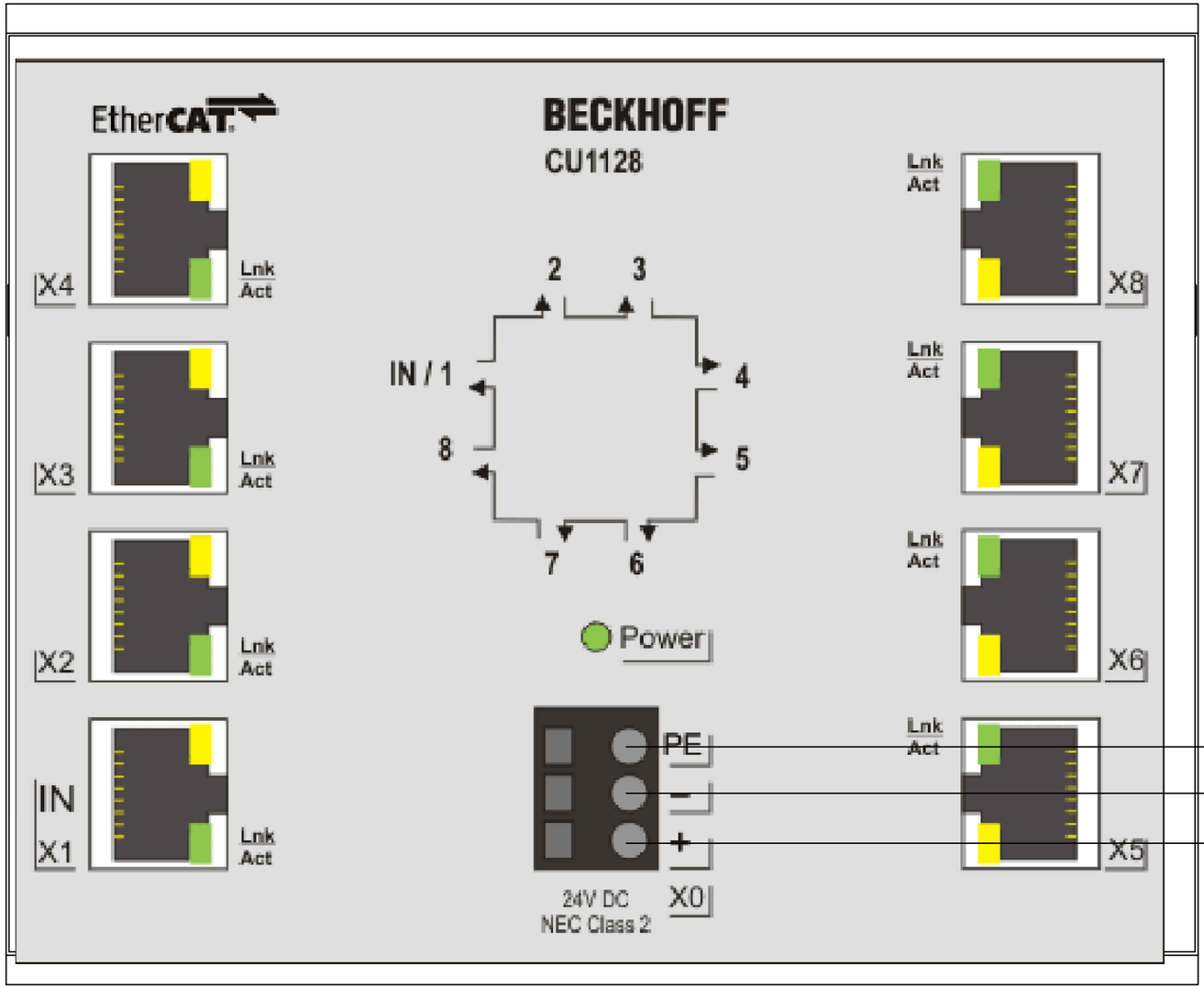
TOTAL PAGES: 175

# 42NS1

# (EtherCAT)




PE
GN/YE
20

0V
WH/BU
20

10L+
BU
20

0V X.X/XX

10L+ X.X/XX

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: NETWORK SWITCH 42NS1 POWER 24VDC

| REV | DESCRIPTION | DATE |
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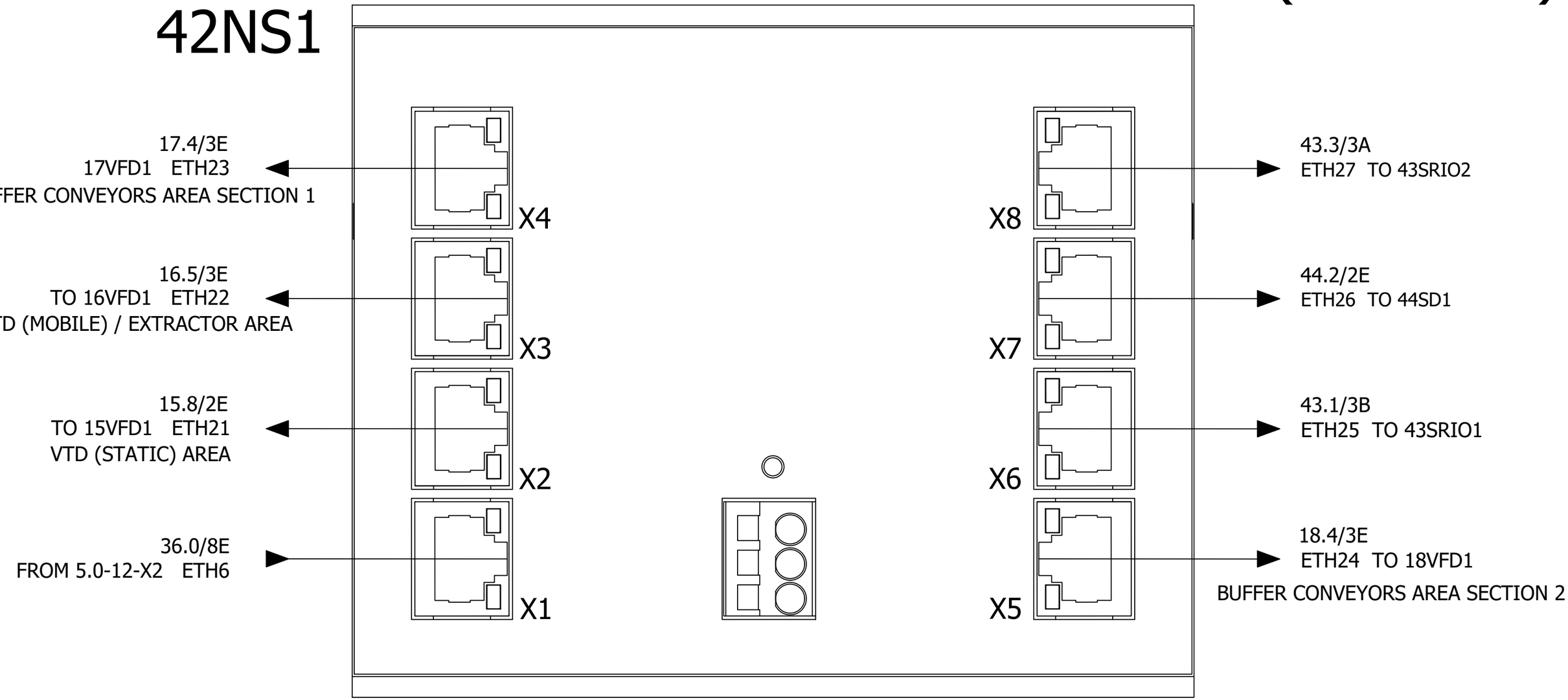
TOTAL PAGES: 175

SHEET 42.0 OF 56

# (EtherCAT)

# 42NS1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: RACK 01 PLC SAFETY DIGITAL IN MODULE 3

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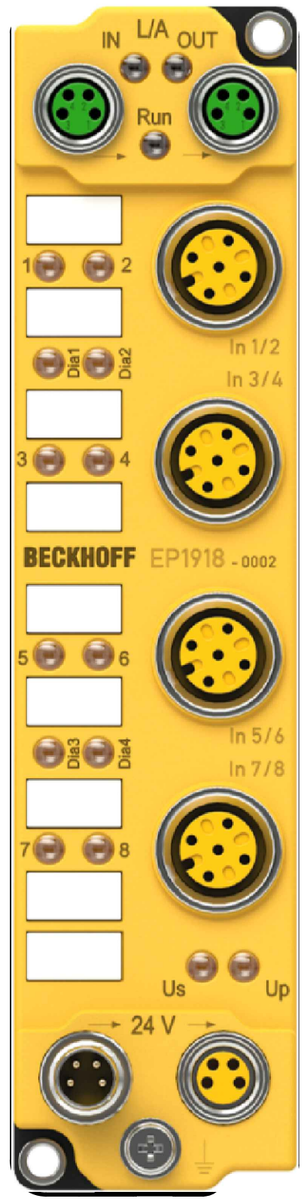
TOTAL PAGES: 175

SHEET 42.1 OF 56

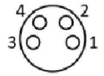
# 43SRIO1

**LED displays**

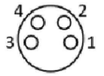
| LED | Display | Meaning |
|---|---|---|
| IN Link/Act | off | no connection to the preceding EtherCAT module |
| | lit | LINK: connection to the preceding EtherCAT module |
| | flashes | ACT: communication with the preceding EtherCAT module |
| OUT Link/Act | off | no connection to the following EtherCAT module |
| | lit | LINK: connection to the following EtherCAT module |
| | flashes | ACT: Communication with the following EtherCAT module |
| Run | off | Status of the EtherCAT module is Init |
| | flashes quickly | Status of the EtherCAT module is pre-operational |
| | flashes slowly | Status of the EtherCAT module is safe-operational |
| | lit | Status of the EtherCAT module is operational |




EtherCAT IN

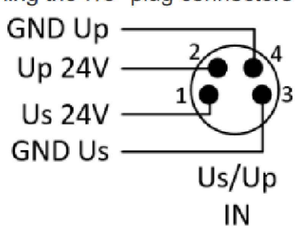


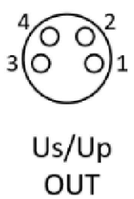

EtherCAT OUT

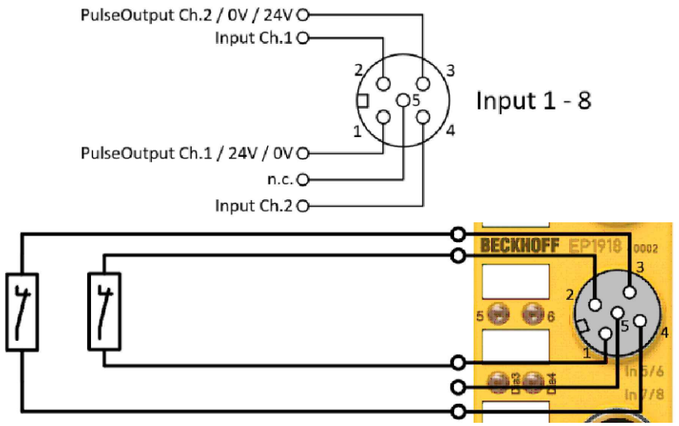



| EtherCAT | | Connector | Cable | | Standard |
|---|---|---|---|---|---|
| **Signal** | **Description** | **M8** | **ZB9010, ZB9020, ZK1090-6292** | **ZB903x, ZK1090-31xx** | **TIA-568B** |
| Tx + | Transmit Data+ | Pin 1 | yellow[1] | orange/white[2] | white/orange |
| Tx - | Transmit Data- | Pin 4 | orange[1] | orange[2] | orange |
| Rx + | Receive Data+ | Pin 2 | white[1] | blue/white[2] | white/green |
| Rx - | Receive Data- | Pin 3 | blue[1] | blue[2] | green |
| Shield | Shield | Housing | Shield | Shield | Shield |

We recommend fastening the 7/8" plug connectors with a torque of **1.5 Nm**.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 43SRIO1 SAFETY DIGITAL IN MODULE

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JOB # C40572

TOTAL PAGES: 175

SHEET 43.0 OF 56

# 43SRIO1

ZK1090-3191-0100

FROM 42NS1-X6 ETH25
42.1/9D

43SC1 TO OVERTRAVEL UPPER-NC

43SC2 TO OVERTRAVEL LOWER-NC

ZK2020-3200-0100

FROM 24L+ & 0VL- 43PC1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 43SRIO1 SAFETY DIGITAL IN MODULE

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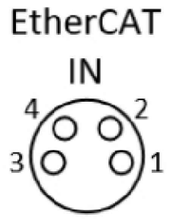
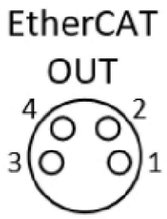
TOTAL PAGES: 175

SHEET 43.1 OF 56

# 43SRIO2




| EtherCAT | | Connector |
|---|---|---|
| **Signal** | **Description** | **M8** |
| Tx + | Transmit Data+ | Pin 1 |
| Tx - | Transmit Data- | Pin 4 |
| Rx + | Receive Data+ | Pin 2 |
| Rx - | Receive Data- | Pin 3 |
| Shield | Shield | Housing |

| LED | Display | Meaning |
|---|---|---|
| IN Link/Act | off | no connection to the preceding EtherCAT module |
| | lit | LINK: connection to the preceding EtherCAT module |
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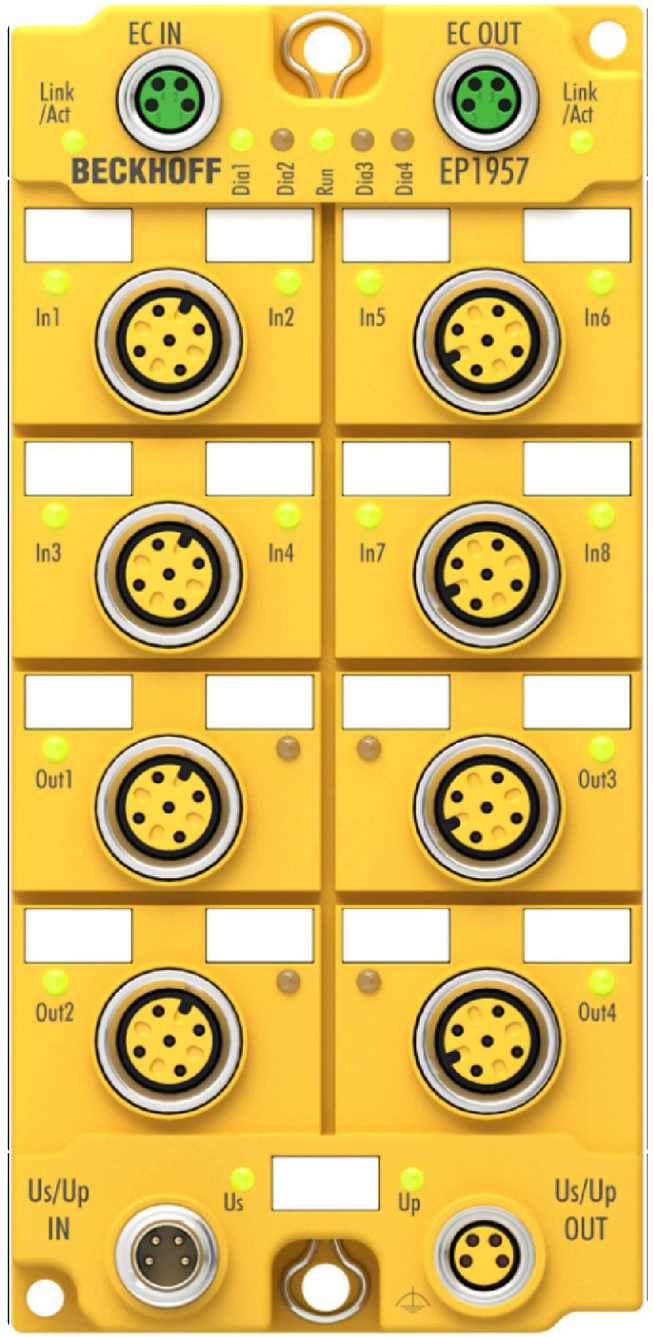


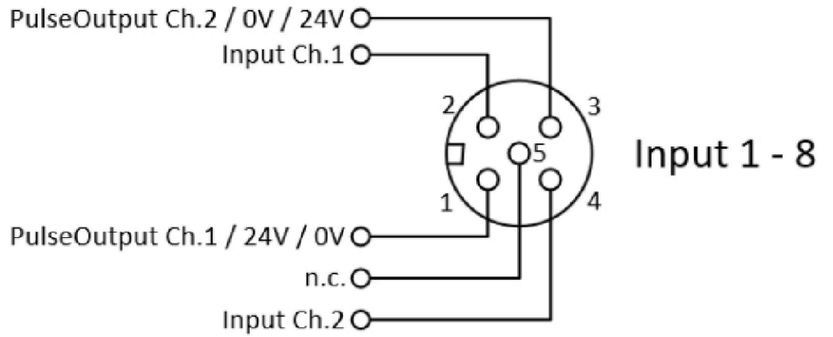


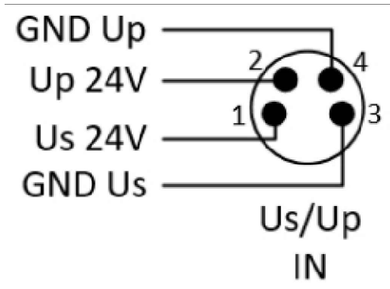


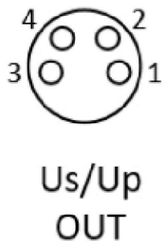


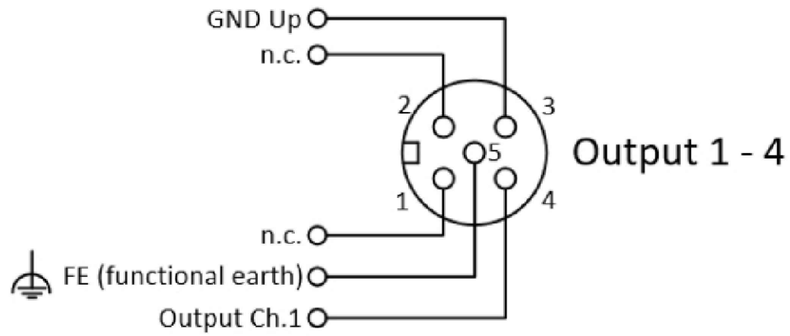


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PAGE DESCRIPTION: 43SRIO2 SAFETY DIGITAL IN/OUT MODULE

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TOTAL PAGES: 175

SHEET 43.2 OF 56

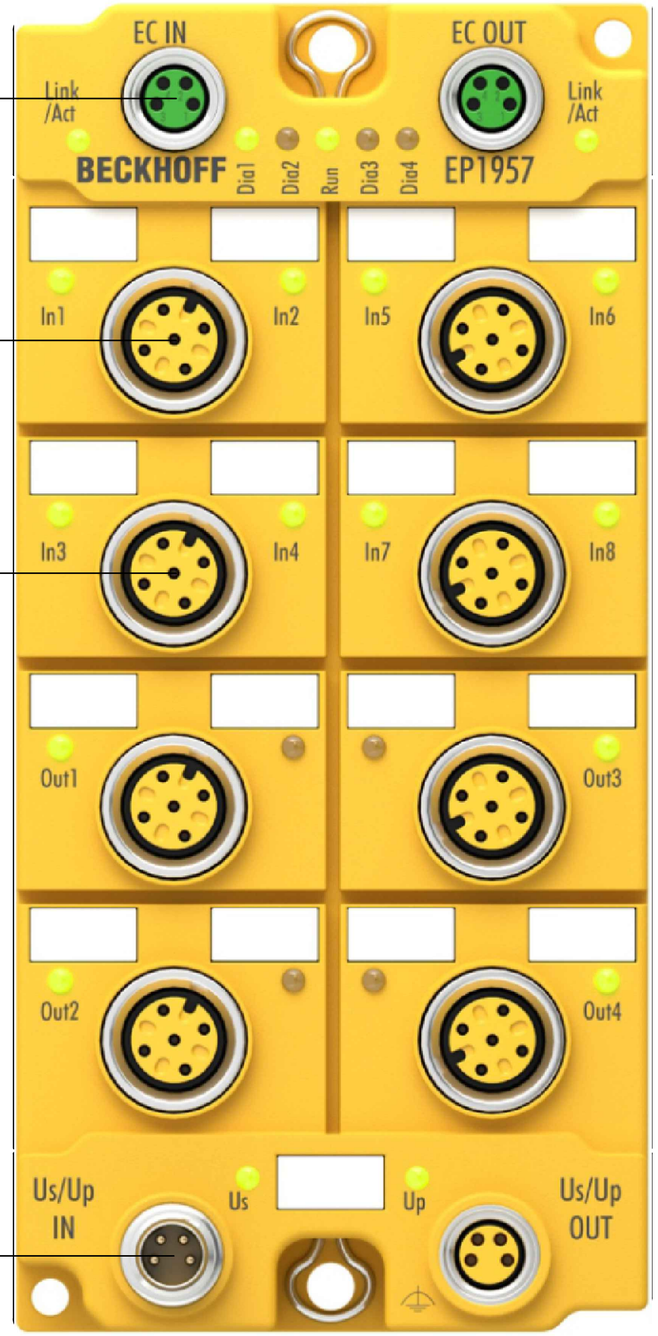
# 43SRIO2

42.1/9B FROM 42NS1-X8 ETH27

43.4/7C TO 43LC1 RECEIVER

43.4/5C TO 43LC2 SENDER

FROM 24L+ & 0VL- 12.2/

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PAGE DESCRIPTION: 43SRIO2 SAFETY DIGITAL IN/OUT MODULE

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SHEET 43.3 OF 56

## System connection (M12, 5-pin)

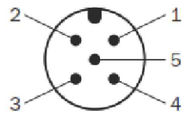


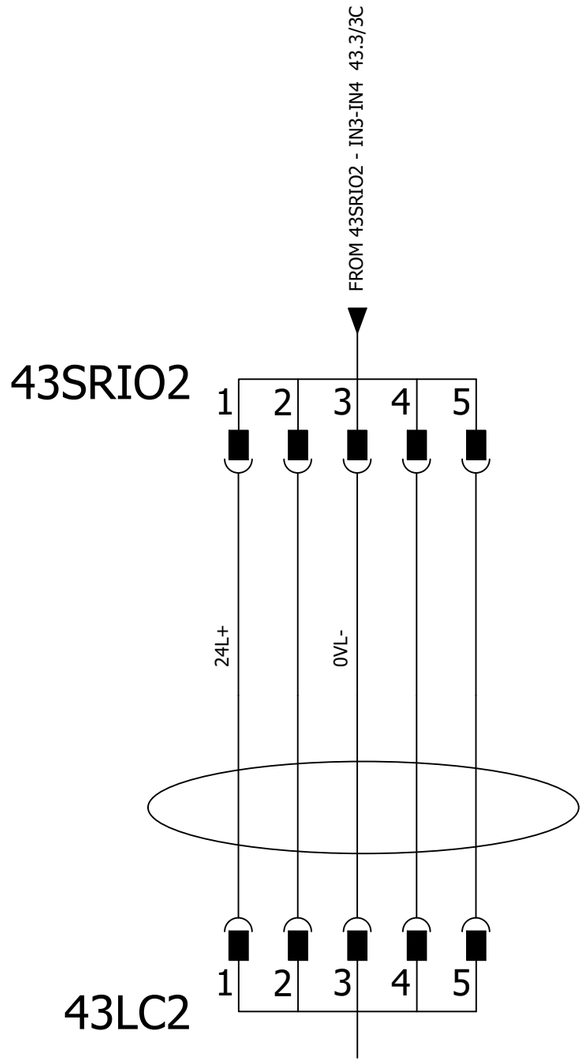

*Figure 26: System connection (male connector, M12, 5-pin)*

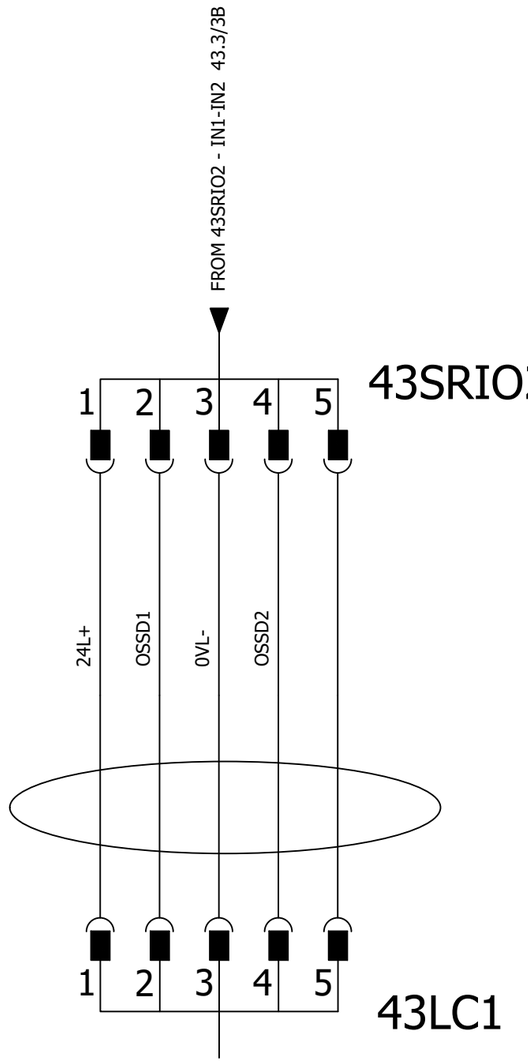
| Pin | Wire color [1] | Sender | Receiver |
|---|---|---|---|
| 1 | Brown | +24 V DC (voltage supply input) | +24 V DC (voltage supply input) |
| 2 | White | Reserved | OSSD1 (switching output 1) |
| 3 | Blue | 0 V DC (voltage supply input) | 0 V DC (voltage supply input) |
| 4 | Black | Reserved | OSSD2 (switching output 2) |
| 5 | Gray | Not assigned | Not assigned |

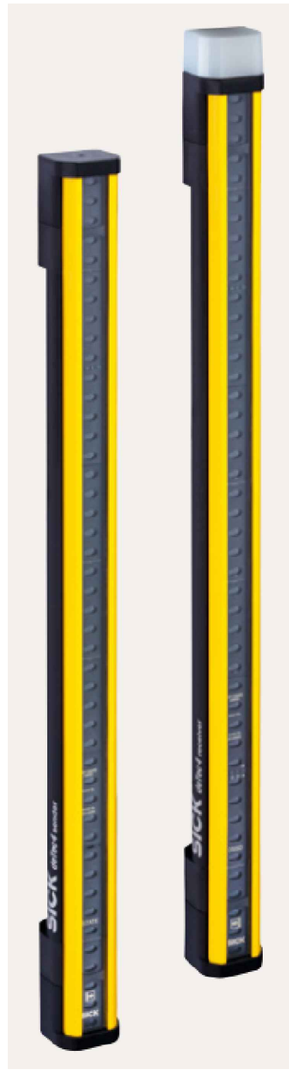
LIGHT CURTAIN - CAROUSEL CONVEYORS

SICK - RECEIVER - C4C-EV13530B10000 PART # 1107661

SICK - SENDER - C4C-SV13530B10000 PART # 1107660



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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LIGHT CURTAIN - CAROUSEL CONVEYORS

| REV | DESCRIPTION | DATE |
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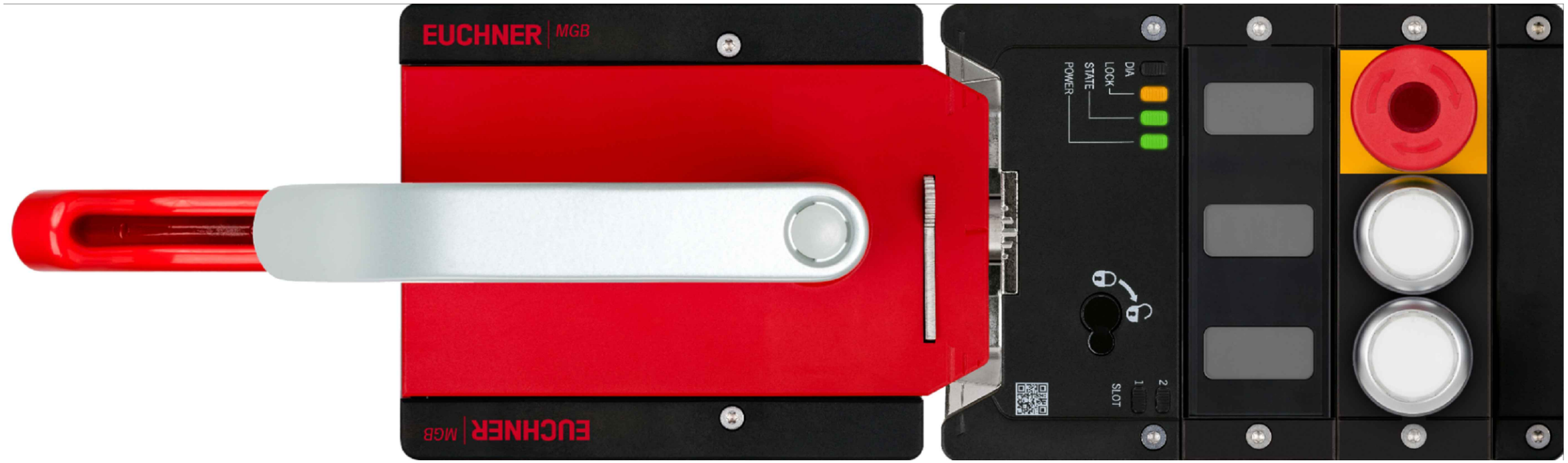
TOTAL PAGES: 175

SHEET 43.4 OF 56

# 44SD1

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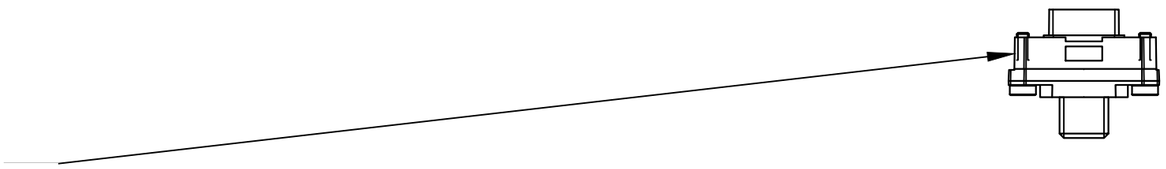

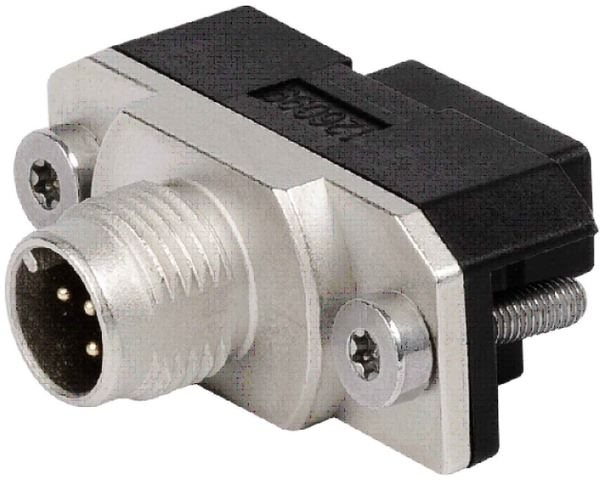

direct plug

AC-MC-SB-M-A-157024

MBM-EC-S1-MLI-3B-173401

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: Guard lock with guard lock monitoring

| REV | DESCRIPTION | DATE |
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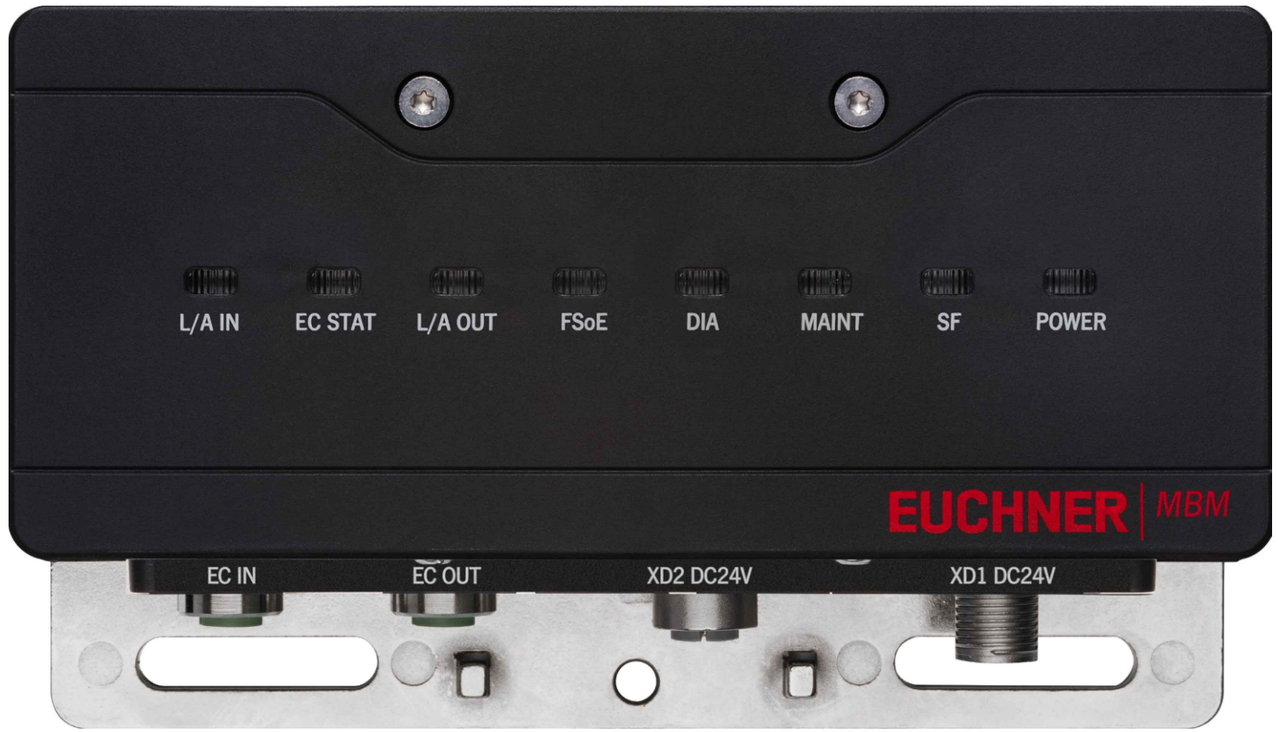
TOTAL PAGES: 175

SHEET 44.0 OF 56

| LED | Description |
|---|---|
| **Bus module MBM** | |
| L/A IN | Connection correct: statically on<br>Data transfer: flashing at 10 Hz<br>Color: green |
| EC STAT | Normal state: statically on<br>Flashing codes according to EtherCAT specification<br>Color: green/red |
| L/A OUT | Connection correct: statically on<br>Data transfer: flashing at 10 Hz<br>Color: green |
| FSoE | Normal state: statically on<br>Color: green<br>FSoE inactive/passivated: statically on<br>Color: yellow<br>FSoE Reset/Init State: flashing<br>Color: yellow<br>FSoE error: statically on<br>Color: red<br>FSoE – no connection to control system: off |
| DIA | Indicates errors<br>Color: red |
| MAINT | Indicates the error flashing code<br>Color: red/green/yellow |
| SF | System fault:<br>statically on (see chapter *12. Diagnostics, troubleshooting and aids*)<br>Color: red |
| Power | Power Up: flashing at 5 Hz<br>Normal state: statically on<br>Color: green |




44SD1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: Guard lock with guard lock monitoring

| REV | DESCRIPTION | DATE |
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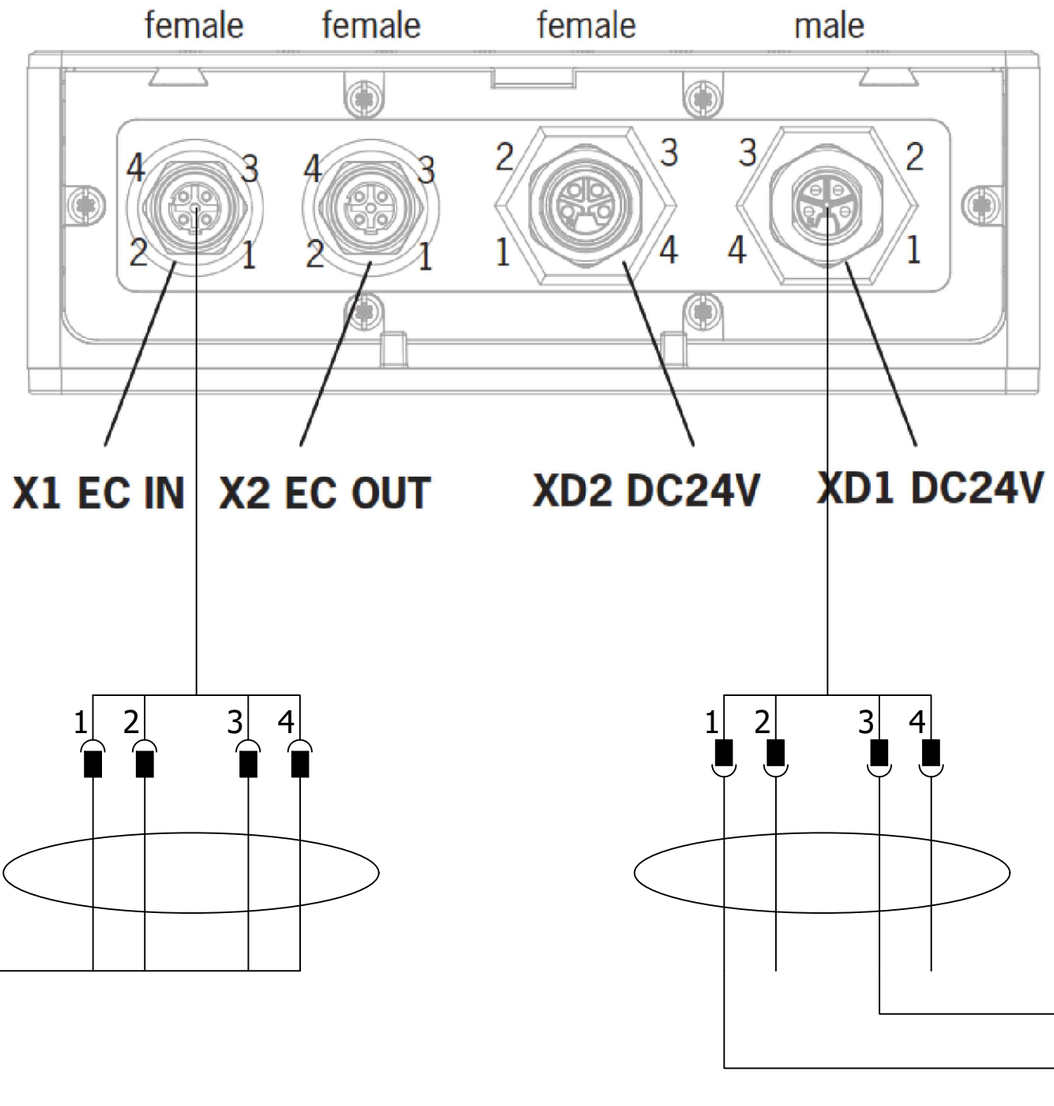
SHEET 44.1 OF 56

# Terminal assignment for version with Power M12 plug (L-coded), 4-pin, and bus M12 plug (D-coded)

| Pin | Description |
|---|---|
| X1.1 | Transmit Data TX+ |
| X1.2 | Receive Data RX+ |
| X1.3 | Transmit Data TX– |
| X1.4 | Receive Data RX– |
| Functional earth on plug housing | |

| Pin | Description |
|---|---|
| X2.1 | Transmit Data TX+ |
| X2.2 | Receive Data RX+ |
| X2.3 | Transmit Data TX– |
| X2.4 | Receive Data RX– |
| Functional earth on plug housing | |




| Pin | Description |
|---|---|
| XD1.1 | L1 operating voltage US DC 24 V |
| XD1.2 | N2 auxiliary voltage UP [1] 0 V |
| XD1.3 | N1 operating voltage US 0 V |
| XD1.4 | L2 auxiliary voltage UP [1] DC 24 V |

| Pin | Description |
|---|---|
| XD2.1 | L1 operating voltage US DC 24 V |
| XD2.2 | N2 auxiliary voltage UP [1] 0 V |
| XD2.3 | N1 operating voltage US 0 V |
| XD2.4 | L2 auxiliary voltage UP [1] DC 24 V |

1) The auxiliary voltage is not required for the MGB2 system.

44SD1

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: Guard lock with guard lock monitoring

| REV | DESCRIPTION | DATE |
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SHEET 44.2 OF 56

# CAROUSEL CONVEYORS AREA SECTION 3

FROM 23.0-9-X1 32.0/7C 32ECP1

45RIO1

45ECP1

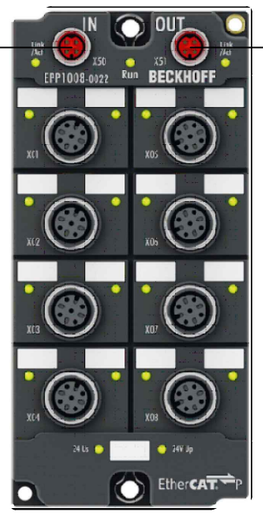
45RIO2

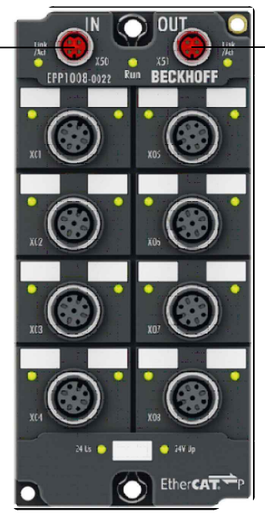
45ECP2

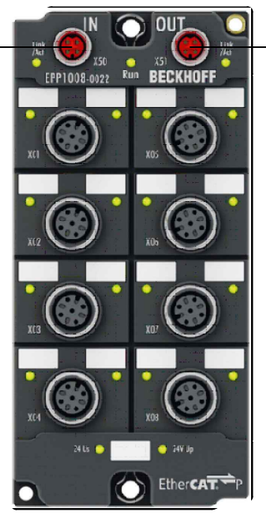
45RIO3

45ECP3

45RIO4



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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

| REV | DESCRIPTION | DATE |
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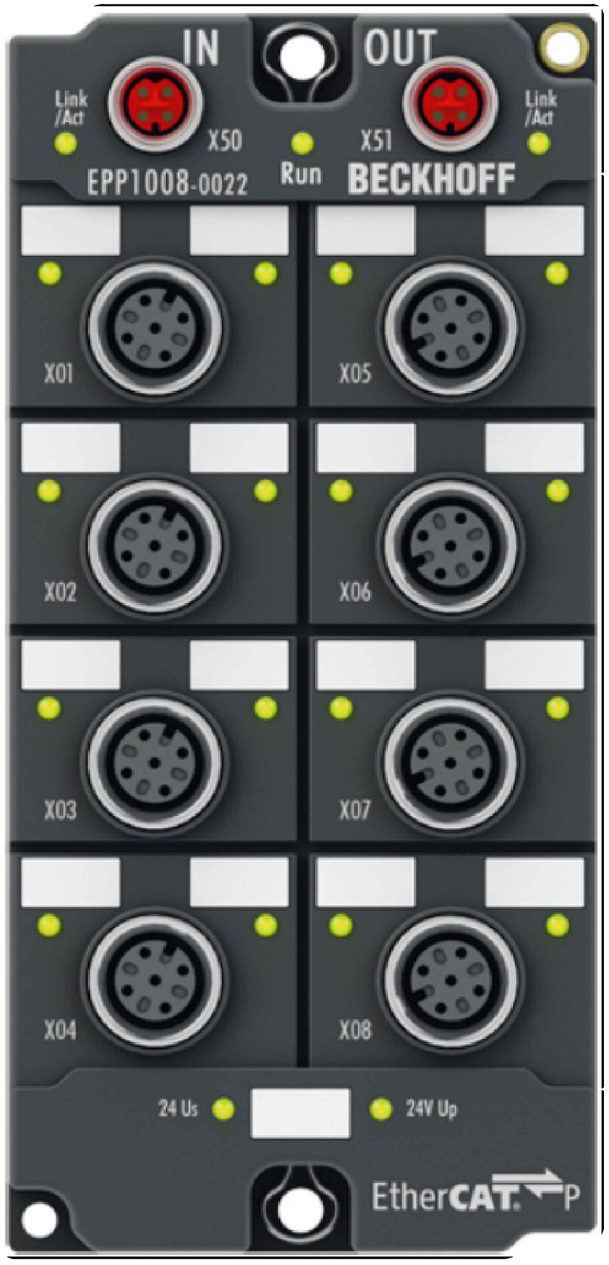
SHEET 45.0 OF 56

# EPP1008-0022




EtherCAT P
M8 socket, 4-pin, P-coded, shielded

1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

DIGITAL INPUTS
M12 socket, 5-pin, A-coded

1 | +24 V $U_S$
2 | n.c.
3 | $GND_S$
4 | Input
5 | n.c.

CAROUSEL
CONVEYOR
POPUP 1-2

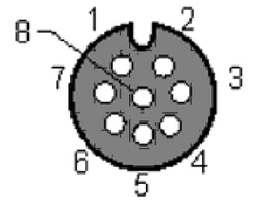
CAROUSEL CONVEYOR
TURNTABLE+ROLLER

# EPP5001-0002

EtherCAT P
M8 socket, 4-pin, P-coded, shielded

1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

SSI encoder input
M12 socket, 8-pin




1 | $GND_S$
2 | Sensor Supply
3 | Clock +
4 | Clock -
5 | Data +
6 | Data -
7 | Set
8 | Direction

CAROUSEL
CONVEYOR
TURNTABLE
EPP5001-0002
SSI ENCODER

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL
PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

| REV | DESCRIPTION | DATE |
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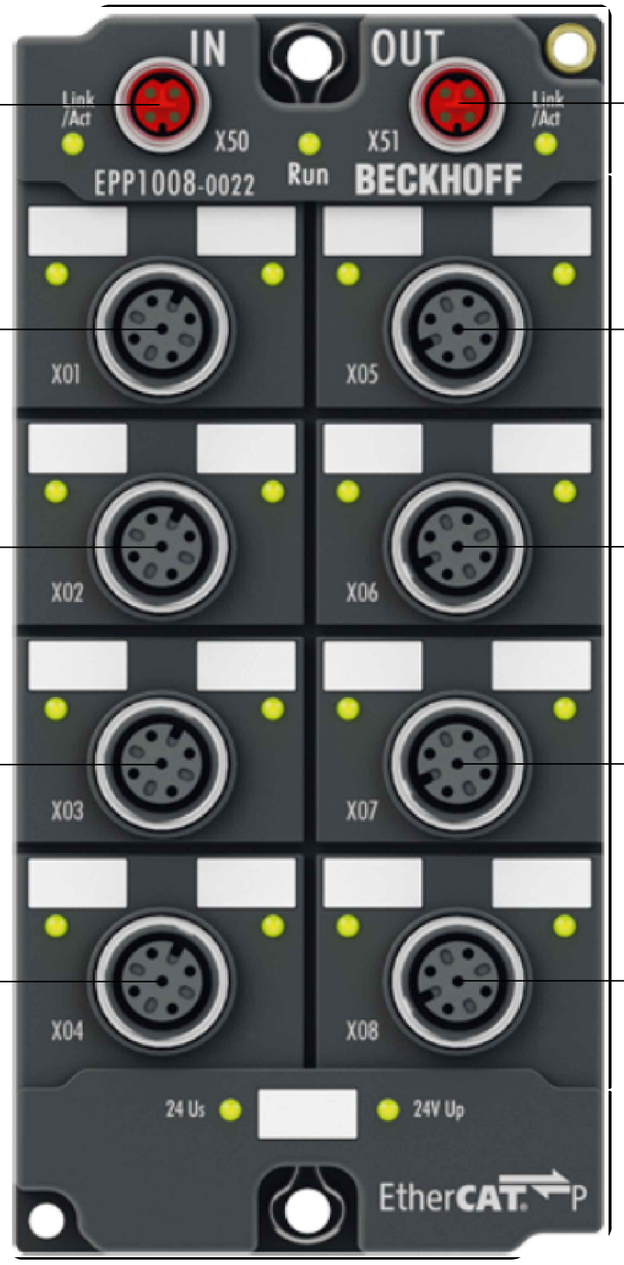
JOB # C40572
TOTAL PAGES: 175
SHEET 45.1 OF 56

# EPP1008-0022

# 45RIO1

FROM 23.0-9-X1 32.0/7C 32ECP1 → 32ECP1

45ECP1 → 45ECP1 45.3/3B TO 45RIO2-X50

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO1 REMOTE I/O INPUT LAYOUT

| REV | DESCRIPTION | DATE |
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TOTAL PAGES: 175

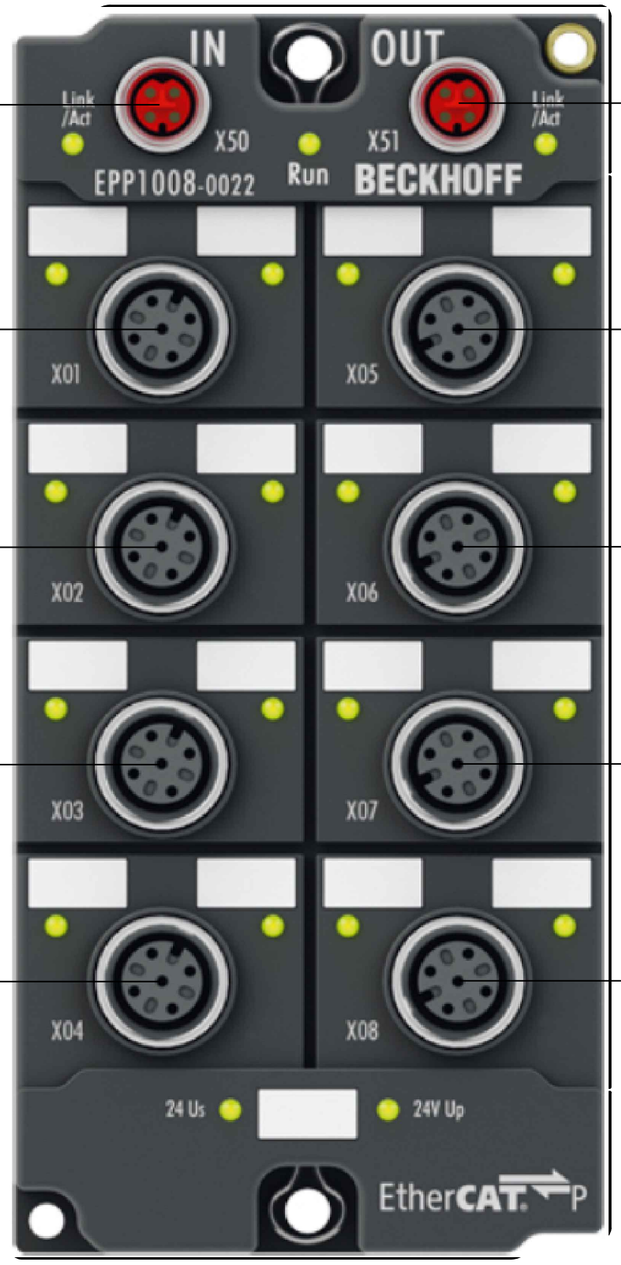
SHEET 45.2 OF 56

# EPP1008-0022

# 45RIO2

FROM 45RIO1-X51 45.1/8B 45RIO1 → 45ECP1

45ECP2 → 45ECP2 45.4/3B TO 45RIO3-X50




????????? → X01

X05 → ?????????

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X06 → ?????????

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X07 → ?????????

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X08 → ?????????

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO2 REMOTE I/O INPUT LAYOUT

| REV | DESCRIPTION | DATE |
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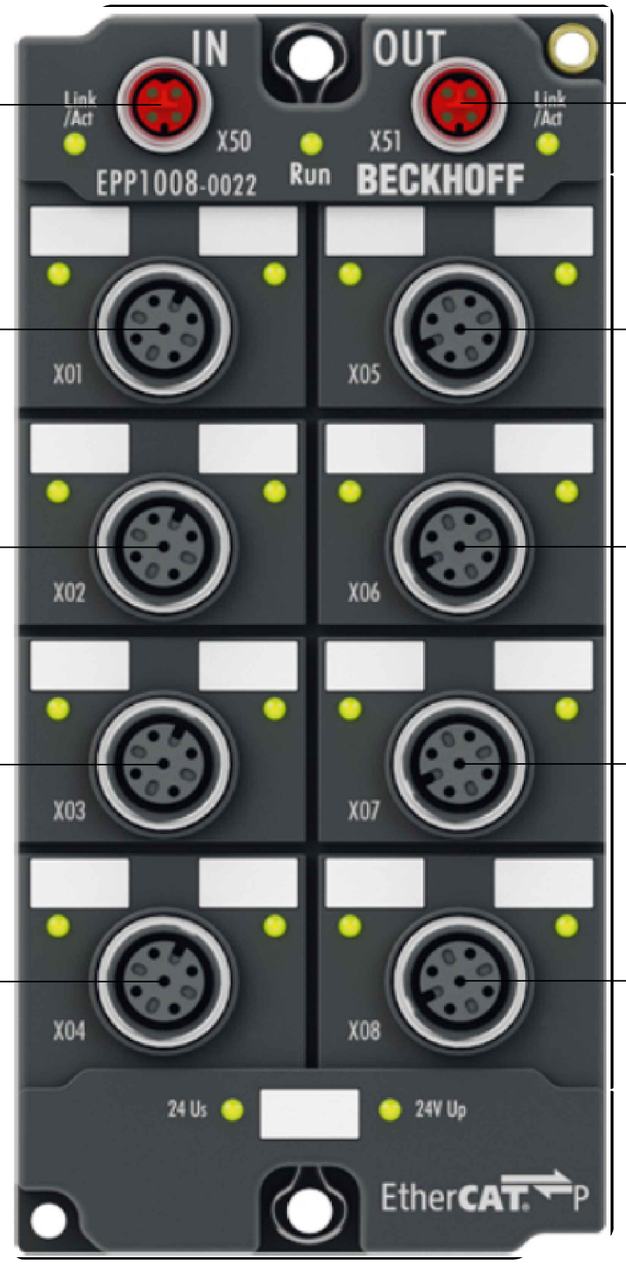
SHEET 45.3 OF 56

# EPP1008-0022

# 45RIO3

FROM 45RIO2-X51 45.2/8B 45ECP2

45ECP2




45ECP3

45ECP3 45.5/3B TO 45RIO4-X50

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PAGE DESCRIPTION: 45RIO3 REMOTE I/O INPUT LAYOUT

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SHEET 45.4 OF 56

# EPP5001-0002

## 45RIO4

FROM 45RIO3-X51 45.4/8B 45ECP3 → 45ECP3

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 45RIO4 REMOTE I/O INPUT LAYOUT

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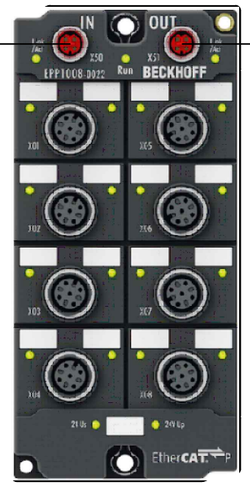
TOTAL PAGES: 175

SHEET 45.5 OF 56

# CAROUSEL CONVEYORS AREA SECTION 1 & 2

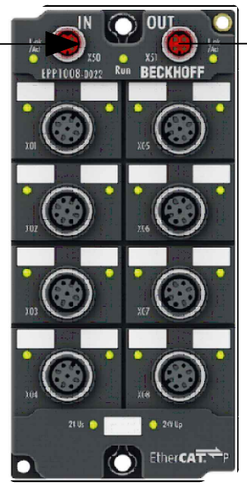
FROM 23.0-10-X1 33.0/7C 33ECP1

## 46RIO1



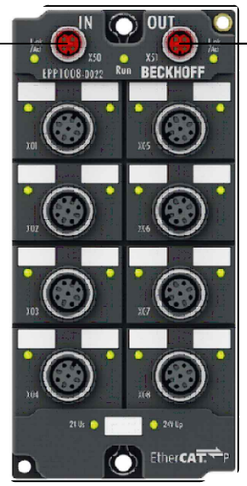

46ECP1

## 46RIO2



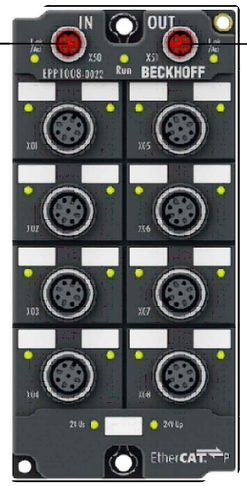

46ECP2

## 46RIO3



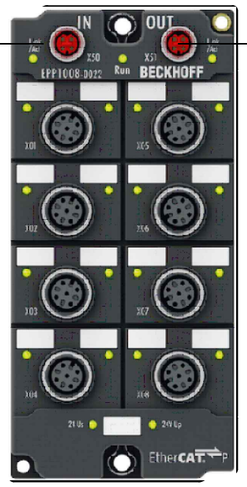

46ECP3

## 46RIO4



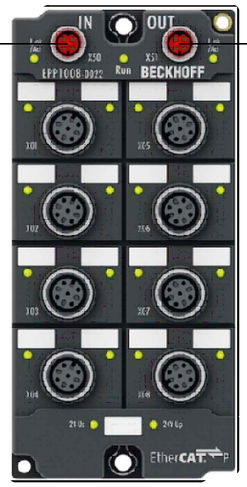

46ECP4

## 46RIO5



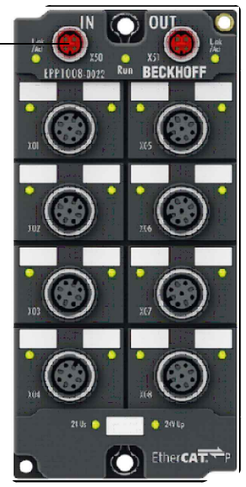

46ECP5

## 46RIO6




46ECP6

## 46RIO7

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

| REV | DESCRIPTION | DATE |
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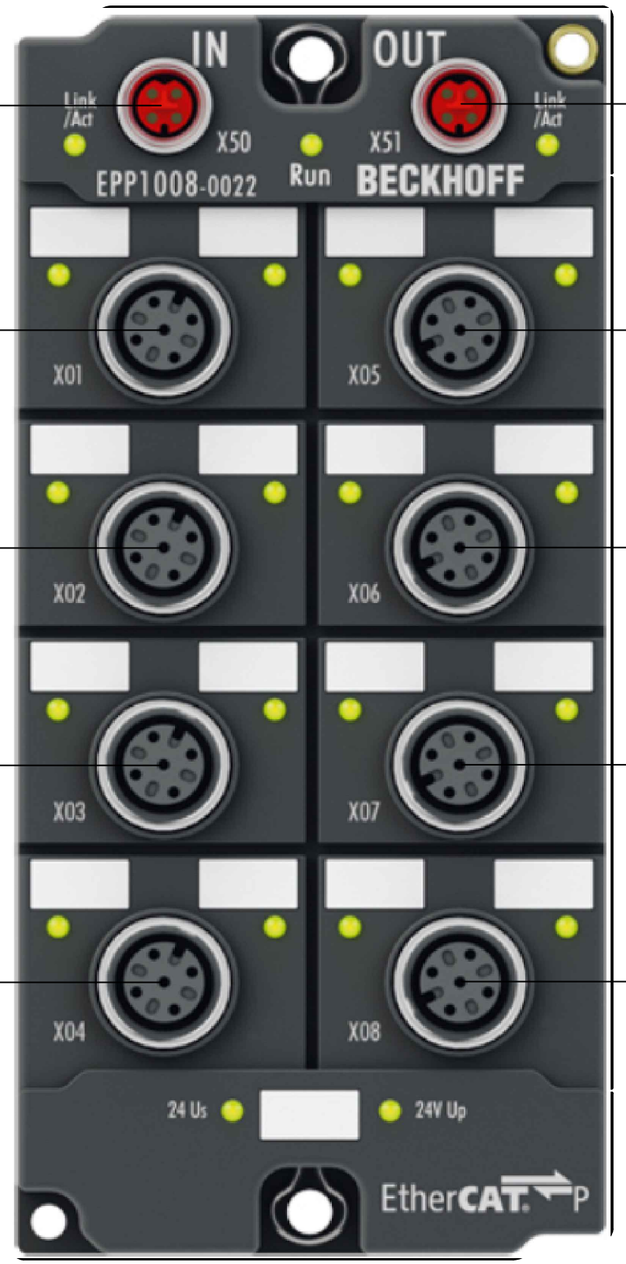
JOB # C40572

TOTAL PAGES: 175

SHEET 46.0 OF 56

# 46RIO1

FROM 23.0-10-X1 33.0/7C 33ECP1 → 33ECP1




46ECP1 → 46ECP1 46.2/3B TO 46RIO2-X50

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 46RIO1 REMOTE I/O INPUT LAYOUT

| REV | DESCRIPTION | DATE |
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JOB # C40572

TOTAL PAGES: 175

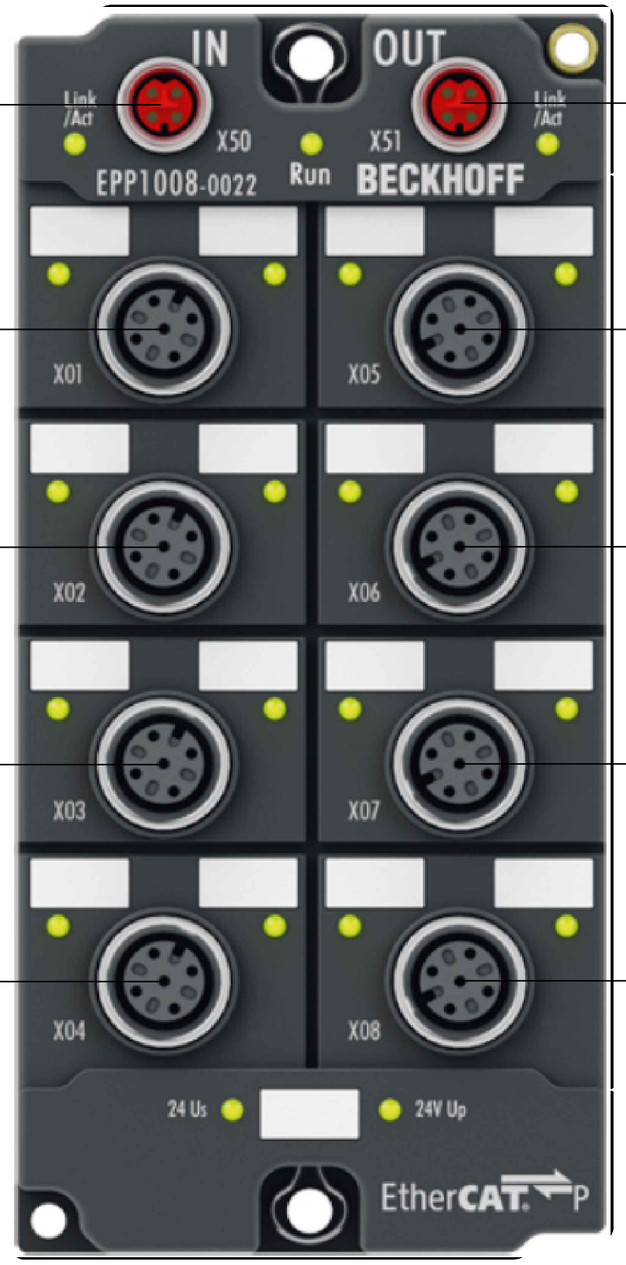
SHEET 46.1 OF 56

# EPP1008-0022

# 46RIO2

FROM 46RIO1-X51 46.1/8B 46ECP1

46ECP1




46ECP2

46ECP2 46.3/3B TO 46RIO3-X50

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PAGE DESCRIPTION: 46RIO2 REMOTE I/O INPUT LAYOUT

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TOTAL PAGES: 175

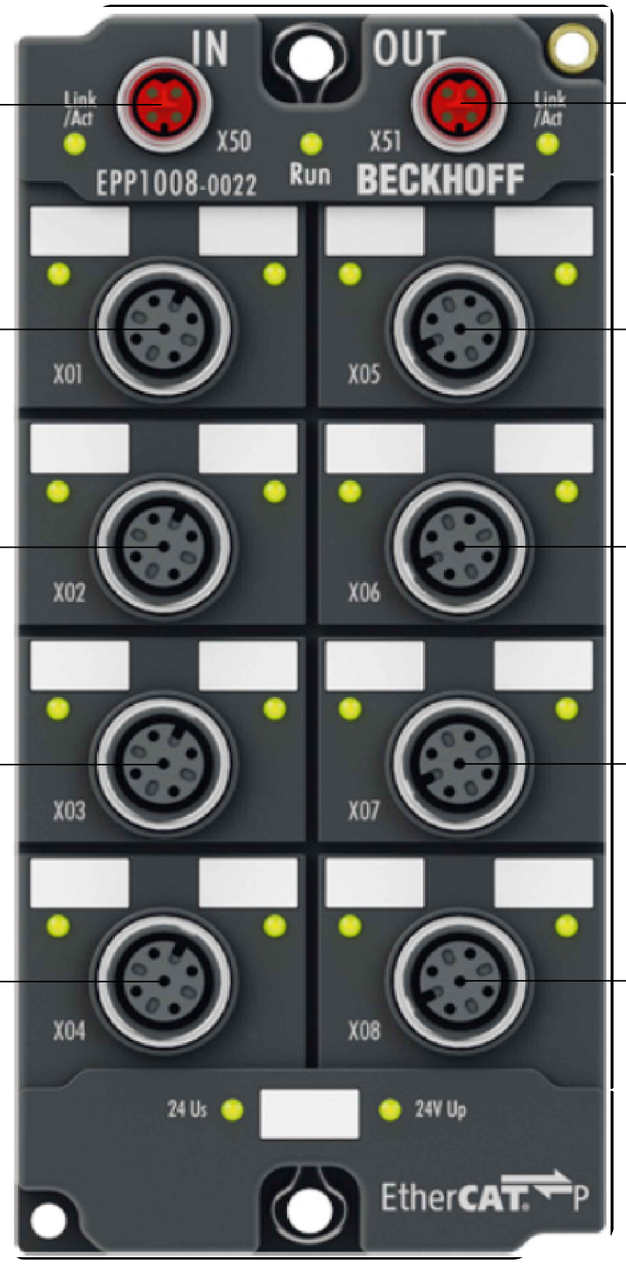
SHEET 46.2 OF 56

# EPP1008-0022

# 46RIO3

FROM 46RIO2-X51 46.2/8B 46ECP2

46ECP2




46ECP3

46ECP3 46.4/3B TO 46RIO4-X50

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PAGE DESCRIPTION: 46RIO3 REMOTE I/O INPUT LAYOUT

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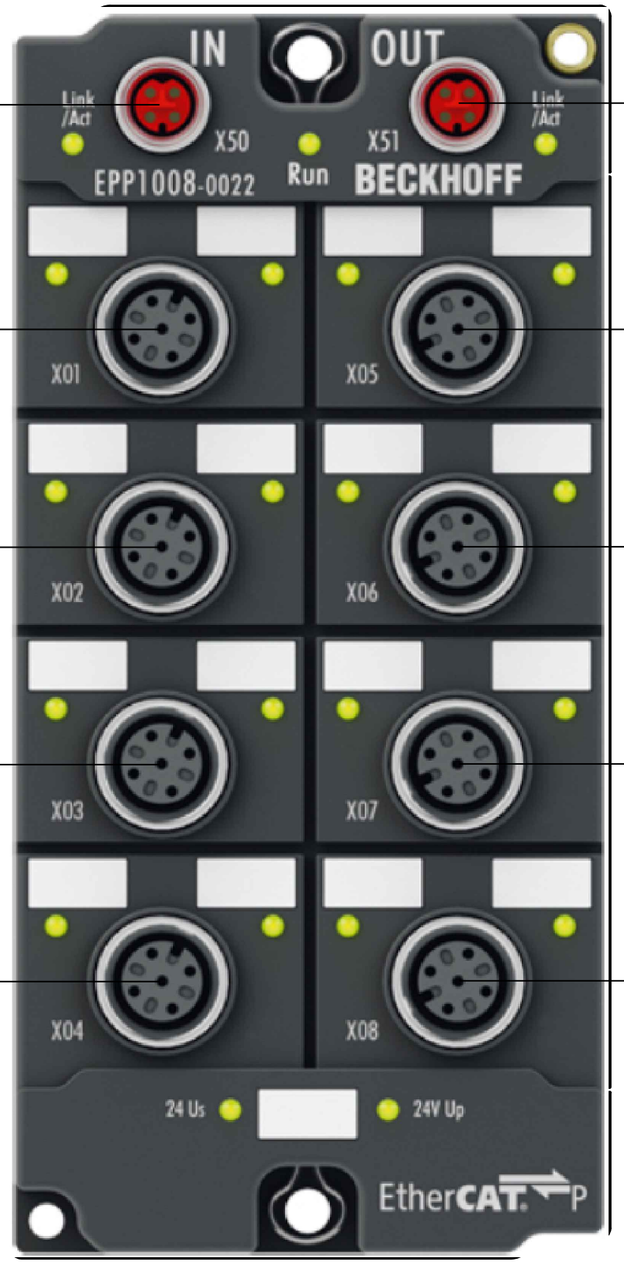
TOTAL PAGES: 175

SHEET 46.3 OF 56

# EPP1008-0022

# 46RIO4

FROM 46RIO3-X51 46.3/8B 46ECP3 → 46ECP3




46ECP4 → 46ECP4 46.5/3B TO 46RIO5-X50

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PAGE DESCRIPTION: 46RIO4 REMOTE I/O INPUT LAYOUT

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TOTAL PAGES: 175

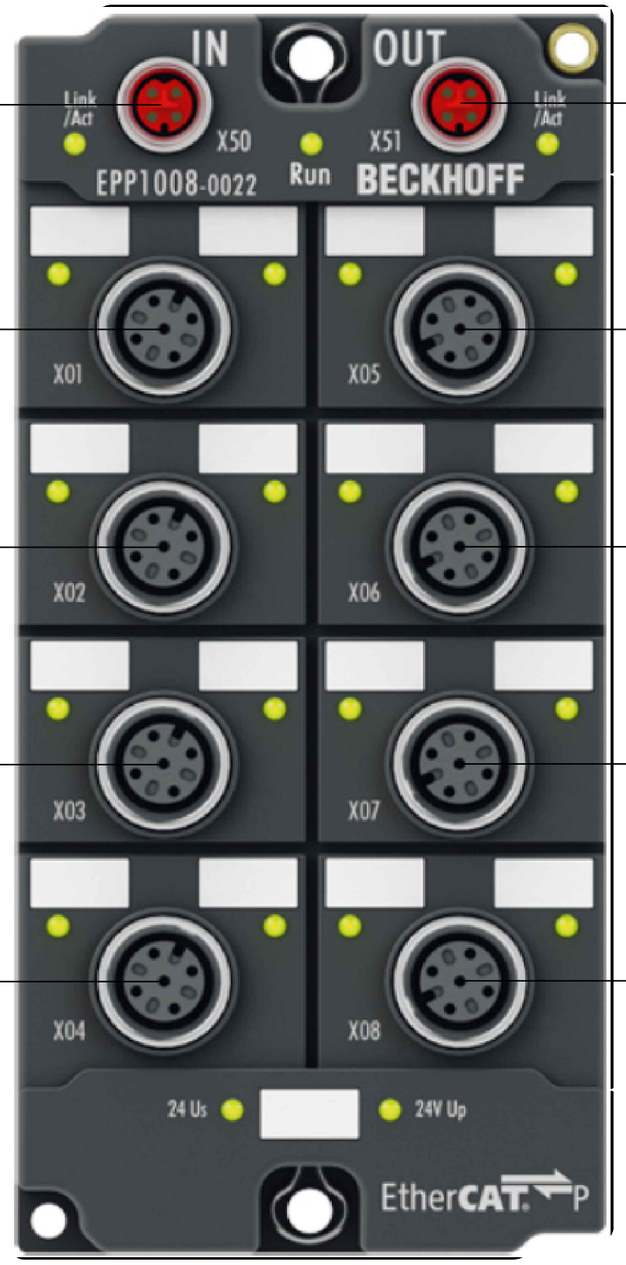
SHEET 46.4 OF 56

# EPP1008-0022

# 46RIO5

FROM 46RIO4-X51 46.4/8B 46ECP4

46ECP4




46ECP5

46ECP5 46.6/3B TO 46RIO6-X50

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 46RIO5 REMOTE I/O INPUT LAYOUT

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TOTAL PAGES: 175

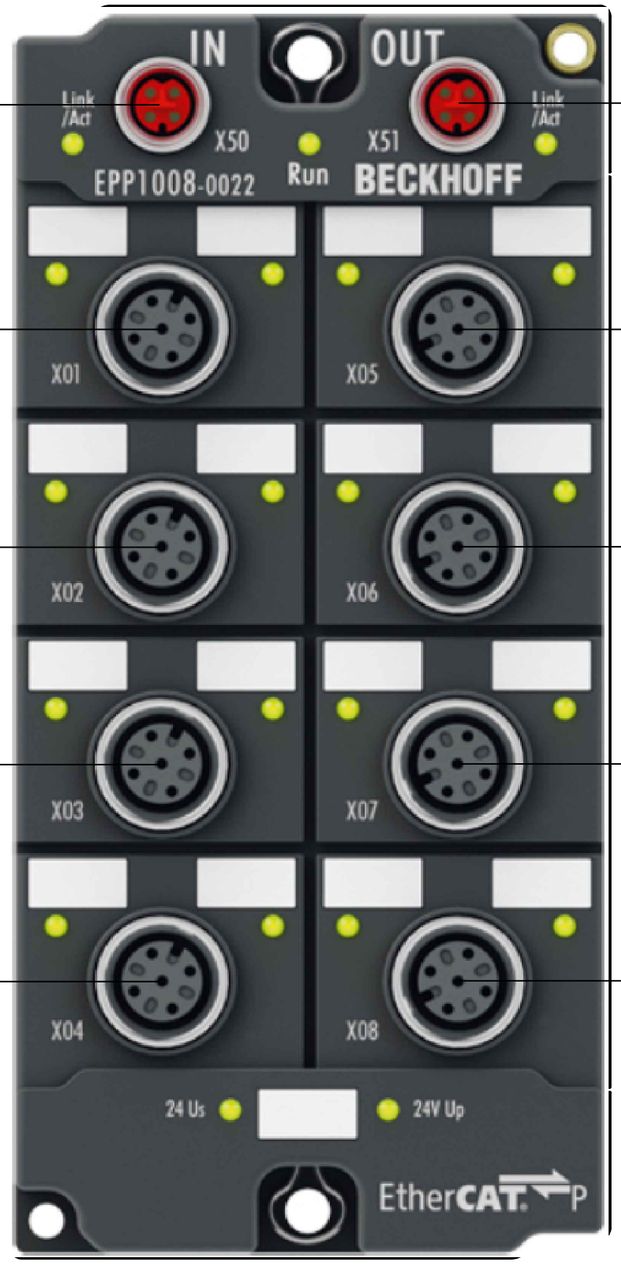
SHEET 46.5 OF 56

# EPP1008-0022

# 46RIO6

FROM 46RIO5-X51 46.5/8B 46ECP5 → 46ECP5

46ECP6 → 46ECP6 46.6/3B TO 46RIO7-X50

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PAGE DESCRIPTION: 46RIO6 REMOTE I/O INPUT LAYOUT

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TOTAL PAGES: 175

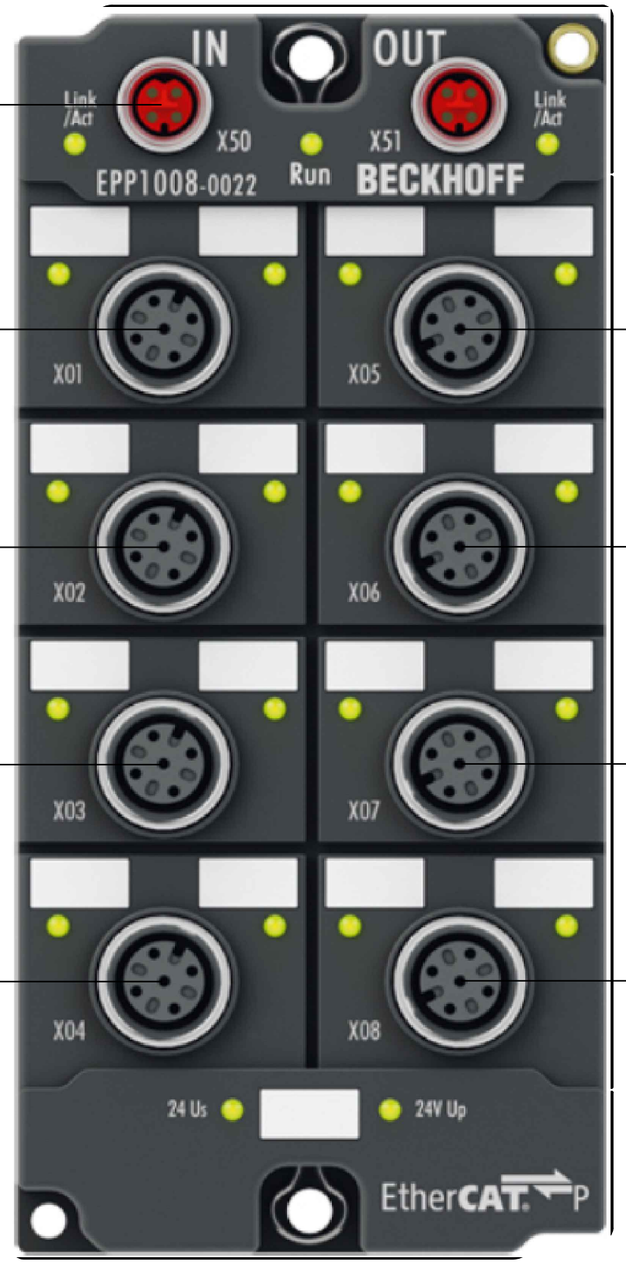
SHEET 46.6 OF 56

# EPP1008-0022

# 46RIO7

FROM 46RIO6-X51 46.6/8B 46ECP6

46ECP6

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 46RIO7 REMOTE I/O INPUT LAYOUT

| REV | DESCRIPTION | DATE |
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TOTAL PAGES: 175

SHEET 46.7 OF 56

# VTD (MOBILE) / EXTRACTOR AREA

FROM 23.0-11-X1 34.0/7C 34ECP1

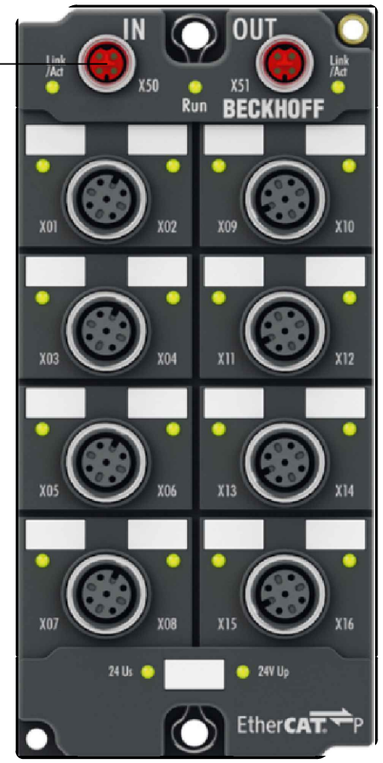
47ECP1

## 47RIO1

EPP2328-0002

EXTRACTOR CONVEYOR LOCKING PIN

## 47RIO2




EPP1809-0022

EXTRACTOR CONVEYOR

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

| REV | DESCRIPTION | DATE |
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JOB # C40572

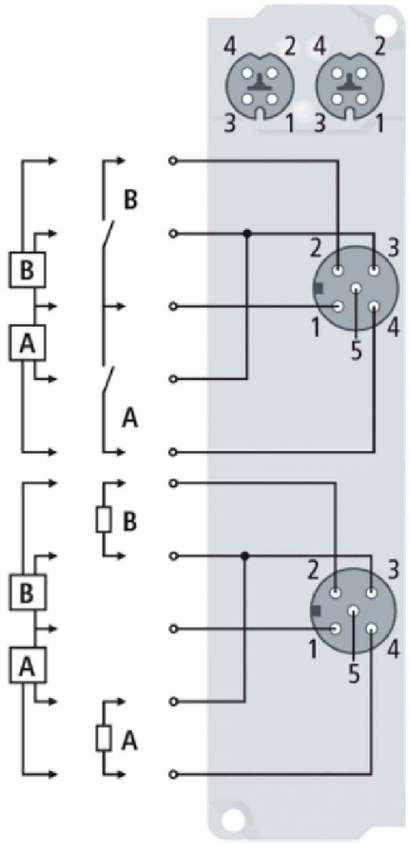
TOTAL PAGES: 175

SHEET 47.0 OF 56

# 47RIO1
# EPP2328-0002

| Contact | Signal | Voltage | Core color [1] |
|---|---|---|---|
| 1 | Tx + | $GND_S$ | yellow |
| 2 | Rx + | $GND_P$ | white |
| 3 | Rx - | $U_P$: peripheral voltage, +24 $V_{DC}$ | blue |
| 4 | Tx - | $U_S$: control voltage, +24 $V_{DC}$ | orange |
| Housing | Shield | Shield | Shield |

2 x M8 socket, 4-pin, P-coded, shielded

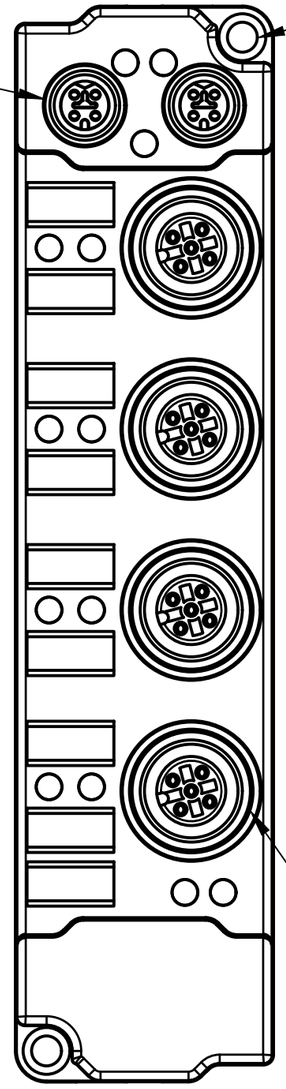



1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

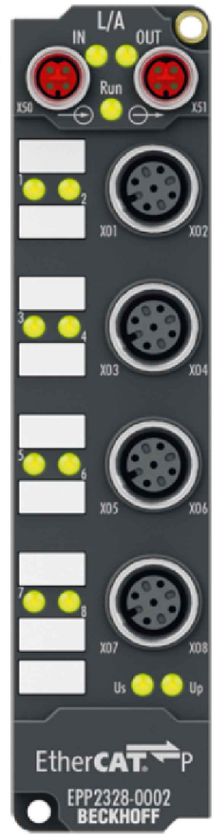
| Connector diameter | Tightening torque |
|---|---|
| M8 | 0.4 Nm |
| M12 | 0.6 Nm |



The upper mounting holes also serve as a connection for functional earth (FE).
Make sure that the box is grounded to low impedance via the functional earth (FE) connection. You can achieve this, for example, by mounting the box on a grounded machine bed.

EtherCAT P input

EtherCAT P downstream connection




Digital inputs

Digital outputs

4 x M12 socket, 5-pin, A-coded

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 47RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

| REV | DESCRIPTION | DATE |
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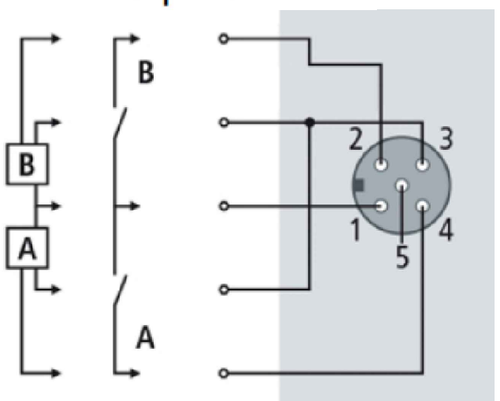
JOB # C40572

TOTAL PAGES: 175

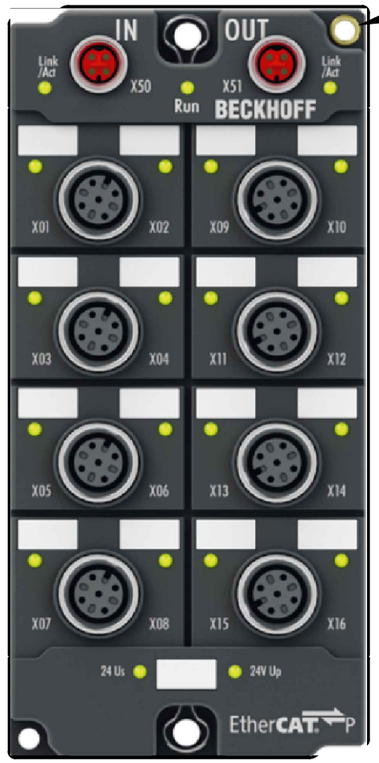
SHEET 47.1 OF 56

1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.




# 47RIO2




EPP1809-0022

The upper mounting holes also serve as a connection for functional earth (FE).
Make sure that the box is grounded to low impedance via the functional earth (FE) connection. You can achieve this, for example, by mounting the box on a grounded machine bed.

**Pin assignment**

| Pin | Function | Wire color [1)] |
|---|---|---|
| 1 | $U_S$ | brown |
| 2 | Input B | white |
| 3 | $GND_S$ | blue |
| 4 | Input A | black |
| 5 | - | gray |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL
PAGE DESCRIPTION: 47RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

| REV | DESCRIPTION | DATE |
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JOB # C40572
TOTAL PAGES: 175
SHEET 47.2 OF 56

# PLEASE REFERENCE DOCKING PINS J-BOX ELECTRICAL DRAWING

## 47RIO1
## EPP2328-0002

FROM 23.0-11-X1 34.0/7C 34ECP1 → 34ECP1

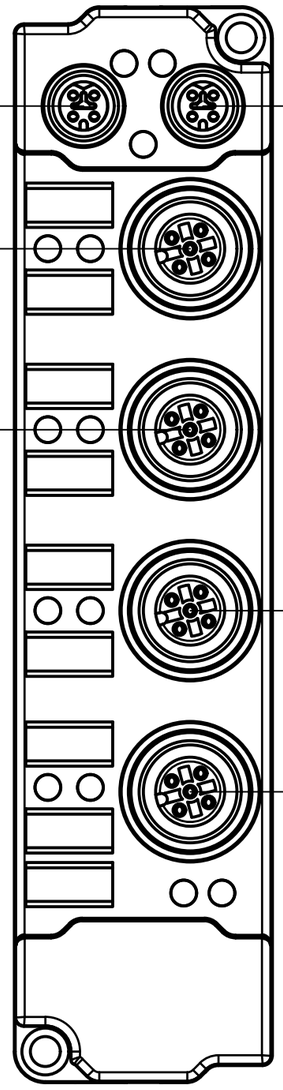
47ECP1 → 47ECP1 47.4/3B TO 47RIO2-X50

FROM LEFT LOCKING PINS EXTENDED PROXIMITY SENSORS — II

FROM RIGHT LOCKING PINS EXTENDED PROXIMITY SENSORS — III

FROM LEFT LOCKING PINS RETRACTED PROXIMITY SENSORS — II

FROM RIGHT LOCKING PINS RETRACTED PROXIMITY SENSORS — III



II — TO 3KA1 IN JUNCTION BOX.

III — TO 3KA2 IN JUNCTION BOX.

II — TO 3KA3 IN JUNCTION BOX.

III — TO 3KA4 IN JUNCTION BOX.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 47RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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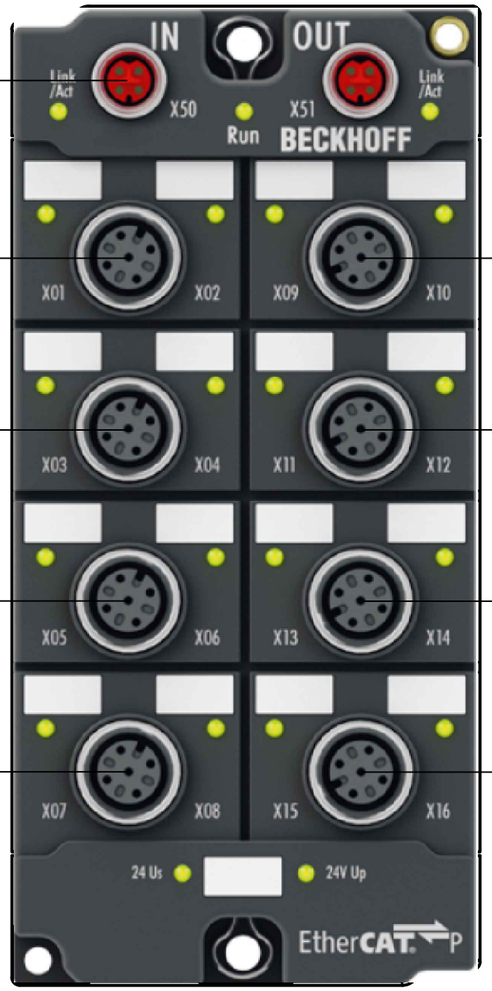
TOTAL PAGES: 175

SHEET 47.3 OF 56

# 47RIO2

FROM 47RIO1-X51 47.3/7B 34ECP1

47ECP1




EPP1809-0022

LEVEL DETECT 1 SENSOR 1

LEVEL DETECT 2 SENSOR 2

CREEP SENSOR 1

SLOWDOWN SENSOR 2

PALLET DETECTION #1

PALLET DETECTION #2

PALLET DETECTION #3

PALLET DETECTION #4

PALLET DETECTION #5

PALLET DETECTION #6

SHUTTLE DETECTION #1

SHUTTLE DETECTION #2

ZS2000-4722

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 47RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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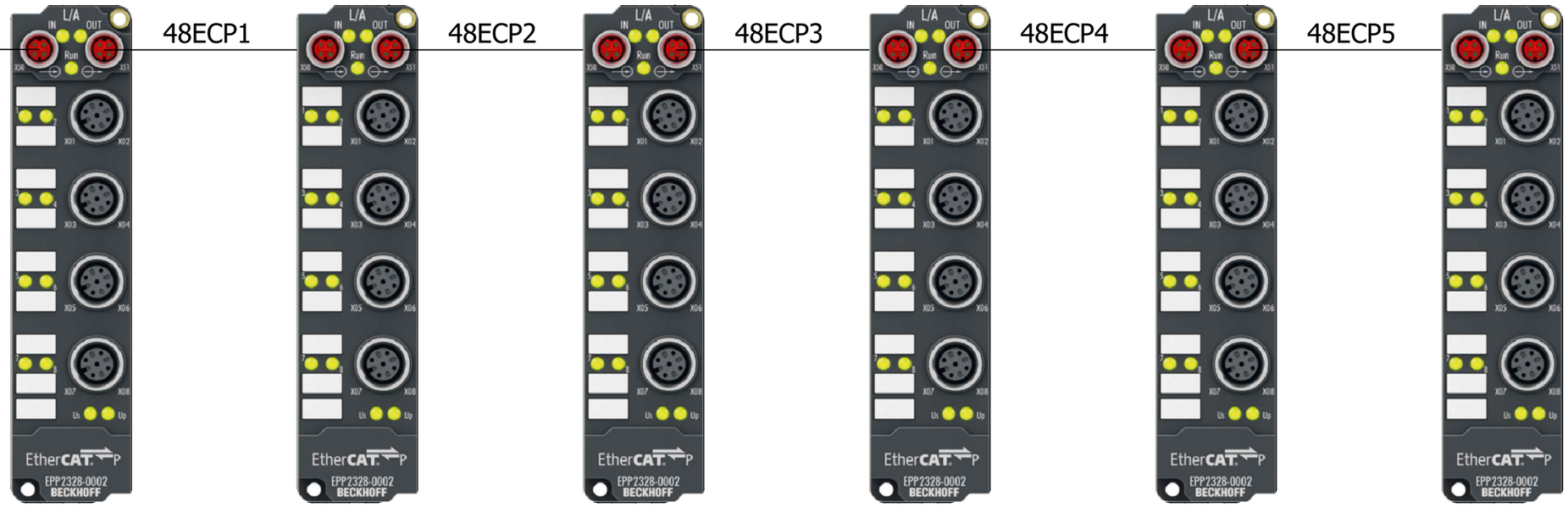
TOTAL PAGES: 175

SHEET 47.4 OF 56

# BUFFER CONVEYORS AREA SECTION 1 & 2

FROM 23.0-11-X2 34.0/7E 34ECP2

48RIO1 — 48ECP1 — 48RIO2 — 48ECP2 — 48RIO3 — 48ECP3 — 48RIO4 — 48ECP4 — 48RIO5 — 48ECP5 — 48RIO6




| 48RIO1 | 48RIO2 | 48RIO3 | 48RIO4 | 48RIO5 | 48RIO6 |
|---|---|---|---|---|---|
| EPP2328-0002 | EPP2328-0002 | EPP2328-0002 | EPP2328-0002 | EPP2328-0002 | EPP2328-0002 |
| CONVEYOR WITH STOPPER LEVEL 1 | CONVEYOR WITH STOPPER LEVEL 2 | CONVEYOR WITH STOPPER LEVEL 3 | CONVEYOR WITH STOPPER LEVEL 4 | CONVEYOR WITH STOPPER LEVEL 5 | CONVEYOR WITH STOPPER LEVEL 6 |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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JOB # C40572

TOTAL PAGES: 175

SHEET 48.0 OF 56

CONVEYOR
WITH
STOPPER
LEVEL 1

# 48RIO1
# EPP2328-0002

FROM 23.0-11-X2 34.0/7E 34ECP2 → 34ECP2

34ECP2 → 48ECP1

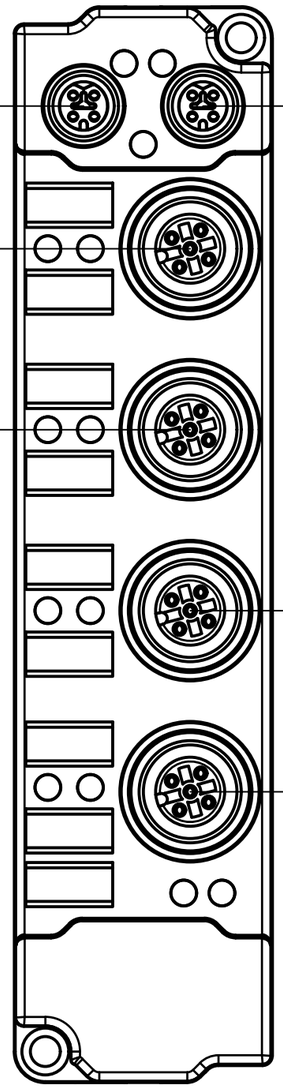
48ECP1 48.2/4B TO 48RIO2-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2



TO 53RR1-(3) PG 53.0/5A

TO 53RR2-(3) PG 53.0/7E

FOR WIRING METHOD, PLEASE REFER TO PAGE 47.1

ZS2000-4722

X2

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO1 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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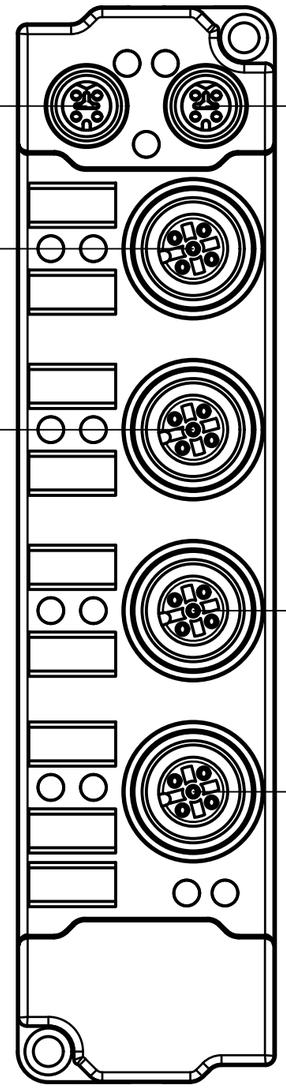
TOTAL PAGES: 175

CONVEYOR
WITH
STOPPER
LEVEL 2

# 48RIO2
# EPP2328-0002



FROM 48RIO1-X51 48.1/7B 48ECP1 → 48ECP1

48ECP2 → 48ECP2 48.3/4B TO 48RIO3-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

TO 53RR3-(3) PG 53.1/5A

TO 53RR4-(3) PG 53.1/7E

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X2

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO2 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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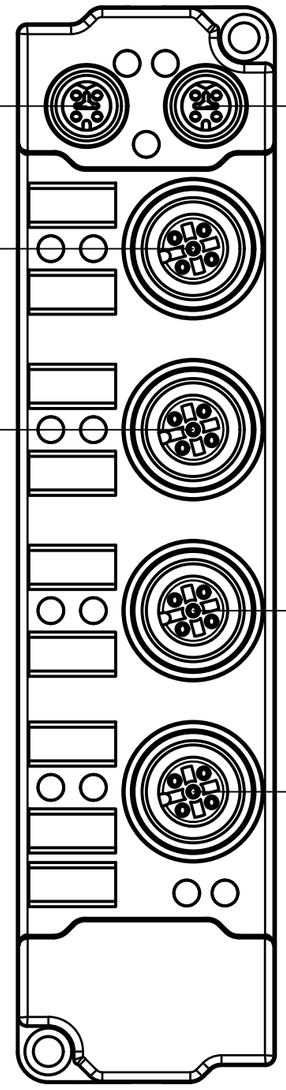
TOTAL PAGES: 175

CONVEYOR
WITH
STOPPER
LEVEL 3

# 48RIO3
# EPP2328-0002



FROM 48RIO2-X51 48.2/7B 48ECP2 → 48ECP2

48ECP3 → 48ECP3 48.4/4B TO 48RIO4-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

TO 53RR5-(3) PG 53.2/5A

TO 53RR6-(3) PG 53.2/7E

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X4

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO3 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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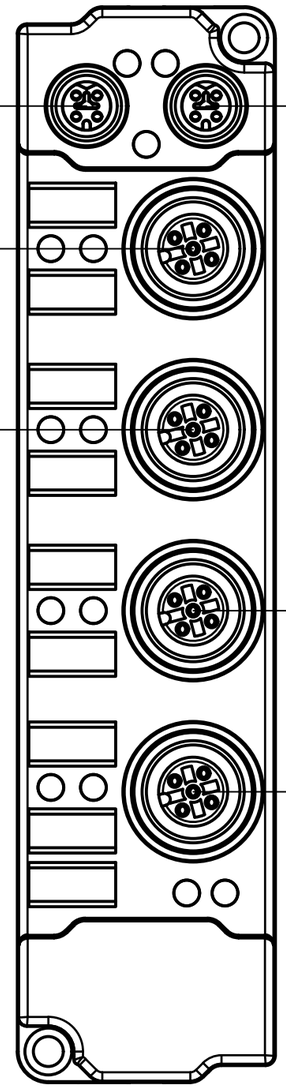
JOB # C40572

TOTAL PAGES: 175

CONVEYOR
WITH
STOPPER
LEVEL 4

# 48RIO4
# EPP2328-0002



FROM 48RIO3-X51 48.3/7B 48ECP3 → 48ECP3

48ECP4 → 48ECP4 48.5/4B TO 48RIO5-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

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ZS2000-4722

X4

TO 53RR7-(3) PG 53.3/5A

TO 53RR8-(3) PG 53.3/7E

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO4 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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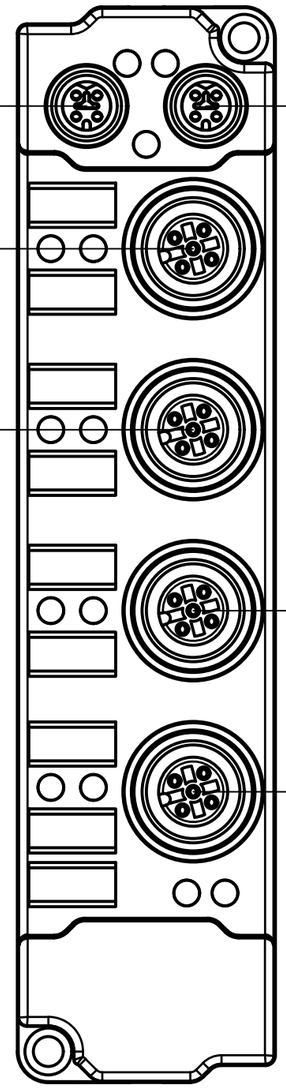
JOB # C40572

TOTAL PAGES: 175

CONVEYOR
WITH
STOPPER
LEVEL 5

# 48RIO5
# EPP2328-0002



FROM 48RIO4-X51 48.4/7B 48ECP4 → 48ECP4

48ECP5 → 48ECP5 48.6/4B TO 48RIO6-X50

SAFETY STOPPER UPPER SENSOR — 1

SAFETY STOPPER LOWER SENSOR — 2

BUFFER CONVEYOR SENSOR 1 — 1

BUFFER CONVEYOR SENSOR 2 — 2

TO 53RR9-(3) PG 53.4/5A

TO 53RR10-(3) PG 53.4/7E

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO5 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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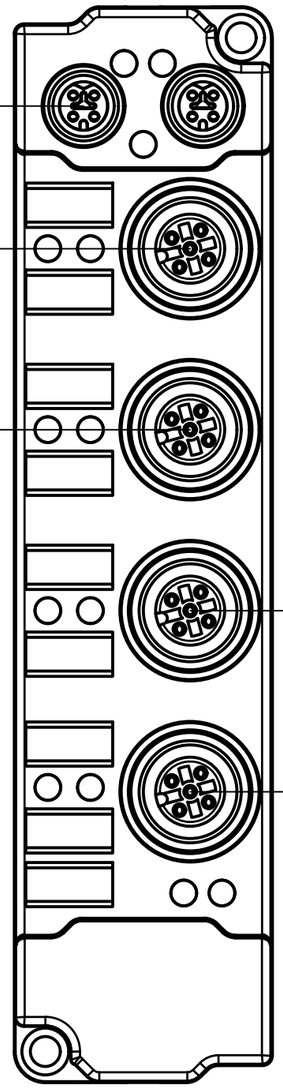
TOTAL PAGES: 175

CONVEYOR
WITH
STOPPER
LEVEL 6

# 48RIO6
# EPP2328-0002



FROM 48RIO5-X51 48.5/7B 48ECP5 ▶ 48ECP5

SAFETY STOPPER UPPER SENSOR ▶ 1

SAFETY STOPPER LOWER SENSOR ▶ 2

BUFFER CONVEYOR SENSOR 1 ▶ 1

BUFFER CONVEYOR SENSOR 2 ▶ 2

TO 53RR11-(3) PG 53.5/5A ◀

TO 53RR12-(3) PG 53.5/7E ◀

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: 48RIO6 EtherCAT P NETWORK INTERFACE REMOTE I/O MODULE LAYOUT

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: SPARE

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| DATE | DATE | CURRENT REVISION: |
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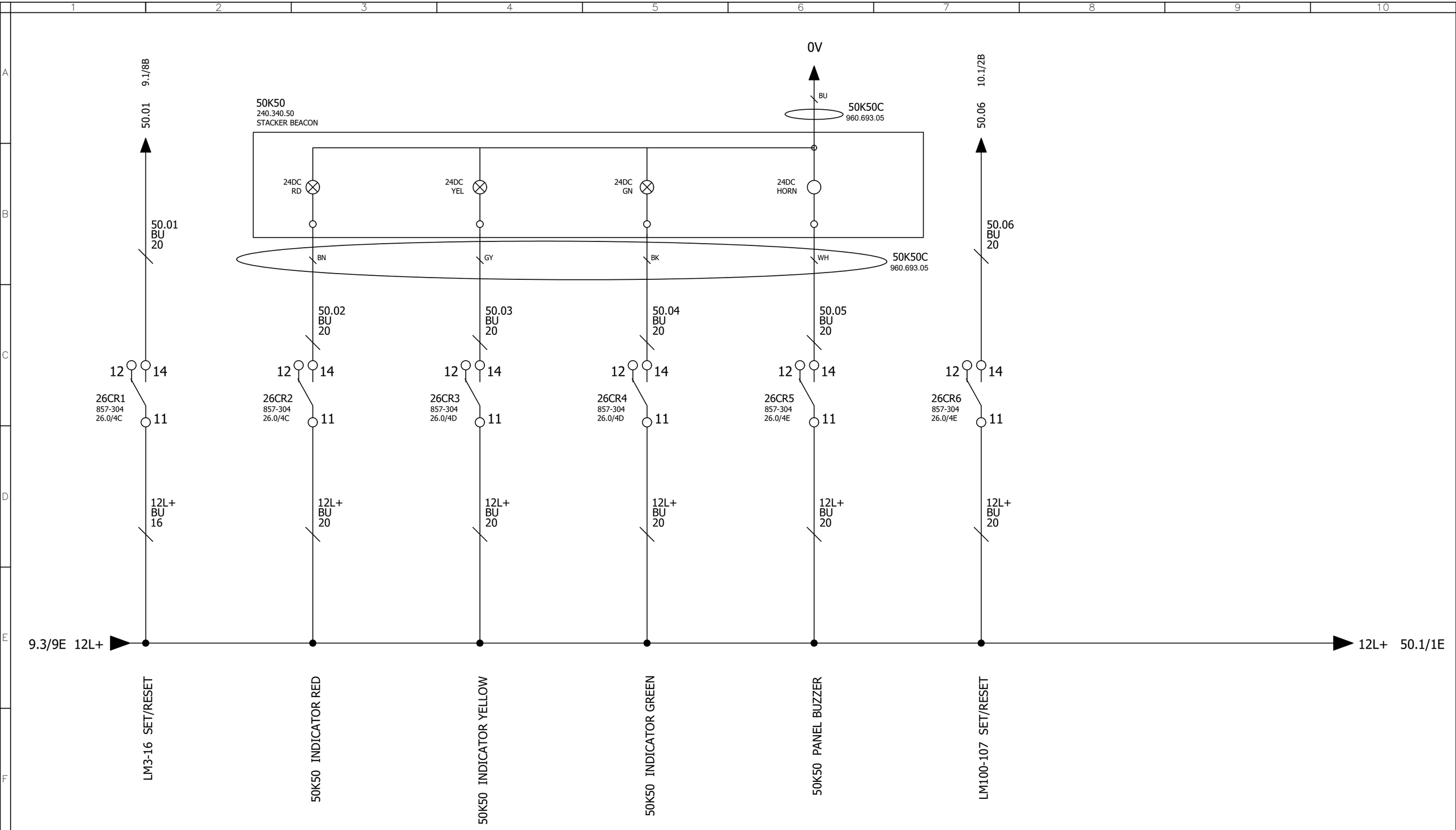
JOB # C40572

TOTAL PAGES: 175

SHEET 49.0 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: RELAYS

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| PAUL | Carlos A | C40572 |
| DATE | DATE | CURRENT REVISION: |
| 9/17/26 | 9/8/26 | D |

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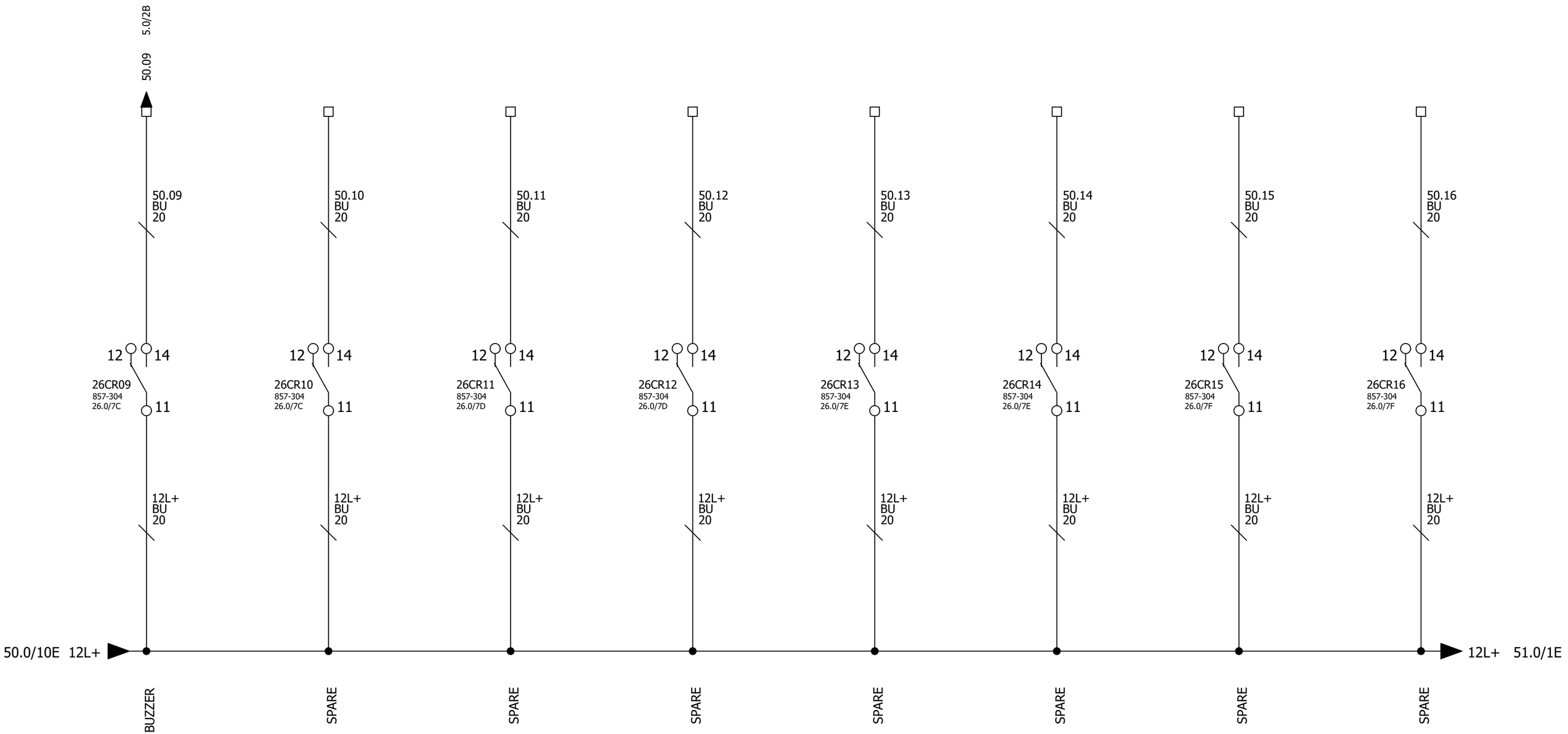
JOB # C40572

TOTAL PAGES: 175

SHEET 50.0 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: RELAYS

| REV | DESCRIPTION | DATE |
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| DRAWN BY: | APPROVED FOR BUILD BY: | DRAWING NUMBER: |
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| PAUL | Carlos A | C40572 |
| DATE | DATE | CURRENT REVISION: |
| 9/17/26 | 9/8/26 | D |

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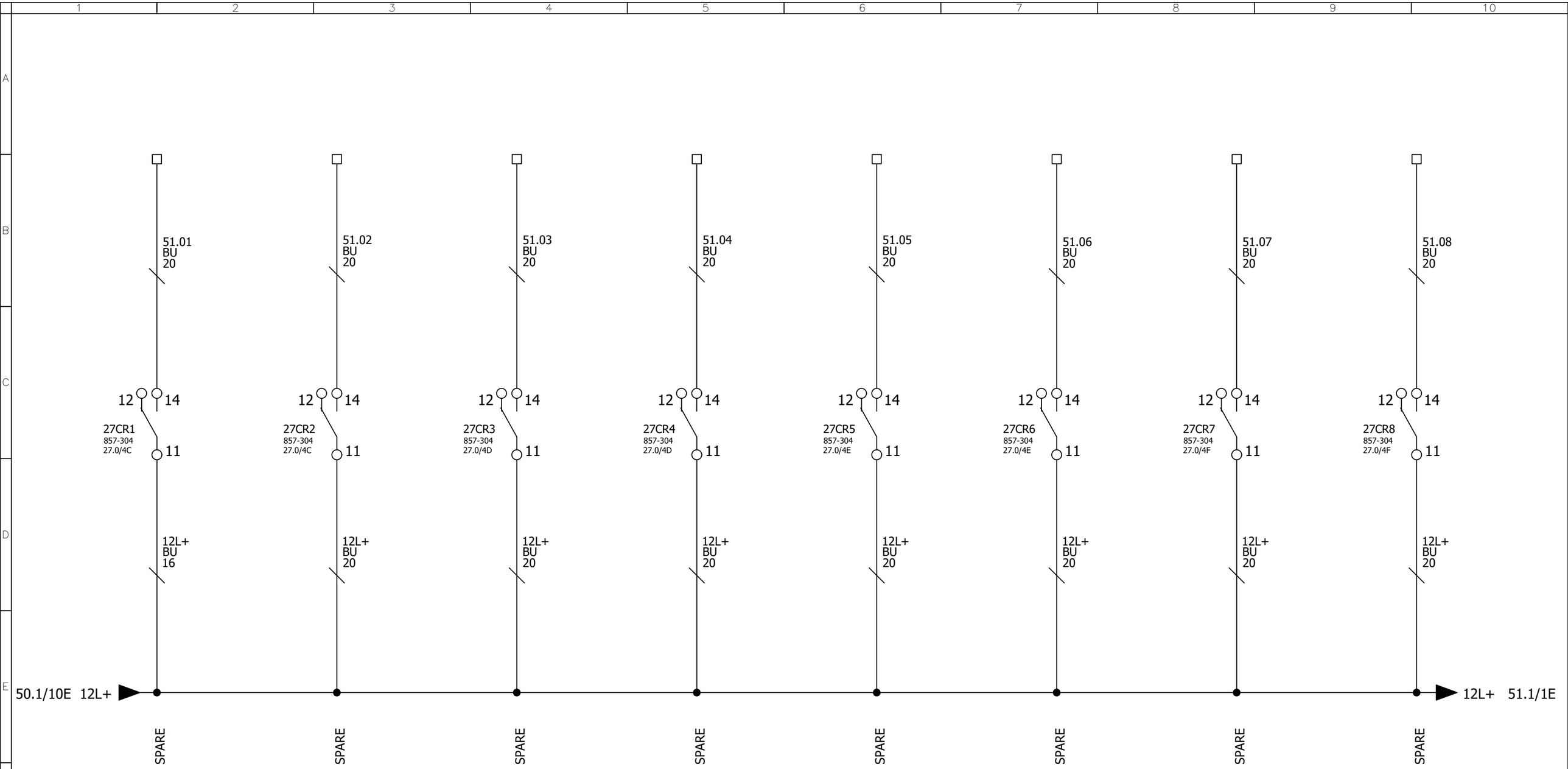
JOB # C40572

TOTAL PAGES: 175

SHEET 50.1 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: RELAYS

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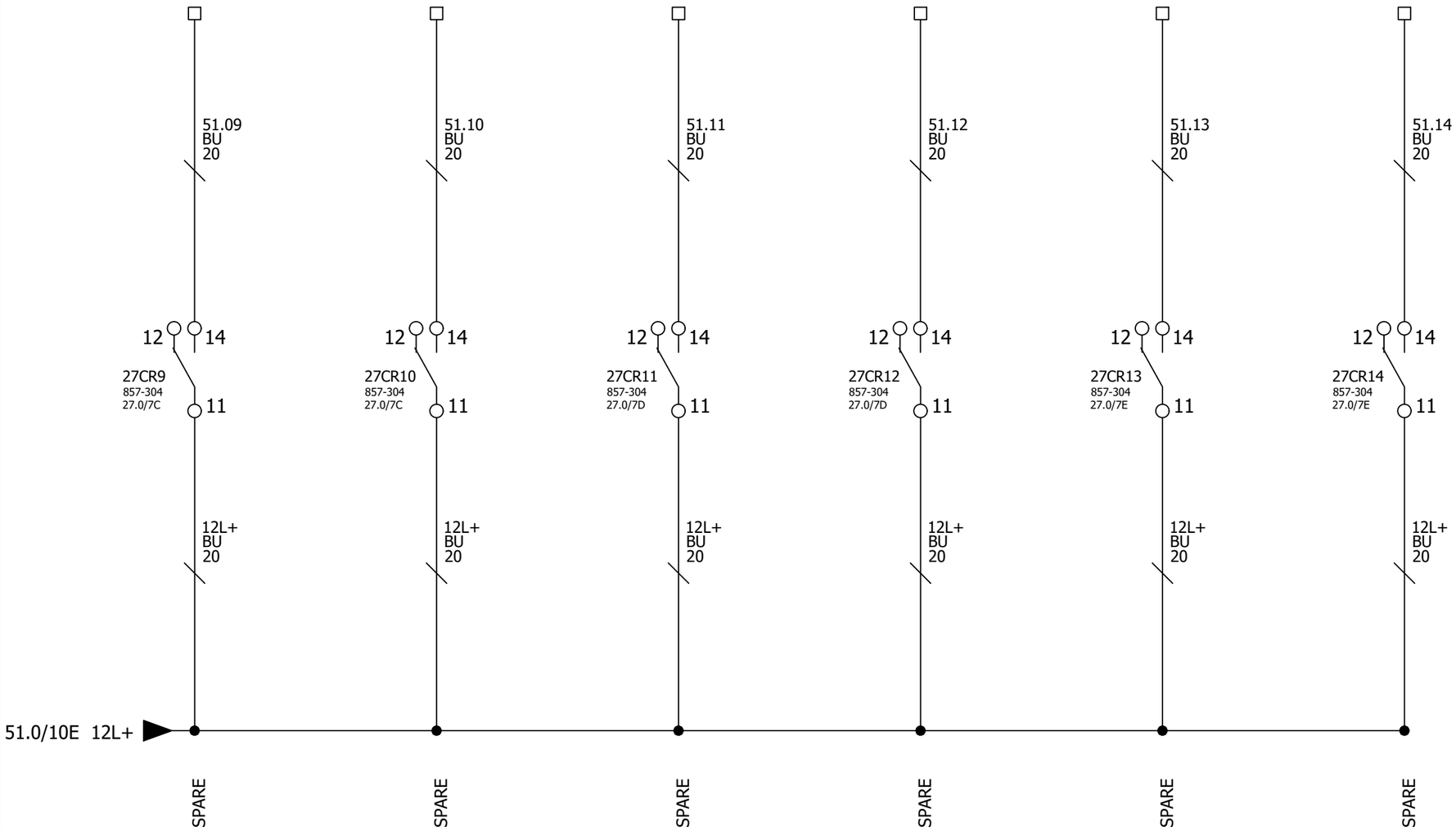
JOB # C40572

TOTAL PAGES: 175

SHEET 51.0 OF 56

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: RELAYS

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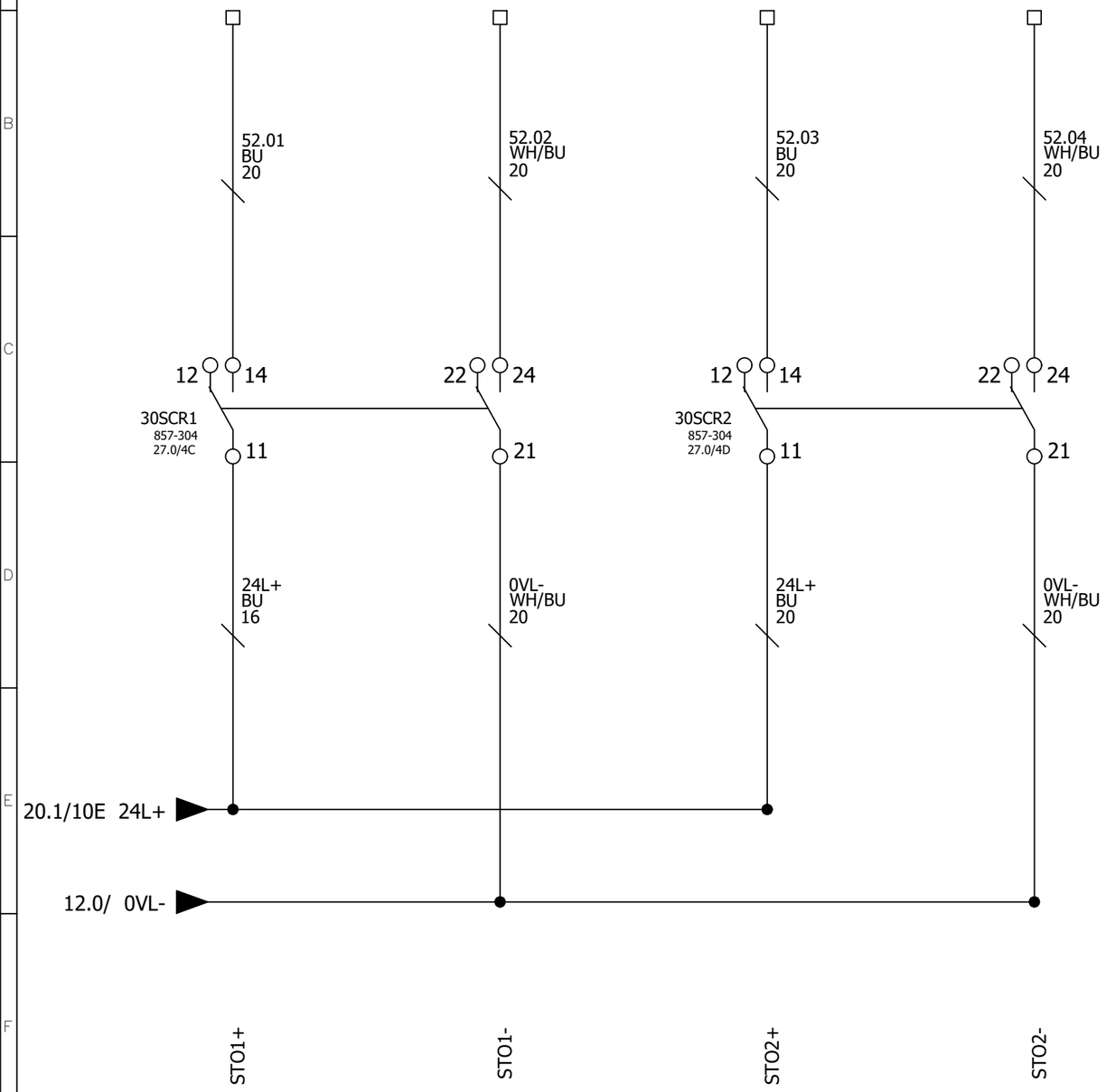
TOTAL PAGES: 175

SHEET 51.1 OF 56




## VFD's STO #1

CAROUSEL CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 5 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

CAROUSEL CONVEYORS AREA VFD's
SECTION 3 TOTAL OF 4 VFD's

## VFD's STO #2

VTD STATIC AREA DRIVE

VTD MOBILE EXTRACTOR CONVEYOR AREA VFD

BUFFER CONVEYORS AREA VFD's
SECTION 1 TOTAL OF 3 VFD's

BUFFER CONVEYORS AREA VFD's
SECTION 2 TOTAL OF 3 VFD's

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

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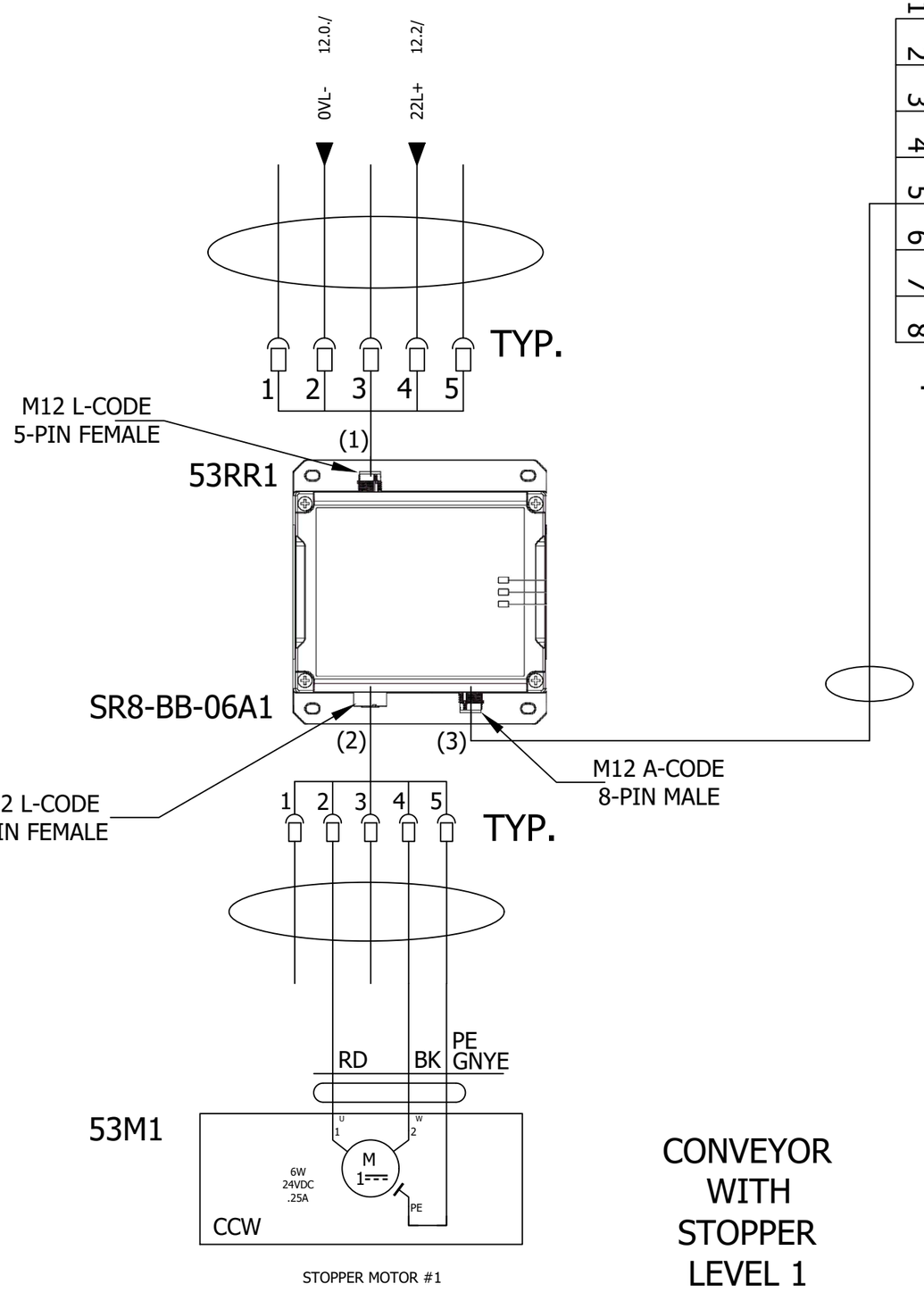
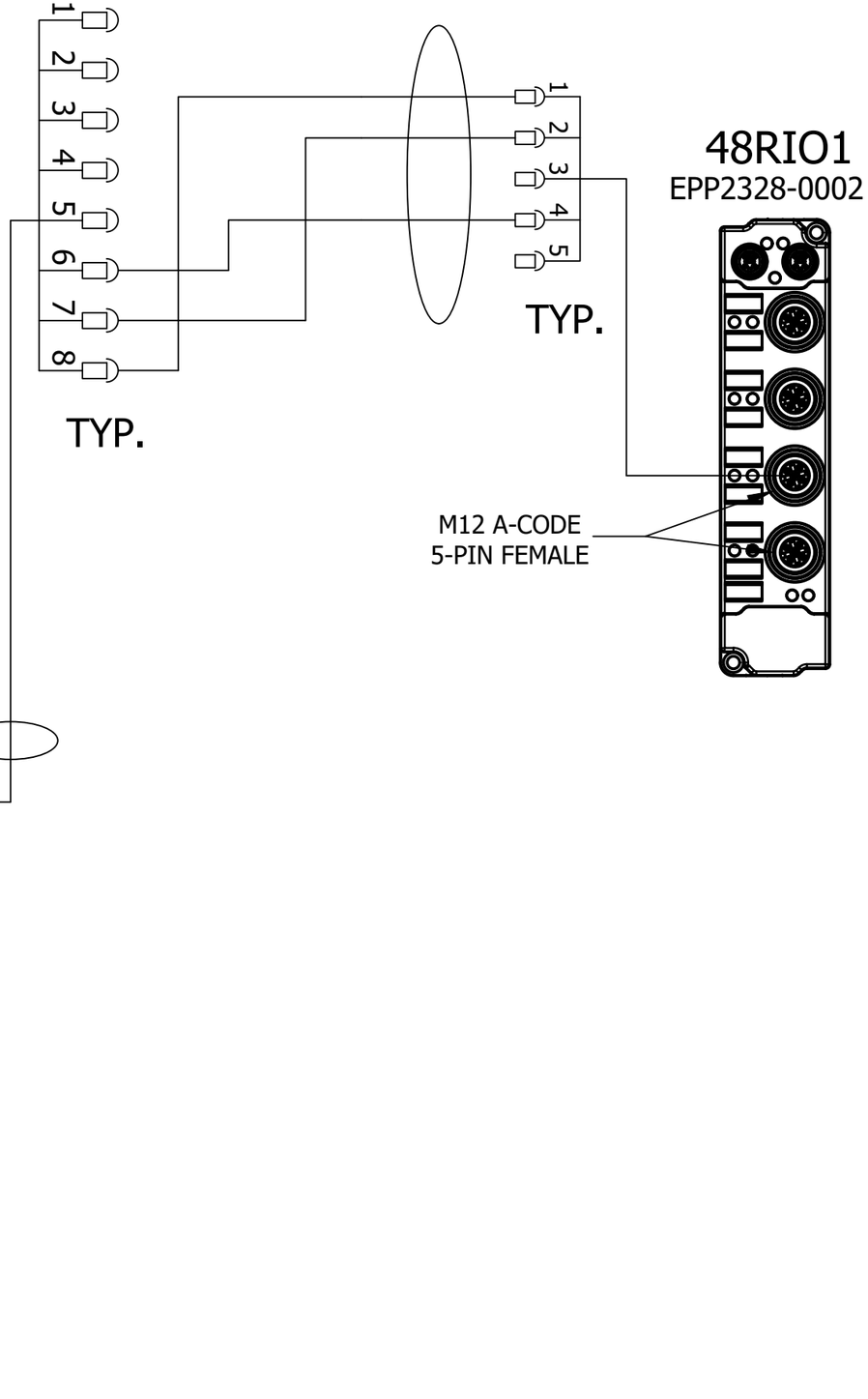
TOTAL PAGES: 175

SHEET 52.0 OF 56

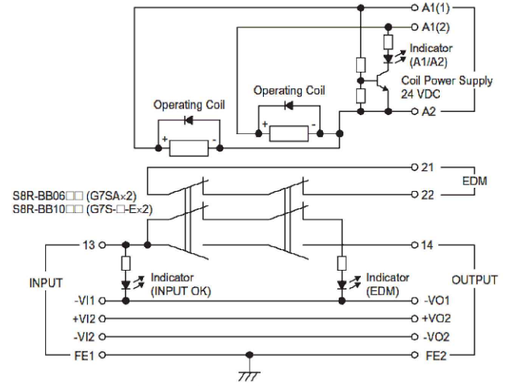



STOPPER MOTOR #1

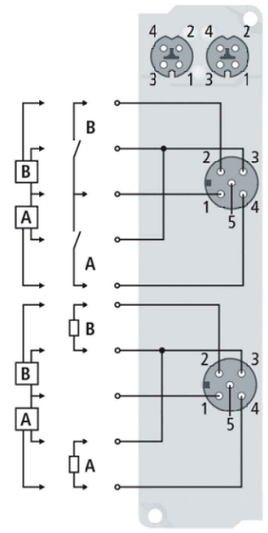



# CONVEYOR WITH STOPPER LEVEL 1

# SR8-BB-06A1




| No. | Connector Name | Shape | Pin Number/Name |
|---|---|---|---|
| (1) | Input Terminal M12-L (Plug) | | FE: FE1 (Functional Earth ⏚)<br>1: +VI2<br>2: -VI1<br>3: -VI2<br>4: 13 |
| (2) | Output Terminal M12-L (Socket) | | FE: FE2<br>1: +VO2<br>2: -VO1<br>3: -VO2<br>4: 14 |
| (3) | EDM/Coil Terminal M12-A (Plug) | | 1: 21<br>2: 22<br>3: N/A<br>4: N/A<br>5: N/A<br>6: A1 (1)<br>7: A1 (2)<br>8: A2 |




1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

# EPP2328-0002

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 1 STOPPER MOTORS

| REV | DESCRIPTION | DATE |
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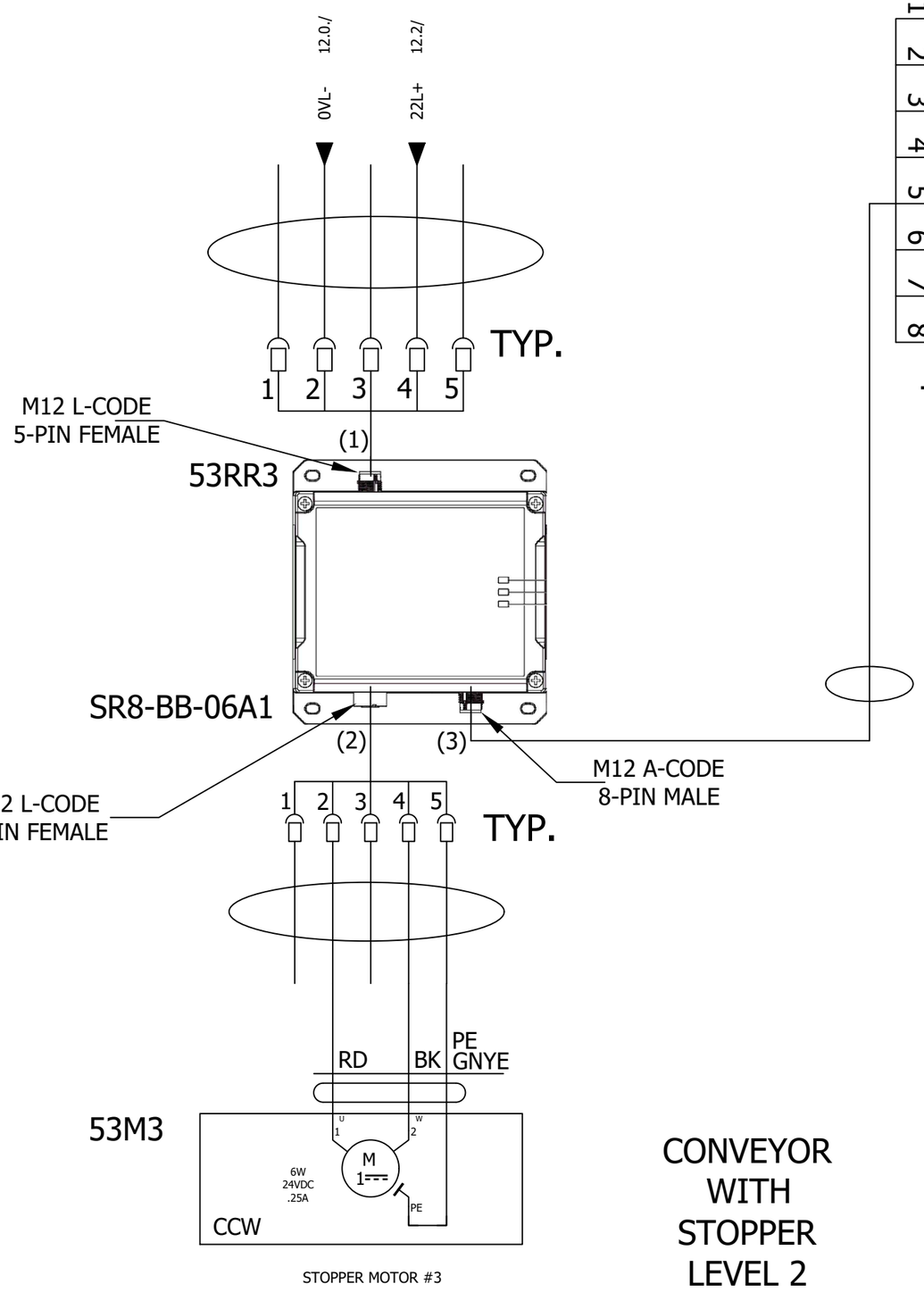
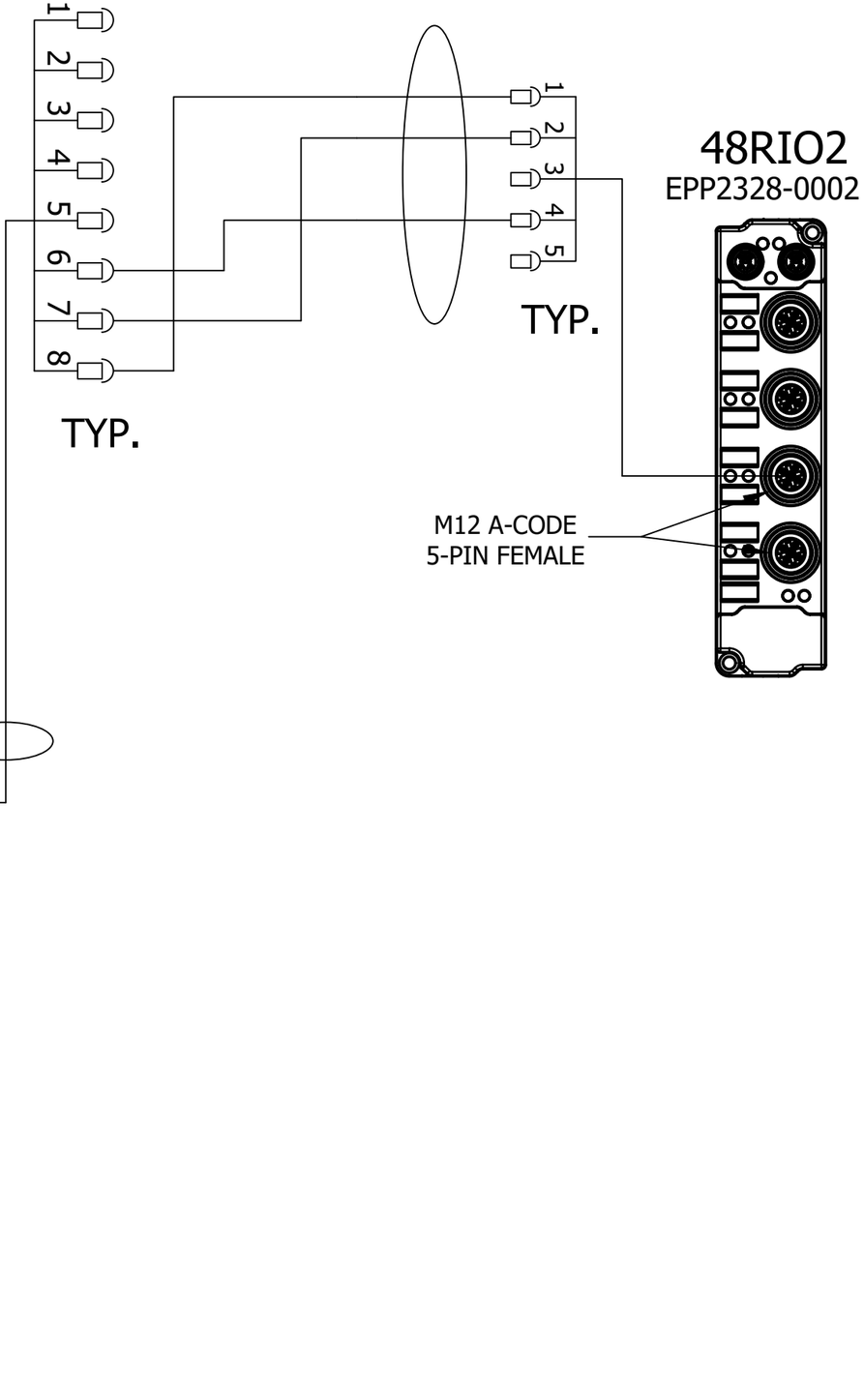
JOB # C40572

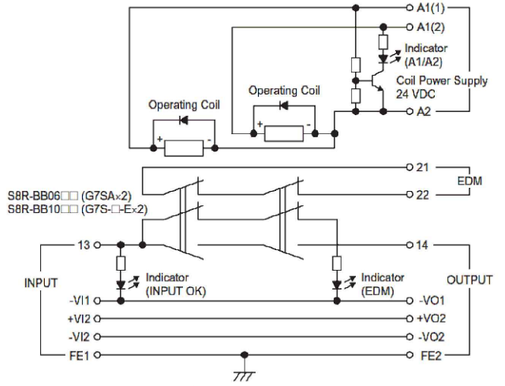
TOTAL PAGES: 175

SHEET 53.0 OF 56

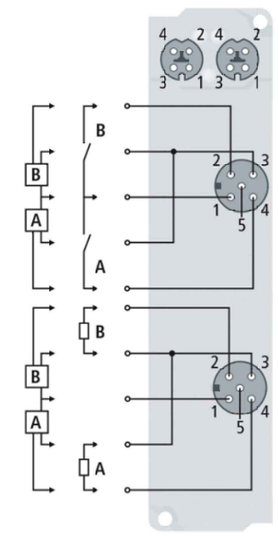



# SR8-BB-06A1




| No. | Connector Name | Shape | Pin Number/Name |
|---|---|---|---|
| (1) | Input Terminal M12-L (Plug) | | FE: FE1 (Functional Earth ⏚)<br>1: +VI2<br>2: -VI1<br>3: -VI2<br>4: 13 |
| (2) | Output Terminal M12-L (Socket) | | FE: FE2<br>1: +VO2<br>2: -VO1<br>3: -VO2<br>4: 14 |
| (3) | EDM/Coil Terminal M12-A (Plug) | | 1: 21<br>2: 22<br>3: N/A<br>4: N/A<br>5: N/A<br>6: A1 (1)<br>7: A1 (2)<br>8: A2 |




1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

# EPP2328-0002

# CONVEYOR WITH STOPPER LEVEL 2

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 2 STOPPER MOTORS

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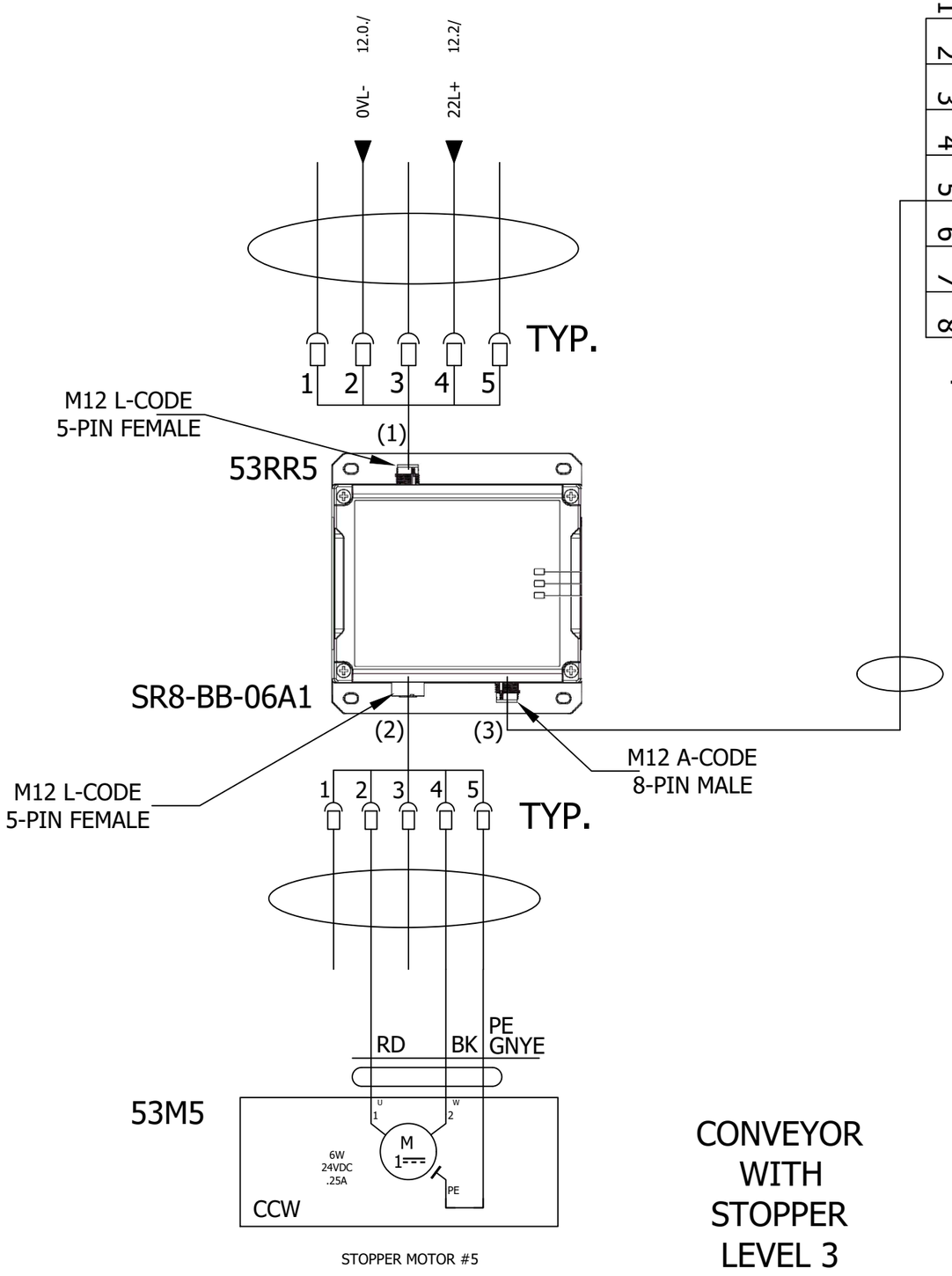
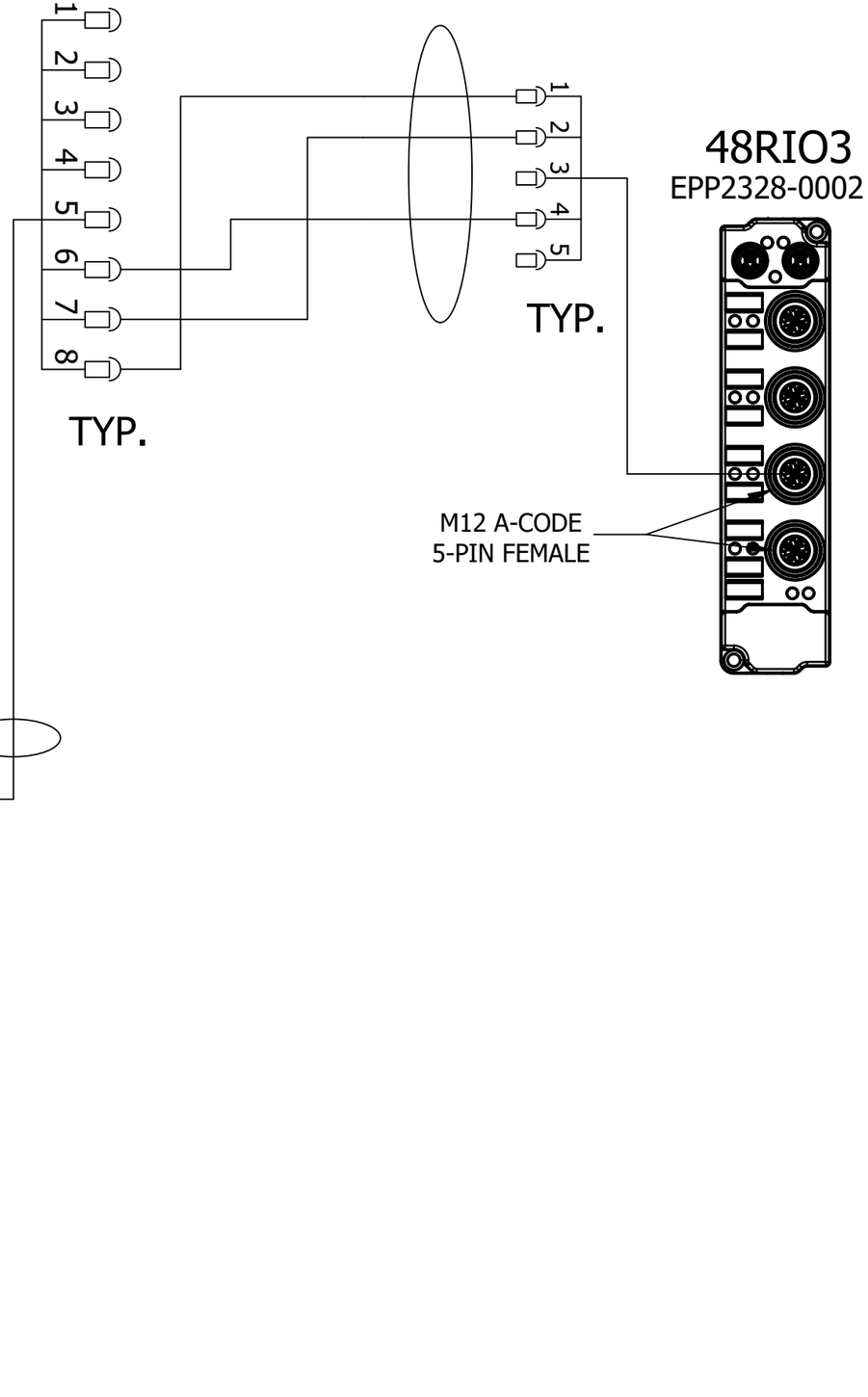
TOTAL PAGES: 175

SHEET 53.1 OF 56

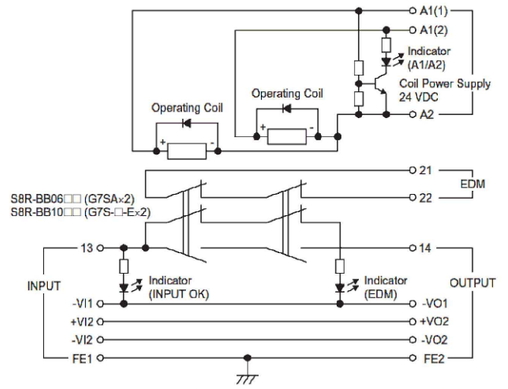



STOPPER MOTOR #5

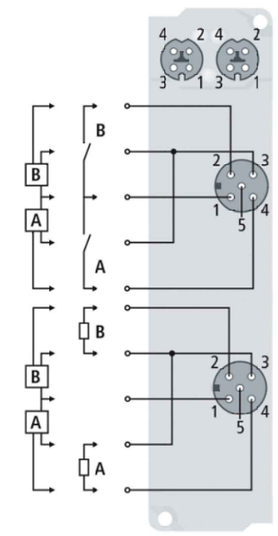



# CONVEYOR WITH STOPPER LEVEL 3

# SR8-BB-06A1




| No. | Connector Name | Shape | Pin Number/Name |
|---|---|---|---|
| (1) | Input Terminal M12-L (Plug) | | FE: FE1 (Functional Earth ⏚)<br>1: +VI2<br>2: -VI1<br>3: -VI2<br>4: 13 |
| (2) | Output Terminal M12-L (Socket) | | FE: FE2<br>1: +VO2<br>2: -VO1<br>3: -VO2<br>4: 14 |
| (3) | EDM/Coil Terminal M12-A (Plug) | | 1: 21<br>2: 22<br>3: N/A<br>4: N/A<br>5: N/A<br>6: A1 (1)<br>7: A1 (2)<br>8: A2 |




1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 3 STOPPER MOTORS

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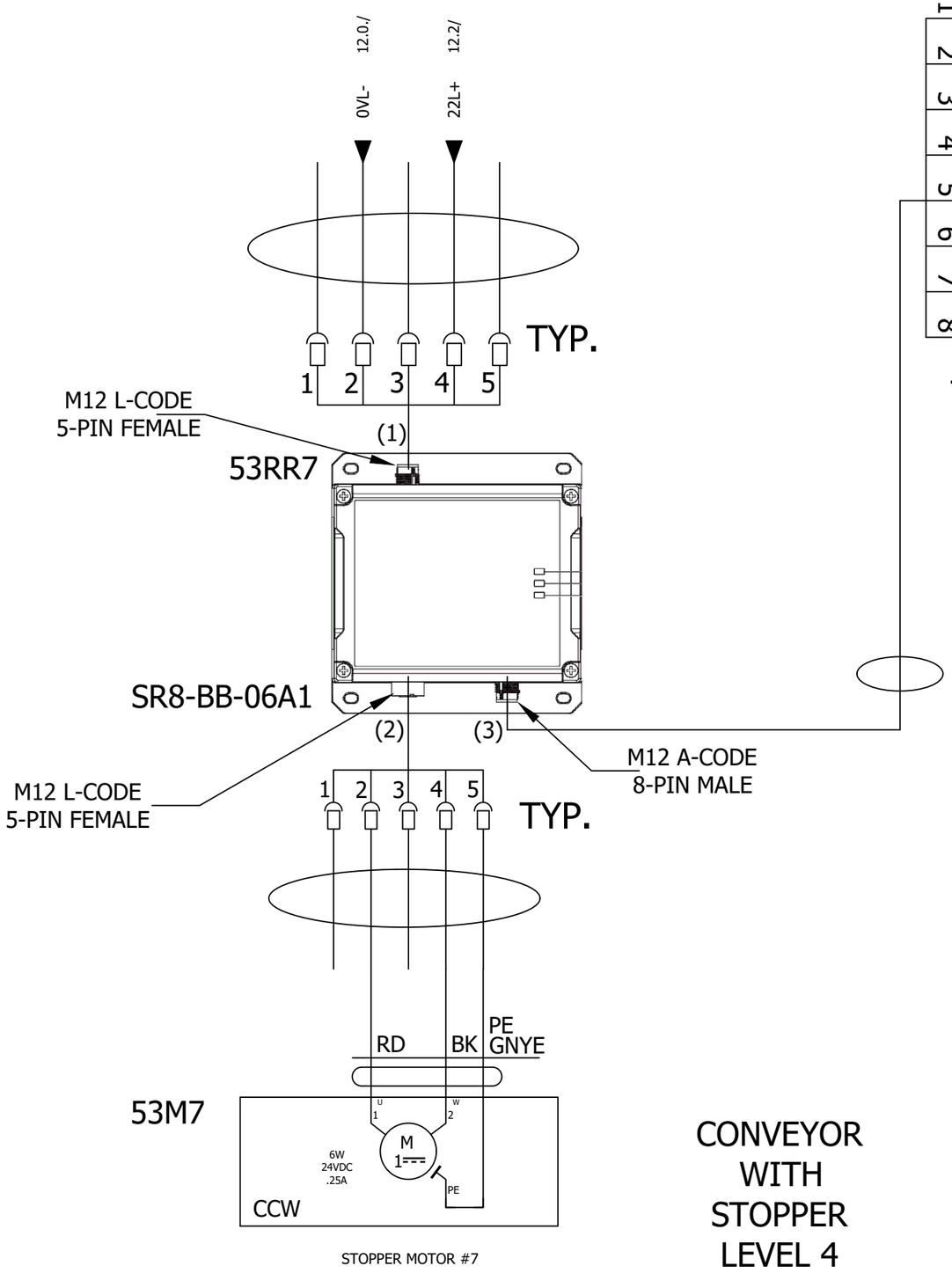
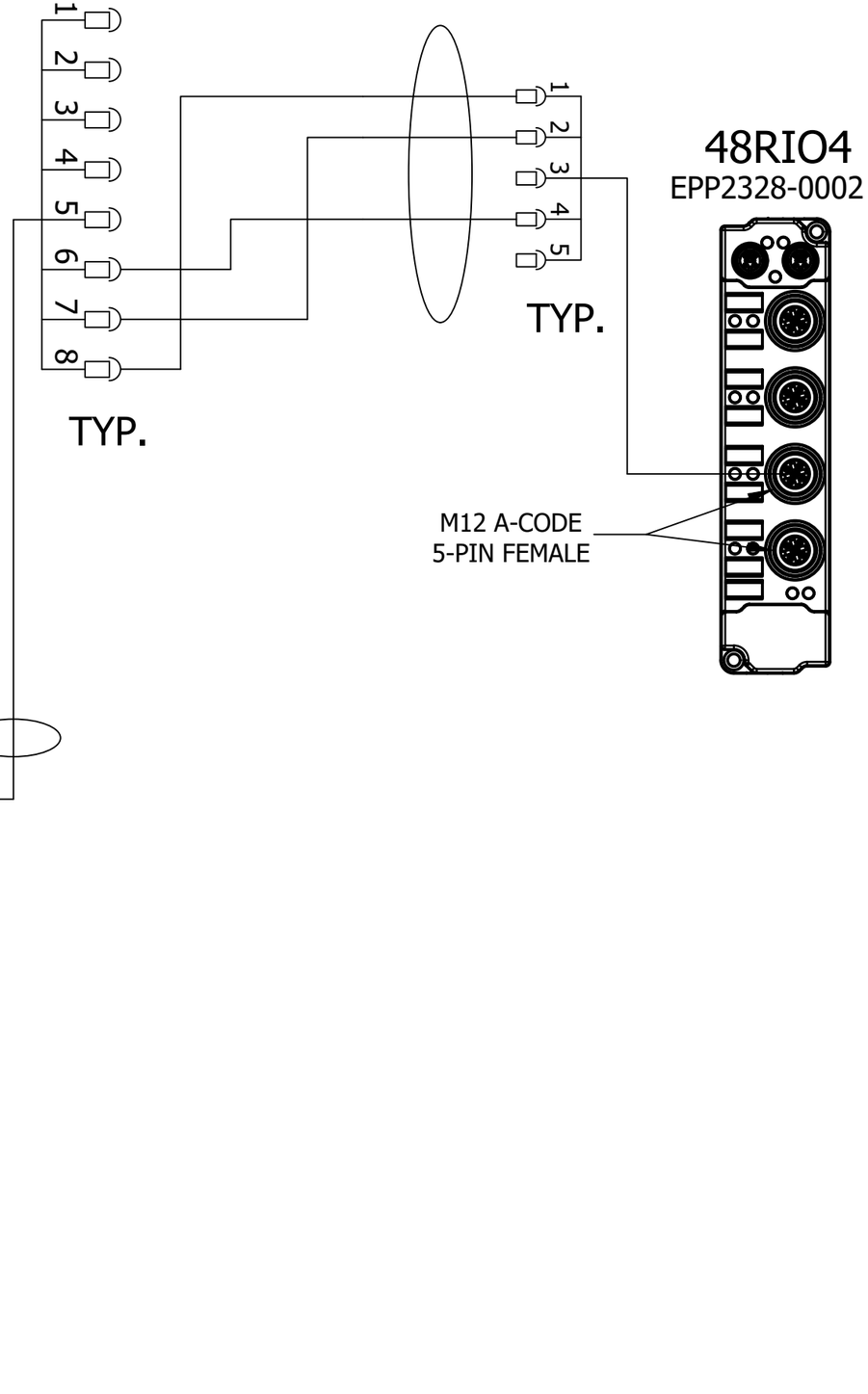
TOTAL PAGES: 175

SHEET 53.2 OF 56

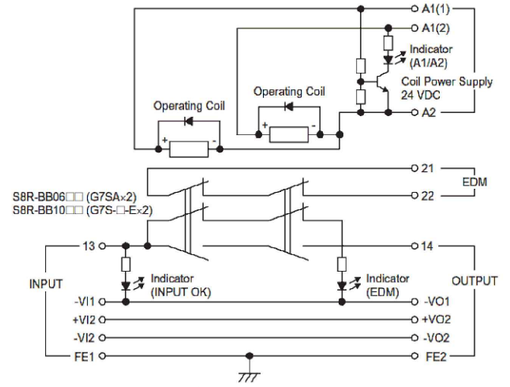



STOPPER MOTOR #7

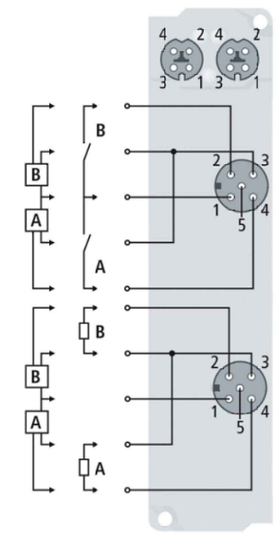



# CONVEYOR WITH STOPPER LEVEL 4

# SR8-BB-06A1




| No. | Connector Name | Shape | Pin Number/Name |
|---|---|---|---|
| (1) | Input Terminal M12-L (Plug) | | FE: FE1 (Functional Earth ⏚)<br>1: +VI2<br>2: -VI1<br>3: -VI2<br>4: 13 |
| (2) | Output Terminal M12-L (Socket) | | FE: FE2<br>1: +VO2<br>2: -VO1<br>3: -VO2<br>4: 14 |
| (3) | EDM/Coil Terminal M12-A (Plug) | | 1: 21<br>2: 22<br>3: N/A<br>4: N/A<br>5: N/A<br>6: A1 (1)<br>7: A1 (2)<br>8: A2 |




1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 4 STOPPER MOTORS

| REV | DESCRIPTION | DATE |
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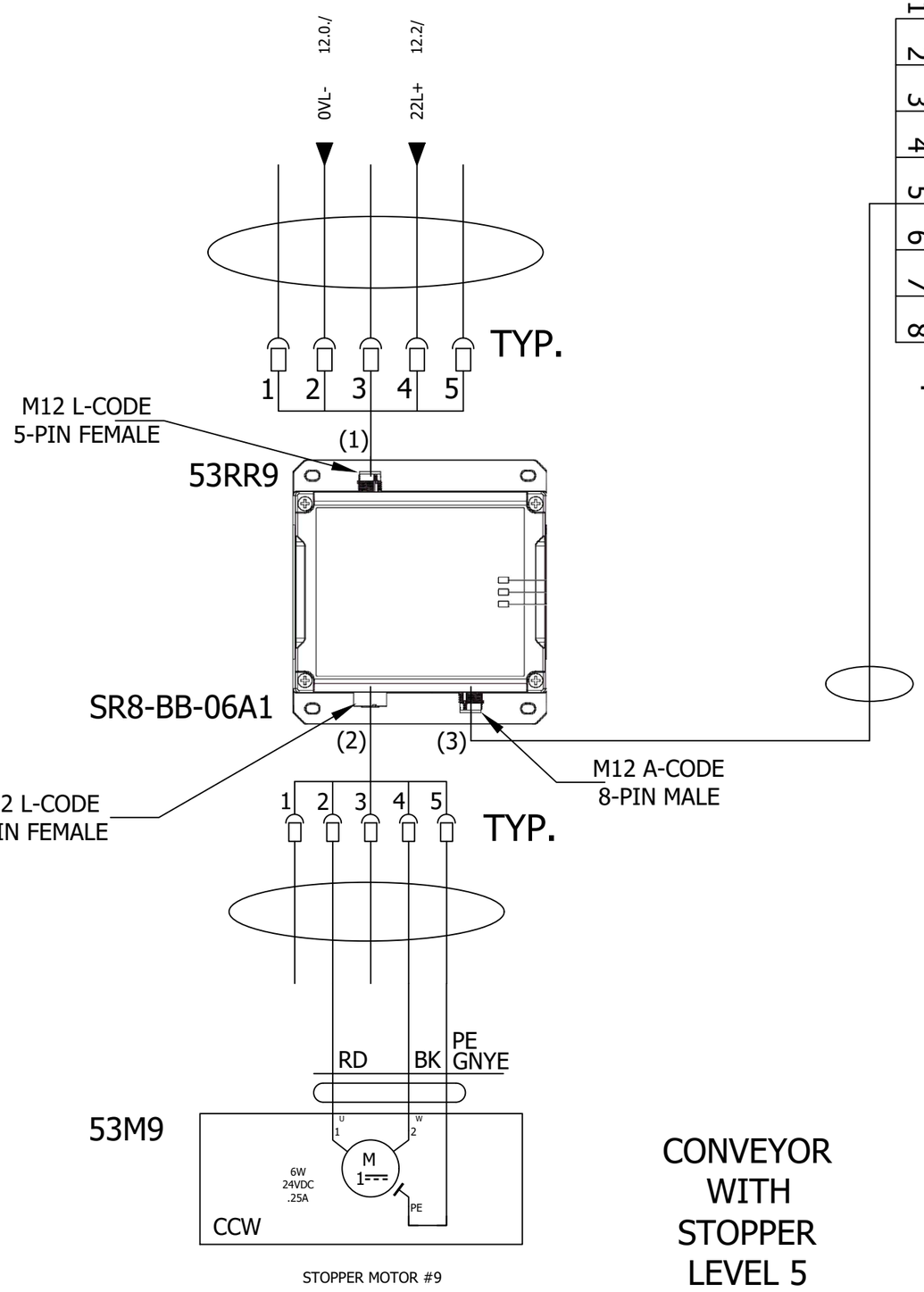
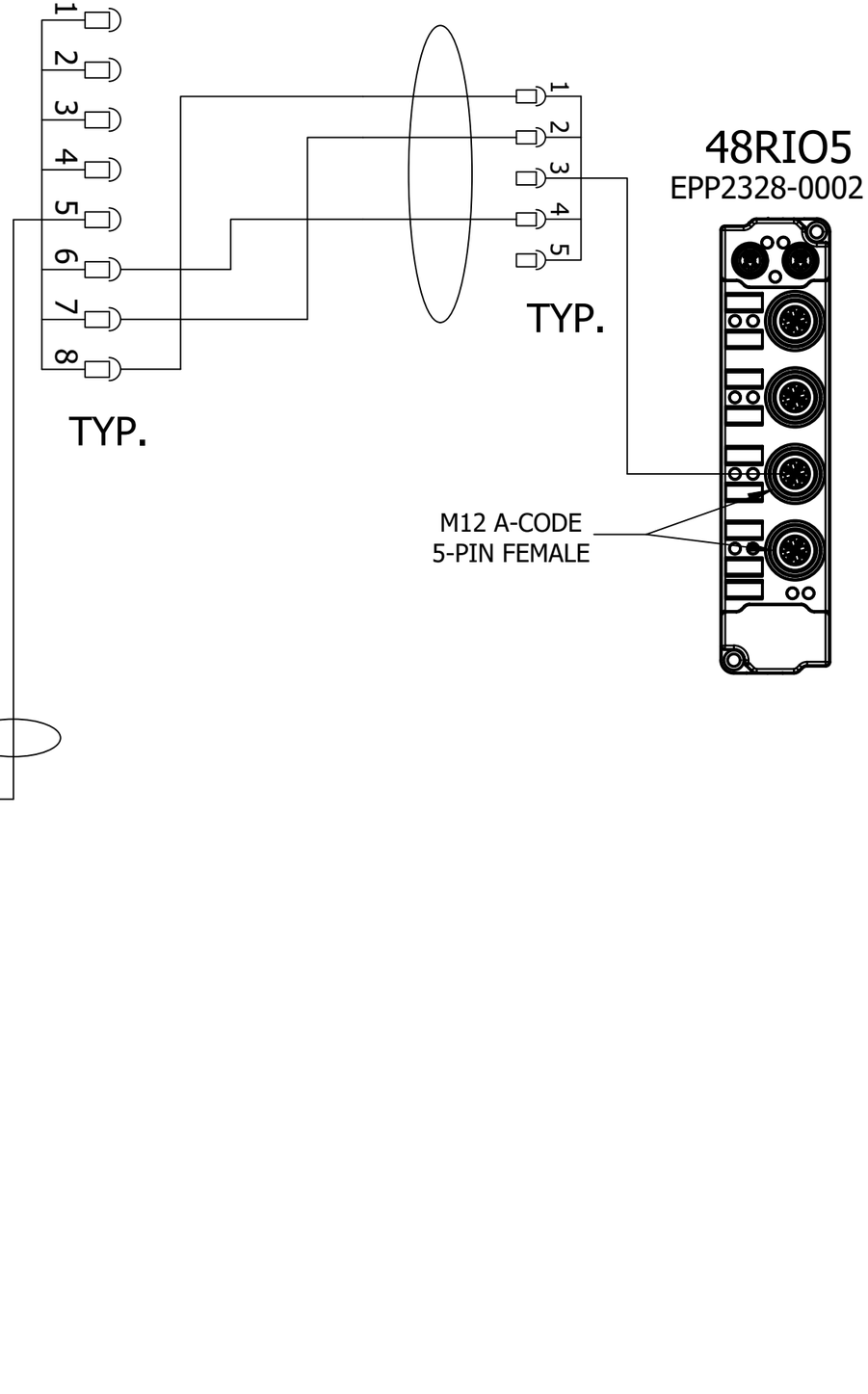
TOTAL PAGES: 175

SHEET 53.3 OF 56

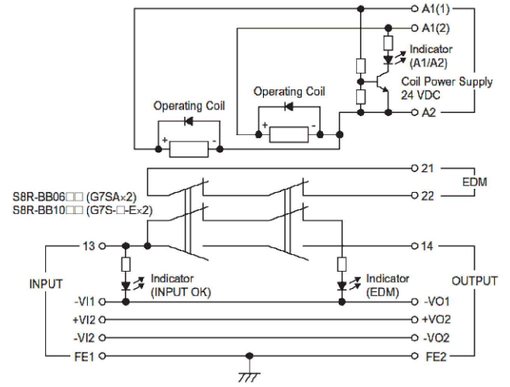



STOPPER MOTOR #9

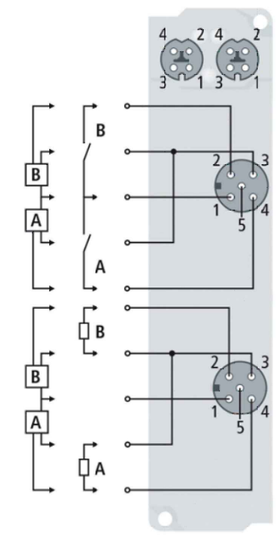



# CONVEYOR WITH STOPPER LEVEL 5

# SR8-BB-06A1




| No. | Connector Name | Shape | Pin Number/Name |
|---|---|---|---|
| (1) | Input Terminal M12-L (Plug) | | FE: FE1 (Functional Earth ⏚)<br>1: +VI2<br>2: -VI1<br>3: -VI2<br>4: 13 |
| (2) | Output Terminal M12-L (Socket) | | FE: FE2<br>1: +VO2<br>2: -VO1<br>3: -VO2<br>4: 14 |
| (3) | EDM/Coil Terminal M12-A (Plug) | | 1: 21<br>2: 22<br>3: N/A<br>4: N/A<br>5: N/A<br>6: A1 (1)<br>7: A1 (2)<br>8: A2 |




1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 5 STOPPER MOTORS

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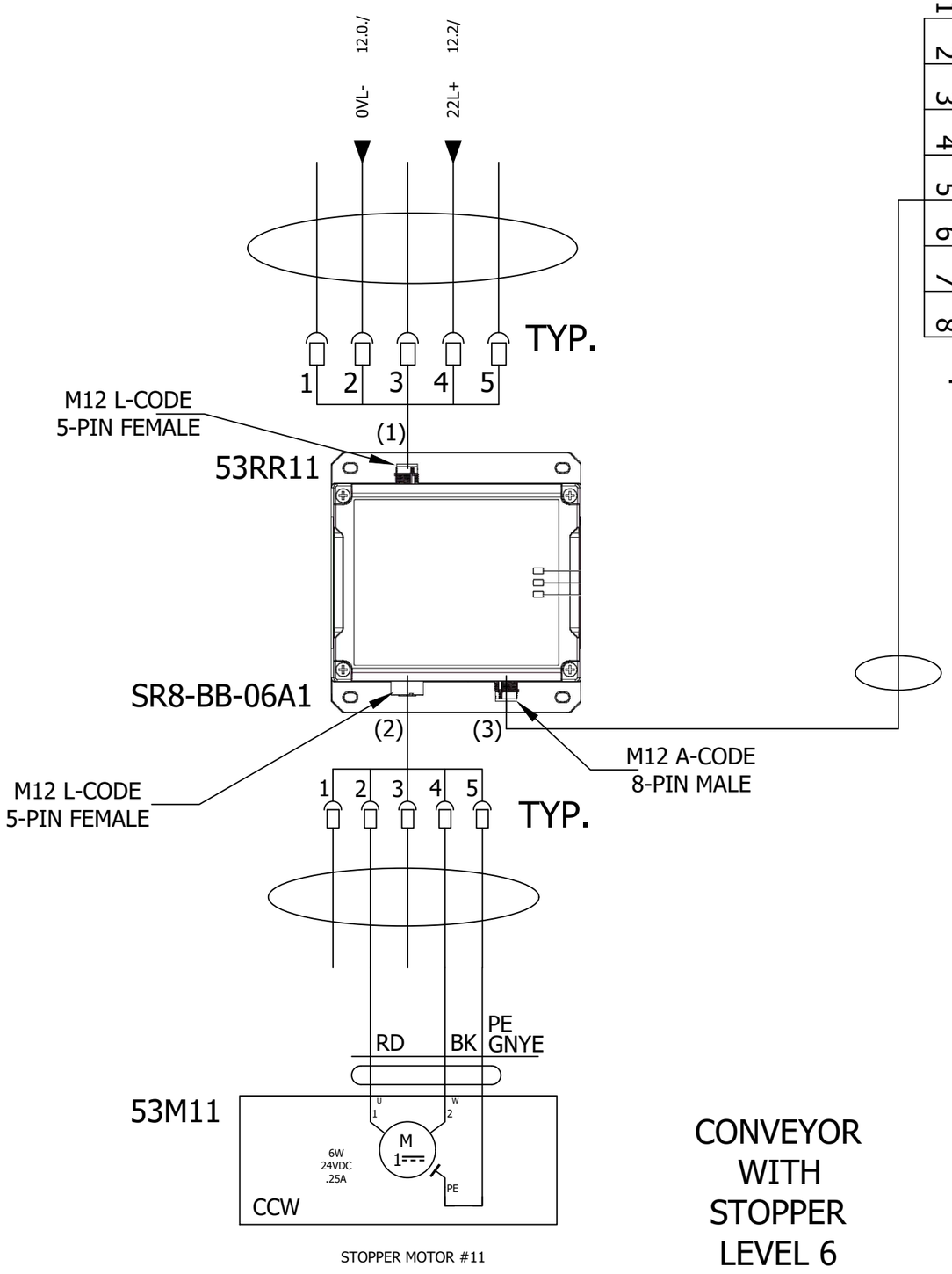
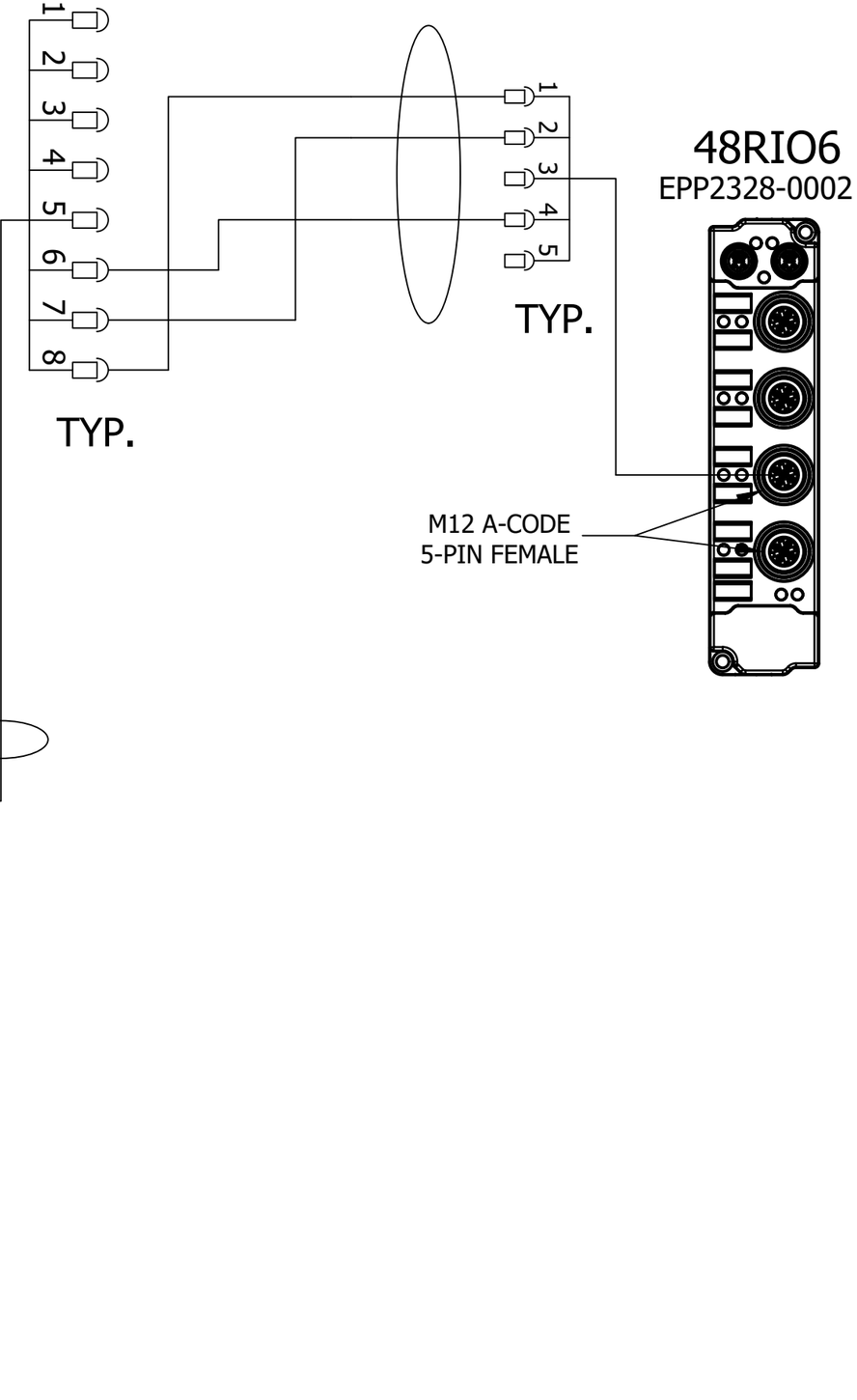
JOB # C40572

TOTAL PAGES: 175

SHEET 53.4 OF 56

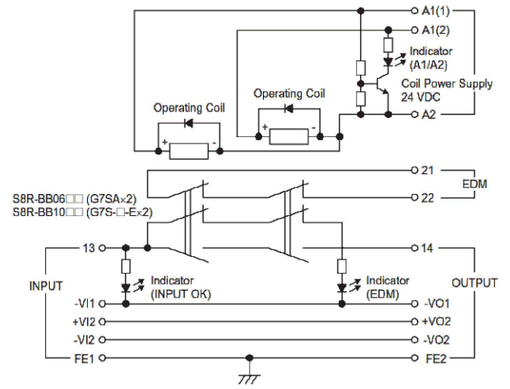



STOPPER MOTOR #11

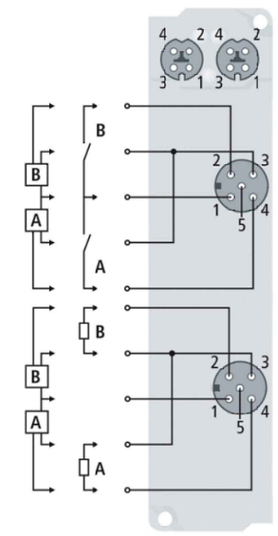



# CONVEYOR WITH STOPPER LEVEL 6

# SR8-BB-06A1




| No. | Connector Name | Shape | Pin Number/Name |
|---|---|---|---|
| (1) | Input Terminal M12-L (Plug) | | FE: FE1 (Functional Earth ⏚)<br>1: +VI2<br>2: -VI1<br>3: -VI2<br>4: 13 |
| (2) | Output Terminal M12-L (Socket) | | FE: FE2<br>1: +VO2<br>2: -VO1<br>3: -VO2<br>4: 14 |
| (3) | EDM/Coil Terminal M12-A (Plug) | | 1: 21<br>2: 22<br>3: N/A<br>4: N/A<br>5: N/A<br>6: A1 (1)<br>7: A1 (2)<br>8: A2 |




1 | Tx+ / $GND_S$
2 | Rx+ / $GND_P$
3 | Rx- / $U_P$
4 | Tx- / $U_S$

1 | +24 V $U_S$
2 | Input B
3 | $GND_S$
4 | Input A
5 | n.c.

1 | +24 V $U_P$
2 | Output B
3 | $GND_P$
4 | Output A
5 | n.c.

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

PAGE DESCRIPTION: LEVEL 6 STOPPER MOTORS

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TOTAL PAGES: 175

SHEET 53.5 OF 56

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SHEET 55.0 OF 56

| Component Tag(s) | Manufacturer / Vendor | Part Number | Description | Qty / Panel | Qty / 5 Panels | Drawing Sheet(s) |
|---|---|---|---|---|---|---|
| 7THM1 | FINDER | 7T.81.0.000.2301 | Compact, DIN rail-mounted panel thermostat, Temperature Setting Range: -20°C to +40°C (-4°F to +104°F), Contact Configuration: SPST-NO (Normally Open), Contact Rating: 10 A at 250 V AC | 1 | 5 | 7.0 |
| 7FAN1 | Hawa | 3156-0291-40-71-00 | Filter Fan, 460 V, UL, cUL, CE, CUT-OUT (mm) 291 x 291, NEMA Type 12, 427 CFM, RAL 7035 (Lt. Grey) | 2 | 10 | 7.0 |
| 7C1 | EATON | XTMC9A10TD | mini contactor, 8.80A, 24 Vdc, 1NO, 9A | 1 | 5 | 7.0 |
| 7CB1 | EATON | FAZ-D1/3-NA | miniature circuit breaker, current-limiting, 2A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC | 1 | 5 | 7.0 |
| 8BPS1 | Omron | S8BA-24D24D960SBF | Main body Uninterruptible Power Supply (UPS) / Control Unit Part 960w | 1 | 5 | 8.0, 8.2 |
| 8PS1, 8PS2, 8PS3 | Omron | S8VK-WB48024 | DIN Rail Power Supplies 3Phs 480V Pwr Sup 20A 24VDC | 2 | 10 | 8.0, 8.1, 8.3 |
| 8PS1BAT | OMRON | S8BA-S960L | Battery unit 7800 mAh | 1 | 5 | 8.0 |
| 8CB1, 8CB3, 8CB5 | EATON | FAZ-D1.5/3-NA | miniature circuit breaker, current-limiting, 1.5A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC | 1 | 5 | 8.0, 8.1, 8.3 |
| 8CB2, 8CB4, 8CB6, 8CB7 | EATON | FAZ-C20/1-NA-DC-SP | 489 Industrial miniature circuit breaker-supplementary protector,Single package,Medium levels of inrush current are expected,20A,10 kAIC,Single-pole,125 Vdc per pole,5-10X | 1 | 5 | 8.0, 8.1, 8.3 |
| LM1-16; LM100-107 | LUTZE | 716401 | Electronic load monitoring up to DC 10 A<br>Single-channel design, Adjustable current range: DC 1 A – 10 A<br>Adjustable characteristics, fast, medium, slow 1, -2, -3 | 24 | 120 | 9.0, 10.0 |
| 15CB1 | EATON | FAZ-D30/3-NA | miniature circuit breaker, current-limiting, 30A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC | 1 | 5 | 15.1 |
| 17CB1, 20CB1, 21CB1 | EATON | FAZ-D15/3-NA | miniature circuit breaker, current-limiting, 15A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC | 3 | 15 | 17.1, 20.1, 21.1 |
| 16CB1 | EATON | FAZ-D5/3-NA | miniature circuit breaker, current-limiting, 5A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC | 1 | 5 | 16.2 |
| 18CB1 | EATON | FAZ-D7/3-NA | miniature circuit breaker, current-limiting, 7A, 480Y / 277 VAC / 96 VDC, 3-pole, D curve, thermal magnetic, 10kA @ 480Y/277 VAC | 2 | 10 | 18.1 |
| 26CR1-26CR16; 27CR1-27CR16 | WAGO | 857-304 | Relay module; Nominal input voltage: 24 VDC; 1 changeover contact; Limiting continuous current: 6 A; Yellow status indicator | 32 | 160 | 26.0, 27.0, 50.0, 50.1, 51.0, 51.1, 52.0 |
| 50K50C | WERMA | 240.340.50 | Multi-Color (RGB) with Buzzer, UL | 1 | 5 | 50.0 |
| 50K50C | Werma | 960.693.05 | 5 Pole, Female Right Angle, 2M, 4A, MDC (A-coded) | 1 | 5 | 50.0 |
| 4PI1 | Panduit | VS2-AVT-3PB-16 | VeriSafe™ Testers VeriSafe 2.0 Absence of Voltage Tester. Includes Three-Phase Battery-powered Indicator Module, 16ft (4.8m) System Cable, and 10ft (3.0m) Sensor Leads | 1 | 5 | 4.0 |
| 5PI2; 5PI3 | EATON | M22-T-W-W | Press-to-Test unit, 22.5 mm, Momentary, Illuminated, Bezel: Silver, IP67, IP69K, NEMA 4X, 13, 12-30 Vac/dc, White button | 2 | 10 | 5.0 |

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PROJECT DESCRIPTION: SpaceX Project INBOUND VTD PANEL

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JOB # C40572

TOTAL PAGES: 175

SHEET 56.0 OF 56

| Component Tag(s) | Manufacturer / Vendor | Part Number | Description | Qty / Panel | Qty / 5 Panels | Drawing Sheet(s) |
|---|---|---|---|---|---|---|
| 8RS1 | Omron | S8VK-R20 | Redundancy Unit | 1 | 5 | 8.3 |
| 4PI1 | PANDUIT | VS2-AVT-3P-ISO | | 1 | 5 | 4.0 |
| 8RS1 | Omron | S8VK-R20 | Redundancy unit | 1 | 5 | 8.3 |
| 15VFD1 | SEW-EURODRIVE | MPX23A-DSI10A-MPI-0460-5E3-CA-C | 15 kW VFD | 1 | 5 | 15.0 |
| | SEW-EURODRIVE | DR2L132L4/BE20HR/TF/AK8W | 11 kW motor; drawing also shows FA77B gear unit | 1 | 5 | 15.2 |
| 15DE1 | SICK | DL50-P2228 | Distance sensor / SSI connection | 1 | 5 | 15.4 |
| 16VFD1; 17VFD1-17VFD4; 18VFD1-18VFD3; 20VFD1-20VFD5; 21VFD1-21VFD2 | SEW-EURODRIVE | MMF31S02-D11-5D3-SFDI0-C/IV/BW1/DSI10A-0025 | 0.75 kW VFD | 15 | 75 | 16.0, 17.0, 18.0, 20.0, 21.0 |
| | Siemens | 6ES71946KA000XA0 | ET200 Y cable | 8 | 40 | 16.7, 16.8, 17.11, 17.12, 18.6, 18.7, 47.3 |
| 16PE1-5; 17PE1-23 | SICK | GL6-P1112 | Sensor | 25 | 125 | 16.8, 17.12, 18.7 |
| 17M1-17M4; 18M1-18M3; 20M1-20M5; 21M1-21M2 | SEW-EURODRIVE | DRN80M4/BE1HR/ISU/TF | 0.75 kW motor | 14 | 70 | 17.3, 17.4, 18.2, 20.3, 20.4, 21.2 |
| | Beckhoff | CX5330-0225 | PLC/controller | 1 | 5 | 23.0 |
| 23.0-13 | Beckhoff | EK1122 | EtherCAT module | 2 | 10 | 23.0, 36.0 |
| 23.0-1, 23.0-2 | Beckhoff | EL1809 | 24 VDC input terminal | 2 | 10 | 23.0, 24.0, 25.0 |
| 23.0-3, 23.0-4 | Beckhoff | EL2809 | 24 VDC output terminal | 2 | 10 | 23.0, 26.0, 27.0 |
| 23.0-8 | Beckhoff | EL9410 | DC supply terminal for network interface cards | 1 | 5 | 23.0, 31.0 |
| 23.0-12 | Beckhoff | EL6633 | PROFINET interface terminal | 1 | 5 | 23.0, 35.0 |
| 23.0-7 | Beckhoff | EL2912 | Safety output terminal | 1 | 5 | 23.0, 30.0 |
| 23.0-5 | Beckhoff | EL6910 | Safety CPU controller | 1 | 5 | 23.0, 28.0 |
| 23.0-6 | Beckhoff | EL1918 | Safety input terminal | 1 | 5 | 23.0, 29.0 |
| | Beckhoff | EL1322 | PLC terminal | 1 | 5 | 23.0, 32.0, 33.0, 34.0 |
| | EBA Automation | CB310-S00 | KeTop component - 107780 | 1 | 5 | 39.3 |
| | EBA Automation | T155-M10-NT0-W10E | KeTop HMI - 140318 | 1 | 5 | 39.3 |
| | | SR-X300 | Barcode reader | 1 | 5 | 39.3 |
| 40RIO1 | TURCK | TBEN-S2-4IOL | I/O-Link master / remote I/O device | 1 | 5 | 40.0 |
| | KEYENCE | SR-300X | Device shown on 40.1 | 1 | 5 | 40.1 |
| | Beckhoff | ZK1090-3191-0100 | Cable / connection item | 1 | 5 | 43.1 |
| | Beckhoff | ZK2020-3200-0100 | Cable / connection item | 1 | 5 | 43.1 |
| 43LC1 | SICK | C4C-EV13530B10000 | Light curtain receiver; drawing Part # 1107661 | 1 | 5 | 43.4 |
| 43LC2 | SICK | C4C-SV13530B10000 | Light curtain sender; drawing Part # 1107660 | 1 | 5 | 43.4 |

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JOB # C40572

TOTAL PAGES: 175

SHEET 56.1 OF 56

| Component Tag(s) | Manufacturer / Vendor | Part Number | Description | Qty / Panel | Qty / 5 Panels | Drawing Sheet(s) |
|---|---|---|---|---|---|---|
| 44SD1 | | MGB2-L1HE-MLI-U-DA-R-165567 | Guard lock device | 1 | 5 | 44.0 |
| | | AC-MC-SB-M-A-157024 | Guard lock associated component | 1 | 5 | 44.0 |
| | | MBM-EC-S1-MLI-3B-173401 | Guard lock associated component | 1 | 5 | 44.0 |
| 47RIO1; 48RIO1-48RIO6 | Beckhoff | EPP2328-0002 | EtherCAT P remote I/O module | 7 | 35 | 47.0, 47.1, 47.3, 48.0, 48.1, 48.2, 48.3, 48.4, 48.5, 48.6, 53.0, 53.1, 53.2, 53.3, 53.4, 53.5 |
| 45RIO1-45RIO3; 46RIO2-46RIO7 | Beckhoff | EPP1008-0022 | EtherCAT P remote I/O module | 9 | 45 | 45.1, 45.2, 45.3, 45.4, 46.2, 46.3, 46.4, 46.5, 46.6, 46.7 |
| 45RIO4 | Beckhoff | EPP5001-0002 | EtherCAT P remote I/O module | 1 | 5 | 45.1, 45.5 |
| 47RIO2 | Beckhoff | EPP1809-0022 | EtherCAT P remote I/O module | 1 | 5 | 47.0, 47.2, 47.4 |
| | Beckhoff | ZS2000-4722 | Connection/accessory shown with remote I/O | 7 | 35 | 47.4, 48.1, 48.2, 48.3, 48.4, 48.5, 48.6 |
| 53RR1, 53RR3, 53RR5, 53RR7, 53RR9, 53RR11 | Omron | SR8-BB-06A1 | Stopper motor interface/device | 6 | 30 | 53.0, 53.1, 53.2, 53.3, 53.4, 53.5 |

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